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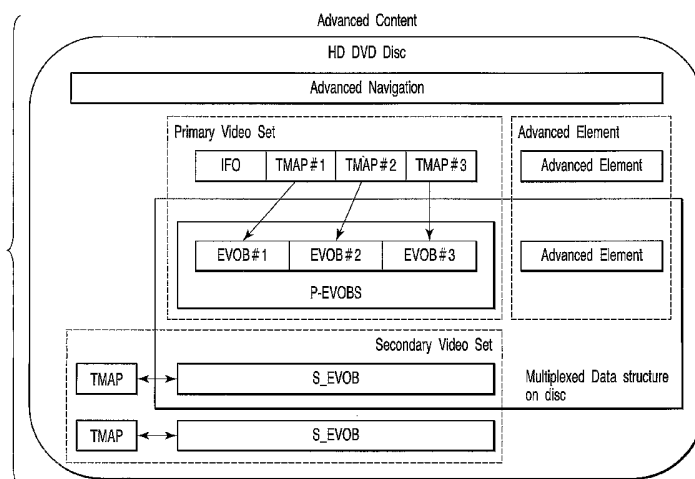
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(54) Title: INFORMATION STORAGE MEDIUM, INFORMATION PLAYBACK METHOD, INFORMATION DECODE
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(57) Abstract: In content recorded in a playback only information storage medium, a video object for playing back a video in a method different from the conventional playback sequence and its playback order are controlled. To achieve the control, the data area (12) stores a primary object set (P-EVOBS) which is a group of at least one primary object (EVOB#1, #2, ...) for managing a relationship between a playback time and a recording position in accordance with at least one time map (TMAP#1, #2, ...; or TMAPIT), and includes a main picture stream, and a secondary object (S-EVOB) in which a relationship between the playback time and the recording position is managed in accordance with an individual time map (TMAP), and includes another picture stream to be played back simultaneously with the main picture stream.

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D E S C R I P T I O N

5 INFORMATION STORAGE MEDIUM, INFORMATION
PLAYBACK METHOD, INFORMATION DECODE
METHOD, AND INFORMATION PLAYBACK APPARATUS

Technical Field

10 The present invention relates to an information
storage medium such as an optical disc or the like, a
method of playing back this information storage medium,
a method of decoding information obtained from this
information storage medium, a communication line, or
the like, and an information playback apparatus for
15 playing back this information storage medium.

Background Art

In recent years, DVD-Video discs having high image
quality and advanced functions, and video players that
play back such discs have prevailed, and the range of
20 choice for peripheral devices and the like used to play
back such multi-channel audio has broadened. An
environment in which users can personally implement a
home theater and allows users to freely enjoy movies,
animations, and the like with high image quality and
25 high sound quality at home has become available.
As described in Jpn. Pat. Appln. KOKAI Publication
No. 10-50036, a playback apparatus which can
superimpose various menus by changing, e.g., text
colors for playback video pictures from a disc has been

proposed.

However, in recent years, along with the improvement of image compression techniques, a demand has arisen for realization of higher image quality from both users and content providers. In addition to realization of higher image quality, the content providers require an environment that can provide more attractive content to users by upgrading and expanding the content (e.g., more colorful menus, improvement of interactiveness, and the like) in content such as menu screens, bonus video pictures, and the like as well as a title itself. Furthermore, some users require to freely enjoy content by playing back still picture data sensed by the user, subtitle text data acquired via Internet connection, and the like by freely designating their playback positions, playback regions, or playback times.

As described above, an environment that can provide more attractive content to users by upgrading and expanding the content (e.g., more colorful menus, improvement of interactiveness, and the like) in content such as menu screens, bonus video pictures, and the like in addition to realization of higher image quality of a title itself is required.

On the other hand, in order to produce content with more colorful menus and high interactiveness, a technique different from conventional content

production is required. Hence, much time must be spent to master such a technique. For this reason, a content providing environment that allows the conventional production technique and can realize high image quality of a title itself (although functions are a little more than the conventional technique) is required at the same time.

In a conventional DVD-Video disc (ROM disc), a video object to be played back (to be referred to as a VOB or EVOB) and/or its playback order is determined on the basis of program chain (PGC) information which is set, determined in advance, and recorded on a disc by content producers. However, a video object to be played back and its playback order are determined in advance upon its production and cannot be changed after the disc is produced. That is, when the above video object to be played back and its playback order are to be changed, the content producers need to make new management information of the DVD-Video disc and record a changed PGC on a new disc, and users need to buy the DVD-Video disc with the changed PGC recorded thereon.

The present invention has been made in consideration of the above situation, and has as one of its subjects to implement video objects to be played back using playback control information implemented by a markup language or the like and control of its playback order with respect to content recorded on a

conventional DVD-video disc.

In other words, an object of the present invention is to provide an environment to implement video objects to be played back by a method different from an existing playback sequence and control of its playback order with respect to content recorded on a read-only information storage medium such as a DVD-video disc or the like.

Disclosure of Invention

10 An information storage medium (1) according to an embodiment of the present invention has a data area (12) in which a video data recording area (20) including a management area (30) for recording management information (HDVMG) and an object area (40, 15 50) for recording object (HDVTS, AHDVTS) managed by the management information, and an advanced content recording area (21) including information (21A - 21E) different from recording content (30 - 50) in the video data recording area (20) are stored, and a file 20 information area (11) in which file information (FIG. 2) corresponding to the recording content in the data area (12) is stored. In this information storage medium, the data area (12) is configured to store a primary object set (P-EVOBS) which is a group of at 25 least one of primary objects (EVOB#1, #2, and the like) for managing a relationship between a playback time (TM_DIFF or the like) and a recording position

(TM_EN_ADR or the like) in accordance with at least one of time maps (TMAP#1, #2, and the like; corresponding to TMAPIT), and includes a main picture stream, and a secondary object (S-EVOB) in which the relationship between the playback time (TM_DIFF) and the recording position (TM_EN_ADR) is managed in accordance with an individual time map (TMAP), and includes another picture stream to be played back simultaneously with the main picture stream.

By practicing this invention, an information storage medium and its playback apparatus, which can implement colorful expressions and can form attractive content, can be provided.

Brief Description of Drawings

FIG. 1 shows an example of the data structure of recording information on disc-shaped information storage medium (optical disc, etc.) 1 according to an embodiment of the present invention;

FIG. 2 is a view for explaining an example of a file system used to manage content recorded on the disc-shaped information storage medium according to the embodiment of the present invention;

FIG. 3 shows an example of the data structure of HD video manager information (HDVMDGI) recorded on an HD video manager (HDVMDG) recording area;

FIG. 4 shows an example of the data structure of an HD video manager information management table

(HDVMGI_MAT) included in the HD video manager information (HDVMGI) and the recording content of category information (HDVMG_CAT) stored in this management table;

5 FIG. 5 shows an example of the data structure of a title search pointer table (TT_SRPT) recorded in the HD video manager information (HDVMGI);

 FIG. 6 shows an example of the data structure of an HD video manager menu PGC unit table
10 (HDVMGM_PGC_UT) recorded in the HD video manager information (HDVMGI);

 FIG. 7 shows an example of the data structure of each HD video manager menu language unit (HDVMGM_LU#n);

 FIG. 8 shows an example of the recording content
15 of an HDVMGM_PGC category (HDVMGM_PGC_CAT);

 FIG. 9 shows an example of the data structure of a parental management information table (PTL_MAIT) recorded in the HD video manager information (HDVMGI);

 FIG. 10 shows an example of the data structure of
20 each parental management information (PTL_MAI#n);

 FIG. 11 shows an example of the data structure of an HD video title set attribute information table (HDVTS_ATTRT) recorded in the HD video manager information (HDVMGI);

25 FIG. 12 shows an example of the data structure of a text data manager (TXTDT_MG) recorded in the HD video manager information (HDVMGI);

FIG. 13 shows an example of the data structure of each text data language unit (TXTDT_LU#n);

FIG. 14 shows an example of the data structure of text data (TXTDT);

5 FIG. 15 shows an example of the data structure of an HD video manager menu cell address table (HDMVGM_C_ADT) recorded in the HD video manager information (HDVMGI);

10 FIG. 16 shows an example of the data structure of an HD video manager menu video object unit address map (HDVMGM_VOBU_ADMAP) recorded in the HD video manager information (HDVMGI);

15 FIG. 17 shows an example of the data structure of an HD menu audio object set information table (HDMENU_AOBSIT) recorded in the HD video manager information (HDVMGI);

FIG. 18 shows an example of the data structure of a menu video object area (HDVMGM_VOBS) recorded in the HD video manager (HDVMG) area;

20 FIG. 19 shows an example of the data structure of a menu audio object area (HDMENU_AOBS) recorded in the HD video manager (HDVMG) area;

25 FIG. 20 shows an example of the data structure of HD video title set information (HDVTSI) recorded on each HD video title set (HDVTS#n) recording area;

FIG. 21 shows an example of the data structure of an HD video title set information management table

(HDVTSI_MAT) recorded in the HD video title set
information (HDVTSI);

FIG. 22 shows an example of the data structure of
an HD video title set part-of-title search pointer
5 table (HDVTS_PTT_SRPT) recorded in the HD video title
set information (HDVTSI);

FIG. 23 shows an example of the data structure of
an HD video title set program chain information table
(HDVTS_PGCIT) recorded in the HD video title set
10 information (HDVTSI);

FIG. 24 shows an example of the recording content
of an HDVTS_PGC category (HDVTS_PGC_CAT);

FIG. 25 shows an example of the data structure of
an HD video title set menu PGC unit table
15 (HDVTSM_PGC_UT) recorded in the HD video title set
information (HDVTSI);

FIG. 26 shows an example of the data structure of
each HD video title set menu language unit
(HDVTSM_LU#n);

20 FIG. 27 shows an example of the recording content
of an HDVTSM_PGC category (HDVTSM_PGC_CAT);

FIG. 28 shows an example of the data structure of
an HD video title set time map table (HDVTS_TMAPT)
recorded in the HD video title set information
25 (HDVTSI);

FIG. 29 shows an example of the data structure of
an HD video title set menu cell address table

(HDVTSM_C_ADT) recorded in HD video title set information (HDVTSI);

FIG. 30 shows an example of the data structure of an HD video title set menu video object unit address map (HDVTSM_VOBU_ADMAP) recorded in HD video title set information (HDVTSI);

FIG. 31 shows an example of the data structure of an HD video title set cell address table (HDVTS_C_ADT) recorded in HD video title set information (HDVTSI);

FIG. 32 shows an example of the data structure of an HD video title set video object unit address map (HDVTS_VOBU_ADMAP) recorded in HD video title set information (HDVTSI);

FIG. 33 shows an example of the data structure of program chain general information (PGC_GI) included in program chain information (PGCI: e.g., corresponding to one of HDVTS_PGCI in FIG. 23), and the recording content of a PGC graphics unit stream control table (PGC_GUST_CTLT) and resume/audio object category (RSM&AOB_CAT) stored in this PGCI;

FIG. 34 shows an example of the data structure of a program chain command table (PGC_CMDT) included in the program chain information (PGCI);

FIG. 35 shows an example of the content of program chain command table information (PGC_CMDTI) and each resume command (RSM_CMD) included in the program chain command table (PGC_CMDT);

FIG. 36 shows an example of the data structure of a program chain program map (PGC_PGMAP) and that of a cell position information table (C_POSIT) included in the program chain information (PGCI);

5 FIG. 37 shows an example of the data structure of a cell playback information table (C_PBIT) included in the program chain information (PGCI);

10 FIG. 38 is a block diagram showing an example of the internal structure of a playback apparatus for the disc-shaped information storage medium (optical disc, etc.) according to the embodiment of the present invention;

15 FIG. 39 is a block diagram for explaining an example of the arrangement of each decoder in the apparatus shown in FIG. 38;

FIG. 40 is a view for explaining the concept of imaginary video access unit IVAU;

20 FIG. 41 is a view for explaining a practical example of system parameters used in the embodiment of the present invention;

FIG. 42 shows an example of a list of commands used in the embodiment of the present invention;

25 FIG. 43 shows practical examples in respective fields of the commands used in the embodiment of the present invention;

FIG. 44 shows an example of allocation of graphics units GU in video objects;

FIG. 45 shows an example of the data structure in each graphics unit;

FIG. 46 shows an example of header information content and general information content in each graphics unit;

FIG. 47 is a view for explaining image examples of mask data and graphics data in each graphics unit;

FIG. 48 is a view showing an example of video composition including mask patterns;

FIG. 49 is a view for explaining an example of button position information in graphics unit GU;

FIG. 50 is a view for explaining an example of the recording content of an advanced content recording area of the information content recorded on disc-shaped information storage medium (optical disc, etc.) 1 according to another embodiment of the present invention;

FIG. 51 is a view for explaining an example of the recording content of an advanced HD video title set (AHDVTS) recording area of the information content recorded on disc-shaped information storage medium (optical disc, etc.) 1 according to still another embodiment of the present invention;

FIG. 52 shows an example of the data structure of advanced HD video title set information (AHDVTSI) recorded on the advanced HD video title set recording area;

FIG. 53 shows an example of the data structure of an advanced HD video title set information management table (AHDVTSI_MAT) recorded in the advanced HD video title set information (AHDVTSI), and the recording
5 content of category information (AHDVTS_CAT) stored in this management table;

FIG. 54 shows an example of the data structure of an advanced HD video title set part-of-title search pointer table (AHDVTS_PTT_SRPT) recorded in the
10 advanced HD video title set information (AHDVTSI);

FIG. 55 shows an example of the data structure of an advanced HD video title set program chain information table (AHDVTS_PGCIT) recorded in the advanced HD video title set information (AHDVTSI);

15 FIG. 56 shows an example of the data structure of program chain general information (PGC_GI) included in program chain information (PGCI: e.g., corresponding to AHDVTS_PGCI in FIG. 55);

FIG. 57 shows an example of the data structure of
20 an advanced HD video title set cell address table (AHDVTS_C_ADT) recorded in the advanced HD video title set information (AHDVTSI);

FIG. 58 shows an example of the data structure of a time map information table (TMAPIT) recorded in the
25 advanced HD video title set information (AHDVTSI);

FIG. 59 shows an example of the data structure of each time map information (TMAPI) included in the time

map information table (TMAPIT), and the recording content of time map generation information (TMAP_GI) stored in this time map information;

FIG. 60 shows an example of the data structure of a time entry table (TM_ENT) included in the time map information (TMAPI) and the recording content of the number of time entries (TM_EN_Ns) and a time entry (TM_EN) stored in this time entry table;

FIG. 61 shows an example of the recording content of a video object unit entry (VOBU_ENT), those of an interleaved unit address entry (ILVU_ADR_ENT), and those of an entry video object number (ENT_VOBN), which are included in the time map information (TMAPI);

FIG. 62 is a flowchart for explaining an example of the playback sequence of an advanced VTS (AHDVTS in FIGS. 51, 74, 79, and the like) according to the content of information (application type) included in the management information (e.g., AHDVTS_CAT in FIG. 53);

FIG. 63 is a view for explaining the configuration of a navigation pack (NV_PCK) allocated at the head of each data unit (EVOBU) used in an expanded video object (a video object in an HDVTS) according to the embodiment of the present invention;

FIG. 64 shows an example of the data structure of playback control information (PCI) in the navigation pack (NV_PCK) used in the expanded video object;

FIG. 65 shows an example of the data structure of data search information (DSI) in the navigation pack (NV_PCK) used in the expanded video object;

FIG. 66 is a view for explaining an example of the configuration of an advanced VTS (AHDVTS);

FIG. 67 is a view for explaining elements which form a time map according to the embodiment of the present invention;

FIG. 68 is a view for explaining practical elements which form the time map;

FIG. 69 shows an example of a case wherein a plurality of objects (e.g., VOB#2 and VOB#3) are to be played back using ILVU data of an interleaved block;

FIG. 70 is a view for explaining a time map of an ILVU interval in the example of FIG. 69;

FIG. 71 is a view for explaining a time map in the interleaved block;

FIG. 72 is a block diagram showing an example of the internal structure of a playback apparatus according to still another embodiment of the present invention;

FIG. 73 is a view for explaining a part (HDVMG_CAT) of the recording content of an HD video manager (HDVMG) recording area of the information content recorded on disc-shaped information storage medium (content type 1 disc) 1 according to still another embodiment of the present invention;

FIG. 74 is a view for explaining the data structure (AHDVMGI is allocated in the HDVMG unlike in the example of FIG. 1) of an HD video manager (HDVMG) recording area of the information content recorded on disc-shaped information storage medium (content type 2 disc example 1) 1 according to still another embodiment of the present invention;

FIG. 75 shows an example of the data structure of advanced HD video manager information (AHDVMGI) recorded on the HD video manager (HDVMG) shown in FIG. 74;

FIG. 76 shows an example of the data structure of an advanced HD video manager information management table (AHDVMGI_MAT) included in the advanced HD video manager information (AHDVMGI), and the recording content of category information (HDVMG_CAT) stored in this management table;

FIG. 77 shows an example of the data structure of an advanced title search pointer table (ADTT_SRPT) included in the advanced HD video manager information (AHDVMGI);

FIG. 78 is a view for explaining a playback model (example 1) of a disc that records an advanced VTS (AHDVTS);

FIG. 79 is a view for explaining the data structure of video data recording area 20 and advanced content recording area 21 of the information content

recorded on disc-shaped information storage medium
(content type 2 disc example 2) 1 according to still
another embodiment of the present invention;

FIG. 80 shows an example of the data structure of
5 advanced HD video manager information (AHDVMGI) that
can be recorded in an HD video manager (HDVMG) shown in
FIG. 79;

FIG. 81 shows an example of the data structure of
an advanced HD video manager information management
10 table (AHDVMGI_MAT) included in the advanced video
manager information (AHDVMGI) in FIG. 80, and the
recording content (the content different from FIG. 76)
of category information (HDVMG_CAT) stored in this
management table;

15 FIG. 82 shows an example of the data structure
(the content different from FIG. 77) of an advanced
title search pointer table (ADTT_SRPT) included in the
advanced video manager information (AHDVMGI) in
FIG. 80;

20 FIG. 83 is a view for explaining the relationship
between the advanced VTS playback state and standard
VTS playback state;

FIG. 84 is a view for explaining a playback
control module shift command on the DVD-Video playback
25 engine side;

FIG. 85 is a flowchart for explaining a switching
algorithm of a user command process;

FIG. 86 is a view for explaining a domain transition model in a content type 2 disc (FIG. 79, etc.) which records the advanced VTS and standard VTS together;

5 FIG. 87 is a view for explaining a playback model (example 2) that records the advanced VTS (AHDVTS) and standard VTS (HDVTS) together;

FIG. 88 is a view for explaining a unique reference model of objects in a disc that records the
10 advanced VTS (AHDVTS) and standard VTS (HDVTS) together;

FIG. 89 is a view for explaining a shared reference model of objects in a disc that records the advanced VTS (AHDVTS) and standard VTS (HDVTS)
15 together;

FIG. 90 is a view for explaining a practical example of loading information included in advanced content;

FIG. 91 is a block diagram for explaining the
20 arrangement of a buffer manager in an interactive engine of the apparatus shown in FIG. 72; and

FIG. 92 is a flowchart for explaining an example of the apparatus operation when the interactive engine of the apparatus shown in FIG. 72 is activated.

25 FIG. 93 is a view for explaining an example of the configuration of an advanced VTS having multiple PGCs;

FIG. 94 is a view for explaining an example of the

configuration of an advanced VTS having one PGC;

FIG. 95 is a view for explaining a description example (an example using the chapter/PTT numbers) of a playback sequence in a playback sequence information file (e.g., file PBSEQ001.XML in FIG. 2);

FIG. 96 is a view for explaining another description example (an example using the cell numbers) of a playback sequence in a playback sequence information file (a PBSEQ001.XML file or the like);

FIG. 97 is a view for explaining still another description example (an example using the PGC number and chapter/PTT numbers) of a playback sequence in a playback sequence information file (file PBSEQ001.XML or the like);

FIG. 98 is a view for explaining yet another description example (an example using the PGC number and cell numbers) of a playback sequence in a playback sequence information file (file PBSEQ001.XML or the like);

FIG. 99 is a flowchart for explaining an example of the processing for initializing the playback sequence of an advanced VTS by a DVD playback engine using a playback sequence information file (e.g., file PBSEQ001.XML in FIG. 2) (so as to initialize to use a playback sequence based on the description of the playback sequence information file in place of that based on existing PGC information);

FIG. 100 is a block diagram for explaining an example of the internal structure of a playback apparatus according to still another embodiment of the present invention;

5 FIG. 101 is a view showing another example of the data structure of an advanced HD video title set program chain information table (AHDVTS_PGCIT) recorded in advanced HD video title set information (AHDVTSI);

10 FIG. 102 is a view showing an example of the plane configuration upon super in posing output frames of respective modules in a video mixer shown in FIG. 100;

FIG. 103 is a view for explaining an example of time map information (TMAPI) including no time entry in a case wherein one TMAPI is stored in one TMAP file;

15 FIG. 104 is a view for explaining an example of time map information (TMAPI) including no time entry in a case wherein one or more pieces (in this example, two pieces) of TMAPI are stored in one TMAP file;

20 FIG. 105 is a view for explaining the configuration of time map information for EVOBs which are allocated in an interleaved block and form angles;

FIG. 106 is a view showing an example of the data structure of a time map information table (TMAPIT) including no time entry;

25 FIG. 107 is a view showing an example of the data structure of time map information (TMAPI) including no time entry;

FIG. 108 is a view showing an example of the data structure of control packs (standard GCI_PCK and advanced GCI_PCK) including general control information (GCI);

5 FIG. 109 is a view showing an example of the data structure of general control information (GCI);

FIG. 110 is a view for explaining another example of the data structure of advanced HD video title set information (advanced VTSI) recorded in the advanced HD video title set recording area;

10 FIG. 111 is a view showing an example of the data structure of an advanced HD video title set attribute information table (AHDVTS_ATTRIT) stored in the advanced VTSI in FIG. 110; and

15 FIG. 112 is a view showing an example of the data structure of an advanced HD video title set EVOB information table (AHDVTS_EVOBIT) stored in the advanced VTSI in FIG. 110.

FIG. 113 shows an example of a case (case 1) wherein one program stream obtained by multiplexing a primary object (movie object) and a secondary object (advanced object) is recorded on a disc, and the advanced object (secondary object) is independently present as a program stream on an external communication line (Web);

25

FIG. 114 is a view for explaining a decoding model in the case 1;

FIG. 115 shows an example of a case (case 2-1) wherein the program streams of the primary object and secondary object (two program streams obtained by multiplexing using pack units) are recorded on the disc, and the advanced object (secondary object) is
5 independently present as the program stream on the external communication line (Web);

FIG. 116 is a view for explaining a decoding model in the case 2-1;

10 FIG. 117 shows an example of a case (case 2-2) wherein the program streams of the primary object and secondary object (two program streams obtained by multiplexing using access units) are recorded on the disc, and the advanced object (secondary object) is
15 independently present as the program stream on the external communication line (Web);

FIG. 118 is a view for explaining a decoding model in the case 2-2;

FIG. 119 is a view for explaining an example (a
20 case wherein private stream 1 is used to identify objects) of a stream ID which is used to identify the content of the primary object and secondary object;

FIG. 120 shows an example of the arrangement of a sub-stream ID for private stream 1 in the stream ID
25 shown in FIG. 119;

FIG. 121 shows an example of the arrangement of a sub-stream ID for private stream 2 in the stream ID

shown in FIG. 119;

FIG. 122 is a view for explaining another example
(a case wherein private stream 3 is newly provided to
identify objects) of a stream ID which is used to
5 identify the content of the primary object and
secondary object;

FIG. 123 shows an example of the arrangement of
the sub-stream ID for private stream 1 in the stream ID
shown in FIG. 122;

10 FIG. 124 shows an example of the arrangement of
the sub-stream ID for private stream 2 in the stream ID
shown in FIG. 122;

FIG. 125 shows an example of the arrangement of a
sub-stream ID for private stream 3 in the stream ID
15 shown in FIG. 122;

FIG. 126 is a flowchart for explaining an example
of a processing sequence when the primary object and/or
secondary object is played back from the disc and/or
external communication line (Web);

20 FIG. 127 is a view for explaining a playback path
of the primary object and secondary object from the
disc;

FIG. 128 is a view for explaining the playback
path of the primary object from the disc and the
25 secondary object from the external communication line
(Web);

FIG. 129 shows an example of the data structure of

a time map information table including a time map type flag (TMAP_TYPE_FL);

FIG. 130 is a view for explaining Markup description example 1;

5 FIG. 131 is a view for explaining Markup description example 2;

FIG. 132 is a view for explaining Markup description example 3;

FIG. 133 shows another example of a case (case 1a) wherein one program stream obtained by multiplexing a primary object (movie object) and a secondary object (advanced object) is recorded on a disc, and the advanced object (secondary object) is independently present as a program stream on an external communication line (Web);

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FIG. 134 shows still another example of a case (case 1b) wherein one program stream obtained by multiplexing a primary object (movie object) and a secondary object (advanced object) is recorded on a disc, and the advanced object (secondary object) is independently present as a program stream on an external communication line (Web);

20

FIG. 135 is a view for explaining a decoding model in the case 1a;

25 FIG. 136 is a view for explaining an example of a smoothing buffer operation in the decoding model in case 1a;

FIG. 137 shows an example of the outline of an advanced content on the disc;

FIG. 138 shows an example of the outline of a playback system model of the advanced content;

5 FIG. 139 is a block diagram for explaining an example of a data flow in the playback system model of the advanced content;

FIG. 140 is a block diagram for explaining another example of a data flow in the playback system model of the advanced content;

10

FIG. 141 is a block diagram for explaining still another example of a data flow in the playback system model of the advanced content;

FIG. 142 is a block diagram for explaining still another example of a data flow in the playback system model of the advanced content;

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FIG. 143 is a block diagram for explaining an example of an image output mixing model in the playback system model of the advanced content;

20 FIG. 144 shows a concrete example of the image output mixing model;

FIG. 145 is a block diagram for explaining an example of an audio output mixing model in the playback system model of the advanced content;

25 FIG. 146 is a block diagram for explaining an example of a user interface process in the playback system model of the advanced content; and

FIG. 147 is a flowchart for explaining an example of a startup process after inserting the disc.

Best Mode for Carrying Out the Invention

Best Mode for Carrying Out the Invention

5 An information storage medium, information playback method, and information playback apparatus according to an embodiment of the present invention will be described hereinafter with reference to the accompanying drawings. FIG. 1 is a view for explaining
10 the information content recorded on a disc-shaped information storage medium according to the embodiment of the present invention. Information storage medium 1 shown in FIG. 1(a) can be configured by a high-density optical disk (a high-density or high-definition digital versatile disc: HD_DVD for short) which uses, e.g., a
15 red laser of a wavelength of 650 nm or a blue laser of a wavelength of 405 nm (or less).

Information storage medium 1 includes lead-in area 10, data area 12, and lead-out area 13 from the inner
20 periphery side, as shown in FIG. 1(b). This information storage medium 1 adopts the ISO 9660 and UDF bridge structures as a file system, and has ISO 9660 and UDF volume/file structure information area 11 on the lead-in side of data area 12.

25 Data area 12 allows mixed allocations of video data recording area 20 used to record DVD-Video content (also called standard content or SD content), another

video data recording area (advanced content recording area used to record advanced content) 21, and general computer information recording area 22, as shown in FIG. 1(c).

5 Video data recording area 20 includes HD video manager (High Definition-compatible Video Manager [HDVMG]) recording area 30 that records management information associated with the entire HD_DVD-Video content recorded in video data recording area 20, HD
10 video title set (High Definition-compatible Video Title Set [HDVTS], also called standard VTS) recording area 40 which are arranged for respective titles, and record management information and video information (video objects) for respective titles together, and advanced
15 HD video title set (advanced VTS) recording area [AHDVTS] 50, as shown in FIG. 1(d).

 HD video manager (HDVMG) recording area 30 includes HD video manager information (High Definition-compatible Video Manager Information [HDVMGI]) area 31
20 that indicates management information associated with overall video data recording area 20, HD video manager information backup (HDVMGI_BUP) area 34 that records the same information as in HD video manager information area 31 as its backup, and menu video object
25 (HDVMGM_VOBS) area 32 that records a top menu screen indicating whole video data recording area 20, as shown in FIG. 1(e).

In the embodiment of the present invention, HD video manager recording area 30 newly includes menu audio object (HDMENU_AOBS) area 33 that records audio information to be output in parallel upon menu display.

5 An area of first play PGC language select menu VOBS (FP_PGCM_VOBS) 35 which is executed upon first access immediately after disc (information storage medium) 1 is loaded into a disc drive is configured to record a screen that can set a menu description language code

10 and the like.

One HD video title set (HDVTS) recording area 40 that records management information and video information. (video objects) together for each title includes HD video title set information (HDVTSI) area

15 41 which records management information for all content in HD video title set recording area 40, HD video title set information backup (HDVTSI_BUP) area 44 which records the same information as in HD video title set information area 41 as its backup data, menu video

20 object (HDVTSM_VOBS) area 42 which records information of menu screens for each video title set, and title video object (HDVTSTT_VOBS) area 43 which records video object data (title video information) in this video title set.

25 FIG. 2 is a view for explaining an example of a file system which manages content recorded on the disc-shaped information storage medium according to the

embodiment of the present invention. The areas (30, 40) shown in FIG. 1 form independent files in the file system having the ISO 9660 and UDF bridge structures. Conventional (standard SD) DVD-Video content are
5 allocated together under a directory named "VIDEO_TS". On the other hand, files according to the embodiment of the present invention have a configuration in which an HVDVD_TS directory for storing information files that handle High-Definition video data, and an ADV_OBJ
10 directory for storing information files that handle advanced object data are allocated under a Root directory, as shown in, e.g., FIG. 2.

The HVDVD_TS directory broadly includes a group of files which belong to a menu group used for a menu, and
15 groups of files which belong to title set groups used for titles. As the group of files that belong to the menu group, an information file (HVI00001.IFO) for a video manager having information used to manage the entire disk, its backup file (HVI00001.BUP), and
20 playback data files (HVM00001.EVO to HVM00003.EVO) of expanded video object sets for a menu used as background frames of a menu are stored.

As the group of files that belong to a title set #n group (e.g., title set #1 group), an information
25 file (HVIxxx01.IFO: xxx = 001 to 999) for a video title set having information used to manage title set #n, its backup file (HVIxxx01.BUP: xxx = 001 to 999), playback

data files (HVTxxxxyy.EVO: xxx = 001 to 999, yy = 01 to 99) of expanded video object sets for title set #n used as a title are stored.

Furthermore, as the group of files that belong to
5 an advanced title set group, an information file
(HVIA0001.IFO) for a video title set having information
used to manage an advanced title set, its backup file
(HVIA0001.BUP), playback data files (HVTAxxyy.EVO: xx =
01 to 99, yy = 01 to 99) of video object sets for
10 advanced title sets used as titles, time map
information files (HVMAxxxxx.MAP: xxxxx = 0001 to 9999)
for advanced title sets, their backup files
(HVMAxxxxx.BUP: xxxxx = 0001 to 9999, not shown), and the
like are stored.

15 The ADV_OBJ directory stores a startup information
file (STARTUP.XML), loading information file
(LOAD001.XML), playback sequence information file
(PBSEQ001.XML), markup language file (PAGE001.XML),
moving picture data, animation data, still picture data
20 file, audio data file, font data file, and the like.
Note that the content of the startup information file
include startup information of data such as moving
picture data, animation data, still picture data, audio
data, font data, a markup language used to control
25 playback of these data, and the like. The loading
information file records loading information (that can
be described using a markup language/script

language/style sheet, and the like), which describes information associated with files to be loaded onto a buffer in a playback apparatus, and the like.

The playback sequence information file (PBSEQ001.XML) records playback sequence information (that can be also described using a markup language or the like), which defines a section to be played back of the playback data files of expansion video object sets for advanced title sets in the advanced title set group, and the like.

Note that the markup language is a language that describes text attributes along commands which are defined in advance, and can give the font type, size, color, and the like to a character string as attributes. In other words, the markup language is a description language which describes structures (headings, hyperlinks, and the like) and modification information (character size, the state of composition, and the like) of sentences in these sentences by partially bounding special character strings called tags.

Since a document written using the markup language becomes a text file, the user can normally read it using a text editor, and can edit that file, of course. As typical markup languages, Standard Generalized Markup Language (SGML), Hypertext Markup Language (HTML) evolved from SGML, TeX, and the like are known.

FIG. 3 shows an example of the detailed data structure in HD video manager information (HDVMGI) area 31 shown in FIG. 1(e). At the head of this area 31, HD video manager information management table (HDVMGI_MAT) 310, which records management information common to the entire HD_DVD-Video content recorded in video data recording area 20 together, is allocated. After this table, title search pointer table (TT_SRPT) 311 that records information helpful to search (to detect the start positions of) titles present in the HD_DVD-Video content, HD video manager menu program chain information unit table (HDVMGM_PGCI_UT) 312 that records management information of a menu screen, which is separately allocated for each menu description language code used to display a menu, parental management information table (PTL_MAIT) 313 that records information for managing pictures fit or unfit for children to see as parental information, HD video title set attribute information table (HDVTS_ATRT) 314 that records attributes of title sets together, text data manager (TXTDT_MG) 315 that records text information to be displayed for the user together, HD video manager menu cell address table (HDVMGM_C_ADT) 316 that records information helpful to search for the start address of a cell that forms the menu screen, HD video manager menu video object unit address map (HDVMGM_VOBU_ADMAP) 317 that records address

information of VOBUs which indicates a minimum unit of video objects that form the menu screen, and HD menu audio object set information table (HDMENU_AOBSIT) 318 are stored in turn. HD menu audio object set information table (HDMENU_AOBSIT) 318 in HD video manager information (HDVMGI) area 31 records management data for objects in menu audio object (HDMENU_AOBS) area 33.

Note that the data structure from HD video manager information management table (HDVMGI_MAT) 310 to HD video manager menu video object unit address map (HDVMGM_VOBU_ADMAP) 317 matches that of the conventional DVD-Video management information.

In the embodiment of the present invention, the field of HD menu audio object set information table (HDMENU_AOBSIT) 318 to be newly added is separately allocated after those which match the conventional DVD-Video management information. With this allocation, a description of a conventional control program using the conventional DVD-Video management information can be utilized upon practicing the present invention (the description of the control program using management information with the same data structure as in the conventional DVD-Video can be commonly used in the prior art and the present invention). In this manner, generation of a control program for an information playback apparatus according to the embodiment of the

present invention can be simplified.

FIG. 4 shows an example of the detailed data structure in HD video manager information management table (HDVMGI_MAT) 310 in FIG. 3. In this management table 310, information of first play PGC (FP_PGC) that records language select menu management information for the user, the start address information (HDMENU_AOBS_SA) of an HDMENU_AOBS, the start address information (HDMENU_AOBSIT_SA) of an HDVGM_AOBS information table, information of the number (HDVGM_GUST_Ns) of HDVGM graphics unit streams, HDVGM graphics unit stream attribute information (HDVGM_GUST_ATR), and the like are allocated.

In addition, HD video manager information management table (HDVMGI_MAT) 310 records various kinds of information: an HD video manager identifier (HDVMG_ID), the end address (HDVMG_EA) of the HD video manager, the end address (HDVMGI_EA) of the HD video manager information, the version number (VERN) of the HD_DVD-Video standard, an HD video manager category (HDVMG_CAT), a volume set identifier (VLMS_ID), an adaptation identifier (ADP_ID), the number (HDVTS_Ns) of HD video title sets, a provider unique identifier (PVR_ID), a POS code (POS_CD), the end address (HDVMGI_MAT_EA) of the HD video manager information management table, the start address (FP_PGC_SA) of first play program chain information, the start address

(HDVMGM_VOBS_SA) of an HDVMGM_VOBS, the start address
 (TT_SRPT_SA) of the TT_SRPT, the start address
 (HDVMGM_PGCI_UT_SA) of the HDVMGM_PGCI_UT, the start
 address (PTL_MAIT_SA) of the PTL_MAIT, the start
 5 address (HDVTS_ATRT_SA) of the HDVTS_ATRT, the start
 address (TXTDT_MG_SA) of the TXTDT_MG, the start
 address (HDVMGM_C_ADT_SA) of the HDVMGM_C_ADT, the
 start address (HDVMGM_VOBU_ADMAP_SA) of the
 HDVMGM_VOBU_ADMAP, an HDVMGM video attribute
 10 (HDVMGM_V_ATR), the number (HDVMGM_AST_Ns) of HDVMGM
 audio streams, an HDVMGM audio stream attribute
 (HDVMGM_AST_ATR), the number (HDVMGM_SPST_Ns) of HDVMGM
 sub-picture streams, and an HDVMGM sub-picture stream
 attribute (HDVMGM_SPST_ATR).

15 In FIG. 4, the HD video manager category
 (HDVMG_CAT) includes RMA#1, RMA#2, RMA#3, RMA#4, RMA#5,
 RMA#6, RMA#7, and RMA#8 which are determined by
 dividing the world countries into predetermined
 regions, and indicate playback availability information
 20 in respective regions, and application type indicating
 the VMG category. Note that application type assumes
 the following values:

Application type = 0000b: including only standard VTS
 = 0001b: including only advanced VTS
 25 = 0010b: including both advanced VTS and standard
 VTS

That is, when application type is "0000b", it indicates

that this information storage medium is the one
(content type 1 disc) including only standard VTS; when
application type is "0001b", it indicates that this
information storage medium is the one (content type 2
5 disc) including only advanced VTS; and when application
type is "0010b", it indicates that this information
storage medium is the one (content type 2 disc)
including both standard VTS and advanced VTS (to be
described in detail later).

10 FIG. 5 shows an example of the internal structure
of title search pointer table (TT_SRPT) 311 shown in
FIG. 3. Title search pointer table (TT_SRPT) 311
includes title search pointer table information
(TT_SRPTI) 311a, and title search pointer (TT_SRP)
15 information 311b. One or a plurality of pieces of
title search pointer (TT_SRP) information 311b in title
search pointer table (TT_SRPT) 311 can be set in
correspondence with the number of titles included in
the HD_DVD-Video content. Title search pointer table
20 information (TT_SRPTI) 311a records common management
information of title search pointer table (TT_SRPT)
311: the number (TT_SRP_Ns) information of title search
pointers included in title search pointer table
(TT_SRPT) 311, and the end address (TT_SRPT_EA)
25 information of title search pointer table (TT_SRPT) 311
in a file (HD_VMG00.HDI in FIG. 2) of the HD video
manager information (HDVMGI) area.

One title search pointer (TT_SRP) information 311b records various kinds of information associated with a title pointed by this search pointer: a title playback type (TT_PB_TY), the number (AGL_Ns) of angles, the number (PTT_Ns) of Part_of_Titles (PTT), title Parental_ID_Field (TT_PTL_ID_FLD) information, an HDVTS number (HDVTSN), an HDVTS title number (HDVTS_TTN), and the start address (HDVTS_SA) of this HDVTS.

FIG. 6 shows an example of the internal structure of HD video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312 shown in FIG. 3. HD video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312 records HD video manager menu program chain information unit table information (HDVMGM_PGCI_UTI) 312a that records common management information in HD video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312, HD video manager menu language units (HDVMGM_LU) 312c which are arranged for menu description language codes used to display a menu, and record management information associated with menu information, and the like. Table 312 has information of HD video manager menu language units (HDVMGM_LU) 312c as many as the number of menu description language codes supported by the HD_DVD-Video content. HD video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312 has information of HD video manager menu language unit search pointers (HDVMGM_LU_SRP) 312b, which have the start address information of respective HD video

manager menu language units (HDVMGM_LU) 312c, as many
as the number of HD video manager menu language units
(HDVMGM_LU) 312c, so as to facilitate access to HD
video manager menu language units (HDVMGM_LU) 312c for
5 respective menu description language codes.

HD video manager menu PGCI unit table information
(HDVMGM_PGCI_UTI) 312a has information of the number
(HDVMGM_LU_Ns) of HD video manager menu language units,
and the end address (HDVMGM_PGCI_UT_EA) of this HD
10 video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312
in a file (HD_VMG00.HDI in FIG. 2) of the HD video
manager information (HDVMGI) area.

Each HD video manager menu language unit search
pointer (HDVMGM_LU_SRP) information 312b has not only
15 differential address information (HDVMGM_UT_SA) from
the start position of HD video manager menu PGCI unit
table (HDVMGM_PGCI_UT) 312 in the file (HD_VMG00.HDI in
FIG. 2) of the HD video manager information (HDVMGI)
area to the head position of corresponding HD video
20 manager menu language unit (HDVMGM_LU) 312c, but also
information of an HD video manager menu language code
(HDVMGM_LCD) indicating the menu description language
code of corresponding HD video manager menu language
unit (HDVMGM_LU) 312c, and information of the
25 presence/absence (HDVMGM_EXST) of an HD video manager
menu indicating if corresponding HD video manager menu
language unit (HDVMGM_LU) 312c has a menu screen to be

displayed for the user as a video object (VOB or EVOB).

FIG. 7 shows an example of the detailed data structure in HD video manager menu language unit #n (HDVMGM_LU#n) 312c (FIG. 6) recorded in HD video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312 shown in FIG. 3. HD video manager menu language unit (HDVMGM_LU) 312c has the following pieces of information: HD video manager menu language unit information (HDVMGM_LUI) 312c1 that records common management information associated with a menu in HD video manager menu language unit (HDVMGM_LU) 312c, HD video manager menu program chain information (HDVMGM_PGCI) 312c3 having a structure shown in FIG. 33, and information 312c2 of HDVMGM_PGCI search pointers (HDVMGM_PGCI_SRP#1 to HDVMGM_PGCI_SRP#n) each indicating a differential address from the head position of HD video manager menu language unit (HDVMGM_LU) 312c to that of each HD video manager menu program chain information (HDVMGM_PGCI) 312c3 in the file (HD_VMG00.HDI in FIG. 2) of the HD video manager information (HDVMGI) area.

HD video manager menu language unit information (HDVMGM_LUI) 312c1 allocated in the first field (group) in HD video manager menu language unit #n (HDVMGM_LU#n) 312c has information associated with the number (HDVMGM_PGCI_SRP_Ns) of HDVMGM_PGCI_SRP data, and the end address (HDVMGM_LU_EA) information of the

HDVMGM_LU. Each information 312c of HDVMGM_PGCI search
pointers (HDVMGM_PGCI_SRP#1 to HDVMGM_PGCI_SRP#n) has
start address (HDVMGM_PGCI_SA) information of the
HDVMGM_PGCI and HDVMGM_PGC category (HDVMGM_PGC_CAT)
5 information.

FIG. 8 shows an example of the recording content
of the HDVMGM_PGC category (HDVMGM_PGC_CAT) shown in
FIG. 7. HDVMGM_PGC category information
(HDVMGM_PGC_CAT) in HDVMGM_PGCI search pointer #n
10 (HDVMGM_PGCI_SRP#n) 312c2 records selection information
of audio information which is to be simultaneously
played back upon displaying an HD content menu in the
embodiment of the present invention on the screen, and
an audio information selection flag (audio selection
15 information) indicating start/end trigger information
of audio information playback. As audio data which is
to be simultaneously played back upon displaying the HD
content menu in the embodiment of the present invention
on the screen, the following audio data can be
20 selected:

<1> audio data (distributed and recorded in audio
packs; not shown) recorded in menu video object area
(HDVMGM_VOBS) 32 shown in FIG. 1(e), or

<2> audio data which exist in menu audio object
25 area (HDMENU_AOBS) 33 shown in FIG. 1(e) as one or more
menu AOB data (HDMENU_AOB) arranged in turn, as shown
in FIG. 19.

When the audio information selection flag (Audio Selection information) = "00b" is selected, audio data <1> are played back, and audio playback is interrupted upon switching menus. When the audio information selection flag (audio selection information) = "10b" or "11b" is selected, audio data <2> of menu AOB (HDMENU_AOB) in menu audio object area (HDMENU_AOBS) 33 are played back. Upon playing back audio data <2>, if the audio information selection flag designates "11b", the audio data begin to be played back from the beginning every time the menu screen is changed; if it designates "10b", playback of the audio data continues irrespective of switching of menu screens.

In the embodiment of the present invention, menu audio object area (HDMENU_AOBS) 33 can store a plurality of types of menu AOB (HDMENU_AOB) data, as shown in FIG. 19. An audio selection number (audio number information) shown in FIG. 8 can be used as selection information of menu AOB (HDMENU_AOB) to be simultaneously played back upon displaying the menu display PGC of interest. This audio information number can be used to "select which menu AOB from the top" of those which are allocated as menu AOB selection candidates, as shown in FIG. 19.

In addition, the HDVMGM_PGC category (HDVMGM_PGC_CAT) information in FIG. 8 can record entry type information used to check if a PGC of interest is

an entry PGC, menu ID information indicating a menu identification (e.g., a title menu or the like), block mode information, block type information, PTL_ID_FLD information, and the like.

5 FIG. 9 shows an example of the data structure in parental management information table (PTL_MAIT) 313 shown in FIG. 3. As shown in, e.g., FIG. 9, parental management information table 313 includes parental management information table information (PTL_MAITI) 10 313a, one or more parental management information search pointers (PTL_MAI_SRP#1 to PTL_MAI_SRP#n) 313b, and a plurality of pieces of parental management information. (PTL_MAI#1 to PTL_MAI#n) 313c as many as the number of search pointers. Note that parental 15 management information table information (PTL_MAITI) 313a records information such as the number (CTY_Ns) of countries, the number (HDVTS_Ns) of HDVTS data, the end address (PTL_MAIT_EA) of the PTL_MAIT, and the like. Each parental management information search pointer 20 (PTL_MAI_SRP) 313b records information such as a country code (CTY_CD), the start address (PTL_MAI_SA) of the PTL_MAI, and the like.

FIG. 10 shows an example of the data structure in parental management information (PTL_MAI) 313c shown in 25 FIG. 9. This parental management information (PTL_MAI) 313c has one or more pieces of parental level information (PTL_LVL1) 313c1. Each parental level

information (PTL_LVL1) 313c1 includes information of parental ID field (PTL_ID_FLD_HDVMG) 313c11 for HDVMG, and parental ID field (PTL_ID_FLD_HDVTs) 313c12 for HDVTs. Information of each parental ID field (PTL_ID_FLD_HDVTs) 313c12 for HDVTs can store parental ID field (PTL_ID_FLD) for PGC selection.

FIG. 11 shows an example of the data structure of HD video title set attribute information table (HDVTs_ATTR) 314 shown in FIG. 3. As shown in FIG. 11, this HD video title set attribute information table 314 includes: HD video title set attribute table information (HDVTs_ATTRI) 314a having information of the number (HDVTs_Ns) of HDVTs data and the end address (HDVTs_ATTR_EA) of the HDVTs_ATTR; HDVTs video title set attribute search pointers (HDVTs_ATTR_SRP) 314b each of which records information of the start address (HDVTs_ATTR_SA) of the HDVTs_ATTR; and HDVTs video title set attributes (HDVTs_ATTR) 314c each having information of the end address (HDVTs_ATTR_EA) of the HDVTs_ATTR, HD video title set category (HDVTs_CAT), and HD video title set attribute information (HDVTs_ATTRI).

FIG. 12 shows an example of the data structure of text data manager (TXTDT_MG) 315 shown in FIG. 3. As shown in FIG. 12, this text data manager 315 includes text data manager information (TXTDT_MGI) 315a having information of a text data identifier (TXTDT_ID), the number (TXTDT_LU_Ns) of TXTDT_LU data, and the end

address (TXTDT_MG_EA) of the text data manager; text
data language unit search pointers (TXTDT_LU_SRP) 315b
each of which records various kinds of information
including a text data language code (TXTDT_LCD), a
5 character set (CHRS), and the start address
(TXTDT_LU_SA) of the TXTDT_LU; and text data language
units (TXTDT_LU) 315c.

FIG. 13 shows an example of the internal data
structure of text data language unit (TXTDT_LU) 315c.
10 As shown in FIG. 13, this text data language unit 315c
includes various kinds of information: text data
language unit information (TXTDT_LUI) 315c1 that
records the end address (TXTDT_LU_EA) information of
the TXTDT_LU; item text search pointer search pointer
15 (IT_TXT_SRP_SRP_VLM) 315c2 for volume that records the
start address (IT_TXT_SRP_SA_VLM) information of the
IT_TXT_SRP for volume; item text search pointer search
pointers (IT_TXT_SRP_SRP_TT) 315c3 for volume each of
which holds the start address (IT_TXT_SRP_SA_TT)
20 information of the IT_TXT_SRP for title; and text data
(TXTDT) 315c4.

FIG. 14 shows an example of the internal data
structure of text data (TXTDT) 315c4. As shown in
FIG. 14, this text data 315c4 records various kinds of
25 information: text data information (TXTDTI) 315c41
having information of the number (IT_TXT_SRP_Ns) of
IT_TXT_SRP data; item text search pointers (IT_TXT_SRP)

315c42 each of which records an item text identifier
code (IT_TXT_IDCD) and the start address (IT_TXT_SA)
information of the IT_TXT; and item text (IT_TXT) data
315c43.

5 FIG. 15 shows an example of the data structure of
HD video manager menu cell address table (HDVMGM_C_ADT)
316 shown in FIG. 3. As shown in FIG. 15, this HD
video manager menu cell address table 316 records
various kinds of information: HD video manager menu
10 cell address table information (HDVMGM_C_ADTI) 316a
having information of the number (HDVMGM_VOB_Ns) of VOB
data in HDVMGM_VOBS and the end address
(HDVMGM_C_ADT_EA) of the HDVMGM_C_ADT; and a plurality
of pieces of HD video manager menu cell piece
15 information (HDVMGM_CPI) 316b each of which records
information of a VOB_ID number (HDVMGM_VOB_IDN) of an
HDVMGM_CP, a Cell_ID number (HDVMGM_C_IDN) of the
HDVMGM_CP, the start address (HDVMGM_CP_SA) of the
HDVMGM_CP, and the end address (HDVMGM_CP_EA) of the
20 HDVMGM_CP ("CP" of HDVMGM_CP indicates a cell piece).

FIG. 16 shows an example of the data structure of
HD video manager menu video object unit address map
(HDVMGM_VOBU_ADMAP) 317 shown in FIG. 3. As shown in
FIG. 16, this HD video manager menu video object unit
25 address map 317 records various kinds of information:
HD video manager menu video object unit address map
information (HDVMGM_VOBU_ADMAPI) 317a having

information of the end address (HDVMGM_VOBU_ADMAP_EA) of the HDVMGM_VOBU_ADMAP; and start addresses (HDVMGM_VOBU_AD#1 to HDVMGM_VOBU_AD#n) 317b of HDVMGM_VOBU data.

5 FIG. 17 shows the management information content for menu audio object (HDMENU_AOB) itself, and shows an example of the internal data structure of HD menu audio object set information table (HDMENU_AOBSIT) 318 shown in FIG. 3 stored in HD video manager information
10 (HDVMGI) area 31 shown in FIG. 1(e). As shown in FIG. 17, HD menu audio object set information table information (HDMENU_AOBSITI) 318a allocated at the first field of HD menu audio object set information table 318 stores HDMENU_AOB_Ns as information of the
15 number of AOB data in HDMENU_AOBS, and the end address information (HDMENU_AOBSIT_EA) of the HDMENU_AOBSIT. In the embodiment of the present invention, a plurality of types of menu audio objects (audio data) can be recorded in information storage medium 1.

20 In HD menu audio object set information table 318 shown in FIG. 17, one or more pieces of HD menu audio object information (HDMENU_AOBI) 318b are allocated after HD menu audio object set information table information 318a. Each HD menu audio object
25 information (HDMENU_AOBI) 318b indicates management information for each individual menu audio object (audio data), and includes playback information

(HDMENU_AOB_PBI) of HDMENU_AOB, attribute information (HDMENU_AOB_ATR) of HDMENU_AOB, the start address information (HDMENU_AOB_SA) of HDMENU_AOB#n (HDMENU_AOB of interest), and the end address information (HDMENU_AOB_EA) of HDMENU_AOB#n (HDMENU_AOB of interest).

FIG. 18 shows an example of the data structure of menu video object area (HDVMGM_VOBS) 32 shown in FIG. 1(e), which is stored together in, e.g., file HD_VMG01.HDV (file HD_VMG01.HDV can be stored as a file in the menu group in FIG. 2; not shown). As shown in FIG. 18, menu screens (video objects) which record an identical menu screen using different menu description language codes are allocated in juxtaposition with this menu video object area 32. In this way, a plurality of menu screens of a plurality of languages are prepared, and a menu screen can be displayed by arbitrarily selecting one of a plurality of them. For example, when only one Japanese menu VOB is selected, a Japanese menu can be displayed; when only one English menu VOB is selected, an English menu can be displayed. Alternatively, when the display screen is configured to display multi-windows, and the Japanese menu VOB and English menu VOB are selected, the Japanese and English menus can be displayed on the multi-windows.

FIG. 19 shows an example of the data structure of menu audio object area (HDMENU_AOBS) 33 recorded in the

HD video manager (HDVMG) recording area. In the embodiment of the present invention, a plurality of types of menu audio objects (audio data) can be recorded in information storage medium 1. Each menu audio object (AOB) is recorded at a location in menu audio object area (HDMENU_AOBS) 33 in HD video manager recording area (HDVMG) 30, as shown in, e.g., FIG. 1. This menu audio object area (HDMENU_AOBS) 33 forms one file with, e.g., file name HD_MENU0.HDA (file HD_MENU0.HAD can be a file in the menu group in FIG. 2; not shown). Respective menu audio objects (AOB) are allocated and recorded in turn in menu audio object area (HDMENU_AOBS) 33 that forms one file with file name HD_MENU0.HAD, as shown in FIG. 19.

FIG. 20 shows an example of the data structure of HD video title set information (HDVTSI) 41 recorded in each HD video title set (HDVTS#n) recording area. This HD video title set information 41 is recorded together in file HVI00101.IFO and/or HVIA0001.IFO shown in, e.g., FIG. 2 (or independent file VTS00100.IFO in the DVD-Video content; not shown). As shown in FIG. 20, the interior of HD video title set information (HDVTSI) 41 shown in FIG. 1(f) is divided into respective fields (management information groups): HD video title set information management table (HDVTSI_MAT) 410, HD video title set PTT search pointer table (HDVTS_PTT_SRPT) 411, HD video title set program chain information table.

(HDVTS_PGCIT) 412, HD video title set menu PGCI unit
able (HDVTSM_PGCI_UT) 413, HD video title set time map
table (HDVTS_TMAPT) 414, HD video title set menu cell
address table (HDVTSM_C_ADT) 415, HD video title set
5 menu video object unit address map (HDVTSM_VOBU_ADMAP)
416, HD video title set cell address table
(HDVTS_C_ADT) 417, and HD video title set video object
unit address map (HDVTS_VOBU_ADMAP) 418.

HD video title set information management table
10 (HDVTSI_MAT) 410 records management information common
to the corresponding video title set. Since this
common management information (HDVTSI_MAT) is allocated
in the first field (management information group) in HD
video title set information (HDVTSI) area 41, the
15 common management information in the video title set
can be immediately loaded (before the beginning of
object playback). Hence, the playback control process
of the information playback apparatus can be
simplified, and the control processing time can be
20 shortened.

FIG. 21 shows an example of the data structure of
the HD video title set information management table
(HDVTSI_MAT) recorded in the HD video title set
information (HDVTSI). Management information
25 associated with graphics units included in the HDVTS
(Video Title Set according to the embodiment of the
present invention) is recorded in HD video title set

information management table (HDVTSI_MAT) 410 (see
FIG. 20), which is allocated in the first field (group)
in HD video title set information (HDVTSI) area 41
shown in FIG. 1(f). Detailed management information
5 content are as shown in FIG. 21. That is, information
of the number of graphics unit streams and attribute
information are separately recorded for a menu screen
and title (display picture) in the HDVTS as information
of the number (HDVTSM_GUST_Ns) of HDVTSM graphics unit
10 streams, HDVTSM graphics unit stream attribute
information (HDVTSM_GUST_ATR), information of the
number (HDVTS_GUST_Ns) of HDVTS graphics unit streams,
and HDVTS graphics unit stream attribute table
information (HDVTS_GUST_ATRT).

15 Also, as shown in FIG. 21, HD video title set
information management table (HDVTSI_MAT) 410 records
various kinds of information: an HD video title set
identifier (HDVTS_ID), the end address (HDVTS_EA) of
the HDVTS, the end address (HDVTSI_EA) of the HDVTSI,
20 the version number (VERN) of the HD_DVD-Video standard,
an HDVTS category (HDVTS_CAT), the end address
(HDVTSI_MAT_EA) of the HDVTSI_MAT, the start address
(HDVTSM_VOBS_SA) of the HDVTSM_VOBS, the start address
(HDVTSTT_VOBS_SA) of the HDVTSTT_VOBS, the start
25 address (HDVTS_PTT_SRPT_SA) of the HDVTS_PTT_SRPT, the
start address (HDVTS_PGCIT_SA) of the HDVTS_PGCIT, the
start address (HDVTSM_PGCI_UT_SA) of the

HDVTSM_PGCI_UT, the start address (HDVTS_TMAP_SA) of
 the HDVTS_TMAP, the start address (HDVTSM_C_ADT_SA) of
 the HDVTSM_C_ADT, the start address
 (HDVTSM_VOBU_ADMAP_SA) of the HDVTSM_VOBU_ADMAP, the
 5 start address (HDVTS_C_ADT_SA) of the HDVTS_C_ADT, the
 start address (HDVTS_VOBU_ADMAP_SA) of the
 HDVTS_VOBU_ADMAP, an HDVTSM video attribute
 (HDVTSM_V_ATR), the number (HDVTSM_AST_Ns) of HDVTSM
 audio streams, an HDVTSM audio stream attribute
 10 (HDVTSM_AST_ATR), the start address (HDVTSM_SPST_Ns) of
 the number of HDVTSM sub-picture streams, an HDVTSM
 sub-picture stream attribute (HDVTSM_SPST_ATR), an
 HDVTS video attribute (HDVTS_V_ATR), the number
 (HDVTS_AST_Ns) of HDVTS audio streams, an HDVTS audio
 15 stream attribute table (HDVTS_AST_ATRT), the number
 (HDVTS_SPST_Ns) of HDVTS sub-picture streams, an HDVTS
 sub-picture stream attribute table (HDVTS_SPST_ATRT),
 and an HDVTS multi-channel audio stream attribute table
 (HDVTS_MU_AST_ATRT).

20 FIG. 22 shows an example of the data structure in
 HD video title set PTT search pointer table
 (HDVTS_PTT_SRPT) 411 shown in FIG. 19. This HD video
 title set PTT search pointer table 411 includes various
 kinds of information: PTT search pointer table
 25 information (PTT_SRPTI) 411a having information of the
 number (HDVTS_TTU_Ns) of HDVTS TTU data and the end
 address (HDVTS_PTT_SRPT_EA) of the HDVTS_PTT_SRPT;

title unit search pointers (TTU_SRP) 411b each of which records information of the start address (TTU_SA) of the TTU; and PTT search pointers (PTT_SRP) 411c having information of a program chain number (PGCN) and
5 program number (PGN).

<Allocation of Information that Manages Resume Information>

FIG. 23 shows an example of the data structure of HD video title set program chain information table
10 (HDVTS_PGCIT) recorded in the HD video title set information (HDVTSI). In the embodiment of the present invention, as shown in FIG. 23, an HDVTS_PGC category in HDVTS_PGCI search pointer 412b stores an update permission flag of resume information (RSM permission
15 flag). Information of HDVTS_PGCI search pointer 412b is allocated in HD video title set program chain information table (HDVTS_PGCIT) 412 (FIG. 20) stored in HD video title set information (HDVTSI) area 41 shown in FIG. 1(f). In addition, as shown in FIG. 23, HD
20 video title set program chain information table (HDVTS_PGCIT) 412 also records information of HD video title set PGCI information table (HDVTS_PGCITI) 412a including information of the number (HDVTS_PGCI_SRP_Ns) of HDVTS_PGCI_SRP data and the end address
25 (HDVTS_PGCIT_EA) of the HDVTS_PGCIT. Also, HDVTS_PGCI search pointer (HDVTS_PGCI_SRP) 412b records information of the start address (HDVTS_PGCI_SA) of the

HDVTS_PGCI together with the aforementioned HDVTS_PGC category (HDVTS_PGC_CAT).

FIG. 24 shows an example of the recording content of the HDVTS_PGC category (HDVTS_PGC_CAT). The update permission flag of resume information (RSM permission Flag) shown in FIG. 24 designates whether or not the content of resume information are to be updated after playback of the HDVTS_PGC of interest starts (whether or not resume information is updated as needed in correspondence with the playback state of the PGC of interest). That is, the following process is made in correspondence with the flag:

When RSM permission flag = "0b", resume information is updated, or

when RSM permission flag = "1b", resume information is not updated, and playback resume information is held in the HDVTS_PGC (program chain in the video title set according to the embodiment of the present invention) played back previously.

In addition, the HDVTS_PGC category (HDVTS_PGC_CAT) can record entry type information used to check if a PGC of interest is an entry PGC, title number information in a VTS (video title set) indicated by the corresponding PGC, block mode information, block type information, PTL_ID_FLD information, and the like.

FIG. 25 shows an example of the data structure in HD video title set menu PGCI unit table

(HDVTSM_PGCI_UT) 413 shown in FIG. 20. This HD video title set menu PGCI unit table 413 includes various kinds of information: HD video title set menu program chain information unit table information

5 (HDVTSM_PGCI_UTI) 413a having information of the number (HDVTSM_LU_Ns) of HD video title set menu language units and the end address (HDVTSM_PGCI_UT_EA) of the HDVTSM_PGCI_UT; HD video title set menu language unit search pointers (HDVTSM_LU_SRP) 413b each of which
10 records information of an HD video title set menu language code (HDVTSM_LCD), the presence/absence (HDVTSM_EXST) of a HD video title set menu, and the start address (HDVTSM_LU_SA) of the HDVTSM_LU; and HD video title set menu language units (HDVTSM_LU) 413c.

15 FIG. 26 shows an example of the data structure in HD video title set menu language unit (HDVTSM_LU) 413c. As shown in FIG. 26, this HD video title set menu language unit 413c includes: HD video title set menu language unit information (HDVTSM_LUI) 413c1 having
20 information of the number (HDVTSM_PGCI_SRP_Ns) of HDVTSM_PGCI_SRP data and the end address (HDVTSM_LU_EA) of the HDVTSM_LU; a plurality of pieces of HD video title set menu program chain information (HDVTSM_PGCI) 413c3 having the same data structure as in FIG. 33; and
25 HDVTSM_PGCI search pointers (HDVTSM_PGCI_SRP) 413c2 each of which records information of the HDVTSM_PGC category (HDVTSM_PGC_CAT) and the start address

(HDVTSM_PGCI_SA) of the HDVTSM_PGCI.

As the setting location of information that refers to (designates) the menu AOB (HDMENU_AOB), in the embodiment of the present invention, as for a menu for
5 each HDVTS, that information is allocated in the HDVTSM_PGC category information (HDVTSM_PGC_CAT) in HDVTSM_PGCI search pointer #n (HDVTSM_PGCI_SRP#n) 413c2, as shown in FIG. 26.

FIG. 27 shows an example of the recording content
10 of the HDVTSM_PGC category (HDVTSM_PGC_CAT). AOB Number information in the HDVTSM_PGC category information (HDVTSM_PGC_CAT) shown in FIG. 27 means AOB number information (AOB Number) which designates AOB number #n (indicating which AOB of menu AOB
15 (HDMENU_AOB) data which are arranged, as shown in FIG. 19 corresponds to) to be played back in HDMENU_AOBU. Also, audio selection information means selection information of audio information which is to be simultaneously played back upon displaying an HD
20 content menu in the embodiment of the present invention on the screen, and an audio information selection flag (audio selection information) indicating start/end trigger information of audio information playback.

When the audio information selection flag (Audio
25 Selection information) = "00b" is selected, audio data recorded in respective menu video objects are played back, and audio playback is interrupted upon switching

menus. When the audio information selection flag (Audio Selection information) = "10b" or "11b" is selected, audio data of menu AOB (HDMENU_AOB) data stored in menu audio object area (HDMENU_AOBS) 33 are
5 played back. Upon playing back the menu audio data (AOB), if the audio information selection flag = "11b" is designated, the audio data begin to be played back from the beginning every time the menu screen is changed; if "10b" is designated, playback of the audio
10 data continues irrespective of switching of menu screens. In the embodiment of the present invention, menu audio object area (HDMENU_AOBS) 33 can store a plurality of types of menu AOB (HDMENU_AOB) data, as shown in FIG. 19.

15 Audio number information shown in FIG. 27 indicates selection information of menu AOB (HDMENU_AOB) data to be simultaneously played back upon displaying the menu display PGC of interest. This Audio Number information as the selection information
20 of menu AOB data is used to "select which menu AOB from the top" of those which are allocated in FIG. 19 using number information. Also, the HDVTSM_PGC category (HDVTSM_PGC_CAT) records entry type information used to check if a PGC of interest is an entry PGC, menu ID
25 information indicating a menu identification (e.g., a title menu or the like), block mode information, block type information, PTL_ID_FLD information, and the like.

FIG. 28 shows an example of the data structure in HD video title set time map table (HDVTS_TMAPT) 414 shown in FIG. 20. This HD video title set time map table 414 includes various kinds of information: HD video title set time map table information (HDVTS_TMAPTI) 414a that describes information of the number (HDVTS_TMAP_Ns) of HDVTS_TMAP data and the end address (HDVTS_TMAPT_EA) of the HDVTS_TMAPT; HD video title set time map search pointer (HDVTS_TMAP_SRP) 414b having information of the start address (HDVTS_TMAP_SA) of the HDVTS_TMAP; and HD video title set time maps (HDVTS_TMAP) 414c each of which records information of the length (TMU) of a time unit (sec) as a reference in a map entry, the number (MAP_EN_Ns) of map entries, and a map entry table (MAP_ENT).

FIG. 29 shows an example of the data structure in HD video title set menu cell address table (HDVTSM_C_ADT) 415 shown in FIG. 20. As shown in FIG. 29, this HD video title set menu cell address table 415 includes various kinds of information: HD video title set menu cell address table information (HDVTSM_C_ADTI) 415a having information of the number (HDVTSM_VOB_Ns) of VOB data in an HDVTM_VOBS and the end address (HDVTSM_C_ADT_EA) of the HDVTSM_C_ADT; and a plurality of pieces of HD video title set menu cell piece information (HDVTSM_CPI) 415b each of which records information of a VOB_ID number (HDVTSM_VOB_IDN)

of an HDVTSM_CP, a Cell_ID number (HDVTSM_C_IDN) of the HDVTSM_CP, the start address (HDVTSM_CP_SA) of the HDVTSM_CP, and the end address (HDVTSM_CP_EA) of the HDVTSM_CP.

5 FIG. 30 shows an example of the data structure of HD video title set menu video object unit address map (HDVTSM_VOB_U_ADMAP) 416 shown in FIG. 20. As shown in FIG. 30, this HD video title set menu video object unit address map 416 includes: HD video title set menu video
10 object unit address map information (HDVTSM_VOB_U_ADMAP_I) 416a that describes the information of the end address (HDVTSM_VOB_U_ADMAP_EA) of the HDVTSM_VOB_U_ADMAP, and information of HD video title set menu video object unit addresses
15 (HDVTSM_VOB_U_AD) 416b each having information of the start address (HDVTSM_VOB_U_SA) of an HDVTSM_VOB_U.

 FIG. 31 shows an example of the data structure in HD video title set cell address table (HDVTS_C_ADT) 417 shown in FIG. 20. As shown in FIG. 31, this HD video
20 title set cell address table 417 includes various kinds of information: HD video title set cell address table information (HDVTS_C_ADT_I) 417a having the information of the number (HDVTS_VOB_Ns) of VOB data in an HDVTS_VOBS and the end address (HDVTS_C_ADT_EA) of the
25 HDVTS_C_ADT; and a plurality of pieces of HD video title set cell piece information (HDVTS_CPI) 417b each including a VOB_ID number (HDVTS_VOB_IDN) of an

HDVTS_CP, a Cell_ID number (HDVTS_C_IDN) of the HDVTS_CP, the start address (HDVTS_CP_SA) of the HDVTS_CP, and the end address (HDVTS_CP_EA) of the HDVTS_CP.

5 FIG. 32 shows an example of the data structure in HD video title set video object unit address map (HDVTS_VOBU_ADMAP) 418 shown in FIG. 20. As shown in FIG. 32, this HD video title set video object unit address map 418 includes various kinds of information:
10 HD video title set video object unit address map information (HDVTS_VOBU_ADMAPI) 418a having information of the end address (HDVTS_VOBU_ADMAP_EA) of the HDVTS_VOBU_ADMAP; and HD video title set video object unit addresses (HDVTS_VOBU_AD) 418b each of which
15 records information of the start address (HDVTS_VOBU_SA) of each HDVTS_VOBU.

 FIG. 33 shows an example of the data structure of program chain general information (PGC_GI) included in program chain information (PGCI: corresponding to one
20 of HDVTS_PGCI in, e.g., FIG. 23), and the recording content of a PGC graphics unit stream control table (PGC_GUST_CTLT) and resume/audio category (RSM&AOB_CAT) stored in this PGCI.

 The information of the update permission flag of
25 resume information (RSM permission flag) and audio information selection flag (audio selection information)/audio information number (audio number

information) as some of characteristic features according to the embodiment of the present invention are stored in PGCI search pointer information in the existing example (see FIGS. 26, 27, etc.). However, 5 the present invention is not limited to this. For example, the PGCI itself can store the update permission flag information of resume information and audio information selection flag/audio information number. This example is FIG. 33. The PGCI information 10 shown in FIG. 33 corresponds to:

a] HD video manager menu program chain information (HDVMGM_PGCI) 312c3 which is shown in FIG. 7 in association with each HD video manager menu language unit (HDVMGM_LU) 312c in FIG. 6 stored in HD 15 video manager menu PGCI unit table (HDVMGM_PGCI_UT) 312 (FIG. 3) in HD video manager information (HDVMGI) area 31 in FIG. 1(e);

b] HD video title set menu program chain information (HDVTSM_PGCI) 413c3 shown in FIG. 26 which 20 is allocated in each HD video title set menu language unit (HDVTSM_LU) 413c in FIG. 25 in HD video title set menu PGCI unit table (HDVTSM_PGCI_UT) 413 in FIG. 20 that shows the data structure in HD video title set information (HDVTSI) area 41 in FIG. 1(f); and

25 c] HDVTS_PGCI 412c (FIG. 23) in HD video title set program chain information table (HDVTS_PGCIT) 412 in FIG. 20 that shows the data structure in HD video

title set information (HDVTSI) area 41 in FIG. 1(f)
(the PGCi information shown in FIG. 33 can be allocated
in one of the above three locations (a) to (c)).

As shown in FIG. 33, the program chain information
5 (PGCI) includes five fields (five management
information groups), i.e., program chain general
information (PGC_GI) 50, program chain command table
(PGC_CMDT) 51, program chain program map (PGC_PGMAP)
52, cell playback information table (C_PBIT) 53, and
10 cell position information table (C_POSIT) 54.

As shown in FIG. 33, RSM & AOB category
information (RSM&AOB_CAT) is recorded at the end of
program chain general information (PGC_GI) 50 allocated
in the first field (management information group) in
15 the PGCi. The RSM & AOB category information
(RSM&AOB_CAT) stores the update permission flag of
resume information (RSM permission information), audio
information selection flag (audio selection
information) and audio information number (audio number
20 information). This RSM permission information have the
same meaning as the content described using FIG. 24.
Also, the content of the audio information selection
flag or audio information number match those described
using FIG. 8 or 27. Furthermore, the RSM & AOB
25 category information (RSM&AOB_CAT) records entry type
information used to check if a PGC of interest is an
entry PGC, block mode information, block type

information, and PTL_ID_FLD information.

Information in the PGC graphics unit stream control table (PGC_GUST_CTLT) that records control information associated with graphics unit streams allocated in the PGC is independently recorded in each of a PGC_GUST_CTL (PGC_GUST#0) field of HD graphics unit stream #0, a PGC_GUST_CTL (PGC_GUST#1) field of SD wide graphics unit stream #1, a PGC_GUST_CTL (PGC_GUST#2) field of 4 : 3 (SD) graphics unit stream #2, and a PGC_GUST_CTL (PGC_GUST#3) field of letterbox (SD) graphics unit stream #3 as independent fields in correspondence with four different types of pictures (an HD picture at 16 : 0, SD picture at 16 : 9, SD picture at 4 : 3, and SD picture at letterbox), as shown in FIG. 33.

In addition to the aforementioned information, program chain general information (PGC_GI) 50 records various kinds of information including PGC content (PGC_CNT), a PGC playback time (PGC_PB_TM), PGC user operation control (PGC_UOP_CTL), a PGC audio stream control table (PGC_AST_CTLT), a PGC sub-picture stream control table (PGC_SPST_CTLT), PGC navigation control (PGC_NV_CTL), a PGC sub-picture palette (PGC_SP_PLT), the start address (PGC_CMDT_SA) of the PGC_CMDT, the start address (PGC_PGMAP_SA) of the PGC_PGMAP, the start address (C_PBIT_SA) of the C_PBIT, and the start address (C_POSIT_SA) of the C_POSIT.

FIG. 34 shows an example of the program chain command table (PGCI_CMDT) included in the program chain information (PGCI). As shown in FIG. 34, a plurality of pieces of command information to be applied to each PGC are allocated together on program chain command table (PGC_CMDT) 51. The allocation of this PGCI information can be one of the three locations (a) to (c), as described using FIG. 33. A resume (RSM) command sequence (or resume sequence) is recorded in program chain command table (PGC_CMDT) 51, as shown in FIG. 34. The information content of the resume sequence (resume command sequence) in the embodiment of the present invention is described in a format in which RSM commands (RSM_CMD) 514 are allocated in juxtaposition with the field of command table 51. One RSM command (RSM_CMD) 514 described in one column in FIG. 34 means one command that can be designated in the HD_DVD-Video content in the present invention, and RSM commands (RSM_CMD) 514 allocated in the resume (RSM) command sequence field are successively (sequentially) executed in turn from the top.

In the embodiment of the present invention, a sequence of cell commands (C_CMD) 513 in FIG. 34 also means a sequential command sequence. That is, command processes are sequentially executed in turn from the top in accordance with the arrangement order of cell commands (C_CMD) 513 shown in FIG. 34. As will be

additionally described with reference to FIG. 37, a structure that can designate some of cell command processing sequences for each cell (the first cell command number at which the sequential process of cell command is to start, and the execution range of the sequential process of cell commands for each cell) in a series of cell command processing sequences designated from cell command #1 (C_CMD#1) to cell command #k (C_CMD#k) is adopted.

Referring to FIG. 34, RSM command (RSM_CMD) 514 indicates a part of a command sequence which is executed "immediately before playback from the middle of a PGC" whose playback was interrupted previously after the control returns from, e.g., a menu screen to the PGC of interest. On the other hand, pre-command (PRE_CMD) 511 means a command executed "immediately before the PGC of interest is to be played back from the beginning". Also, a command to be executed after playback of the PGC of interest is post command (POST_CMD) 512. The number of pre-commands (PRE_CMD) 511, that of post commands (POST_CMD) 512, that of cell commands (C_CMD) 513, and that of RSM commands (RSM_CMD) 514 that can be allocated in one program chain command table (PGC_CMDT) 51 in FIG. 34 can be freely set (any of the numbers of commands to be described may be "0"). In the embodiment of the present invention, the upper limit of a total value

obtained by adding the number of pre-commands (PRE_CMD) 511, that of post commands (POST_CMD) 512, that of cell commands (C_CMD) 513, and that of RSM commands (RSM_CMD) 514 that can be allocated in one program chain command table (PGC_CMDT) 51 is specified to be 1023. Therefore, when all of the number of pre-commands (PRE_CMD) 511, that of post commands (POST_CMD) 512, and that of RSM commands (RSM_CMD) 514 are "0", a maximum of 1023 cell commands (C_CMD) 513 can be set.

FIG. 35 shows an example of the content of program chain command table information (PGC_CMDTI) and those of each resume command (RSM_CMD) included in the program chain command table (PGCI_CMDT). As shown in FIG. 35, program chain command table information (PGC_CMDTI) 510 records PRE_CMD_Ns as information indicating the number of pre-commands (PRE_CMD) 511, POST_CMD_Ns as information indicating the number of post commands (POST_CMD) 512, C_CMD_Ns as information indicating the number of cell commands (C_CMD) 513, and RSM_CMD_Ns as information indicating the number of RSM commands (RSM_CMD) 514, which can be allocated in one program chain command table (PGC_CMDT) 51.

A detailed data structure in RSM command (RSM_CMD) 514 recorded in program chain command table (PGC_CMDT) 51 will be described below. The detailed data structure in RSM command (RSM_CMD) 514 will be

described in this paragraph, but the data structures in pre-command (PRE_CMD) 511, post command (POST_CMD) 512, and cell command (C_CMD) 513 are the same as the detailed data structure in RSM command (RSM_CMD) 514.

5 In the detailed data structure in RSM command (RSM_CMD) 514, an "8-byte" field is merely assigned to each command, as shown in FIG. 35. In this "8-byte" field, any of command content that will be additionally explained with reference to FIG. 43 are selected and

10 recorded. This command stores "command ID-1" data shown in FIG. 42 in its MSB to the third bit in 8 bytes. The data content of the following bits are different depending on the value of "command type" shown in FIG. 42, but they commonly have information of

15 "comparison I-flag", "compare field", and the like shown in FIG. 42 independently of the command type.

FIG. 36 shows an example of the data structures in program chain program map (PGC_PGMAP) 52 and cell position information table (C_POSIT) 54 allocated in

20 the program chain information (PGCI). In program chain program map (PGC_PGMAP) 52, a plurality of pieces of program entry cell number 520 information that record entry cell numbers (EN_CN) indicating the cell numbers corresponding to entries are allocated in

25 correspondence with the number of entries. Cell position information table (C_POSIT) 54 has a structure in which a plurality of pieces of cell position

information (C_POSI) 540 each including a pair of a cell VOB_ID number (C_VOB_IDN) and cell ID number (C_IDN) are allocated in turn.

In the description of FIG. 34, the structure that
5 can designate some of cell command processing sequences for each cell (the first cell command number at which the sequential process of cell command is to start, and the execution range of the sequential process of cell commands for each cell) in a series of cell command
10 processing sequences designated from cell command #1 (C_CMD#1) to cell command #k (C_CMD#k) is adopted. FIG. 37 shows execution range information of the sequential process of cell commands which can be set for each cell. As has been explained in FIG. 33, the
15 PGCI information can be allocated at the three locations (a) to (c). Management information associated with individual cells that form a PGC is recorded in cell playback information (C_PBI) 530 in cell playback information table (C_PBIT) 53 in the PGCI
20 as the management information of the PGC of interest, as shown in FIG. 37.

Information associated with the first cell command number, at which the sequential process of cell command is to start, designated for each cell in a series of
25 cell command processing sequences designated from cell command #1 (C_CMD#1) to cell command #k (C_CMD#k) is recorded in cell command start number information

(C_CMD_SN) in cell playback information (C_PBI) 530, as shown in FIG. 37. At the same time, cell command continuous number information (C_CMD_C_Ns) indicating the number of commands, the command processes of which are to be continuously executed as well as cell command (C_CMD) 513 designated by the cell command start number information (C_CMD_SN) is recorded in cell playback information (C_PBI) 530. Based on these two pieces of information, the execution range of the sequential process of cell command to be executed by the cell of interest is designated. In the embodiment of the present invention, after completion of playback of the cell of interest, a command sequence of the range designated by the cell command start number information (C_CMD_SN) and cell command continuous number information (C_CMD_C_Ns) in FIG. 37 can be executed.

FIG. 37 shows an example of the data structure of the cell playback information table (C_PBIT) included in the program chain information (PGCI). Referring to FIG. 37, cell playback information (C_PBI) can store the following information: a cell category (C_CAT) indicating if a cell of interest corresponds to the start or last cell of an interleaved block when the cell of interest forms an interleaved block corresponding to multi-angle playback, a part of a general continuous block, or a part of an interleaved block corresponding to multi-angle playback; a cell

playback time (C_PBTM) indicating a playback time required to play back the entire cell of interest; the start address position information (C_FVOBU_SA) of the first VOB of the cell; the end address position information (C_FILVU_EA) of the first ILVU of the cell; the start address position information (C_LVOBU_SA) of the last VOB of the cell; the end address position information (C_LVOBU_EA) of the last VOB of the cell, and the like.

FIG. 38 is a block diagram for explaining an example of the internal structure of a playback apparatus of the disc-shaped information storage medium (optical disc, etc.) according to the embodiment of the present invention. Referring to FIG. 38, information storage medium 1 records HD_DVD-Video content according to the embodiment of the present invention. Disc drive unit 1010 plays back the HD_DVD-Video content from this information storage medium 1, and transfers them to data processor unit 1020. A Video Object (VOB) as picture data in the HD_DVD-Video content includes a group of Video Object Unit (VOBU) data as a basic unit shown in FIG. 44(c), and navi pack a3 is allocated at the head in each VOB. Video data, audio data, and sub-picture data are respectively distributed and allocated in video packs a4, audio packs a6, and sub-picture (SP) packs a7, thus forming a multiplexed structure.

The embodiment of the present invention newly have graphics unit data, which is distributed and recorded in graphics unit (GU) packs a5. Demultiplexer 1030 in FIG. 38 demultiplexes a VOB formed by multiplexing these kinds of data into packets. Demultiplexer 1030 transfers video data recorded in video packs a4 to video decoder unit 1110, sub-picture data recorded in sub-picture packs a7 to sub-picture decoder unit 1120, graphics data recorded in graphics unit packs a5 to graphics decoder unit 1130, and audio data recorded in audio packs a6 to audio decoder unit 1140. Respective kinds of incoming data are decoded by decoder units 1110 to 1140, and are combined as needed in video processor unit 1040. Then, the combined data is converted into an analog signal via digital-to-analog converters 1320 and 1330, and the analog signal is output. MPU unit 1210 systematically manages a series of these processes, and temporarily stores data, which is required to be temporarily saved during processing, in memory unit 1220. ROM unit 1230 records processing programs to be processed by MPU unit 1210 and permanent data set in advance. In FIG. 38, information which is input from the user to the information playback apparatus is input via key inputs at key input unit 1310. However, the present invention is not limited to this, and key input unit 1310 may comprise a general remote controller.

FIG. 39 is a block diagram for explaining the internal structure of graphics decoder unit 1130 shown in FIG. 38 in detail. Graphics unit data demultiplexed and extracted by demultiplexer 1030 is temporarily
5 saved in graphics unit input buffer 1130a. The graphics unit data includes highlight information and graphics data and/or mask data, as will be described later with reference to FIG. 45. This highlight information is transferred to highlight decoder 1130b,
10 and is decoded. The graphics data and mask data are decoded to 256-color screen information in graphics decoder 1130e.

Furthermore, after selection of color palettes and a highlight process (e.g., a process for changing a
15 part of graphics data to be highlighted to a striking color) are applied to the decoded graphics data and/or mask data as needed, the graphics data and/or mask data are/is mixed with the decoded highlight data (e.g., picture data which has emphasized frame pixels at
20 positions to be highlighted, and transparent pixels at other positions) by mixer 1130d, and the decoded graphics data and/or mask data modified by the highlight data as needed are/is sent to mixer 1140a. This mixer 1140a mixes the decoded graphics data and/or
25 mask data with video data from video decoder unit 1110 and sub-picture data from sub-picture decoder unit 1120, thus forming a video output. Note that mixer

1140a in FIG. 39 is included in video processor unit 1040 in FIG. 38.

In the arrangement shown in FIG. 39, the decoded output of highlight decoder 1130b may control palette selector 1130g and/or highlight processor 1130h, so
5 that the highlight modification may be directly applied to the decoded output of graphics decoder 1130e (in this case, mixer 1130d can be omitted).

FIG. 40 is a view for explaining the concept of
10 imaginary video access unit (IVAU). An IVAU according to the embodiment of the present invention will be described below using FIG. 40. Each VOB of a movie in the conventional SD DVD-Video content is divided into Video Access Unit (VAU) data, as shown in FIG. 40(a).
15 By matching the boundary position of neighboring VOB data with that of neighboring VAU data, seamless playback between different VOB data can be attained.

In the HD_DVD-Video according to the embodiment of the present invention, as shown in FIG. 40(b),
20 "imaginary access units" IVAU2 to IVAUn (imaginary video access units 2 to n) are set in a period between VAU1 which includes I-picture that records a still picture, and VAU1 including I-picture that records a next still picture to be displayed. This is a
25 characteristic feature of this invention. As the setting method of access units, an interval between (VAU1 including) I-picture from which a still picture

starts and (VAU1 including) the next I-picture is
imaginarily finely time-divided for respective periods
of access units using as a unit the video frame time or
a time in an integer multiple of the video frame. A
5 Decoding Time Stamp (DTS) indicating the input timing
of a still picture to the decoder, and a Presentation
Time Stamp (PTS) indicating the display timing of a
still picture are set in advance for each still
picture. Since one video frame period is determined in
10 National Television System Committee (NTSC) and Phase
Alternation by Line (PAL), the timing of a boundary
position of the "imaginary access units" is calculated,
and the calculated timing is set as an imaginary PTS,
as shown in FIG. 40(c). Then, it can be (imaginarily)
15 considered as if a still picture is repetitively played
back and displayed for respective virtual access units.

In the embodiment of the present invention, as
shown in FIG. 40(d), one VOB is formed of an integer
number of "virtual access units". As a result, in the
20 embodiment of the present invention, a VOB display
time of each still picture becomes an integer multiple
of a video frame. In FIG. 40(c), a VAU (Video Access
Unit) includes one I-picture indicating a still
picture, but an IVAU does not include any I-picture.
25 Hence, no video data is included in the IVAU. That is,
each of a VOB formed by VAU1 to IVAU15 and that formed
by VAU16 to IVAU30 includes only one I-picture. By

contrast, a VOBV formed by IVAU30 to IVAU45 does not include any video data (I-picture).

Note that the embodiment of the present invention allows to define a VOBV having no video data. Also, the embodiment of the present invention inhibits one VOBV from having a plurality of I-picture data, and limits (constrains) so that one VOBV has one or less (including zero) I-picture. As can be seen from comparison of the positions in (c) and (d) of FIG. 40, one VOBV adopts a structure in which a VAU is (imaginarily) allocated ahead of an IVAU. As shown in FIG. 40(e), the first VOBV in an Interleaved Unit (ILVU) always has video data (I-picture that records a still picture).

FIG. 41 is a view for explaining a practical example of system parameters used in the embodiment of the present invention. In the system block diagram in the information playback apparatus shown in FIG. 38, memory unit 1220 is assigned fields for storing system parameters "0" to "23" shown in FIG. 41. Current menu language code information during playback (a language code that can be changed/set by the user and/or a command) is recorded in "SPRM0", and initial menu language code information (a setting language code of the playback apparatus which can be changed/set by only the user) is recorded in "SPRM21". Other kinds of information to be stored in other system parameters

are: audio stream number (ASTN) for TT_DOM in SPRM(1);
 sub-picture stream number (SPSTN) and on/off flag for
 TT_DOM in SPRM(2); angle number (AGLN) for TT_DOM in
 SPRM(3); title number (TTN) for TT_DOM in SPRM(4); VTS
 5 title number (VTS_TTN) for TT_DOM in SPRM(5); title PGC
 number (TT_PGCN) for TT_DOM in SPRM(6); Part_of_Title
 number (PTTN) for One_Sequential_PGC_Title in SPRM(7);
 Highlighted Button number (HL_BTNN) for Selection state
 in SPRM(8); Navigation Timer (NV_TMR) in SPRM(9);
 10 TT_PGCN for NV_TMR in SPRM(10); Player Audio Mixing
 Mode (P_AMXMD) for Karaoke in SPRM(11); Country (or
 Region) Code (CTY_CD) for Parental Management in
 SPRM(12); Parental Level (PTL_LVL) in SPRM(13); Player
 Configuration (P_CFG) for Video in SPRM(14); P_CFG for
 15 Audio in SPRM(15); Initial Language Code (INI_LCD) for
 AST in SPRM(16); Initial Language Code extension
 (INI_LCD_EXT) for AST in SPRM(17); INI_LCD for SPST in
 SPRM(18); INI_LCD_EXT for SPST in SPRM(19); and Player
 Region Code in SPRM(20).

20 FIG. 42 shows an example of a list of commands
 used in the embodiment of the present invention.
 Commands with command ID-1 = "000" to "110" are the
 same as those used in the conventional DVD-Video, but a
 command "Call INTENG" with command ID-1 = "111" is the
 25 one which is newly introduced in the embodiment of the
 present invention and uses an interactive engine.

FIG. 43 shows an example of a command list used in

the HD_DVD-Video content in the embodiment of the present invention. "Compare Field" shown in FIG. 43(a) is used to compare a value in a navigation parameter with a specific value specified by an operand of a
5 command. If this comparison result is true, a subsequent instruction is executed; if it is false, a subsequent instruction is skipped. This instruction is used in combination with other instruction groups. In FIG. 43(a), EQ means Equal; NE, Not Equal; GE, Greater
10 than or equal to; GT, Greater than; LE, Less than or equal to; LT, Less than; and BC, Bitwise Compare.

"Go To Option" in "Branch Field" shown in FIG. 43(b) is used to change the execution order of navigation commands in a pre-command area or post
15 command area, or a resume command area or cell command area. In FIG. 43(b), GoTo means transition to another navigation command, and Break means the end of execution of a navigation command in the pre-command area or post command area, or the resume command area
20 or cell command area. Also, SetTmpPML means confirmation of a temporal change in parental level, a change in parental level, and transition to a specific navigation command if possible.

"Link Option" in "Branch Field" shown in
25 FIG. 43(c) is used to start playback specified in one domain. In FIG. 43(c), LinkPGCN means the start of playback of a PGC of interest by directly designating a

program chain number (PGCN). LinkPTTN means the start of playback of a PTT of interest (or a chapter of interest) by directly designating a part_of_title number (PTTN). LinkPGN means the start of playback of a PG of interest by directly designating a program number (PGN). LinkCN means the start of playback of a cell of interest by directly designating a cell number (CN).

"Jump Option" in "Branch Field" shown in FIG. 43(d) is used to start specific playback after space movement. In FIG. 43(d), Exit means the end of playback. JumpTT means title playback start (when title number TTN is used). JumpVTS_TT means title playback start in a single VTS. CallSS means PGC playback start in a system space that stores resume information. JumpSS means playback start of a part_of_title included in a specific title in a single VTS. CallINTENG represents transfer of the control from a DVD-Video playback engine to an interactive engine (details are shown in FIG. 83).

"SetSystem Field" shown in FIG. 43(e) is used to set a system parameter value, and a mode and value of a general parameter. In FIG. 43(e), SetSTN means setting of a stream number (parameters to be set are SPRM(1), SPRM(2), and SPRM(3)). SetNVTMR means condition setting of the navigation timer (parameters to be set are SPRM(9) and SPRM(10)). SetHL_BTNN means setting of

the highlighted button number for a selection state (a parameter to be set is SPRM(8)). SetAMXMD means setting of an audio mixing mode of the playback apparatus for Karaoke (a parameter to be set is SPRM(11)). SetGPRMMD means setting of modes and values of general parameters (parameters to be set are GPRM(0) and GPRM(15)). SetM_LCD means setting of a menu description language code (a parameter to be set is SPRM(0)). SetRSMI means updating of resume information (parameters to be set are a CN, NV_PCK address, PGC control state, VTSN (Video Title Set Number), SPRM(4), SPRM(5), SPRM(6), SPRM(7), and SPRM(8)).

"Set Field" shown in FIG. 43(f) is used to execute a calculation on the basis of a specific value specified by an operand and a general parameter. The calculation includes the following two types:

- 1) Arithmetic operation
- 2) Bitwise operation

The calculation result is re-stored as a general parameter. In FIG. 43(f), Exp means an exponential calculation; Div, division; and Add, addition.

FIG. 44 shows the allocation of graphics units GU in a video object. The HD_DVD-Video content used in the embodiment of the present invention comply with the multiplexing rule of the MPEG system layer. That is, graphics unit data is segmented into every 2048-byte packs, and these packs are separately allocated. Upon

playback, graphics unit (GU) packs which are distributed and allocated in information storage medium 1 are collected to re-form a single graphics unit stream, as shown in (c) and (d) of FIG. 44. Graphics units can support graphics data corresponding to an HD picture at 16 : 9, SD picture at 16 : 9, SD picture at 4 : 3, and SD picture at letter box, and independent streams are formed in correspondence with the four types of pictures (HD picture at 16 : 9, SD picture at 16 : 9, SD picture at 4 : 3, and SD picture at letter box), as shown in FIG. 44(d).

FIG. 45 shows an example of the data structure in a graphics unit. As shown in FIG. 45, the data structure in the graphics unit includes header information b1, highlight information b2, mask data b3, and graphics data b4. Highlight information b2 includes general information b21, color palette information b22, and button information b23.

FIG. 46 shows an example of the header information content and general information content in the graphics unit. As shown in FIG. 46, the content of the header information include graphics unit size (GU_SZ) information, the start address (HLI_SA) information of the highlight information, and the start address (GD_SA) information of the graphics data. Of these content, the graphics unit size (GU_SZ) information indicates the overall size of the graphics unit shown

at the lower left position in FIG. 45. The start address (HLI_SA) information of the highlight information means an address to the start position of highlight information b2 with reference to the head position (that of header information b1) of the graphics unit shown at the lower left position in FIG. 45. Also, the start address (GD_SA) information of the graphics data means an address to the head position of graphics data b4 with reference to the head position (that of header information b1) of the graphics unit shown at the lower left position in FIG. 45.

Referring to FIG. 45, general information b21 in highlight information b2 has graphics unit playback end time (GU_PB_E_PTM) information, button offset number (BTN_OFN) information, information of the number (BTN_Ns) of buttons, information of the number (NSL_BTN_Ns) of numeral selection buttons, forced selection button number (FOSL_BTNN) information, forced determination button number (FOAC_BTNN) information, and the like. The graphics unit area is distributed and allocated as graphics unit (GU) packs, as described above using FIG. 44. This graphics unit pack (strictly speaking, a packet header in a graphics unit packet included in that pack) records in advance PTS (Presentation Time Stamp) information at which playback of the graphics unit starts. Using this PTS

information and the graphics unit playback end time (GU_PB_E_PTM) information, a graphics unit display time and effective time that allows execution (of a command) (both the start/end times completely match) are set.

5 Since the start/end time information uses a PTS/PTM, the time range can be set with a very high precision.

FIG. 47 is a view for explaining an image example of mask data and graphics data in the graphics unit. As the graphics data, as shown in FIG. 47, picture
10 information (bitmap data or compressed data of that bitmap) for one screen which allows 256-color expression by assigning 8 bits per pixel is recorded. The mask data indicates a position range on the screen where the user can designate command execution, and
15 sets only a screen region by assigning 1 bit per pixel. Since the mask data designates a region in the bitmap format using pixels, not only a plurality of regions located at positions separate from each other can be simultaneously set by masking, but also an arbitrary
20 shaped region can be finely set as a masking screen region using pixels, as shown in FIG. 47. This is also a characteristic feature of this embodiment. A plurality of mask data can be set, and a plurality of menu choices to the user can be supplied (FIG. 47
25 exemplifies three user's choices).

FIG. 48 shows an example of video composition including mask patterns. A screen to be presented to

the user can be generated by compositing main picture
(A) recorded in video packs a4 in FIG. 44(c), graphics
pattern (B) recorded as the graphics data, and mask
data (c) that can set a plurality of patterns, as shown
5 in FIG. 48.

In the embodiment of the present invention, as
shown in FIG. 45, the number n of mask data in a single
graphics unit matches the number n of pieces of button
information recorded in the highlight information, and
10 each mask data # n and button information # n have one-
to-one correspondence. That is, in m that satisfies $1 \leq m \leq n$, the m -th mask data from the top corresponds to
the m -th button information from the top. For example,
when the user highlights (designates) a region
15 designated by the m -th mask data on the screen by
operating a cursor key or the like on a remote
controller (not shown), button command b234 recorded in
 m -th button information b23 is executed in response to
that action. In this manner, each button information
20 # n links with each individual mask data # n . In order
to further facilitate access control to mask data,
button information # n records start address (address
from the head position of the header information to the
 n -th mask data start position in the lower left view of
25 FIG. 45) information b231 and data size information
b232 of corresponding mask data # n . In addition,
button information b23 records neighboring button

position information b233.

The data structure in color palette information b22 in highlight information b2 in FIG. 45 will be described below. Normal color palette b221 stores
5 color information of buttons when the menu screen is presented to the user first (before user selection). When the user selects (designates) a specific button, the display color of that button changes on the screen. Selection color palette b222 records the changed
10 display color of the button. Furthermore, when that button is set, and button command b234 corresponding to the button is about to be executed, the display color of the button can be set to be changed to a color indicating "set". Set color palette b223 has the set
15 display color of the button.

FIG. 49 shows another embodiment associated with the data structure of the graphics unit. The embodiment of FIG. 49 is characterized in that hot spot information is used in place of mask data. In
20 correspondence with this feature, in the example of FIG. 49, a plurality of normal color palettes e221, selection color palettes e222, and set color palettes e223 can be set. As the region designation method of each button information e23 on the screen, a region on
25 the screen can be designated by hot spot position information e233 in place of mask data. Furthermore, in the example of FIG. 49, a plurality of pieces of hot

spot position information e233 can be set for one button information e23, so that a plurality of regions which are separate from each other on the screen can correspond to one button information e23.

5 FIG. 50 is a view for explaining an example of the recording content of an advanced content recording area of the information content recorded on disc-shaped information storage medium (optical disc, etc.) 1 according to another embodiment of the present
10 invention. As shown in FIG. 50(d), advanced content recording area 21 in FIG. 50(c) is configured to include moving picture recording area 21B for recording moving picture data, animation/still picture recording area 21C for recording animation data and still picture
15 data, audio recording area 21D for recording audio data, font recording area 21E for recording font data, and markup/script language recording area 21A for recording information for controlling playback of these data (such information is described using a markup
20 language/script language/StyleSheet, and the like) (the area 21A is the head of the recording order of these areas as shown in FIG. 50).

 The information for controlling playback (recording content in area 21A) describes a playback
25 method (display method, playback sequence, playback switching sequence, selection of objects to be played back, etc.) of advanced content (including audio, still

picture, font/text, moving picture, animation, and the like) and/or DVD-Video content using a markup language, script language, and StyleSheet. For example, markup languages such as HTML (Hyper Text markup
5 Language)/XHTML (eXtensible Hyper Text markup Language), SMIL (Synchronized Multimedia Integration Language), and the like, script languages such as an ECMA (European Computer Manufacturers Association) script, Javascript (Java is the registered trade name),
10 and the like, StyleSheets such as CSS (Cascading Style Sheet), and the like, and so forth, may be used in combination.

markup/script language recording area 21A includes startup recording area 210A for recording startup
15 information, loading information recording area 211A for recording information of files to be loaded onto a buffer in a playback apparatus (see FIG. 90), playback sequence information recording area 215A for defining the playback order of video pictures for playing back
20 the HD_DVD video pictures stored in the expansion video object sets of the advanced title sets using a markup language or script language, markup language recording area 212A for recording the aforementioned markup languages, script recording area 213A for recording the
25 aforementioned script languages, and StyleSheet recording area 214A for recording the aforementioned StyleSheets.

FIG. 51 is a view for explaining an example of the recording content of an advanced HD video title set recording area of the information content recorded on disc-shaped information storage medium (optical disc, etc.) 1 according to still another embodiment of the present invention. An advanced HD video title set (AHDVTS: advanced VTS) shown in FIG. 51(d) is a video object which is specialized to be referred to from a markup language as one of the aforementioned advanced content.

As shown in FIG. 51(e), advanced HD video title set (AHDVTS) recording area 50 includes advanced HD video title set information (AHDVTSI) area 51 that records management information for all the content in advanced HD video title set recording area 50, advanced HD video title set information backup area (AHDVTSI_BUP) 54 that records the same information as in HD video title set information area 51 as backup data, and advanced title video object area (AHDVTSTT_VOBS) 53 that records video object (title picture information) data in an advanced HD video title set.

FIG. 52 shows an example of the data structure of advanced HD video title set information recorded in the advanced HD video title set recording area. This information is recorded together in file HVIA0001.IFO (or VTSA0100.IFO; not shown), and advanced HD video

title set information (AHDVTSI) area 51 shown in
FIG. 51(e) is divided into respective fields
(management information groups): advanced HD video
title set information management table (AHDVTSI_MAT)
5 510, advanced HD video title set PTT search pointer
table (AHDVTS_PTT_SRPT) 511, advanced HD video title
set program chain information table (AHDVTS_PGCIT) 512,
advanced HD video title set cell address table
(AHDVTS_C_ADT) 517, and time map information table
10 (TMAPIT) 519, as shown in FIG. 52.

Note that time map information table (TMAPIT) 519
is one field of advanced HD video title set information
(AHDVTSI) area 51, but it can be recorded in the same
file (HVIA0001.IFO in FIG. 2) as advanced HD video
15 title set information area 51 or in a file (e.g.,
HVM00000.MAP) independent from advanced HD video title
set information area 51.

Advanced HD video title set information management
table (AHDVTSI_MAT) 510 records management information
20 common to the corresponding video title set. Since
this common management information is allocated in the
first field (management information group) in advanced
HD video title set information (AHDVTSI) area 51, the
common management information in the video title set
25 can be immediately loaded. Hence, the playback control
process of the information playback apparatus can be
simplified, and the control processing time can be

shortened.

FIG. 53 shows an example of the data structure of the advanced HD video title set information management table (AHDVTSI_MAT) recorded in the advanced HD video title set information (AHDVTSI), and the recording content of category information (AHDVTS_CAT) stored in this management table. Advanced HD video title set information management table (AHDVTSI_MAT) 510 can store the following information as the common management information in the video title set. That is, as shown in FIG. 53, the advanced HD video title set information management table can store various kinds of information: an advanced HD video title set identifier (AHDVTS_ID), the end address (AHDVTS_EA) of the advanced HDVTS, the end address (AHDVTSI_EA) of the advanced HDVTSI, the version number (VERN) of the HD_DVD-Video standard, an AHDVTS category (AHDVTS_CAT), the end address (AHDVTSI_MAT_EA) of the AHDVTSI_MAT, the start address (AHDVTSTT_VOBS_SA) of the AHDVTSTT_VOBS, the start address (AHDVTS_PTT_SRPT_SA) of the AHDVTS_PTT_SRPT, the start address (AHDVTS_PGCIT_SA) of the AHDVTS_PGCIT, the start address (AHDVTS_C_ADT_SA) of the AHDVTS_C_ADT, the number (ATR1_AGL_Ns) of angles of a video object having attribute information 1 (ATR1), a video attribute (ATR1_V_ATR) of the video object having attribute information 1 (ATR1), the number (ATR1_AST_Ns) of audio

streams of the video object having attribute information 1 (ATR1), an audio stream attribute table (ATR1_AST_ATRT) of the video object having attribute information 1 (ATR1), the number (ATR1_SPST_Ns) of sub-
5 picture streams of the video object having attribute information 1 (ATR1), a sub-picture stream attribute table (ATR1_SPST_ATRT) of the video object having attribute information 1 (ATR1), a multi-channel audio stream attribute table (ATR1_MU_AST_ATRT) of the video
10 object having attribute information 1 (ATR1), and the like (attribute information 2 and attribute information 3 follow).

Of the information that can be stored in the management table (AHDVTSI_MAT) in FIG. 53, the start
15 address (HDVTSM_VOBS_SA) of an HDVTSM_VOBS included in a standard VTS need not exist since the advanced VTS does not include any HDVTSM_VOBS (or may be used as a reserved area). The start address (HDVTSM_PGCI_UT_SA) of the HDVTSM_PGCI_UT included in the standard VTS need
20 not exist since the advanced VTS does not include any HDVTSM_VOBS (or may be used as a reserved area). The start address (HDVTSM_C_ADT_SA) of the HDVTSM_C_ADT included in the standard VTS need not exist since the advanced VTS does not include any HDVTSM (or may be
25 used as a reserved area). The start address (HDVTSM_VOBU_ADMAP_SA) of the HDVTSM_VOBU_ADMAP included in the standard VTS need not exist since the

advanced VTS does not include any HDVTSM (or may be used as a reserved area). Furthermore, the start address (HDVTS_VOBU_ADMAP_SA) of the HDVTS_VOBU_ADMAP included in the standard VTS need not exist since the
5 advanced VTS includes the substitute time map information table (or may be used as a reserved area).

Note that the information (AHDVTS_CAT) indicating categories of the advanced VTS stored in advanced HD video title set information management table
10 (AHDVTSI_MAT) 510 in FIG. 53 is defined as follows:

AHDVTS_CAT = 0000b: no AHDVTS category is specified

AHDVTS_CAT = 0001b: reserved

AHDVTS_CAT = 0010b: advanced VTS with advanced
15 content

AHDVTS_CAT = 0011b: advanced VTS without advanced content

AHDVTS_CAT = other: reserved

The "advanced VTS with advanced content" whose
20 category is indicated by "AHDVTS_CAT = 0010b" basically represents an advanced VTS which is configured with the markup language. That is, in this category, the content producer assumes an "advanced VTS controlled by the markup language", and playback is permitted only
25 according to the control of the markup language but playback of the advanced VTS alone is not permitted. For example, when the content producer describes a

markup language that permits playback of an advanced VTS in a given period only under a specific condition, if playback of the advanced VTS alone is permitted, this period can be undesirably played back under a condition other than the specific condition. Such playback is inhibited for the advanced VTS of the category "AHDVTS_CAT = 0010b".

The "advanced VTS without advanced content" whose category is indicated by "AHDVTS_CAT = 0011b" basically represents an advanced VTS that allows playback of the advanced VTS alone without any markup language. This assumes an advanced VTS which maintains playback compatibility between other recording standards (to be referred to as a VR standard) such as DVD-VR/HDDVD-VR and the playback dedicated standard (to be referred to as a video standard) in the embodiment of the present invention. The video and VR standards have different standard content due to their different use applications (the video standard places an emphasis on interactiveness, and the VR standard places an emphasis on edit functions). By communizing a structurally simplified advanced VTS between the two standards, playback compatibility can be assured between the two standards having different purposes. For example, an information storage medium recorded in an advanced VTS mode in a recorder according to the VR standard can be played back by all playback apparatuses that can play

back the video standard.

FIG. 54 shows an example of the data structure of advanced HD video title set PTT search pointer table (AHDVTS_PTT_SRPT) 511 shown in FIG. 52. Advanced HD video title set PTT search pointer table (AHDVTS_PTT_SRPT) 511 includes various kinds of information: PTT search pointer table information (PTT_SRPTI) 511a having information of the end address (AHDVTS_PTT_SRPT_EA) of the AHDVTS_PTT_SRPT; and PTT search pointers (PTT_SRP) 511c having information of a program number (PGN).

Note that HDVTS_TTU_Ns indicating the number of TTU data of an HDVTS which is included in the standard VTS need not exist since the number of TTU data in the advanced VTS is fixed, i.e., 1 (or if it exists, a fixed value is recorded). The advanced VTS can be configured to include only one title (TT). In this case, "title unit search pointers (TTU_SRP) 411b each of which records information of the start address (TTU_SA) of a TTU (see FIG. 22)" need not exist since there is only one TTU (or if it exists, a fixed value is recorded).

FIG. 55 shows an example of the data structure of advanced HD video title set program chain information table (AHDVTS_PGCIT) recorded in the advanced HD video title set information (AHDVTSI). As shown in FIG. 55, advanced HD video title set program chain information

table (AHDVTS_PGCIT) 512 also records information of advanced HD video title set PGCi information table (AHDVTS_PGCITI) 512a including information of the number (AHDVTS_PGCi_SRP_Ns) of AHDVTS_PGCi_SRP data and the end address (AHDVTS_PGCIT_EA) of the AHDVTS_PGCIT. Also, AHDVTS_PGCi search pointer (AHDVTS_PGCi_SRP) 512b records information of the start address (AHDVTS_PGCi_SA) of the AHDVTS_PGCi together with the aforementioned AHDVTS_PGC category (AHDVTS_PGC_CAT).

Note that a plurality of PGCs can be prepared in the advanced VTS, but there is no function to control the connection relationship upon playback using navigation commands. For this reason, basically, there is only one PGC and one sequential playback of the advanced VTS is managed. In this case, the value of AHDVTS_PGCi_SRP_Ns is fixed, i.e., 1, and one each of search pointer (AHDVTS_PGCi_SRP) 512b and PGC information (AHDVTS_PGCi) 512c are present.

FIG. 56 shows an example of the data structure of program chain general information (PGC_GI) included in program chain information (corresponding to AHDVTS_PGCi in, e.g., FIG. 55). As shown in FIG. 56, the program chain information (PGCI) recorded in PGC information (AHDVTS_PGCi) 512c includes four fields (four management information groups), i.e., program chain general information (PGC_GI) 50, program chain program map (PGC_PGMAP) 52, cell playback information table

(C_PBIT) 53, and cell position information table (C_POSIT) 54. Note that program chain command table (PGC_CMDT) 51 included in the PGCI of the standard VTS (FIG. 34) need not exist in the advanced VTS (or may be
5 used as a reserved area).

As shown in FIG. 56, program chain general information (PGC_GI) 50 records various kinds of information including PGC content (PGC_CNT), a PGC playback time (PGC_PB_TM), a PGC audio stream control
10 table (PGC_AST_CTLT), a PGC sub-picture stream control table (PGC_SPST_CTLT), PGC navigation control (PGC_NV_CTL), a PGC sub-picture palette (PGC_SP_PLT), the start address (PGC_PGMAP_SA) of the PGC_PGMAP, the start address (C_PBIT_SA) of the C_PBIT, and the start
15 address (C_POSIT_SA) of the C_POSIT.

Note that the PGC user operation control (PGC_UOP_CTL) included in the standard VTS does not exist since the user operation control in the advanced VTS is made based on the markup language (or if it
20 exists, PGC_UOP_CTL records a fixed value "00...00b". Also, the PGC graphics unit stream control table (PGC_GUST_CTLT) included in the standard VTS does not exist since no graphics unit is used in the advanced VTS (or may be used as a reserved area). The start
25 address (PGC_CMDT_SA) of the PGC_CMDT included in the standard VTS does not exist since no command table (PGC_CMDT) exists in the advanced VTS (or used as a

reserved area).

Note that the example of the PGC_GI shown in FIG. 56 exemplifies RSM&AOB_CAT at its end. However, RSM&AOB category information (RSM&AOB_CAT) included in the standard VTS, i.e., RSM permission information, Audio selection information, and Audio Number information need not exist since the RSM information is controlled by the markup language and no Audio information is available in the advanced VTS (or may be used as a reserved area).

FIG. 57 shows an example of the data structure in advanced HD video title set cell address table (AHDVTS_C_ADT) 517 shown in FIG. 52. Advanced HD video title set cell address table (AHDVTS_C_ADT) 517 includes various kinds of information: advanced HD video title set cell address table information (AHDVTS_C_ADTI) 517a having the number (AHDVTS_VOB_Ns) of VOB data in an AHDVTS_VOBS and the end address (AHDVTS_C_ADT_EA) of the AHDVTS_C_ADT; and a plurality of pieces of advanced HD video title set cell piece information (AHDVTS_CPI) 517b each including a VOB_ID number (AHDVTS_VOB_IDN) of an AHDVTS_CP, a Cell_ID number (AHDVTS_C_IDN) of the AHDVTS_CP, the start address (AHDVTS_CP_SA) of the AHDVTS_CP, and the end address (AHDVTS_CP_EA) of the AHDVTS_CP.

FIG. 58 shows an example of the data structure in time map information table (TMAPIT) 519 shown in

FIG. 52. Time map information table (TMAPIT) 519 includes time map information table information (TMAPITI) 519a, time map information search pointers (TMAPI_SRP) 519b, and a plurality of pieces of time information (TMAPI) 519c. Time map information table information (TMAPITI) 519a includes the number of pieces of time map information (TMAPI) 519c included in this time map information table (TMAPIT) 519, and the end address information of this time map information table (TMAPIT) 519. Time map information search pointers (TMAPI_SRP) 519b exist as many as the number of pieces of time map information (TMAPI) 519c, and each pointer records the start address where corresponding time map information (TMAPI) 519c is recorded.

FIG. 59 shows an example of the data structure of time map information (TMAPI) 519c shown in FIG. 58. Time map information (TMAPI) 519c includes time map general information (TMAP_GI) 519c1, time entry table (TM_ENT) 519c2, VOB entry table (VOBU_ENTT) 519c3, ILVU_ADR entry table (ILVU_ADR_ENTT) 519c4, and ENT_VOBN table (ENT_VOBNT) 519c5.

Time map general information (TMAP_GI) 519c1 includes TMAP_TYPE indicating the type of blocks which form this time map information (TMAPI) 519c, BLK_ADR indicating the start address of a contiguous or interleaved block, TMU indicating the time duration of

a time entry, VOB_Ns indicating the number of VOB data to be referred to by this time map information (TMAPI) 519c, ILVU_Ns indicating the number of ILVU data per VOB to be referred to by this time map information (TMAPI) 519c, and VOBU_ENT_Ns indicating the number of all VOBU data to be referred to by this time map information (TMAPI) 519c.

In the TMAP_GI in FIG. 59, when blocks that form time map information TMAPI include a contiguous block, "0b" is recorded in TMAP_TYPE; when blocks that forms time map information TMAPI include an interleaved block, "1b" is recorded in TMAP_TYPE. The time duration of the time entry is constant in the time map information, and can be set to be a value, e.g., TMU = 10 sec.

Furthermore, VOB_Ns indicating the number of VOB data to be referred to by the TMAPI indicates the number of VOB data formed by contiguous blocks when blocks that form the TMAPI are contiguous blocks (i.e., TMAP_TYPE = 0b). On the other hand, VOB_Ns indicating the number of VOB data to be referred to by the TMAPI indicates the number of VOB data that form interleaved blocks when blocks that form the TMAPI are interleaved blocks (i.e., TMAP_TYPE = 1b).

FIG. 60 shows an example of the data structure of time entry table (TM_ENT) 519c2 shown in FIG. 59. Time entry table (TM_ENT) 519c2 includes one or more time

entry numbers (TM_EM_Ns) 519c21, and one or more time entries (TM_EN) 519c22. Note that the time entries are allocated for each VOB. More specifically, in the example of FIG. 60, the time entries are allocated in ascending order of VOB#p like time entry (TM_EN) 519c22 group of VOB#1, time entry (TM_EN) 519c22 group of VOB#2, ..., time entry (TM_EN) 519c22 group of VOB#p.

Each time entry number (TM_EM_Ns) 519c21 records TM_EN_Ns indicating the number of time entries (TM_EN) 519c22. Each time entry 519c22 includes VOBU_ENTN indicating the number of VOBU entry (VOBU_ENT) 519c31 designated by the time entry, TM_DIFF indicating the time difference between the time of the time entry calculated based on TMU and the start time of the VOBU designated by the time entry, and TM_EN_ADR indicating an offset address of a Block (a VOB period with valid TMAPI) from the head position.

FIG. 61 shows an example of the data structures of VOBU entry table (VOBU_ENTT) 519c3, ILVU_ADR entry table (ILVU_ADR_ENTT) 519c4, and ENT_VOBN table (ENT_VOBNNT) 519c5 shown in FIG. 59. As shown in FIG. 61, VOBU entry table (VOBU_ENTT) 519c3 includes VOBU entries (VOBU_ENT) 519c31. Each VOBU entry (VOBU_ENT) 519c31 includes 1STREF_SZ indicating the size (which can be indicated by the number of packs) of 1st Reference Picture data (i.e., first I-picture or equivalent data) included in a VOBU, VOBU_PB_TM

indicating the VOB playback time, and VOBU_SZ indicating the size (which can be indicated by the number of packs) of the VOBU.

ILVU_ADR entry table (ILVU_ADR_ENTT) 519c4 includes ILVU_ADR entries (ILVU_ADR_ENT) 519c41. Each ILVU_ADR entry (ILVU_ADR_ENT) 519c41 includes ILVU_ADR indicating an offset address from the head of an Interleaved block for each ILVU address.

ENT_VOBN table (ENT_VOBN_T) 519c5 which indicates a list of VOB data that refer to time map information (TMAPI) 519c includes entry VOB numbers (ENT_VOBN) 519c51. Each entry VOB number (ENT_VOBN) 519c51 includes ENT_VOBN indicating a VOB number to be referred to. Note that ENT_VOBN is described in the order of VOB data that refer to time map information (TMAPI) 519c, and correspondence between the time map and VOB is indicated using the VOB number.

FIG. 62 is a flowchart for explaining an example of the playback sequence of an advanced VTS (AHDVTS in FIGS. 51, 74, 79, and the like) according to the content of information (Application Type) included in management information (e.g., AHDVTS_CAT in FIG. 53). When playback of an advanced VTS is designated, the playback apparatus (FIG. 72, etc.) checks the value of AHDVTS_CAT stored in AHDVTSI_MAT 510. If AHDVTS_CAT = 0011b (YES in step ST620), since this advanced VTS to be played back is a video object without any advanced

content, i.e., playback is controlled based on only data in advanced HD video title set recording area 50 (AHDVTS) in place of the markup/script language, playback can be done based on data of this AHDVTS (a
5 sole playback process of the advanced VTS).

If the value of AHDVTS_CAT is other than "0011b" (e.g., "0010b") (NO in step ST620), since this advanced VTS is a video object with advanced content, playback must be done on the basis of the markup/script language
10 required to control this video object. If not, playback of this video object becomes different from that the content producer intended. Hence, the playback apparatus (FIG. 72, etc.) searches for a markup/script language file associated with this video
15 object. If such file is found (YES in step ST622), the video object is played back on the basis of the description of the markup/script language of that file (an execution process of the markup/script language). If no markup/script language file associated with the
20 video object is found (NO in step ST622), since data required to control playback are not sufficiently prepared, the process ends without playback.

FIG. 63 shows the configuration of a navigation pack (NV_PCK) allocated at the head of each EVOBU in an
25 enhanced video object (EVOB) which is to be referred to by an advanced VTS according to the embodiment of the present invention. The navigation pack includes a

presentation information packet (PCI_PKT) and data search information packet (DSI_PKT), and respective packets store information shown in FIGS. 64 and 65.

FIG. 64 shows an example of the content of the presentation control information (PCI) as playback control information. The presentation control information includes playback control general information (PCI_GI), non-seamless angle position information (NSML_AGLI) which includes the start position information of each angle and does not require any seamless playback upon angle switching, and recording information (RECI). Note that the recording information (RECI) can record specific codes such as a country code, copyright holder code, recording date, recording number, and the like in association with the content of recorded video, audio, and sub-picture data.

The playback control general information (PCI_GI) includes control pack position information (NV_PCK_LBN) indicated by a logical block number (LBN) from the head of a VOBS, EVOBU category information (EVOBU_CAT) including analog copy control information, information (EVOBU_S_PTM) indicating the playback start time and information (EVOBU_E_PTM) indicating the playback end time of an EVOBU, EVOBU playback sequence end time information (EVOBU_SE_E_PTM) indicating information of the playback end time when video playback ends in response to a sequence end code in the EVOBU, and cell

elapsed time information (C_ELTM) indicating an elapsed time in a cell of the EVOBU.

Note that "EVOBU playback start time information (EVOBU_S_PTM)", "EVOBU playback end time information (EVOBU_E_PTM)", and "cell elapsed time information (C_ELTM)" in parentheses in the playback control general information (PCI_GI) are option information, and can be omitted depending on embodiments.

FIG. 65 shows the content of the data search information (DSI) as data search information. The data search information includes data search general information (DSI_GI), seamless playback information (SML_PBI) as information required to make seamless playback without interrupting interleaved units (ILVU) which are interleaved, seamless angle position information (SML_AGLI) that describes a jump address of an interleaved unit of each angle as information required to switch angles without interrupting playback, and sync information (SYNCI) indicating position information of audio and sub-picture packs to be played back synchronously with video data.

The data search general information (DSI_GI) includes control pack playback time information (NV_PCK_SCR) indicated by system clock reference (SCR)-based time information, control pack position information (NV_PCK_LBN) indicated by a logical block number (LBN) from the head of a VOBS, EVOBU adaptation

information (EVOBU_ADP_ID) as information indicating if
a disc to which the standard is applied is a read-only
disc (DVD-ROM) or a writable disc (DVD-R or the like),
EVOBU_EVOB number information (EVOBU_EVOB_IDN: not
5 shown) indicating an ID number of an EVOB that includes
the DSI of interest, EVOBU cell number information
(EVOBU_C_IDN) indicating an ID number of a cell that
includes the DSI of interest, EVOBU attribute number
information (EVOBU_ATRN) indicating the number of
10 attribute information of an EVOB to which the EVOBU of
interest belongs, and cell elapsed time information
(C_ELTM) indicating an elapsed time in a cell of the
EVOBU.

Note that "cell elapsed time information (C_ELTM)"
15 in parentheses in the data search general information
(DSI_GI) is option information, and can be omitted
depending on embodiments.

FIG. 66 is a view for explaining an example of the
configuration of an advanced VTS (AHDVTS). Since the
20 advanced VTS is basically controlled by a markup
language, it requires a simple structure that allows
easy control by the markup language. FIG. 66 shows an
example of such structure. The advanced VTS includes
only one VTS. This VTS includes only one Title. This
25 Title includes only one PGC, which includes one or more
PTT data and one or more Cells. Video object VTS_EVOBS
is referred to by Cells in one-to-one correspondence.

Note that no navigation commands that can be recorded in VTSI and NV_PCK are available in the advanced VTS. The content production process which is complicated due to coexistence of control based on the markup language and that based on navigation commands in the advanced VTS, and the load on the manufacture of the playback apparatus can be avoided.

Furthermore, the standard VTS accesses a video object using VOB search information included in NV_PCK. The advanced VTS does not use any VOB search information in NV_PCK (which need not exist), and newly adds time map information. Upon accessing a video object in accordance with an instruction of the markup language, precise access can be done from an arbitrary location using the time map information.

Note that an attribute number "#n" which identifies an attribute (Attribute #n) assigned to a plurality of EVOBU data corresponding to each EVOB in FIG. 66 can be designated by the EVOBU attribute number information (EVOBU_ATRN) shown in FIG. 65.

FIG. 67 shows time map elements according to the embodiment of the present invention. That is, as a time element of a time map, a starting point of a description (time map unit) is available. The head of a PGC can be defined as a starting point for the PGC, and the head of a VOB can be defined as a starting point for the VOB. A time map time interval may be

fixed to 600 video fields (corresponding to 10 sec) in NTSC, or the time map time interval can be set in the time unit (e.g., the range of 1 to 255 sec in increments of 1 sec). Furthermore, upon forming ILVU data, a time map may be described in only the path of the first ILVU (e.g., only the path of angle number 1 in a multi-angle block) or time maps may be described in all ILVU data.

As for an offset address of a time map, the start address of each VOB can be described. More specifically, the offset address can be described using a relative logical block number from the first logical block of a VTSTT_VOBS, or the offset address can be described using a relative logical block number from the first logical block number of the file of interest (In this case, the file at the current timing may be divided into a plurality of files as needed according to the set time maps). Furthermore, a VOB number quoted by a time map can be associated with a VOB entry, which can be used as acquisition information of corresponding I-picture data and/or time information of this I-picture data.

FIG. 68 shows an example of practical elements of the time map according to the embodiment of the present invention. A block address (BLK_ADR) designates the start address of a contiguous or interleaved block using an offset address from the head of a VTSTT_VOBS.

A time entry address (TM_EN_ADR) of a contiguous block (single VOB) can be designated using an offset address from the head of a block. Also, a time entry address of an interleaved block (a plurality of VOB data) can be designated using an offset address from the head of a block (by the same method as in a single VOB) or time entry tables can be described as many as the number of VOB data. A time unit (TMU) is fixed to a constant value (e.g., 10 sec) in a single VTSTT_VOBS.

An interleaved unit address (ILVU_ADR) can designate the address of each ILVU using an offset address from the head of an Interleaved block. Furthermore, a VOBU size (VOBU_SZ) can describe the size of each VOBU using the number of packs in that VOBU. A first reference picture size (1STREF_SZ) can describe the size of I-picture data of each VOBU using the number of packs.

FIG. 69 shows a case having different playback paths so as to explain the time map according to the embodiment of the present invention. As shown in FIG. 69, disc 1 records two different playback paths (A) and (B). (A) is, for example, the director's cut version of a movie, and (B) is, for example, the theatrical release version. In this example, (A) and (B) include the same introductory chapter (VOB#1) and ending chapter (VOB#4), but have different main chapters (VOB#2 or VOB#3). At this time, in order to

improve the recording efficiency on the disc in practice, the introductory chapter (VOB#1) and ending chapter (VOB#4) are used as a common playback path, and objects (VOB#2 and VOB#3) of different playback paths are independently recorded. However, if these objects are recorded intact, one of (A) and (B) cannot be read out in time upon playback depending on the manner of recording, thus interrupting its playback. In order to solve this problem, as shown in the lowermost column in FIG. 69, respective VOB data (VOB#2 and VOB#3) are broken up into smaller units, and these units are recorded alternately (i.e., so-called interleaved recording), thus implementing seamless playback. A unit of this interleaved recording is an interleaved unit (ILVU).

Note that an interval in which playback data of VOB#1 or VOB#4 are contiguously allocated is defined as a contiguous block, and an interval in which playback data of VOB#2 and VOB#3 are alternately allocated is defined as an interleaved block.

FIG. 70 is a view for explaining the time map of the ILVU interval. In order to form the time map of the interleaved-recorded ILVU interval (the interval of VOB#2 and VOB#3 in FIG. 70(a)) described using FIG. 69, time entries (2-1, 2-2,... of VOB#2, and 3-1, 3-2,... of VOB#3) are assigned to VOB#2 and VOB#3 as playback paths at predetermined time intervals (e.g., 10-sec

time intervals) (FIG. 70(b)), and hold designated addresses. After interleaved allocation, the addresses of the respective time entries are re-designated as offset addresses from the head of the interleaved block (FIG. 70(c)).

FIG. 71 shows an example that generalizes the time map including the interleaved block interval that has been explained using FIG. 70. As shown in FIG. 71, a VTSTT_VOBS of a playback object includes a contiguous block of VOB#p, an interleaved block formed by VOB#q and VOB#r, and a contiguous block of VOB#s (in the example of FIGS. 70 to 72, VOB#p = VOB#1, VOB#q = VOB#2, VOB#r = VOB#3, and VOB#s = VOB#4).

This time map is configured for each block. For this purpose, the start addresses of respective blocks are designated as offset addresses (BLK_ADR) from the head of the VTSTT_VOBS. With this configuration, a time map of each block describes position information to have the head of that block as a starting point, and information that forms the time map is completed in the block.

The address of each time entry (TM_EN#) designated by a predetermined time interval (TMU) (e.g., 10 sec) is indicated by an offset address (TM_EN_ADR) from the head of each block, and is stored as a time entry table (not shown). At this time, if the block of interest is an interleaved block, time entries (TM_EN#q1,

TM_EN#q2,..., and TM_EN#r1, TM_EN#r2,... in this case) as many as the number of VOB data that form the block are separately stored in respective time entry tables (not shown).

5 When the block that forms the time map is an interleaved block, the start addresses (ILVU_ADR) of interleaved units alternately allocated in the interleaved block are designated by offset addresses from the head of the block. With this information, the
10 start position of each ILVU can be easily detected, and ILVU data to be contiguously played back can be seamlessly switched and played back (each ILVU size (ILVU_SZ) can be described in, e.g., TMAP_GI in FIG. 59 (not shown)).

15 As information of a VOB data that stores actual playback information, each time map includes the number of all VOB data (VOBU_Ns; not shown) stored in each block, the size (VOBU_SZ) and playback time (VOBU_PB_TM; not shown) of each VOB, the end address
20 information (1STREF_SZ) of first reference picture (first I-picture) data, and the like. With this information, target data is accessed. The time map may have the end address information (2NDREF_SZ, 3RDREF_SZ; neither are shown) of each of second reference picture
25 (I- or P-picture other than the first reference picture) data and third reference picture (I- or P-picture other than the first and second reference

pictures) data in addition to the first reference picture.

FIG. 72 is a block diagram for explaining an example of the internal structure of a playback apparatus (advanced VTS compatible DVD-Video player) according to another embodiment of the present invention. This DVD-Video player plays back and processes the recording content from information storage medium 1 shown in FIGS. 1, 50, 51, 73, 74, 79, and the like, and downloads and processes advanced content from a communication line (e.g., the Internet or the like).

The DVD-Video player shown in FIG. 72 comprises DVD-Video playback engine (DVD_ENG) 100, interactive engine (INT_ENG) 200, disc unit (disc drive) 300, user interface unit 400, and the like. DVD-Video playback engine 100 plays back and processes an MPEG2 program stream (DVD-Video content) recorded on information storage medium 1. Interactive engine (INT_ENG) 200 plays back and processes advanced content. Disc unit 300 reads out the DVD-Video content and/or advanced content recorded on information storage medium 1. User interface unit 400 supplies an input by the user of the player (user operation) to the DVD-Video player as a user trigger.

Basically, when a standard VTS is to be played back (standard VTS playback state), the user input is

supplied to the DVD-Video playback engine; when an advanced VTS is to be played back (advanced VTS playback state), the user input is supplied to the interactive engine. Even when the advanced VTS is to be played back, a predetermined user input can be directly supplied to the DVD-Video playback engine.

Interactive engine (INT_ENG) 200 comprises an Internet connection unit. This Internet connection unit serves as communication means that connects server unit 500 or the like via a communication line (Internet or the like). Furthermore, interactive engine (INT_ENG) 200 is configured to include buffer unit 209, parser 210, XHTML/SVG/CSS layout manager 207, ECMAscript interpreter/DOM manipulator/SMIL interpreter/timing engine/object (interpreter unit) 205, interface handler 202, media decoders 208a/208b, AV renderer 203, buffer manager 204, audio manager 215, network manager 212, system block 214, persistent storage 216, and the like.

In the block arrangement of FIG. 72, DVD-Video playback controller 102, DVD-Video decoder 101, DVD system block 103, interface handler 202, parser 210, interpreter unit 205, XHTML/SVG/CSS layout manager 207, AV renderer 203, media decoders 208a/208b, buffer manager 204, audio manager 215, network manager 212, system clock 214, and the like can be implemented by a microcomputer (and/or hardware logic) which serves as

the functions of respective blocks by an installed program (firmware; not shown). A work area used upon executing this firmware can be assured using a semiconductor memory (and a hard disc as needed; not shown) in the block arrangement.

DVD-Video playback engine (DVD_ENG) 100 is a device for playing back DVD-Video content recorded on information storage medium 1 shown in FIG. 1 and the like, and is configured to include DVD-Video decoder 101 for decoding the DVD-Video content loaded from disc unit 300, DVD-Video playback controller 102 for making playback control of the DVD-Video content, DVD system clock 103 for determining the decode and output timings in the DVD-Video decoder, and the like.

DVD-Video decoder 101 has a function of decoding main picture data, audio data, and sub-picture data read out from information storage medium 1 shown in FIG. 1 and the like, and outputting the decoded video data (obtained by mixing the main picture data and sub-picture data, etc.) and audio data. That is, the player shown in FIG. 72 can play back video data, audio data, and the like with the MPEG2 program stream structure in the same manner as a normal DVD-Video player.

In addition, DVD-Video playback controller 102 can control playback of the DVD-Video content in accordance with a "DVD control signal" output from interactive

engine (INT_ENG) 200. More specifically, when a given event (e.g., menu call or title jump) has occurred in DVD-Video playback engine 100 upon DVD-Video playback, DVD-Video playback controller 102 can output a "DVD
5 trigger" signal indicating the playback condition of the DVD-Video content to interactive engine (INT_ENG) 200. In this case (simultaneously with output of the DVD trigger signal or at an appropriate timing before and after the output), DVD-Video playback controller
10 102 can output a "DVD status" signal indicating property information (e.g., an audio language, sub-picture subtitle language, playback operation, playback position, various kinds of time information, disc content, and the like set in the player) of the DVD-
15 Video player to interactive engine (INT_ENG) 200.

Interface handler 202 receives a "user trigger" corresponding to a user operation (menu call, title jump, play start, play stop, play pause, or the like) from user interface unit 400. Interface handler 202
20 transmits the received user trigger to interpreter unit 205 as a corresponding "event". For example, the markup language describes the following instructions for this "event".

1 (issue a "command" corresponding to a user
25 operation. That is, the same command as the user operation is transmitted to the DVD-Video playback engine as a DVD control signal.

2 (issue a "command" different from a user operation. That is, the user action is substituted by another operation in accordance with an instruction of the markup language.

5 3 (ignore user trigger. That is, a user event is inhibited since, for example, the user may designate a DVD-Video playback process which is not designed by the content provider.

Note that the content of the user trigger signal transmitted to interface handler 202 may be transmitted to AV renderer 203 as an "AV output control" signal. As a result, for example, when the user has changed the content or window size or has shifted its display position using a cursor key of a remote controller (not shown), a user trigger signal based on this operation is output to AV renderer 203 as a corresponding AV output control signal. In addition, when a user trigger signal which indicates switching between a video/audio output from DVD-Video playback engine 100 and that from interactive engine 200 is sent to AV renderer 203, the video/audio output can be switched in response to the user operation.

Interface handler 202 exchanges a "DVD status" signal, "DVD trigger" signal, and/or "DVD control" signal with DVD-Video playback controller 102, or exchanges a "user trigger" signal with user interface unit 400. Furthermore, interface handler 202 exchanges

"event", "property", "command", and "control" signals with interpreter unit 205.

That is, interface handler 202 can do the following.

- 5 1. Interface handler 202 transmits a "DVD trigger" signal which indicates the operation of DVD-Video playback engine 100 from DVD-Video playback engine 100, or a "user trigger" which indicates the user operation from user interface unit 400 to
10 interpreter unit 205 as an "event".
2. Interface handler 202 transmits a "DVD status" signal which indicates the playback status of DVD-Video playback engine 100 from DVD-Video playback engine 100 to interpreter unit 205 as a "property". At this time,
15 DVD status information is saved in property buffer 202a of interface handler 202 as needed.
3. Interface handler 202 outputs a "DVD control" signal to control playback of DVD-Video playback engine 100 to DVD-Video playback engine 100, an "AV output
20 control" signal to switch video and audio data to AV renderer 203, a "buffer control" signal to load/erase the content of buffer 209 to buffer manager 204, an
 "update control" signal to download update audio data to audio manager 215, and a "media control" signal to
25 instruct decoding of various media to media decoders 208a/208b, in accordance with the content of a
 "command" signal from Interpreter unit 205.

4. Interface handler 202 measures information of DVD system clock 103 in DVD-Video playback engine 100 using its DVD timing generator 202b, and transmits the measurement result to media decoders 208a/208b as a
5 "DVD timing" signal. That is, media decoders 208a/208b can decode various media in synchronism with system clock 103 of DVD-Video playback engine 100.

As described above, interface handler 202 has a function of parsing and interpreting advanced content,
10 and then exchanging control signals and the like between DVD-Video playback engine 100 and interactive engine 200.

Interface handler 202 is configured to exchange a first signal and also a second signal on the basis of
15 the content which are parsed by parser 210 and are interpreted by interpreter unit 205, or a user trigger from an input device (e.g., a remote controller; not shown). In other words, interface handler 202 controls the output states of video and audio signals by AV
20 renderer 203 on the basis of at least one of the first signal exchanged with DVD-Video playback controller 102, and the second signal exchanged with interpreter unit 205.

Note that the first signal pertains to the
25 playback status of information storage medium 1, and corresponds to the "DVD control" signal, "DVD trigger" signal, "DVD status" signal, and the like. The second

signal pertains to the content of the advanced content, and corresponds to the "event" signal, "command" signal, "property" signal, "control" signal, and the like.

5 Interface handler 202 is configured to execute processes corresponding to user triggers in accordance with the markup language. AV renderer 203 is configured to mix video/audio data generated by media decoders 208a/208b with that played back by DVD-Video
10 playback engine 100 on the basis of the execution results of the processes corresponding to user triggers, and to output mixed data. Alternatively, AV renderer 203 is configured to select one of video/audio data generated by media decoders 208a/208b and that
15 played back by DVD-Video playback engine 100 on the basis of the execution result of the "command" in interface handler 202, and to output the selected video/audio data.

 Generally speaking, parser 210 parses the markup
20 language indicating playback control information, which is included in advanced content acquired from information storage medium 1 or advanced content downloaded from the Internet or the like. The markup language is configured by a combination of markup
25 languages such as HTML/XHTML, SMIL, and the like, script languages such as ECMAScript, Javascript, and the like, and stylesheets such as CSS and the like, as

described above. Parser 210 has a function of transmitting an ECMAScript module to an ECMAScript interpreter, a SMIL module to a SMIL interpreter of interpreter unit 205, and an XHTML module to
5 XHTML/SVG/CSS layout manager 207 in accordance with the parsing result.

The ECMAScript interpreter interprets the aforementioned ECMAScript module and follows its instructions. That is, the ECMAScript interpreter has
10 a function of issuing a "command" signal used to control respective functions in interactive engine 200 to interface handler 202 in correspondence with an "event" signal sent from interface handler 202 or a "property" signal read from property buffer 202a of
15 interface handler 202. At this time, the ECMAScript interpreter issues a "command" signal to DVD-Video playback engine 100 or a "media control" signal to media decoders 208a/208b at the timings designated by the markup language in accordance with the time
20 measured by system clock 214. In this manner, the control operation of DVD-Video playback engine 100 and various media control operations (decode control of audio, still picture/animation, text/font, and movies, etc.) can be achieved.

25 The SMIL timing engine interprets the aforementioned SMIL module and follows its instructions. That is, the SMIL timing engine has a

function of issuing a "control" signal to interface handler 202 or media decoders 208a/208b in correspondence with an "event" signal sent from interface handler 202 or a "property" signal read from property buffer 202a of interface handler 202 in accordance with system clock 214. With this function, control of the DVD-Video playback engine 100 and decoding of various media (audio, still picture/animation, text/font, movie) can be achieved at desired timings. That is, the SMIL timing engine can operate based on system clock 214 in accordance with the description of the markup language, or can operate on the basis of DVD system clock 103 from DVD timing generator 202b.

XHTML/SVG/CSS layout manager 207 interprets the aforementioned XHTML module and follows its instructions. That is, XHTML/SVG/CSS layout manager 207 outputs a "layout control" signal to AV renderer 203. The "layout control" signal includes information associated with the size and position of a video screen to be output (this information often includes information associated with a display time such as display start, end, or continuation), and information associated with the level of audio data to be output (this information often includes information associated with an output time such as output start, end, or continuation). Also, text information to be displayed,

which is included in the XHTML module, is sent to media decoders 208a/208b, and is decoded and displayed using desired font data.

Practical methods of parsing and interpreting
5 markup and script languages can adopt the same methods as parsing/interpretation in state-of-the-art techniques such as HTML/XHTML, SMIL, and the like or ECMAScript, Javascript, and the like (the hardware used is the microcomputer that has been mentioned at the
10 beginning of the description of FIG. 72). Note that commands and variables described in scripts are different since objects to be controlled are different. The markup language used upon practicing the present invention uses unique commands and variables associated
15 with playback of the DVD-Video content and/or advanced content. For example, a command that switches the playback content of the DVD-Video content or advanced content in response to a given event is unique to the markup or script language used in the embodiment of the
20 present invention.

As another example of commands and variables unique to the markup or script language, those which are used to change the video size from DVD-Video playback engine 100 and/or interactive engine 200 and
25 to change the layout of that video data are available. A change in video size is designated using a size change command and a variable that designates the size

after change. A change in video layout is designated by a display position change command and a variable that designates the coordinate position or the like after change. When objects to be displayed overlap on the screen, variables that designate z-ordering and transparency upon overlaying are added.

As still another example of commands and variables unique to the markup or script language, those which are used to change the audio level from DVD-Video playback engine 100 and/or interactive engine 200 or to select an audio language to be used are available. A change in audio level is designated by an audio level change command and a variable that designates an audio level after change. An audio language to be used is selected by an audio language change command and a variable that designates the type of language after change. As yet another example, those which are used to control user triggers from user interface unit 400 are available.

On the basis of the commands/variables of the markup and script languages, as exemplified above, a "layout control" signal is sent from XHTML/SVG/CSS layout manager 207 (some functions are often implemented by the SMIL timing engine 206) to AV renderer 203. The "layout control" signal controls the layout on the screen, size, output timing, and output time of video data to be displayed on, e.g., an

external monitor device or the like (not shown), and/or the tone/loudness, output timing, and output time of audio data to be played back from an external loudspeaker (not shown).

5 Media decoders 208a/208b decode data of the advanced content such as audio data, still picture (including a background picture)/animation, text/font data, movie data, and the like included in the advanced content. That is, each of media decoders 208a/208b
10 includes an audio decoder, still picture/animation decoder, text/font decoder, and movie decoder in correspondence with objects to be decoded. For example, audio data in the advanced content, which is encoded by, e.g., MPEG, AC-3, or DTS is decoded by the
15 audio decoder and is converted into non-compressed audio data. Still picture data or background picture data, which is encoded by JPEG, GIF, or PNG, is decoded by the still picture decoder, and is converted into non-compressed picture data. Likewise, movie or
20 animation data, which is encoded by MPEG2, MPEG4, Macromedia Flash, or Scalable Vector Graphics (SVG), is decoded by the movie or animation decoder, and is converted into non-compressed movie/animation data. Text data included in the advanced content is decoded
25 by the text/font decoder using font data (e.g., OpenType format) included in the advanced content, and is converted into text picture data which can be

superimposed on a movie or still picture. Video/audio data, which includes these decoded audio data, picture data, animation/movie data, and text picture data as needed, is sent from media decoders 208a/208b to AV
5 renderer 203. This advanced content is decoded in accordance with an instruction of a "media control" signal from interface handler 202 and in synchronism with a "DVD timing" signal from interface handler 202 and a "timing" signal from system clock 214.

10 AV renderer 203 has a function of controlling a video/audio output. More specifically, AV renderer 203 controls, e.g., the video display position and size (often including the display timing and display time together), and the audio level (often including the
15 output timing and output time together) in accordance with the "layout control" signal output from XHTML/SVG/CSS layout manager 207. Also, AV renderer 203 executes pixel conversion of video data in accordance with the type of designated monitor and/or
20 the type of video data to be displayed. The video/audio outputs to be controlled are those from DVD-Video playback engine 100 and media decoders 208a/208b. Furthermore, AV renderer 203 has a function of controlling mixing and switching of the DVD-Video
25 content and advanced content in accordance with an "AV output control" signal output from interface handler 202.

Note that interactive engine 200 in the DVD-Video player in FIG. 72 comprises an interface for sending the markup language in the advanced content read from information storage medium 1 to parser 210 via buffer unit 209, and an interface for sending data (audio data, still picture/animation data, text/font data, movie data, and the like) in the read advanced content to media decoders 208a/208b via buffer unit 209. These interfaces form an interface (first interface) independent from the Internet connection unit in FIG. 72.

Also, the DVD-Video player in FIG. 72 comprises an interface for receiving advanced content from a communication line such as the Internet or the like, and sending the markup language in the received advanced content to parser 210 via buffer unit 209, and an interface for sending data (audio data, still picture/animation data, text/font data, movie data, and the like) in the received advanced content to media decoders 208a/208b via buffer unit 209. These interfaces form the Internet connection unit (second interface) shown in FIG. 72.

Buffer unit 209 includes a buffer that stores the advanced content downloaded from server unit 500, and also stores the advanced content read from information storage medium 1 via disc unit 300. Buffer unit 209 reads the advanced content stored in server unit 500,

and downloads them via the Internet connection unit under the control of buffer manager 204 based on the markup language/script language.

Also, buffer unit 209 loads the advanced content recorded on information storage medium 1 under the control of buffer manager 204 based on the markup language/script language. At this time, if disc unit 300 is a device that can access the disc at high speed, disc unit 300 can read out the advanced content from information storage medium 1 while playing back the DVD-Video content, i.e., reading out DVD-Video data from information storage medium 1.

If disc unit 300 is not a device that can make high-speed access, or if the playback operation of the DVD-Video content is to be perfectly guaranteed, playback of the DVD-Video content must not be interrupted. In such case, the advanced content is read out from information storage medium 1 and are stored in the buffer in advance prior to the beginning of playback. In this way, since the advanced content is read out from the buffer simultaneously when the DVD-Video content are read out from information storage medium 1, the load on disc unit 300 can be reduced. Hence, the DVD-Video content and advanced content can be simultaneously played back without interrupting playback of the DVD-Video content.

In this manner, since the advanced content

downloaded from server unit 500 is stored in buffer unit 209 in the same manner as those recorded on information storage medium 1, the DVD-Video content and advanced content can be simultaneously read out and played back.

Buffer unit 209 has a limited storage capacity. That is, the data size of the advanced content that can be stored in buffer unit 209 is limited. For this reason, it is possible to erase the advanced content with low necessity and to save those with high necessity under the control of buffer manager 204 (buffer control). Buffer unit 209 can automatically execute such save and erase control.

Furthermore, buffer unit 209 has a function (preload end trigger, load end trigger) of loading content requested by buffer manager 204 from disc unit 300 or server unit 500 into buffer unit 209, and informing buffer manager 204 that the advanced content designated by buffer manager 204 have been loaded into the buffer.

Buffer manager 204 can send the following instructions as "buffer control" to buffer unit 209 in accordance with an instruction of the markup language (even during playback of DVD video content).

(load all or part of a specific file from a server;

(load all or part of a specific file from a disc;

and

(erase all or part of a specific file from a buffer.

Furthermore, buffer manager 204 instructs buffer
5 unit 209 to load the advanced content in accordance
with loading information, which is described in the
markup language (or in a file designated by the markup
language). Buffer manager 204 has a function (buffer
control) of requesting to inform that specific advanced
10 content described in loading information have been
loaded into buffer unit 209.

Upon completion of loading of the specific
advanced content into buffer unit 209, buffer unit 209
informs buffer manager 204 of it, and the buffer
15 manager informs interface handler 202 of it (preload
end trigger, load end trigger).

Audio manager 215 has a function of issuing an
instruction for loading update audio data (audio
commentary data) from information storage medium 1 in
20 disc unit 300 or server unit 500 into buffer unit 209
in accordance with an instruction of the markup
language (update control).

Network manager 212 controls the operation of the
Internet connection unit. That is, network manager 212
25 switches connection/disconnection of the Internet
connection unit when the markup language designates
connection or disconnection to or from the network as a

"command". Also, network manager 212 has a function of checking the connection state to the network, and allows the markup language to download the advanced content in accordance with the connection state to the network.

Persistent storage 216 is an area for recording information (information set by the user and the like) associated with information storage medium 1, and comprises a nonvolatile storage medium such as a hard disc, flash memory, or the like. That is, even after the power supply of the DVD player is turned off, this information is held.

As information associated with the information storage medium to be played back, information such as the playback position of the DVD-Video content or advanced content, user information required in user authentication implemented by the advanced content, a game score of a game implemented by the advanced content, and the like are recorded in accordance with an instruction of the markup language (storage control). As a result, when the information storage medium is played back next time, playback can be continued from the previous position. When the advanced content downloaded from the server into the buffer are recorded in this persistent storage 216 upon playing back the information storage medium, the information storage medium can be played back without

connecting the network from the next time.

The building components of Interactive engine 200 in FIG. 72 can also be summarized as follows. That is, interactive engine 200 comprises:

5 (Parser 210

Parser 210 parses the content of the markup language.

(Interpreter unit 205, XHTML/SVG/CSS layout manager 207

10 Interpreter unit 205 which comprises the ECMAscript interpreter, SMIL timing engine, and the like, and XHTML/SVG/CSS layout manager 207 respectively interpret the parsed modules.

(Interface handler 202

15 Interface handler 202 handles control signals from interpreter unit 205, and those from DVD-Video playback controller 102.

(Media decoders 208a/208b

20 Media decoders 208a/208b generate video/audio data corresponding to audio data, still picture data, text/font data, movie data, and the like included in the advanced content in synchronism with system clock 103 of DVD playback engine 100 or system clock 214 of Interactive engine 200.

25 (AV renderer 203

AV renderer 203 outputs data obtained by mixing video/audio data generated by media decoders 208a/208b

to that played back by DVD-Video playback engine 100 on the basis of the execution result of the "command" in interface handler 202. Or AV renderer 203 selectively outputs one of video/audio data generated by media decoders 208a/208b and that played back by DVD-Video playback engine 100 on the basis of the execution result of the "command" in interface handler 202.

(Buffer unit 209

Buffer unit 209 temporarily stores the advanced content acquired from disc unit 300 or from server unit 500 via the Internet connection unit.

(Buffer manager 204

Buffer manager 204 loads or erases advanced content data to or from buffer unit 209 in accordance with an instruction from interface handler 202 (an instruction of the markup language), or the description of loading information (FIG. 90).

(Network manager

The network manager controls connection or disconnection to or from the network and checks the connection state in accordance with an instruction of the markup language.

(Persistent storage 216

The persistent storage holds information associated with the information storage medium such as the playback position of the content, user information, and the like, and also the advanced content downloaded

from server unit 500.

FIG. 73 shows an example of an information storage medium that records only content (standard content) which can be produced by the conventional production technique and aim at achieving high image quality of a title itself. Note that this information storage medium is called a "content type 1 disc". The content type 1 disc includes HD video manager recording area 30 (at this time, Application Type in HDVMG_CAT in area 30 records "0000b" indicating that information storage medium 1 includes only standard VTS data), and one or more HD video title set recording areas 40, which are recorded in video data recording area 20. In addition, this information storage medium includes neither advanced HD video title set recording area 50 recorded in video data recording area 20 nor the advanced content recorded in advanced content recording area 21.

Upon playing back this information storage medium 1, FP_PGCI recorded in HD video manager information management table 310 is referred to, and playback starts in accordance with the description of the FP_PGCI. This procedure is the same as that of the conventional DVD-Video.

Also, upon playing back this information storage medium 1, in FIG. 72 that shows an example of the arrangement of the DVD player, data supplied from information storage medium 1 is processed by only

DVD-Video playback engine 100, but does not undergo any processing in interactive engine 200. That is, video/audio data processed by DVD-Video playback engine 100 is output while passing through AV renderer 203.

5 FIG. 74 shows an example of an information storage medium that records only content (advanced content) which aim at providing colorful menus, improving interactiveness, and so forth even in content of menu screens, bonus video pictures, and the like in addition
10 to realization of high image quality of a title itself. Note that this information storage medium is called a "content type 2 disc (including only advanced VTS data)". The content type 2 disc (including only advanced VTS data) includes one HD video manager
15 recording area 30 and one advanced HD video title set recording area 50 recorded in video data recording area 20, and advanced content recorded in advanced content recording area 21. In addition, this information storage medium does not include any HD video title set
20 recording area 40 recorded in video data recording area 20.

 Note that since the advanced VTS does not require any menu objects, HD video manager recording area 30 of the content type 2 disc includes advanced HD video
25 manager information recording area (AHDVMGI) 35 and advanced HD video manager information backup area (AHDVMGI_BUP) 36. At this time, Application Type in

HDVMG_CAT in area 30 records "0001b" indicating that information storage medium 11 includes only advanced VTS data.

Upon playing back information storage medium 1 of this "content type 2 disc", startup information (STARTUP.XML) recorded in the markup/script language recording area is referred to, and a "markup language file serving as a start point" described in this information is executed, thus starting playback.

FIG. 75 shows an example of the detailed data structure in advanced HD video manager information (AHDVMGI) area 35 in information storage medium 1 in FIG. 74. Advanced HD video manager information (AHDVMGI) area 35 stores advanced HD video manager information management table (AHDVMGI_MAT) information 350 which records management information common to the entire HD_DVD-Video content recorded in video recording area 20 together, and advanced title search pointer table (ADTT_SRPT) information 351 that records information helpful to search (to detect the start positions of) titles present in the HD_DVD-Video content.

FIG. 76 shows an example of the detailed data structure in advanced HD video manager information management table (AHDVMGI_MAT) 350 in FIG. 75. Advanced HD video manager information management table (AHDVMGI_MAT) 350 records various kinds of information

including an HD video manager identifier (HDVMG_ID),
the end address (HDVMG_EA) of the HD video manager, the
end address (HDVMGI_EA) of the HD video manager
information, the version number (VERN) of the HD_DVD-
5 Video standard, an HD video manager category
(HDVMG_CAT) (in this information storage medium,
Application Type in the HDVMG_CAT records "0001b"), a
volume set identifier (VLMS_ID), an adaptation
identifier (ADP_ID), the number (HDVTS_Ns) of HD video
10 title sets (which records "0" since this information
storage medium stores no standard VTS), a provider
unique identifier (PVR_ID), a POS code (POS_CD), the
end address (AHDVMGI_MAT_EA) of the advanced HD video
manager information management table, and the start
15 address (TT_SPRT_SA) of the TT_SPRT.

Note that this information storage medium does not
store the start address (FP_PGCI_SA) of first play
program chain information, the start address
(HDVMGM_VOBS_SA) of an HDVMGM_VOBS, the start address
20 (HDVMGM_PGCI_UT_SA) of the HDVMGM_PGCI_UT, the start
address (PTL_MAIT_SA) of the PTL_MAIT, the start
address (HDVTS_ATRT_SA) of the HDVTS_ATRT, the start
address (TXTDT_MG_SA) of the TXTDT_MG, the start
address (HDVMGM_C_ADT_SA) of the HDVMGM_C_ADT, the
25 start address (HDVMGM_VOBU_ADMAP_SA) of the
HDVMGM_VOBU_ADMAP, an HDVMGM video attribute
(HDVMGM_V_ATR), the number (HDVMGM_AST_Ns) of HDVMGM

audio streams, an HDVMGM audio stream attribute (HDVMGM_AST_ATR), the number (HDVMGM_SPST_Ns) of HDVMGM sub-picture streams, an HDVMGM sub-picture stream attribute (HDVMGM_SPST_ATR), first play PGCI (FP_PGCI) 5 that records management information for language selection menus, the start address information (HDMENU_AOBS_SA) of an HDMENU_AOBS, the start address information (HDMENU_AOBSIT_SA) of the HDVMGM_AOBS information table, and information of the number 10 (HDVMGM_GUST_Ns) of HDVMGM graphics unit streams, which are stored in the content type 1 disc (or these areas are used as reserved areas).

FIG. 77 shows an example of the internal structure of advanced title search pointer table (ADTT_SRPT) 351 15 shown in FIG. 75. Advanced title search pointer table (ADTT_SRPT) 351 includes advanced title search pointer table information (ADTT_SRPTI) 351a and advanced title search pointer table (ADTT_SRP) information 351c. Only one piece of advanced title search pointer table 20 (ADTT_SRP) information 351c in advanced title search pointer table (ADTT_SRPT) 351 is present in an information storage medium including an advanced VTS but it does not exist in another information storage media.

25 Advanced title search pointer table information (ADTT_SRPTI) 351a records common management information of advanced title search pointer table (ADTT_SRPT) 351,

and records information of the number (ADTT_SRP_Ns) of
title search pointers included in advanced title search
pointer table (ADTT_SRPT) 351 ("1" is recorded since
there is only one advanced VTS in this information
5 storage medium), and the end address (ADTT_SRPT_EA)
information of this advanced title search pointer table
(ADTT_SRPT) 351 (a fixed value is recorded since there
is only one advanced VTS in this information storage
medium) in a file of the advanced HD video manager
10 information (AHDVMGI) area.

One advanced title search pointer (ADTT_SRP)
information 351c records various kinds of information
including the number (PTT_Ns) of Part_of_Titles (PTT),
and the start address (HDVTS_SA) of the HDVTS of
15 interest, in association with a title indicated by this
search pointer. (This medium does not include a title
playback type (TT_PB_TY), the number (AGL_Ns) of
angles, title Parental_ID_Field (TT_PTL_ID_FLD)
information, an HDVTS number (HDVTSN), and an HDVTS
20 title number (HDVTS_TTN), which are stored in the
content type 1 disc, or these areas are used as
reserved areas.)

FIG. 78 is a view for explaining a playback model
(example 1) of a disc that records an advanced VTS
25 (AHDVTS). A playback example of typical content type 2
disc (including only an advanced VTS) will be described
below using FIG. 78.

When playback of the content type 2 disc (including only an advanced VTS) starts, interactive engine (INT_ENG) 200 parses a menu page XML file which is stored in the advanced content recording area used to playback a menu screen described in the markup/script language.

For example, when a menu screen which prompts the user to execute a button selection process while repetitively playing back an impressive scene in movie video picture data is to be formed, the menu page XML file describes a control process for controlling DVD-Video playback engine (DVD_ENG) 100 to repetitively play back video data of the advanced VTS using the markup/script language. Interactive engine (INT_ENG) 200 issues a playback command (arrow a) to DVD-Video playback engine (DVD_ENG) 100 in accordance with the description.

At the same time, the page menu XML file stores a description for forming a menu screen using button images stored in the animation/still picture recording area and font data stored in the font recording area in advanced content recording area 21. Interactive engine (INT_ENG) 200 controls AV renderer 203 to mix the output that forms the screen according to these descriptions, and the video output of the advanced VTS by aforementioned DVD-Video playback engine (DVD_ENG) 100, thus implementing playback of the menu screen.

Next, the user selects a button used to execute playback of a video title itself of menu select buttons laid out on the screen using a remote controller or the like. The menu page XML file describes a script
5 process associated with the selected button, and a jump event to a DVD playback engine control page is generated (arrow b).

The DVD playback engine control page describes a control process for playing back the starting part of
10 the video title itself using the markup/script language. Interactive engine (INT_ENG) 200 issues a playback command to DVD-Video playback engine (DVD_ENG) 100 in accordance with the description (arrow c). The DVD playback engine control page also stores
15 descriptions used to form a menu screen that can be displayed during playback of the video title itself (e.g., a menu is formed using a screen smaller than the video title itself, and is superimposed on the video title itself by seeing through the menu screen) and to
20 superimpose a subtitle, using button images stored in the animation/still picture recording area and font data stored in the font recording area in the advanced content recording area 21. Interactive engine
(INT_ENG) 200 controls AV renderer 203 to mix the
25 output that forms the screen and the video output of the advanced VTS by aforementioned DVD-Video playback engine (DVD_ENG) 100 in accordance with these

descriptions, thus implementing playback of the menu screen and subtitle.

Upon completion of playback of the video title itself, interactive engine (INT_ENG) 200 controls the XML file to be processed to jump to the menu page XML file so as to play back the menu screen again in accordance with the description in the DVD-Video playback engine control page XML file (arrow d). Note that a broken arrow marked with a circle with an oblique line in FIG. 78 indicates that a jump event based on a navigation command in the advanced VTS is inhibited.

FIG. 79 shows an example of an information storage medium which records both content (standard content) which can be produced by the conventional production technique and aim at realizing high image quality of a title itself, and content (advanced content) which aim at providing colorful menus, improving interactiveness, and so forth even in content of menu screens, bonus video pictures, and the like in addition to realization of high image quality of the title itself. Note that this information storage medium is called a "content type 2 disc (including both advanced and standard VTS data)".

The content type 2 disc including both advanced and standard VTS data includes one HD video manager recording area 30, one or more HD video title set

recording areas 40, and one advanced HD video title set
recording area 50, which are recorded in video data
recording area 20, and advanced content (21A to 21E)
recorded in advanced content recording area 21. Since
5 the disc including the advanced VTS does not require
any menu objects, this HD video manager recording area
30 includes advanced HD video manager information
recording area (AHDVMGI) 35 and advanced HD video
manager information backup area (AHDVMGI_BUP) 36. At
10 this time, Application Type in the HDVMG_CAT in area 30
records "0010b" indicating that information storage
medium 1 includes both standard and advanced VTS data.

Upon playing back this information storage medium
(content type 2 disc) 1, startup information
15 (STARTUP.XML) recorded in the markup/script language
recording area is referred to, and a "markup language
file serving as a start point" described in this
information is executed, thus starting playback.

FIG. 80 shows an example of the detailed data
20 structure in advanced HD video manager information
(AHDVMGI) area 35 in the information storage medium in
FIG. 79. Advanced HD video manager information
(AHDVMGI) area 35 stores advanced HD video manager
information management table (AHDVMGI_MAT) information
25 350 which records management information common to the
entire HD_DVD-Video content recorded in video data
recording area 20 together, and advanced title search

pointer table (ADTT_SRPT) information 351 that records information helpful to search (to detect the start positions of) titles present in the HD_DVD-Video content.

5 FIG. 81 shows an example of the detailed data structure in advanced HD video manager information management table (AHDVMGI_MAT) 350 in FIG. 80. Advanced HD video manager information management table (AHDVMGI_MAT) 350 records various kinds of information including an HD video manager identifier (HDVMG_ID),
10 the end address (HDVMG_EA) of the HD video manager, the end address (AHDVMGI_EA) of the advanced HD video manager information, the version number (VERN) of the HD_DVD-Video standard, an HD video manager category (HDVMG_CAT: in this information storage medium,
15 Application Type in the HDVMG_CAT records "0010b"), a volume set identifier (VLMS_ID), an adaptation identifier (ADP_ID), the number (HDVTS_Ns) of HD video title sets, a provider unique identifier (PVR_ID), a
20 POS code (POS_CD), the end address (AHDVMGI_MAT_EA) of the advanced HD video manager information management table, and the start address (TT_SPRT_SA) of the TT_SPRT.

 Note that this information storage medium (content
25 type 2 disc) does not store the start address (FP_PGCI_SA) of first play program chain information, the start address (HDVMGM_VOBS_SA) of an HDVMGM_VOBS,

the start address (HDVMGM_PGCI_UT_SA) of the
HDVMGM_PGCI_UT, the start address (PTL_MAIT_SA) of the
PTL_MAIT, the start address (HDVTS_ATRT_SA) of the
HDVTS_ATRT, the start address (TXTDT_MG_SA) of the
5 TXTDT_MG, the start address (HDVMGM_C_ADT_SA) of the
HDVMGM_C_ADT, the start address (HDVMGM_VOBU_ADMAP_SA)
of the HDVMGM_VOBU_ADMAP, an HDVMGM video attribute
(HDVMGM_V_ATR), the number (HDVMGM_AST_Ns) of HDVMGM
audio streams, an HDVMGM audio stream attribute
10 (HDVMGM_AST_ATR), the number (HDVMGM_SPST_Ns) of HDVMGM
sub-picture streams, an HDVMGM sub-picture stream
attribute (HDVMGM_SPST_ATR), first play PGCI (FP_PGCI)
that records management information for language
selection menus, the start address information
15 (HDMENU_AOBS_SA) of an HDMENU_AOBS, the start address
information (HDMENU_AOBSIT_SA) of the HDVMGM_AOBS
information table, and information of the number
(HDVMGM_GUST_Ns) of HDVMGM graphics unit streams, which
are stored in the content type 1 disc (or these areas
20 are used as reserved areas).

FIG. 82 shows an example of the internal structure
of advanced title search pointer table (ADTT_SRPT) 351
shown in FIG. 80. Advanced title search pointer table
(ADTT_SRPT) 351 includes advanced title search pointer
25 table information (ADTT_SRPTI) 351a, standard title
search pointer (SDTT_SRP) 351b, and advanced title
search pointer table (ADTT_SRP) information 351c. Only

one piece of advanced title search pointer table
(ADTT_SRP) information 351c in advanced title search
pointer table (ADTT_SRPT) 351 is present in an
information storage medium including an advanced VTS
5 but it does not exist in other information storage
media. Also, standard title search pointer (SDTT_SRP)
315b is present only when an information storage medium
records standard VTS data.

Advanced title search pointer table information
10 (ADTT_SRPTI) 351a records, as common management
information of advanced title search pointer table
(ADTT_SRPT) 351, information of the number
(ADTT_SRP_Ns) of title search pointers included in
advanced title search pointer table (ADTT_SRPT) 351,
15 and the end address (ADTT_SRPT_EA) information of this
advanced title search pointer table (ADTT_SRPT) 351 in
a file of the advanced HD video manager information
(AHDVMGI) area.

Only one advanced title search pointer (ADTT_SRP)
20 information 351c records various kinds of information
including the number (PTT_Ns) of Part_of_Titles (PTT),
the start address (HDVTS_SA) of the HDVTS of interest,
and the like, in association with a title indicated by
this search pointer

25 The information storage medium (content type 2
disc) with the structure shown in FIGS. 79 to 82 does
not include a title playback type (TT_PB_TY), the

number (AGL_Ns) of angles, title Parental_ID_Field (TT_PTL_ID_FLD) information, an HDVTS number (HDVTSN), and an HDVTS title number (HDVTS_TTN) (or these areas are used as reserved areas).

5 One standard title search pointer (SDTT_SRP) information 351b records various kinds of information including a title playback type (TT_PB_TY), the number (AGL_Ns) of angles, the number (PTT_Ns) of Part_of_Titles (PTT), title Parental_ID_Field
10 (TT_PTL_ID_FLD) information, an HDVTS number (HDVTSN), an HDVTS title number (HDVTS_TTN), and the start address (HDVTS_SA) of the HDVTS of interest, in association with a title indicated by this search pointer.

15 FIG. 83 is a view for explaining the relationship between the playback states of an advanced VTS and standard VTS. FIG. 83 shows an example a state machine that indicates transition of a playback control module of the content type 2 disc. In a playback process of
20 the content type 2 disc (of a type including both advanced and standard VTS data), playback starts from an initial state when interactive engine (INT_ENG) 200 interprets startup information (STARTUP.XML) recorded in markup/script language recording area 21A, and the
25 control transits to an advanced VTS playback state.

 In the advanced VTS playback state, interactive engine (INT_ENG) 200 generates text information, button

images, and the like, which form a menu screen, and issues a video playback start instruction command to DVD-Video playback engine (DVD_ENG) 100. Interactive engine 200 controls AV renderer 203 to mix the output
5 that forms the screen with the video output of DVD-Video playback engine (DVD_ENG) 100, thus implementing playback of the menu screen.

A markup/script language file that describes a menu page to be interpreted in the advanced VTS
10 playback state describes a script which defines the behaviors of event handlers which are associated with events such as button clicking and the like by the user. For example, an event handler associated with a button image that indicates playback of a movie video
15 title itself describes a command required to shift the control to a standard VTS playback state. When the user selects and executes the title playback button by a remote controller operation or the like, interactive engine (INT_ENG) 200 executes the command required to
20 shift the control to the standard VTS playback state, and the state machine makes the video playback control transit to the standard VTS playback state executed by DVD-Video playback engine (DVD_ENG) 100.

In the standard VTS playback state, DVD-Video
25 playback engine (DVD_ENG) 100 interprets a cell playback information table (C_PBIT), program chain command table (PGC_CMDT), and the like in a program

chain (PGC) stored in a PGC and the like in the standard VTS, and executes playback control of the standard VTS in accordance with their description content. In the standard VTS playback state,
5 interactive engine (INT_ENG) 200 halts, and never instructs DVD-Video playback engine (DVD_ENG) 100 to execute playback control.

The program chain command table (PGC_CMDT) and the like of the standard VTS can describe a shift command
10 ("CallINTENG" or the like in FIG. 43(d)) to the advanced VTS playback state. With such command, DVD-Video playback engine (DVD_ENG) 100 can execute the shift command to the advanced VTS playback state when it executes a command interpretation process upon
15 completion of a series of video playback processes, or DVD-Video playback engine (DVD_ENG) 100 can shift the video playback control to the advanced VTS playback state executed by interactive engine (INT_ENG) 200 upon reception of an event of a user command such as menu
20 call or the like.

Upon shifting from the standard VTS playback state to the advanced VTS playback state, DVD-Video playback engine 100 may temporarily store information such as the video playback position of the standard VTS or the
25 like immediately before the playback control transits to prepare for a resume playback process from interactive engine (INT_ENG) 200, so as to implement a

temporary call process of a menu screen or the like.

Table A below shows a practical example of commands used to shift from the advanced VTS playback state to the standard VTS playback state in the markup/script language file to be interpreted by interactive engine (INT_ENG) 200 (commands other than those in this example may be adopted as needed).

<Table A>

(Command Name)

10 (Argument)

CallDVDENG_TT

title number

15 CallDDVENG_PTT

title number, chapter number

CallDVDENG_TM

title number, playback start time position

20

CallDVDENG_RSM

no argument

25 In table A, CallDVDENG_TT is a command that designates the title number of a standard VTS upon shifting from the advanced VTS playback state to the standard VTS playback state. DVD-Video playback engine

(DVD_ENG) 100 loads a standard VTS including the designated title, and starts playback from the head of the title.

5 CallDDVENG_PTT is a command that designates the title number and chapter number (PTT number) of a standard VTS upon shifting from the advanced VTS playback state to the standard VTS playback state. DVD-Video playback engine (DVD_ENG) 100 loads a standard VTS including the designated title, and starts
10 playback from the head of the designated chapter number (PTT number).

15 CallDVDENG_TM is a command that designates the title number and an offset of the playback start time from the head of the title video of a standard VTS upon shifting from the advanced VTS playback state to the standard VTS playback state. DVD-Video playback engine (DVD_ENG) 100 loads a standard VTS including the designated title, and starts playback from the designated playback time position.

20 CallDVDENG_RSM is a command that designates execution of a resume process upon shifting from the advanced VTS playback state to the standard VTS playback state. Upon reception of this command, DVD-Video playback engine (DVD_ENG) 100 resumes playback in
25 accordance with the temporarily stored playback position information when the control transits from the immediately preceding standard VTS playback state to

the advanced VTS playback state.

FIG. 84 shows an example of argument definition of a command (CallINTENG command) required to shift from the standard VTS playback state to the advanced VTS playback state of navigation commands to be interpreted by the DVD-Video playback engine (DVD_ENG). In the entire command bit sequence, a command code is stored in bits b63 to b48, and b15 to b0 are assigned to a reserved area for future expansion.

A 16-bit control parameter storage area is assigned to b47 to b32. At a specific playback position or in an event of a standard VTS to be interpreted by DVD-Video playback engine (DVD_ENG) 100, this area can store an arbitrary value which is used to select an arbitrary process in the description of the markup/script language file to be interpreted by interactive engine (INT_ENG) 200 after the control transits to the advanced VTS playback state. That is, this data area can be used for an arbitrary purpose upon producing video content. An area for storing the playback start cell number in the resume process is assigned to b31 to b23.

An area for storing a menu identifier is assigned to b19 to b16, and is used to designate the type of menu to be called upon calling a menu especially when the control transits from the standard VTS playback state to the advanced VTS playback state. The type of

menu identifier that can be called includes:

0010b (title menu

0011b (root menu

0100b (sub-picture menu

5 0101b (audio menu

0110b (angle menu

0111b (chapter menu, etc.

Also, more detailed behavior differences may be expressed based on the aforementioned control parameter or by combining the control parameter and menu identifier.

FIG. 85 is a flowchart for explaining the switching algorithm of a user command process. This flowchart exemplifies a process for switching a module that handles a process when a user command is generated. Upon playing back the content type 2 disc (of a type including both advanced and standard VTS data), when an event of a user command associated with button depression on a remote controller or front panel (not shown) is generated, a user operation module confirms the current playback state (step ST850), and switches a module which is to be notified of the user event. If the current state is the advanced VTS playback state (YES in step ST850), the user operation module notifies interactive engine (INT_ENG) 200 of the user event; if the current state is the standard VTS playback state (NO in step ST850), the user operation

module notifies DVD-Video playback engine (DVD_ENG) 100 of the user event, thus executing the process of the user command.

FIG. 86 shows an example of domain transition of the content type 2 disc. In a typical content type 2 disc (of a type including both advanced and standard VTS data), a VMG menu domain (VMGM_DOM) and VTS menu domain (VTSM_DOM) are formed of an advanced VTS and an XML file described in the markup/script language, and a title domain (TT_DOM) such as a video title itself is formed of a standard VTS.

Menu video picture data in the VMG menu domain and VTS menu domain is realized by playing back video picture data stored in the advanced VTS in accordance with the description of the "XML file" in addition to text information and button images rendered in accordance with the description of the "menu XML file" described in the markup/script language.

Transition between the VMG menu domain and VTS menu domain is implemented by executing a hyperlink process between menu XML files described in these menu XML files. At this time, playback of the advanced VTS may stop in correspondence with a change in page, and playback may start from a new position or may be continued from the previous position.

Transition from the VMG menu domain (VMGM_DOM), VTS menu domain (VTS_DOM), or the like to the title

domain (TT_DOM) is implemented by executing a playback start command of a standard VTS (e.g., a CallDVDENG_xxx command listed in table A above) described in an XML file, and transferring the DVD playback control to DVD-
5 Video playback engine 100.

On the other hand, transition from the title domain (TT_DOM) to the VMG menu domain (VMGM_DOM) may be implemented by defining a new command such as the aforementioned CallINTENG command and storing this new
10 command in the program chain command table (PGC_CMDT) in the standard VTS. Alternatively, transition from the title domain (TT_DOM) to the VMG menu domain (VMGM_DOM) may take place when an argument of a CallSS command indicates VMGM_DOM. Also, an event generated
15 upon depression of a root menu button arranged on a remote controller or the like (not shown) may be acquired, and transition from the title domain (TT_DOM) to the VMG menu domain (VMGM_DOM) may take place upon acquisition of this event.

20 Likewise, transition from the title domain (TT_DOM) to the VTS menu domain (VTSM_DOM) may be implemented by defining a new command such as the aforementioned CallINTENG command or the like, and storing this new command in the program chain command
25 table (PGC_CMDT) in the standard VTS, or this domain transition may take place when an argument of a CallSS command indicates VTSM_DOM. Also, an event generated

upon depression of a title menu button arranged on a remote controller or the like (not shown) may be acquired, and transition from the title domain (TT_DOM) to the VTS menu domain (VTSM_DOM) may take place upon acquisition of this event.

FIG. 87 is a view for explaining a playback model (example 2) of a disc that records both an advanced VTS (AHDVTS) and standard VTS (HDVTS). A playback example of a typical content type 2 disc (of a type including both advanced and standard VTS data) will be explained using FIG. 87.

When playback of the content type 2 disc (including both advanced and standard VTS data) starts, interactive engine (INT_ENG) 200 parses a menu page XML file which is stored in the advanced content recording area and is required to play back a menu screen described in the markup/script language.

For example, when a menu screen which prompts the user to execute a button selection process while repetitively playing back an impressive scene in movie video picture data is to be formed, the menu page XML file describes a control process for controlling DVD-Video playback engine (DVD_ENG) 100 to repetitively play back video data of the advanced VTS using the markup/script language. Interactive engine (INT_ENG) 200 issues a playback command (arrow a) to DVD-Video playback engine (DVD_ENG) 100 in accordance with the

description.

At the same time, the menu page XML file stores a description for forming a menu screen using button images stored in the animation/still picture recording area and font data stored in the font recording area in
5 advanced content recording area 21. Interactive engine (INT_ENG) 200 controls AV renderer 203 to mix the output that forms the screen according to these descriptions, and the video output of the advanced VTS
10 by aforementioned DVD-Video playback engine (DVD_ENG) 100, thus implementing playback of the menu screen.

Next, the user selects a button used to execute playback of a video title itself of menu select buttons laid out on the screen using a remote controller or the
15 like. The menu page XML file describes a script process associated with the selected button, and a jump event to a DVD playback engine control page is generated (arrow b).

The DVD playback engine control page describes a
20 CallDVDENG_TT command which has the title number indicating the head of a video title itself as an argument. When interactive engine (INT_ENG) 200 executes this command, transition from the advanced VTS playback state to the standard VTS playback state takes
25 place (arrow c).

After transition to the standard VTS playback state, DVD-Video playback engine (DVD_ENG) 100 executes

playback of the standard VTS that stores the video title itself. Depending on video content, a playback position jump process to a playback position of another VTS may be taken place in accordance with the description of a playback control command stored in the VTS (arrow d). Note that a broken arrow marked with a circle with an oblique line in FIG. 87 indicates that a jump event based on a navigation command in the advanced VTS is inhibited. On the other hand, a jump event based on a navigation command is allowed in the standard VTS (arrow d', d'', or the like).

Upon completion of playback of the video title itself, a "CallINTENG command" described in the program chain command table in the program chain (PGC) is executed, thus causing transition from the standard VTS playback state to the advanced VTS playback state (arrow e).

Interactive engine (INT_ENG) 200 controls the XML file to be processed to jump to the menu page XML file so as to play back the menu screen again in accordance with a script description described in a handler of a CallINTENG command generating event in the DVD-Video playback engine control page XML file (arrow f).

FIG. 88 shows the relationship among an advanced VTS, standard VTS, and video objects (called EVOB or VOB data) in the content type 2 disc including both advanced and standard VTS data. In FIG. 88, an

advanced VTS that forms a menu and two standard VTSS which form a title (video title) are present.

Respective VTSS refer to independent video objects. In this example, video picture data required to form a

5 menu is quite different from that which forms a title.

With the configuration shown in FIG. 88, when a "menu screen which prompts the user to execute a button

selection process while repetitively playing back an impressive scene in movie video picture data is to be

10 formed", two video objects must be prepared although

the video title and menu video picture data are the

same. In order to avoid such duplicate preparation of

"two video objects", a "shared reference model of objects" shown in FIG. 89 can be referred to.

15 FIG. 89 is a view for explaining a shared reference model of objects in a disc that records an advanced VTS (AHDVTS) and standard VTS (HDVTS)

together. Since each of the advanced VTS side and

standard VTS side stores a time map, the advanced VTS

20 and standard VTS can refer to the same video objects,

and an arbitrary period of a given scene in the video

title can be extracted and used as a background picture

of a menu screen. In this way, the content producer

can reduce the number of processes for producing two

25 video objects to one (in association with a shared

object to be referred to). Also, since the two objects

are reduced to one, the required capacity of the

information storage medium can be reduced, and improvement of the image quality of the video title itself, addition of a new bonus picture, and the like can be realized accordingly.

5 When a video object (VOB) to be shared by the advanced VTS and standard VTS is played back as the advanced VTS, PCI/DSI often includes information which is not required as the standard VTS, as shown in FIGS. 64 and 65. When such video object is played back
10 as the standard VTS, playback is made using such information. However, when the video object is played back as the advanced VTS, playback is made while skipping such information, i.e., ignoring it.

FIG. 90 is a view for explaining a practical
15 example of loading information included in advanced content. The loading information includes a file name & location field, file size field, content type field, reference start time field, reference end time field, and the like. The file name & location field describes
20 the URL address and file name of a file when that file is present on the server unit 500, or describes the directory on a disc and file name of a file when that file is present on the disc. The file size field describes the file size of a file (unit: bytes). The
25 content type field describes the type of content using MIME types. The reference start time field describes a reference start time of a file from the markup language

or the like, and the reference end time field describes a reference end time of that file from the markup language or the like (that is, when this time has elapsed, the file loaded on the buffer may be immediately erased).

Basically, a file with the reference start time = "0" must be loaded onto the buffer (209 in FIGS. 72 and 91 or the like) before playback starts (i.e., before the beginning of execution of the markup language ("preload")). For other data, the playback apparatus determines the loading start times of all files using the reference start times, reference end times, and file sizes which are described in the loading information, and information associated with a communication rate acquired by the playback apparatus. In this way, the user wait time until the beginning of display of the advanced content/the beginning of playback of the DVD-Video content can be minimized.

FIG. 91 shows the arrangement of buffer manager 204 and its peripheral units, and FIG. 92 shows the flow upon loading data onto V buffer 209. When interactive engine 200 is started up, a startup information file (STARTUP.XML) as one of advanced content recorded on information storage medium 1 in the disc unit is loaded (step ST10). Parser 210 parses this startup information (step ST12). Interpreter unit 205 interprets the parsed startup information.

Interpreter unit 205 registers an operation upon generation of a "preload end" event (trigger) (for example, loading/execution of markup language file INDEX.XML indicating the default screen configuration starts), and an operation upon generation of a "load end" event (trigger) (for example, execution of a user operation which is inhibited so far is permitted) (step ST14).

Note that the control of "user operation" can be made by PGC user operation control (PGC_UOP_CTL) in the standard VTS, and can be made by the markup language in the advanced VTS.

Furthermore, loading information (see FIG. 90) is loaded (step ST16). This loading information may be described in the aforementioned startup file, may be recorded as one file on disc 1, or may be recorded as one file on server 500. When the loading information is recorded on disc 1 or server 500, the recording location and file name are described in the startup file. The loading information is loaded by interactive engine (INT_ENG) 200 in accordance with this description, and is parsed by parser 210 (step ST18). Interpreter unit 205 interprets the parsed loading information, and buffer manager 204 loads the advanced content onto buffer 209 (step ST20).

The loading information describes the file name and location (a place where a file exists), file size,

content type or MIME type (the type of data), reference start and end times (data reference duration), and the like of each file to be downloaded.

Buffer manager 204 loads advanced content with the
5 reference start time = "0" (i.e., files that must be stored in the buffer before the beginning of display of the advanced content/the beginning of playback of the DVD-Video content) in accordance with this description (step ST22). At this time, files to be loaded are
10 loaded from disc 1 or server unit 500 in accordance with the description order of the loading information. In this case, for example, the loading information designates advanced content (INDEX.XML file and its related files) that form the first page as those to be
15 "preloaded".

After all advanced content to be "preloaded" are loaded onto buffer 209 (YES in step ST24), buffer 209 sends a "preload end trigger" signal to buffer manager 204 (step ST26). Upon reception of the "preload end
20 trigger" signal from buffer 209, buffer manager 204 sends a "preload end trigger" signal to interface handler 202. Upon reception of the "preload end trigger" signal from buffer manager 204, interface handler 202 sends a "preload end event" signal as an
25 event to interpreter unit 205.

Interpreter unit 205 has registered the operation upon generation of the "preload end event", as

described above, and executes the registered operation (step ST28). For example, as the operation, execution of loading of INDEX.XML which has been loaded onto buffer 209 and forms the first page is registered.

5 Also, INDEX.XML designates start of playback of DVD-Video content. In this manner, upon completion of preloading of the advanced content (upon generation of the "preload end event"), display of the advanced content/playback of the DVD-Video content starts.

10 In order to quicken this playback start time, only advanced content which form the first page may be designated as those to be "preloaded". However, since advanced content other than the first page are not stored in buffer 209 at the beginning of playback, user
15 operations such as fastforwarding, skip, time search, and the like must be inhibited.

While display of the advanced content/playback of the DVD-Video content is performed, buffer manager 204 loads remaining advanced content (files to be stored in
20 the buffer after the beginning of display of the advanced content/the beginning of playback of the DVD-Video content) in accordance with the description of the loading information (step ST30). At this time, the playback apparatus determines the loading start times
25 and order of all advanced content using the reference start times, reference end times, and file sizes which are described in the loading information, and

information associated with a communication rate acquired by the playback apparatus (e.g., using a value given by $\text{priority} = \text{reference start time} - \text{file size} / \text{communication rate}$).

5 For example, the loading information describes that a "preload end trigger" is generated upon completion of loading of advanced content that form the first page, and a "load end trigger" is generated upon completion of loading of advanced content which form
10 the second page.

 If advanced content which form the second page are loaded onto buffer 209 (YES in step ST32), buffer 209 sends a "load end trigger" signal to buffer manager 204. Upon reception of the "load end trigger" signal
15 from buffer 209, buffer manager 204 sends a "load end trigger" signal to interface handler 202. Upon reception of the "load end trigger" signal from buffer manager 204 (step ST34), interface handler 202 sends a "load end event" signal as an event to interpreter
20 unit 205.

 Interpreter unit 205 has registered the operation upon generation of the "load end event", as described above, and executes the registered operation (step ST36). For example, when user operations such as
25 fastforwarding, skip, time search, and the like are inhibited, the operation for permitting the inhibited user operations is registered. That is, since all

advanced content are stored in buffer 209, the user operations need not be inhibited.

FIG. 93 is a view for explaining the configuration of an advanced VTS (AHDVTS) which exceptionally has multiple PGCs. A VTS_EVOBS of the advanced VTS in FIG. 93 includes one interleaved block. This interleaved block is used to implement playback of the director's cut version and theatrical release version, as shown in, e.g., FIG. 69. In many cases, EVOBs in the interleaved block of such VTS_EVOBS have different playback time durations. In case of such advanced VTS, VTSI may manage information associated with video playback in a plurality of PGCs.

A playback sequence is defined by the cell playback information table (C_PBIT; 53 in FIG. 56) stored in a PGC. The cell position information table (C_POSIT; 54 in FIG. 56) associates cells used in playback and actual cells using EVOB numbers (EVOB#1, etc.) and cell numbers (Cell#1 to Cell#3, etc.) in the VTS_EVOBS. Furthermore, of the cells in the VTS_EVOBS, cells that form the interleaved block are segmented for respective interleaved units (ILVUs), and are allocated at separate positions in the interleaved block on the HD_DVD disc.

In the example of FIG. 93, the cell playback information table is configured as follows. That is, for example, PGC#1 as the director's cut version plays

back a contiguous block formed by EVOB#1, then plays
back EVOB#3 formed by an interleaved block, and finally
plays back a contiguous block formed by EVOB#4 all in
the VTS_EVOBS. For example, PGC#2 as the theatrical
5 release version plays back a contiguous block formed by
EVOB#1, then plays back EVOB#2 formed by an interleaved
block, and finally plays back a contiguous block formed
by EVOB#4 all in the VTS_EVOBS.

In this manner, upon describing the playback
10 sequences of the director's cut version and theatrical
release version, respective cells (EVOBs) in the
interleaved block period have different playback time
durations. In such case, the playback sequences are
defined by dividing PGCs, as in the example of FIG. 93.
15 In this way, accesses to playback positions in time
units can be easily managed.

FIG. 94 is a view for explaining the configuration
of an advanced VTS (AHDVTS) which includes an
interleaved block but has one PGC. This example is
20 convenient, e.g., when the interleaved block forms an
angle block.

In the example of FIG. 94, the cell playback
information table is configured as follows. That is, a
playback sequence defined by PGC#1 plays back a
25 contiguous block formed by EVOB#1, then plays back
EVOB#2 formed by an interleaved block, and finally
plays back a contiguous block formed by EVOB#4 all in

the VTS_EVOBS.

In this case, it appears that no sequence for playing back EVOB#3 that forms another angle is defined. However, in practice, in the angle block, 5 EVOBs which form respective angles, and their cells and ILVU boundaries have equal playback times, and the same multiplexed audio data is used. Hence, angles are configured to be seamlessly switched in ILVU boundary units. Therefore, a parameter that indicates the 10 current playback angle is defined, and the angle can be switched based on the value of this parameter.

To summarize the above description, when the playback sequence of the interleaved block that forms the angle block is to be defined, the playback time is 15 uniquely defined by the cell playback information table given by one PGC, and cells to be actually played back of a VTS_EVOBS can be specified in combination with the aforementioned parameter indicating the playback angle.

In the embodiment of the present invention, using 20 playback control information which is stored in the ADV_OBJ in FIG. 2 and is described using the markup/script language, playback of identical DVD content can be flexibly configured. That is, using the description of the aforementioned playback sequence 25 information file (PBSEQ001.XML in FIG. 2, etc.), a function of freely configuring the playback order of a DVD video picture stored in a VTS_EVOBS in

predetermined units (independently of PGC information and navigation information in navigation packs which are originally recorded on disc 1) can be implemented.

FIG. 95 shows a description example of a playback sequence in the playback sequence information file. Assume that the configuration in FIG. 93 is newly defined using the description of the above playback sequence information file (PBSEQ001.XML in FIG. 2, etc.). In the first line of the description in FIG. 95, "directors_cut" is defined as a name for uniquely defining the playback sequence, and it is defined that this playback sequence is described based on PGC information of PGC#1 and title#1.

In the second to fourth lines of the description in FIG. 95, three chapters (PTT numbers) that form the playback sequence of "directors_cut" are defined, and names that uniquely define these chapters are defined using an attribute "id". The playback order of the chapters (PTT numbers) is defined using attribute information "order", and associations between these chapters and those in PGC#1 in FIG. 93 are defined using an attribute "pgc" (in this example, the playback order is described in three entries, e.g., chapter "order" "1", "2", and "3").

The definition of the playback sequence using the markup language description in the aforementioned playback sequence information file is convenient for a

case wherein "an advanced VTS is defined as DVD video picture materials divided into respective chapters (PTTs), which are re-defined in correspondence with use purposes like a playback sequence used in a menu screen, that used in title playback, and that used in bonus content". Since this playback sequence is defined using the markup language, it can be easily edited later. For example, such playback sequence can be advantageously applied to, e.g., a case wherein a different sequence is to be defined later using movie content (divided into a plurality of chapters) already printed on a DVD-Video disc as a material (reordering of the playback order of a plurality of chapters including repetitive playback of a specific chapter and/or playback skip of a specific chapter).

FIG. 96 shows an example in which the same playback sequence as that in FIG. 95 is described using cell units with respect to the advanced VTS shown in FIG. 71. When each of the chapters (PTT numbers) shown in FIG. 95 corresponds to one cell, the playback sequence is configured by three entries (chapter "order" = "1", "2", and "3") as in FIG. 95.

FIG. 97 shows an example of the playback sequence upon expressing a playback sequence across a plurality of PGCs. For example, the playback sequence in FIG. 97 is configured to continuously play back different video parts of the director's cut version and theatrical

release version, which are formed by the interleaved block. Such configuration is effective to create content that give an explanation about the difference between the director's cut version and theatrical release version in DVD bonus content or the like.

As the differences of the markup language used to describe the playback sequences, in the examples shown in FIGS. 95 and 96, the first line describes the PGC number that uniquely designates the chapters (PTT numbers) or cell numbers, while in the example of FIG. 97, the markup description that designates the chapter number includes the PGC number. With such description, one playback sequence can be configured across a plurality of PGCs (across PGC#1 and PGC#2 in this example).

FIG. 98 shows an example in which the same playback sequence as that in FIG. 97 is described using cell units with respect to the advanced VTS shown in FIG. 93. Since PTT = 3 (PTT#3) of PGC number = 2 (PGC#2) are configured by two cells (corresponding to cell#1 and cell#2 of EVOB#2 in the example of FIG. 93), the number of cell entries required to express the same playback sequence is increased to two ("cell" = "4" of chapter "order" = "3" and "cell" = "5" of chapter "order" = "4" in the example of FIG. 98).

Since the description method in the aforementioned playback sequence information file (PBSEQ001.XML in

FIG. 2, etc.) allows flexible definitions, a more complicated, detailed playback sequence can be described using a definition different from the above example. By defining the playback sequences as
5 exemplified above, the flow of playback of the advanced VTS stored in a DVD disc can be flexibly changed (after distribution of the disc). For example, after a given DVD disc was released, a movie company contrives a new method of enjoying the DVD video picture, and delivers
10 a new playback sequence via the Internet. Then, users enjoy playback of the DVD video picture using the new playback sequence.

Likewise, the use method that allows the user to edit an arbitrary playback sequence by himself or
15 herself and to enjoy video picture playback by joining his or her favorite scenes can be provided (in this case, information obtained by editing the playback sequence by the user himself or herself can be saved in, e.g., persistent storage 216 in FIG. 72 or 100).

FIG. 99 is a flowchart showing an example of the processing for initializing the playback sequence of the advanced VTS (e.g., for re-setting the settings based on the default playback sequence to those of another playback sequence described in the playback
20 sequence information file) in DVD playback engine 100 in, e.g., FIG. 72 or 100 using the playback sequence information file (PBSEQ001.XML in FIG. 2, etc.) prior
25

to playback of the advanced VTS.

Upon starting playback of the advanced VTS, interactive engine 200 begins to initialize the DVD-Video player (definition of a playback sequence of
5 objects to be played back) in accordance with a predetermined procedure described in, e.g., startup information recording area 210A in FIG. 50.

If it is determined in a condition determination part in step ST100 that the described initialization
10 procedure describes a playback sequence setting command of the advanced VTS based on playback sequence information (YES in step ST100), interactive engine 200 registers playback sequence information (e.g., the description of PBSEQ001.XML in FIG. 2) in DVD playback
15 engine 100 (step ST102). DVD playback engine 100 resets the playback sequence of the advanced VTS in accordance with the playback sequence information registered by interactive engine 200 in step ST102 (step ST104).

20 If it is determined in a condition determination part in step ST100 that no playback sequence setting command of the advanced VTS based on playback sequence information is described (NO in step ST100), DVD
playback engine 100 determines a playback sequence in
25 accordance with cell playback information (C_PBIT) in program chain information (PGCI) recorded in the advanced VTS (step ST106).

DVD playback engine 100 controls playback of the advanced VTS in accordance with the playback sequence set based on the cell playback information (C_PBIT) in step ST106, or controls playback of the advanced VTS in accordance with a playback command from interactive engine 200 on the basis of the playback sequence set based on the description of the playback sequence information file or the like in step ST104 (step ST108). After execution of playback using all advanced VTSSs, the playback process ends.

In other words, FIG. 99 executes the following processing. That is, it is checked if a playback sequence definition based on playback sequence information (playback sequence information acquired from, e.g., the Internet if it is not stored in the playback sequence information recording area) is available (ST100). If no playback sequence definition based on playback sequence information is available (NO in step ST100), expanded video objects (EVOBs) are played back (ST108) on the basis of management information (PGCI) in the management area (ST106); if the playback sequence definition based on playback sequence information is available (YES in step ST100; initialize the playback sequence), expanded video objects are played back (ST108) on the basis of the playback sequence information (ST102 to ST104).

Alternatively, the processing in FIG. 99 is

executed as follows. It is checked if a playback sequence definition based on playback sequence information is available (ST100). If the playback sequence definition based on playback sequence information is available (YES in step ST100; initialize the playback sequence), expanded video objects are played back (ST108) on the basis of at least one of a sequence of the program chain numbers, a sequence of the cell numbers, and a sequence of the chapter numbers, which are defined by the playback sequence information (ST102 to ST104).

FIG. 100 is a system block diagram for explaining an example of the internal structure of a playback apparatus (advanced VTS compatible DVD-Video player: another example of the apparatus shown in FIG. 72) according to another embodiment of the present invention. This DVD-Video player plays back and processes the recording content (DVD-Video content and/or advanced content) from information storage medium 1 (which records the VTSI and VTS_EVOBS shown in, e.g., FIGS. 93, 94, and the like) shown in FIGS. 1, 50, 73, 74, 79, and the like, and downloads and processes advanced content from a communication line (e.g., the Internet/home network or the like).

In the system arrangement of the embodiment shown in FIG. 100, interactive engine 200 comprises parser 210, advanced object manager 610, data cache 620,

streaming manger 710, event handler 630, system clock
214, interpreter unit 205 including a layout engine,
style engine, script engine, and timing engine, media
decoder unit 208 including moving picture/animation,
5 still picture, text/font, and sound decoders, graphics
superposing unit 750, secondary picture/streaming
playback controller 720, video decoder 730, audio
decoder 740, and the like.

On the other hand, DVD playback engine 100
10 comprises DVD playback controller 102, DVD decoder unit
101 including an audio decoder, main picture decoder,
sub-picture decoder, and the like, and so forth.

The DVD-Video player comprises, as functional
modules to be provided to interactive engine 200 and
15 DVD playback engine 100, persistent storage 216, DVD
disc 1, file system 600, network manager 212,
demultiplexer 700, video mixer 760, audio mixer 770,
and the like. Also, as modules which are the functions
of the DVD-Video player and are mainly used by
20 interactive engine 200 to perform information
acquisition and operation control via system manager
800, the player comprises an NIC, disc drive
controller, memory controller, FLASH memory controller,
remote controller, keyboard, timer, cursor, and the
25 like.

The recording locations and formats of advanced
content other than DVD-Video data to be handled by

interactive engine 200 are as follows (note that a disc described as a DVD disc includes not only a normal DVD-Video disc but also a next-generation HD_DVD disc).

1. file format data on the DVD disc
- 5 2. multiplexed divided data in an EVOB on the DVD disc
3. file format data in the persistent storage of the DVD-Video player
4. file format data or streaming data on a
10 network server on the Internet/home network

"File format data on the DVD disc" of "1." is stored in advanced content recording area 21 in FIG. 79. Interactive engine 200 loads an advanced content file on the DVD disc via the file system.

15 "Multiplexed divided data in an EVOB on the DVD disc" of "2." has a data format which is multiplexed and recorded in a VTS_EVOBS recorded in advanced HD video title set recording area (AHDVTS) 50 in FIG. 79. As the multiplexed data, data redundant to "file format data on the DVD disc" of "1." are recorded. Such data
20 is loaded to demultiplexer 700 in correspondence with loading of the VTS_EVOBS, and if the demultiplexed data are divided data of advanced content, they are sent to advanced object manager 610.

25 Advanced object manager 610 temporarily stores the divided data of the advanced content received from demultiplexer 700, and stores them as file format data

of the advanced content in data cache 620 at the reception timing of data that can form one file.

As multiplexed advanced content data in an EVOB on the DVD disc, file data obtained by compressing one or
5 a plurality of advanced content files in accordance with a predetermined method may be divisionally stored, so as to improve the efficiency of data upon multiplexing. In this case, advanced object manager 610 temporarily stores divided data until the
10 compressed data can be decompressed, and stores decompressed advanced content data in data cache 620 at a timing at which the advanced content data can be handled as a file format.

"File format data in persistent storage 216 of the
15 DVD-Video player" of "3." corresponds to, e.g., introduction movie data of a new film or the like which is downloaded from the Internet and is stored at a predetermined position on persistent storage 216 while interactive engine 200 is playing back a DVD title
20 including advanced content created by a given movie company.

For example, when a DVD title including other advanced content created by that movie company is played back, the following use method may be adopted.
25 That is, "interactive engine 200 searches the predetermined position on persistent storage 216 in accordance with the description of the markup/script

language of advanced content. If interactive engine 200 finds the saved introduction movie data of the new film there, it jumps to an XML page required to refer to/play back that data. If the playback process is selected by a user operation, interactive engine 200 plays back the introduction movie data of the new film stored in persistent storage 216."

An example of file format data of "file format data or streaming data on a network server on the Internet/home network" of "4." corresponds to the aforementioned introduction movie data of the new film or the like. As an example of streaming data, the following use method may be adopted. That is, "when DVD-Video data of a movie on a DVD disc includes only Japanese and English audio data, a movie company creates Chinese audio data, and a DVD-Video player connected to the Internet plays back the Chinese audio data in synchronism with video picture data on the DVD disc while sequentially downloading it."

In the system block diagram of FIG. 100, file system 600, parser 210, interpreter unit 205, media decoder unit 208, data cache 620, network manager 212, streaming manager 710, graphics superposing unit 750, secondary picture/streaming playback controller 720, video decoder 730, audio decoder 740, demultiplexer 700, DVD playback controller 102, DVD decoder unit 101, and the like can be implemented by a microcomputer

and/or hardware logic which implement/implements
respective module functions by parsing built-in
programs (firmware; not shown). A work area (including
a temporary buffer used in a decoding process) used
5 upon executing this firmware can be assured using a
semiconductor memory (not shown) (and a hard disc
device as needed) of each module. Furthermore, the
system includes communication means for control signals
(not shown) between respective modules so as to attain
10 data supply and a synchronization process, and
operation control between required modules can be
managed. The communication means include signal lines
of the hardware logic, event/data notification
processes between software programs, and the like.

15 The behaviors for respective functions of the DVD-
Video player will be described below using the system
block diagram of FIG. 100. The DVD-Video player that
plays back advanced content implements richly
expressive menus and more interactive playback control,
20 which are difficult to attain in the conventional DVD,
using an XML file and style sheet described using the
markup/script language or the like. An example in
which a menu page including a button selection that
outputs an animation effect or effect sound upon
25 selection of the user is to be configured will be
examined.

The configuration and functions of the menu page

are defined by a menu XML page described using the markup/script language. The menu XML page is stored in a DVD disc, and interpreter unit 205 passes the content of the menu XML page parsed by parser 210 to the layout engine, style engine, script engine, timing engine, and the like in accordance with their description content.

The timing engine receives time events from system clock 214 at predetermined intervals, and instructs processing instructions to the layout engine, style engine, and script engine on the basis of the description of the menu XML page arranged in the timing engine. These engines refer to configuration information of the menu XML page managed by them, and issue decode process instructions to media decoder unit 208 as needed.

Media decoder unit 208 loads media data from the advanced object save area such as data cache 620 or the like as needed in accordance with instructions from interpreter unit 205, and executes decode processes.

Of data decoded by media decoder unit 208, moving picture/animation, still picture, and text/font output results associated with graphics display are sent to graphics superposing unit 750, which generates frame data of a graphics plane to be output in accordance with the descriptions of the layout and style sheet of interpreter unit 205, and outputs it to video mixer 760.

Video mixer 760 mixes the output frame of graphics superposing unit 750, an output frame of the video decoder which is output in accordance with an instruction from secondary picture/streaming playback controller 720, output frames of the main picture decoder and sub-picture decoders in DVD decoder unit 101 which are output in accordance with an instruction from DVD playback controller 102, an output frame of the cursor function of the DVD-Video player, and the like in accordance with a predetermined superposing rule while synchronizing these output frames. Video mixer 760 converts the mixed output frame data into a television output signal, and outputs it onto a video output signal line.

The behavior of the secondary picture/streaming playback controller 720 which is output in synchronism with the output frame of the graphics frame will be described below. As a main storage destination of secondary picture data, a DVD disc and streaming server on the Internet or home network are assumed.

Upon playback of secondary picture data stored on the DVD disc, IFO/VOBS (including an EVOBS) data is loaded from the DVD disc to demultiplexer 700. Demultiplexer 700 identifies various types of multiplexed data, and demultiplexes and sends data associated with main picture playback control to DVD playback controller 102, data associated with main

picture, sub-picture, and audio of the DVD-Video to DVD decoder unit 101, and data associated with secondary picture playback control to secondary picture/streaming playback controller 720. If advanced object data are
5 multiplexed and stored in this data, these data are sent to advanced object manager 610.

Secondary picture/streaming playback controller 720 executes playback control of secondary picture data on the DVD disc on the basis of a playback control
10 signal from interpreter unit 205. For example, when interpreter unit 205 instructs not to execute playback of stored secondary picture data, all data are discarded here. When a playback instruction is issued, secondary picture/streaming playback controller 720
15 outputs data shaped to a format and data size suited to decode processes to video decoder 730 and audio decoder 740. Video decoder 730 and audio decoder 740 execute decode processes while synchronizing their output timings with the output from DVD decoder unit 101, in
20 accordance with an instruction from secondary picture/streaming playback controller 720.

Control signals instructed by secondary picture/streaming playback controller 720 include instructions of the video position, the degree of
25 scaling, that of a transparency process, a chroma color process, and the like to video decoder 730, and a volume control instruction, channel mixing instruction,

and the like to audio decoder 740.

When the user designates fastforwarding, jump, or the like via a remote controller or the like, event handler 630 acquires an event from the remote controller, and notifies the script engine of interpreter unit 205 of that event. The script engine runs in accordance with the markup/script description of an XML file used to execute playback control, and confirms the presence/absence of an event handler of the remote controller process. If the XML file used to execute the playback control defines an explicit behavior, the script engine executes a process according to the description; if nothing is defined, it executes a predetermined process.

When fastforwarding is to be executed as a result of the user's remote controller process, interpreter unit 205 instructs DVD playback controller 102 and secondary picture/streaming playback controller 720 to execute fastforwarding. DVD playback controller 102 re-configures a read schedule of VOBS data to change a data read process from the DVD disc in accordance with the fastforwarding instruction from interpreter unit 205. In this way, control is made to supply required data to fastforwarding playback of DVD playback controller 102 and DVD decoder unit 101 without causing any underflow. Since data to be supplied to secondary picture/streaming playback controller 720 are stored in

correspondence with the main picture data allocation, secondary picture data suited to fastforwarding playback are supplied from demultiplexer 700 in synchronism with the data read process required for fastforwarding executed by DVD playback controller 102.

Upon playing back stream data based on the secondary picture/streaming playback control, secondary picture/streaming playback controller 720 instructs streaming manager 710 to read streaming data on a predetermined network server and to supply the read data to itself on the basis of a playback control signal from interpreter unit 205.

Streaming manager 710 requests network manager 212 to execute a protocol control process of actual streaming data reception, and acquires data from the network server. At this time, for example, when the bit rate of the streaming data is high, look-ahead cashing of streaming data is made using a streaming buffer area on data cache 620 which is set in advance based on startup information, thus making control for broadening, e.g., an allowance of reception bit rate variations of streaming data.

In this case, streaming manager 710 temporarily stores streaming data from the network server in the streaming buffer on data cache 620, and supplies data stored in the streaming buffer on data cache 620 in response to a streaming data read request from

secondary picture/streaming playback controller 720.
When no streaming buffer is assured on data cache 620,
streaming manager 710 sequentially outputs streaming
data acquired from the network server to secondary
5 picture/streaming playback controller 720.

When secondary picture/streaming playback
controller 720 performs playback control of streaming
data on the network, it need not always perform
playback in synchronism with video picture playback of
10 DVD playback engine 100. For this reason, secondary
picture/streaming playback controller 720 need not play
back any streaming data even when DVD playback engine
100 does not perform any video picture playback, or it
need not synchronize the playback state of streaming
15 data with that (e.g., a special playback state such as
a fastforwarding state or pause state) of DVD playback
engine 100.

Upon executing the playback process of streaming
data read from a streaming server on the network, data
20 supply underflow is likely to occur. In this case, a
priority process can be designated in the description
of the markup/script language of advanced content to
flexibly define behaviors as follows. For example, the
playback process of DVD playback engine 100 is
25 preferentially executed, and DVD-Video playback is
continued even when streaming data is interrupted.
Alternatively, playback of streaming data is

preferentially executed, and DVD-Video playback is interrupted when streaming data is interrupted. Data to be played back by secondary picture/streaming playback controller 720 may be video data alone or
5 audio data alone.

An example of the functions of respective modules which form the system block diagram of FIG. 100 will be explained below.

Persistent storage 216: It stores generated file data, file data downloaded from the Internet/home
10 network, and the like in accordance with an instruction from interpreter unit 205. Data stored in persistent storage 216 are held even when the ON/OFF event of the power switch of the DVD-Video player occurs.

15 Interpreter unit 205 can erase data in persistent storage 216.

DVD disc 1: It stores advanced content and DVD-Video data. Sector data on the DVD disc are read in accordance with read requests from the file system and
20 demultiplexer.

File system 600: It manages the file system for respective recording modules/devices, and provides a file access function to file data read/write requests from the advanced object manager and the like. As an
25 example of the file system for respective recording modules/devices, when persistent storage 216 comprises a FLASH memory, a file system for the FLASH memory is

used to control to average memory rewrite accesses.
DVD disc 1 is accessed using a UDF or ISO9660 file
system. As for files on the network, network manager
212 executes actual protocol control such as HTTP,
5 TCP/IP, and the like, and the file system itself relays
the file access function to network manager 212. The
file system manages data cache 620 as, e.g., a RAM
disc.

Network manager 212: It provides a read (write as
10 needed) function of file data provided on an HTTP
server on the network to the file system. It also
executes actual protocol control in accordance with a
sequential read request of stream data from streaming
manager 710, acquires the requested data from the
15 streaming server on the network, and passes the
acquired data to streaming manager 710.

Demultiplexer 700: It reads data on the DVD disc
in accordance with a read instruction of sector data
that store IFO/VOBS data from DVD playback controller
20 102 (and the secondary picture/streaming playback
controller when secondary picture data alone is played
back). As for multiplexed data of the read data,
demultiplexer 700 supplies demultiplexed data to
appropriate processing units. Demultiplexer
25 700 supplies IFO data to the DVD playback
controller and secondary picture/streaming playback
controller 720. Demultiplexer 700 outputs main

picture/sub-picture/audio data associated with DVD-Video stored in a VOBS to DVD decoder unit 101, and control information (NV_PCK) to DVD playback controller 102. Demultiplexer 700 outputs control information and
5 picture/audio data associated with secondary picture data to secondary picture/streaming playback controller 720. When advanced objects are multiplexed in a VOBS, these data are output to advanced object manager 610.

Parser 210: It parses the markup language
10 described in an XML file and outputs the parsed result to interpreter unit 205.

Advanced object manager 610: It manages an advanced object file to be handled by interactive engine 200. Upon reception of an access request to an
15 advanced object file from parser 210, interpreter unit 205, media decoder unit 208, and the like, advanced object manager 610 confirms the storage state of file data on data cache 620 managed by manager 610. If the requested file data is stored in data cache 620,
20 advanced object manager 610 reads data from data cache 620, and outputs the file data to a module that issued the read request. If the requested data is not stored in data cache 620, advanced object manager 610 reads file data from the DVD disc, a network server on the
25 Internet/home network, or the like, which stores corresponding data, onto data cache 620, and simultaneously outputs the file data to a module that

issued the read request. As for data stored in persistent storage 216, advanced object manager 610 does not normally execute any cache process to data cache 620.

5 As another principal function of advanced object manager 610, when multiplexed advanced object data is stored in VOBS data loaded by demultiplexer 700, advanced object manager 610 temporarily stores these data output from demultiplexer 700, and stores them in
10 data cache 620 at a timing at which they can be stored as file data. When an advanced object file is stored in VOBS data in a format that compresses one or a plurality of files together, advanced object manager 610 temporarily stores divided data to a size that
15 allows decompression, and then decompresses and stores data in data cache 620 as file data.

Advanced object manager 610 stores advanced object data in data cache 620, and timely deletes a file, which becomes unnecessary in playback of the advanced
20 content of interactive engine 200, from data cache 620, in accordance with an instruction from interpreter unit 205 or a predetermined rule. With this delete process, the data cache area having a limited size can be effectively used in accordance with the progress of
25 playback of the advanced content.

Interpreter unit 205: This is a module for controlling the behavior of entire interactive engine

200. It initializes data cache 620 and DVD playback controller 102 in accordance with startup information, loading information, or playback sequence information parsed by parser 210. In the playback process of the advanced content, interpreter unit 205 passes layout information, style information, script information, and timing information parsed by parser 210 to respective processing modules, sends control signals to media decoder unit 208, secondary picture/streaming playback controller 720, DVD playback controller 102, and the like in accordance with their descriptions, and executes playback control among modules.

Layout engine: The layout engine (one of internal components of interpreter unit 205) handles information associated with objects used in graphics output of the advanced content. It manages definitions, attribute information, and layout information on the screen of moving picture/animation, still picture, text/font, sound objects, and the like, and also manages association information with style information about modifications upon rendering.

Style engine: The style engine (one of internal components of interpreter unit 205) manages information associated with detailed modifications upon rendering of rendering objects managed by the layout engine.

script engine: The script engine (one of internal components of interpreter unit 205) manages

descriptions associated with handler processes that pertain to button depression events from a user interface device (U/I device) such as a remote controller or the like and event messages from the system manager. Event handler 630 defines processing content upon occurrence of a corresponding event, and the script engine changes parameters of graphics rendering objects, and control of DVD playback controller 102, secondary picture/streaming playback controller 720, and the like in accordance with its description.

Timing engine: The timing engine (one of internal components of interpreter unit 205) controls scheduled processes associated with the behavior of graphics rendering objects and playback of secondary picture/streaming data. The timing engine refers to system clock 214, and when system clock 214 matches the timing of the scheduled control process, the timing engine controls respective modules to execute the playback process of the advanced content.

Media decoder unit 208: It executes the decode process of advanced objects in accordance with a control signal from interpreter unit 205. Media to be handled by media decoder unit 208 include cell animation that successively plays back still images of PNG/JPEG or the like as moving picture data, vector animation that successively renders vector graphics,

and the like. Media decoder unit 208 can handle JPEG, PNG, GIF, and the like as still picture data. Upon rendering text data, media decoder unit 208 mainly refers to font data such as vector font (open font) and the like and executes rendering of text data designated by interpreter unit 205. As sound data, those which have relatively short playback times such as PCM, MP3, and the like are assumed. Such sound data is mainly used a sound effect involved in an event such as button clicking or the like. Of the decode results of media decoder unit 208, the outputs associated with graphics are output to graphics superposing unit 750. Also, sound outputs are output to audio mixer 770.

Graphics superposing unit 750: It superposes the outputs of graphics rendering objects output from media decoder unit 208 in accordance with the descriptions of the layout engine and style engine, and generates output image frame data. Most of rendering objects have transparency process information, and graphics superposing unit 750 also executes a transparency calculation process of these objects. The generated output image frame data is output to video mixer 760.

Data cache 620: It is mainly used in two use applications. In one use application, data cache 620 is used as a file cache of an advanced object file, and temporarily stores an advanced object file on the DVD disc or network. In the other use application, data

cache 620 is used as a buffer of streaming data, and is managed by streaming manager 710. The allocations and sizes of the data cache used as the file cache and streaming buffer may be described in startup

5 information or the like and may be managed for respective advanced content, or the data cache may be used to have predetermined allocations.

Streaming manager 710: It manages supply of streaming data between secondary picture/streaming
10 playback controller 720 and network manager 212. When the bit rate of streaming data is relatively small and the streaming buffer need not be used, streaming manager 710 controls network manager 212 to sequentially supply streaming data acquired from a
15 streaming server to secondary picture/streaming playback controller 720.

When the bit rate of streaming data is relatively large, streaming manager 710 can control supply of streaming data using the streaming buffer which is
20 explicitly assured by the producer of advanced content. Streaming manager 710 stores data to be supplied to secondary picture/streaming playback controller 720 in the streaming buffer assured on data cache 620 in accordance with instructions of the streaming buffer
25 size and read-ahead size interpreted by interpreter unit 205. When the data of the instructed read-ahead size is stored in the stream buffer, streaming manager

710 begins to supply streaming data to secondary picture/streaming playback controller 720. At the same time, as soon as a free space of a given size is assured on the streaming buffer, streaming manager 710
5 issues a data acquisition request to the streaming server, thus efficiently managing the streaming buffer.

Secondary picture/streaming playback controller 720: It executes playback control of streaming data supplied from streaming manager 710 and secondary
10 picture data supplied from demultiplexer 700 in accordance with a playback control signal from interpreter unit 205.

Video decoder 730: It plays back video picture data supplied from secondary picture/streaming playback controller 720 in accordance with a control signal from
15 secondary picture/streaming playback controller 720. When video picture data is secondary picture data supplied from demultiplexer 700 or when it is instructed to synchronize streaming data with DVD video
20 picture playback, video decoder 730 decodes data to synchronize the output timing of DVD decoder unit 101 with its output timing, and outputs decoded data to video mixer 760.

Video decoder 730 has a chroma color process
25 function for video picture data as its characteristic function. It manages a chroma color area designated by a specific one color or a plurality of colors as a

transparent area to form output frame data of video mixer 760.

Audio decoder 740: It plays back audio data supplied from secondary picture/streaming playback controller 720 in accordance with a control signal from secondary picture/streaming playback controller 720. When audio data is that of secondary picture data supplied from demultiplexer 700 or when it is instructed to synchronize streaming data with DVD video picture playback, audio decoder 740 decodes data to synchronize the output timing of DVD decoder unit 101 with its output timing, and outputs decoded data to audio mixer 770.

DVD playback controller 102: It acquires playback control data of DVD-Video from demultiplexer 700 on the basis of a playback control signal from interpreter unit 205, and executes playback control of main picture/sub-picture/audio data of DVD decoder unit 101.

DVD decoder unit 101: It comprises an audio decoder, main picture decoder, sub-picture decoder, and the like, and manages decode processes and output processes while synchronizing respective decoder outputs in accordance with a control signal from DVD playback controller 102.

Audio decoder: The audio decoder in DVD decoder unit 101 decodes audio data supplied from demultiplexer 700 and outputs the decoded data to audio mixer 770 in

accordance with a control signal from DVD playback controller 102.

Main picture decoder: The main picture decoder in DVD decoder unit 101 decodes main picture data supplied from demultiplexer 700 and outputs the decoded data to video mixer 760 in accordance with a control signal from DVD playback controller 102.

Sub-picture decoder: The sub-picture decoder in DVD decoder unit 101 decodes sub-picture data supplied from demultiplexer 700 and outputs the decoded data to video mixer 760 in accordance with a control signal from DVD playback controller 102.

Video mixer 760: It receives output frames from graphics superposing unit 750, video decoder 730, the main picture decoder and sub-picture decoder in DVD decoder unit 101, and the cursor module, generates an output frame in accordance with a predetermined superposing rule, and outputs a video output signal. In general, each frame data has transparency information as the whole frame data or at an object or pixel level, and video mixer 760 superposes output frames from respective modules using such transparency information.

Audio mixer 770: It receives audio data from media decoder unit 208, audio decoder 740, and the audio decoder in DVD decoder unit 101, and generates and outputs an output audio signal in accordance with a

predetermined mixing rule.

System manager 800: It can provide an interface for status and control of respective modules in the DVD-Video player. Interpreter unit 205 acquires the status of DVD-Video player or can change the behavior via an application interface (API) or the like provided by the system manager.

Network connection controller (NIC): This is a module that implements a network connection function, and corresponds to an Ethernet controller (Ethernet is the registered trade name) or the like. The NIC provides information such as connection status of a network cable and the like via the system manager.

Disc drive controller: It corresponds to a reading device of a DVD disc, and provides status information such as the presence/absence of a DVD disc on a disc tray, disc type, and the like.

Memory controller: It manages the system memory: it provides an area to be used as data cache 620, and executes access management of a work memory used by respective software (firmware) modules.

FLASH memory controller: It provides an area used as persistent storage 216, and executes access management to the FLASH memory that stores execution codes and the like of respective software (firmware) modules.

Remote controller: It executes remote control of

the DVD-Video player, and generates a button depression event of the user to event handler 630.

Keyboard: It executes keyboard control of the DVD-Video player, and generates a keyboard depression event of the user to event handler 630.

Timer: It supplies system clocks, and provides a timer function used by the DVD playback time.

Cursor: It generates a pointer image of the remote controller or the like, and changes the position of the pointer image upon depression of direction keys and the like.

Interpreter unit 205 in FIG. 100 outputs a playback control signal to DVD playback controller 102. In this playback control signal, a new command is added to the conventional DVD playback control command, thus allowing more flexible playback control. That is, in order to define playback sequence information of an advanced VTS using the aforementioned playback sequence information (which corresponds to the PBSEQ001.XML file in FIG. 2, and is information stored in playback sequence information recording area 215A in FIG. 50, playback sequence information externally fetched via the Internet or the like, or playback sequence information which is generated by the system firmware when the user freely re-arranges chapter icons and is stored in persistent storage 216), a command for initializing using the playback sequence information

must be issued from interactive engine 200 to DVD playback engine 100.

An "InitPBSEQ()" command" is a command which is newly defined for the aforementioned purpose, and
5 allows interpreter unit 205 to notify DVD playback controller 102 of the playback sequence information of an advanced VTS to be played back and to initialize it. As an argument of the "InitPBSEQ command", sequence information of the PGC number, PTT numbers, and the
10 like as a basis of the playback sequence is given (see FIGS. 95 to 98). If the advanced VTS includes a plurality of PGCs, the PGC number specifies a PGC to be selected. The PTT numbers can define the order of chapters to be played back with reference to the
15 PGC_PGMAP number in the PGC designated by the PGC number. Since only one advanced VTS is stored on the DVD disc, and includes only one title, they need not be designated.

Note that the playback order can be described
20 using cell units, as described above. In this case, the argument of the "InitPBSEQ command" is sequence information of the PGC number and cell numbers. The cell numbers can define the order of cells to be played back with reference to the C_PBIT number in the PGC
25 designated by the PGC number. If the advanced VTS includes only one PGC, the argument of the PGC number in an "InitPBSEQ function" need not be used.

To summarize, the apparatus in FIG. 100 is configured to include the following elements. That is, the apparatus is configured to comprise a video playback engine (100) which plays back expanded video objects (EVOBs) from an information storage medium (disc 1); and an interactive engine (200) which acquires advanced content as information (e.g., 21A to 21E in FIG. 50) different from the recording content of a video data recording area from the information storage medium or an external server, and outputs an AV output corresponding to at least one of the playback output of the video playback engine and the content of the advanced content in accordance with the description of a markup language. The processing that "outputs an AV output corresponding to at least one of the playback output of the video playback engine and the content of the advanced content in accordance with the description of a markup language" can correspond to ST102 to ST104 + ST108 or ST106 + ST108 in FIG. 99.

FIG. 101 shows an example of the data structure of an advanced HD video title set program chain information table (AHDVTS_PGCIT) recorded in advanced HD video title set information (AHDVTSI). As shown in FIG. 101, advanced HD video title set program chain information table (AHDVTS_PGCIT) 512 records information of advanced HD video title set PGCI information table (AHDVTS_PGCITI) 512a including

information of the number (AHDVTS_PGCIT_SRP_Ns) of
AHDVTS_PGCIT_SRP data and the end address
(AHDVTS_PGCIT_EA) of the AHDVTS_PGCIT. In addition,
the advanced HD video title set information (AHDVTSI)
5 includes AHDVTS_PGCIT search pointers (AHDVTS_PGCIT_SRP)
512b and PGC information (AHDVTS_PGCIT) 512c as program
chain information in correspondence with the number
indicated by AHDVTS_PGCIT_SRP_Ns. Each AHDVTS_PGCIT
search pointer (AHDVTS_PGCIT_SRP) 512b includes
10 information of an AHDVTS_PGC category (AHDVTS_PGC_CAT)
indicating the type of AHDVTS_PGC, and the start
address (AHDVTS_PGCIT_SA) of AHDVTS_PGCIT. Note that the
AHDVTS_PGC category can have the same content as in
FIG. 24.

15 FIG. 102 shows an example of the plane
configuration upon superposing the output frames of
respective modules in video mixer 760 in FIG. 100. In
this example, main picture plane MVX output from the
main picture decode in DVD decoder unit 101 is arranged
20 at the lowermost position of the superposed planes.
Main picture plane MVX normally does not have
transparency information.

Secondary picture plane SVX is arranged on main
picture plane MVX. The output of this secondary
25 picture plane SVX includes video picture data of
streaming data (in this embodiment, video picture
decoding processes of secondary picture and streaming

data are exclusive, and these data are never decoded at the same time). Secondary picture plane SVX can have a transparency value of the entire plane as the superposing process with main picture data, and a
5 chroma color process can be applied to a non-transparent pixel region.

This chroma color process may be executed by video decoder 730, and may be implemented in a format including transparency information as output data of
10 video decoder 730. In this case, the transparency information of, e.g., a chroma color region is 0% (full transparency), and the remaining region has a transparency value applied to secondary picture data. The chroma color process may be executed by video mixer
15 760. In this case, for example, output data from video decoder 730 includes image frame data including a chroma color and chroma color information, and transparency value information for the secondary picture plane. Video mixer 760 applies a transparency
20 process to a region designated by the chroma color to be fully transparent and the remaining region to have an input transparency value on the basis of the input image frame data.

Sub-picture plane SPX arranged on secondary
25 picture plane SVX is the output from the sub-picture decoder in DVD decoder unit 101. On sub-picture plane SPX, a transparency value can be applied to sub-picture

rendering objects (text and highlight information).

Graphics plane GRX arranged on sub-picture plane SPX is the output frame of the graphics superposing unit, and a transparency value is applied to this plane at a pixel level. A transparency value of the entire object is generally designated for an advanced object using the markup language. When a rendering object itself can describe a transparency value at a pixel level like PNG data, a transparency value obtained by multiplying that for each pixel of the object itself and that for the entire object becomes the transparency value of the object image at the pixel level. Graphics superposing unit 750 executes superposing and transparency processes of a plurality of rendering objects, and outputs the final color values and transparency values of graphics plane GRX as output data to video mixer 760.

Cursor plane CUX arranged on graphics plane GRX is a plane of a pointer image of the remote controller, mouse, or the like, and is arranged at the uppermost position of all the image planes. In general, cursor plane CUX uses a transparency value for the entire pointer image.

Video mixer 760 executes the superposing process of the output image frames of respective modules in accordance with superposing models defined as described above. Note that the above definition is an example of

the superposing rule in video mixer 760, and a different superposing order of planes may be used or another transparency value process may be applied.

Another embodiment (an example without entry time
5 entry) of time map information (TMAPI) shown in
FIGS. 58 to 61 will be described below with reference
to FIGS. 103 to 107. FIGS. 103 and 104 show an example
of the time map configuration for EVOBs allocated in a
contiguous block. FIG. 103 shows example 1 in which
10 one TMAPI is stored in one TMAP file, and FIG. 104
shows example 2 in which one or more pieces of TMAPI
are stored in one TMAP file. As shown in FIGS. 103 and
104, one EVOB corresponds to one TMAPI, and a structure
that allows time to each EVOB (address conversion
15 using each TMAPI stored in a file is adopted. Each
TMAPI includes one or more pieces of EVOBU entry
information, and EVOBUS in each EVOB can be accessed
using this information.

FIG. 105 shows an example of the time map
20 configuration for EVOBs which are allocated in an
interleaved block and form angles, so as to allow the
user to attain seamless angle switching. As shown in
FIG. 105, an EVOB for one angle corresponds to one
TMAPI, and a structure that allows time to each EVOB
25 (address conversion using each TMAPI stored in the
file is adopted as in the EVOB time map allocated in
the contiguous block. Each TMAPI includes one or more

pieces of EVOBU entry information and one or more
pieces of ILVU entry information, and the head of each
ILVU in each EVOB and each EVOBU in that ILVU, which
are allocated in the interleaved block, can be
5 accessed.

Since each EVOB allocated in the interleaved block
is stored in one file, all pieces of time map
information required to play back that angle period can
be acquired, and required files need not be searched
10 for each time, thus improving the processing
efficiency.

FIGS. 106 and 107 show an example of the data
structure of a time map including no time entry. As
shown in FIG. 106, a time map information (TMAPI) table
15 includes TMAP information table information (TMAPITI)
indicating the configuration of TMAPI stored in a file,
a TMAP information search pointer group (TMAPI_SRPs)
that gives a search pointer to each stored TMAPI, and a
TMAP information group (TMAPIs) that stores EVOBU entry
20 information of each TMAPI.

Time map information table information TMAPITI
includes information (TMAPI_Ns) indicating the number
of pieces of TMAPI stored in a TMAP file, block type
information (TMAP_TYPE) indicating whether the block
25 type of an EVOB stored in the TMAP file is a contiguous
block (= 0) or interleaved block (= 1), angle type
information (AGL_TYPE) indicating whether the angle

type of an EVOB stored in the TMAP file is no angle (= 0), non-seamless angle (= 1), or seamless angle (= 2), and information (TMAPIT_EA) indicating the end address of the table.

5 Each time map information search pointer TMAPI_SRP includes information (TMAPI_SA) indicating the start address of target TMAPI, information (EVOB_IDN) indicating the identification number of an EVOB designated by the target TMAPI, information (EVOB_ADR) indicating the start address of the EVOB designated by the target TMAPI, information (EVOB_PB_TM) indicating the playback time of the EVOB designated by the target TMAPI using, e.g., the number of fields, information (EVOBU_ENTI_Ns) indicating the number of pieces of EVOBU entry information stored in the target TMAPI, information (ILVU_ENTI_Ns: if no interleaved block is formed, ILVU_ENTI_Ns = 0) indicating the number of pieces of ILVU entry information stored in the target TMAPI, and information (AGLN: if no angle block is formed, AGLN = 0) indicating the angle number of the EVOB of the target TMAPI.

 As shown in FIG. 107, each time map information TMAPI includes an EVOBU_ENTI group and ILVU_ENTI group. The EVOBU_ENTI group includes one or more pieces of EVOBU entry information (EVOBU_ENTI). Each EVOBU_ENTI includes a size (EVOBU_SZ) of each EVOBU stored in an EVOB, which is indicated by, e.g., the number of packs,

a playback time (ESOBU_PB_TM) indicated by, e.g., the number of fields, and a size (1STREF_SZ) of first reference picture data, which is indicated by, e.g., the number of packs).

5 The ILVU_ENTI group includes one or more ILVU entry information (ILVU_ENTI). Each ILVU_ENTI includes the start address (ILVU_ADR) of each ILVU stored in an EVOB, and a size (ILVU_SZ) of each ILVU, which is indicated by, e.g., the number of EVOBUs.

10 FIG. 108 shows an example of the structure which is different from that of a navigation pack (NV_PCK) shown in FIG. 63. As in NV_PCK, a general control information pack (GCI_PCK) allocated at the head of an EVOBU uses standard GCI_PCK shown in FIG. 108(a) in an
15 EVOB in a standard VTS. This pack includes general control information (GCI) stored in a general control information packet (GCI_PKT), presentation control information (PCI) stored in a presentation control packet (PCI_PKT), and data search information (DSI)
20 stored in a data search information packet (DSI_PKT).

 Also, in an EVOB in an advanced VTS, an advanced GCI_PCK shown in FIG. 108(b) is used. This pack includes general control information (GCI) stored in a general control information packet (GCI_PKT) and data
25 search information (DSI) stored in a data search information packet (DSI_PKT).

 FIG. 109 shows information stored in the general

control information (GCI). The general control information includes information (GCI_GI) associated with the entire EVOBU and pack in which that information is stored, information (DCI_CCI_SS) indicating the states of copy control information and display control information in the EVOBU and pack, display control information (DCI) indicating the aspect ratio and the like, copy control information (CCI) such as CGMS information, analog copy control information, and the like, recording information (RECI) that gives copyright information such as ISRC data or the like, and so forth.

FIG. 110 shows another embodiment of the data structure of advanced VTS 151a. As shown in FIG. 110, advanced HD video title set information (AHDVTSI) area 51 shown in FIG. 51(e) is divided into areas (management information groups) including advanced HD video title set information management table (AHDVTSI_MAT) 510a including no attribute information of video data, audio data, and the like, advanced HD video title set search pointer table (AHDVTS_PTT_SRPT) 511a used to search for the head of a part of title (PTT) corresponding to a chapter part of a title, advanced HD video title set program chain information table (AHDVTS_PGCIT) 512a that gives the playback sequence of a title, advanced HD video title set attribute information table (AHDVTS_ATTRIT) 515a that

gives attribute information of each EVOB, and advanced HD video title set expanded video object set information table (AHDVTS_EVOBIT) 516a that gives information of each EVOB.

5 FIG. 111 shows an example of the data structure which shows the content of the advanced HD video title set attribute information table (AHDVTS_ATTRIT). As shown in FIG. 111, AHDVTS_ATTRIT 515a includes advanced HD video title set attribute information table
10 information (AHDVTS_ATTRITI), one or more advanced HD video title set attribute information search pointers (AHDVTS_ATTRI_SRP), and one or more pieces of advanced HD video title set attribute information (AHDVTS_ATTRI).

 The advanced HD video title set attribute
15 information table information (AHDVTS_ATTRITI) has (AHDVTS_ATTRI_SRP_Ns) indicating the number of pieces of attribute information stored in the table (the number of AHDVTS_ATTRI_SRPs), and (AHDVTS_ATTRIT_EA) indicating the end address of the table. The advanced HD video
20 title set attribute information search pointer (AHDVTS_ATTRI_SRP) has (AHDVTS_ATTRI_SA) indicating the start address of each attribute information. The advanced HD video title set attribute information (AHDVTS_ATTRI) indicates attribute information for a
25 corresponding EVOB.

 More specifically, the AHDVTS_ATTRIT has information (AHDVTS_V_ATTR) indicating video attribute information

such as MPEG-2, MPEG-4 AVC (H.264), SMPTE VC-1, and the like stored in an EVOB, information (AHDVTS_AST_Ns) indicating the number of audio streams, audio stream attribute information (AHDVTS_AST_ATR) such as DD+,
5 DTS++, MLP, LPCM, and the like (all of DD for Dolby Digital, DTS for Digital Theater System, and MLP for Meridian Lossless Packing are the registered trade names) stored in an EVOB, multi-channel audio stream attribute information (AHDVTS_MU_AST_ATR), information
10 (AHDVTS_SPST_Ns) indicating the number of sub-picture streams, sub-picture stream attribute information (AHDVTS_SPST_ATR) indicating the SD size (2 bits/pixel), HD size (2 bits/pixel), SD/HD size (8 bits/pixel), or the like stored in an EVOB,
15 information (AHDVTS_SPST_SDPLT) indicating a color palette for sub-picture SD, information (AHDVTS_SPST_HDPLT) indicating a color palette for sub-picture HD, and the like.

FIG. 112 shows an example of the data structure
20 that shows the content of the advanced HD video title set EVOB information table (AHDVTS_EVOBIT). As shown in FIG. 112, AHDVTS_EVOBIT 516a includes advanced HD video title set EVOB information table information (AHDVTS_EVOBITI), one or more advanced HD video title
25 set EVOB information search pointers (AHDVTS_EVOBI_SRP), and one or more pieces of advanced HD video title set EVOB information (AHDVTS_EVOBI).

The advanced HD video title set EVOB information table information (AHDVTS_EVOBITI) has information (AHDVTS_EVOBI_SRP_Ns) indicating the number of pieces of EVOB information stored in the table. (the number of AHDVTS_EVOBI_SRPs) and information (AHDVTS_EVOBIT_EA) indicating the end address of the table. Note that the advanced HD video title set EVOB information search pointer (AHDVTS_EVOBI_SRP) has information (AHDVTS_EVOBI_SA) indicating the start address of each EVOBI. The advanced HD video title set EVOB information (AHDVTS_EVOBI) has information (EVOB_IDN) of an EVOB identification number used to identify each EVOB, information (EVOB_ATRN) of an EVOB attribute information number indicating an attribute corresponding to each EVOB, information (TMAP_FILE_NAME) indicating the time map file name that stores time map information used to access each EVOB, and the like. Note that the number described in EVOB_ATRN is the number indicated by the advanced HD video title set attribute information search pointer (AHDVTS_ATRI_SRP#) of the advanced HD video title set attribute information table (AHDVTS_ATRIT).

Next, still another embodiment of the present invention will be described. The object of this embodiment is to provide a method of reading out data in order to play back a Movie Object (to be referred to as a Primary hereinafter) serving as a playback

subject, and an Advanced Object (to be referred to as a Secondary hereinafter) which can be played back simultaneously with playback of the Movie Object. In order to achieve this object, two methods are
5 available. One is a method of multiplexing (MUXing) the Primary and the Secondary as one program stream (PS), and the other is a method of multiplexing the Primary and the Secondary as two PSs. When the Primary and the Secondary are multiplexed as the two PSs, the
10 Primary and the Secondary can be multiplexed using Pack units or Access Units (AUs) in an Angle format.

FIG. 113 shows an example of a case (case 1) wherein one program stream (1PS) obtained by multiplexing the Primary Object (Movie Object) and the
15 Secondary Object (Advanced Object) is recorded on a disc, and the Advance Object (Secondary Object) is independently present as another program stream on an external communication line (Web).

In the case 1, the Primary/Secondary may be
20 multiplexed (MUXed) as one PS using Pack units. In this case, as shown in FIG. 113, the Objects of the Primary and Secondary are managed in accordance with video title set information (VTSI: corresponding to an HVDVD_TS file shown in FIG. 2, AHDVTSI in FIG. 52, and
25 the like), and the Secondary information is managed in accordance with advanced object information (AOBI: corresponding to an ADV_OBJ file in FIG. 2). FIG. 114

shows a Decoding model for this arrangement.

FIG. 114 is a view for explaining the decoding model in case 1. The PS sent from the Disc (corresponding to disc 1 shown in FIGS. 1, 50, 51, 73, 74, and 79) is demultiplexed by first demultiplexer (DeMUX1) 114a to send the PS to Decoders 114n to 114s of the Primary and Secondary. The demultiplexed PSs are stored in Input Buffers 114g to 114m. The Secondary Contentent from the Web is temporarily stored in Buffer 114f for playing back in synchronism with the Disc, and then sent to Input Buffers 114k and 114m via second demultiplexer (DeMUX2) 114b and switches SW1 and SW2. By mixing Data decoded by Decoders 114n to 114s, the Objects of the Primary and Secondary can be displayed simultaneously (synchronously).

Next, a case wherein two PSs including the Primary as one program stream (one PS) and the Secondary as one PS are used will be described. FIG. 115 shows an example of a case (case 2-1) wherein the PSs of the Primary and Secondary Objects are recorded as two program streams (PS-1 VOB and PS-2 VOB) obtained by multiplexing these objects using pack units, and the Advanced Object (Secondary Object) is independently present as another PS on the external communication line (Web). In case 2-1, two objects are multiplexed (MUX) using Pack units.

FIG. 116 is a view for explaining the decoding

model in case 2-1. The content on the Disc is divided into the Primary and Secondary streams by first demultiplexer (DeMUX1) 116a. The divided streams are sent to second demultiplexer (DeMUX2) 116b and third demultiplexer (DeMUX3) 116c in order to send the divided streams to the corresponding Decoders. Since the Secondary content is sent from the network (Web), the DeMUX3 116c selectively receives the content from the Web (if there is content on the Web) or the Disc (if there is no content on the Web), via switch SW3.

FIG. 117 shows an example of a case (case 2-2) wherein the PSs of the Primary and Secondary Objects are multiplexed and recorded as the two program streams using access units (AUs), and the Advanced Object (Secondary Object) is independently present as another PS on the external communication line (Web). In case 2-2, the two PSs are multiplexed using AUs. In this case, in the arrangement similar to that of an ILVU on a currently used DVD-Video, the recorded content of the Primary and Secondary can be displayed simultaneously (synchronously) by simultaneously displaying a plurality of Angles.

Note that in case 2-2, in comparison with case 2-1, the size of each access unit is increased. (The pack size in case 2-1 is only 2 kB. However, in case 2-2, since the access unit includes the plurality of packs, the size of the access unit becomes relatively large.)

Hence, upon storage of the object data in Input Buffer (e.g., 116g shown in FIG. 116), the object data is started to be supplied to the Decoder. After that, the data loading rate of the Input Buffer cannot catch up with the consuming rate (data readout rate of Input Buffer) of the buffering data. A countermeasure for this problem will be described below.

FIG. 118 is a view for explaining the decoding model in case 2-2. In this model, Buffers 118d to 118f for stably supplying data to second demultiplexer (DeMUX2) 118b and third demultiplexer (DeMUX3) 118c are connected to the output of first demultiplexer (DeMUX1) 118a. (The maximum data amount to be buffered to these Buffers, i.e., buffer size to be used can be determined on the basis of the simulation result of the disc or Web to be actually used. More specifically, Buffer 118f for the Web is preferably large enough to avoid the complete consumption of the buffering data even when the data transmission from the external communication line is unstable.)

Next, a method of using a stream_id to indicate the Content of the Primary or Secondary will be described. Each of these Demultiplexers (DeMUX1 to DeMUX3) demultiplexes the stream by using this stream_id (and sub_stream_id as needed). This demultiplexing process is performed to send the Data to Input Buffers 118g to 118m which respectively output

the demultiplexed data to Decoders 118n to 118s.

As the setting method of the stream_id and sub_stream_id, two methods are available. One is a method of defining an identifier (id) for the Secondary Content in a private_stream1 in the currently used DVD-Video standard (see FIGS. 119 to 121), and the other is a method of newly providing a private_stream3 to have a Secondary id (see FIGS. 122 to 125).

FIG. 119 is a view for explaining an example of the stream_id which is used to identify the content of the Primary and Secondary Objects (when the private_stream1 is used to identify the objects). As needed, this stream_id includes "110x0***b" indicating an MPEG audio stream *** corresponding to a decoding audio stream number, "11100000b" indicating a video stream, "10111101b" indicating the private_stream1, "10111111b" indicating the private_stream2, and others (e.g., an area which is not currently used).

FIG. 120 shows an example of the arrangement of the sub_stream_id for the private_stream1 in the stream_id shown in FIG. 119. As needed, this sub_stream_id includes "001*****b" indicating the sub-picture stream, "01001000b" for reservation, "011*****b" for reservation for an expanded sub-picture, "10000***b" indicating the Dolby AC-3 (registered trademark), "10001***b" optionally indicating the DTS (registered trademark) audio stream,

"10010***b" optionally indicating the SDDS (registered trademark) audio stream, "10100***b" indicating a linear PCM audio stream, "11111111b" indicating a stream defined by a content provider, "10010001b" indicating the MPEG2 picture stream of the Secondary Content, "10010010b" indicating the MPEG4/AVC stream of the Secondary Content, "10010011b" indicating the VC-1 stream of the Secondary Content, "11000***b" indicating the Dolby Digital+ (registered trademark) stream of the Secondary Content, "11001***b" indicating the DTSHD (registered trademark) stream of the Second Content, "11010***b" indicating the SDDS (registered trademark) audio stream of the Secondary Content, "11100***b" indicating the linear PCM audio stream of the Secondary Content, and others (for future presentation data).

FIG. 121 shows an example of the arrangement of the sub_stream_id for the private_stream2 in the stream_id shown in FIG. 119. As needed, this sub_stream_id includes "00000000b" indicating the stream of a Presentation Control Information (PCI), "00000001b" indicating the stream of a Data Search Information (DSI), "11111111b" indicating the stream defined by the content provider, and others (for future navigation data).

FIG. 122 is a view for explaining another example of the stream_id used to identify the content of the Primary and Secondary Objects (when the private_stream3

is newly provided to identify the objects). As needed, this stream_id includes "110x0***b" indicating an MPEG audio stream *** corresponding to the decoding audio stream number, "11100000b" indicating the video stream, 5 "10111101b" indicating the private_stream1, "10111111b" indicating the private_stream2, "10110000b" indicating the private_stream3, and others (e.g., an area which is not currently used).

FIG. 123 shows an example of the arrangement of 10 the sub_stream_id for the private_stream1 in the stream_id shown in FIG. 122. As needed, this sub_stream_id includes "001*****b" indicating the sub-picture stream, "01001000b" for reservation, "011*****b" for reservation for the expanded sub- 15 picture, "10000***b" indicating the Dolby AC-3 (registered name), "10001***b" optionally indicating the DTS (registered name) audio stream, "10010***b" optionally indicating the SDDS (registered name) audio stream, "10100***b" indicating the linear PCM audio 20 stream, "11111111b" indicating the stream defined by the content provider, and others (for future presentation data). The sub_stream_id for the private_stream1 shown in FIG. 123 has the content obtained by excluding the content pertaining to the 25 "Secondary Content" from the sub_stream_id for the private_stream1 shown in FIG. 120.

FIG. 124 shows an example of the arrangement of

the sub_stream_id for the private_stream2 in the stream_id shown in FIG. 122. As in the case shown in FIG. 121, as needed, this sub_stream_id includes "00000000b" indicating a PCI stream, "00000001b"

5 indicating a DSI stream, "11111111b" indicating the stream defined by the content provider, and others (for future navigation data).

FIG. 125 shows an example of the arrangement of the sub_stream_id for the private_stream3 in the stream_id shown in FIG. 122. As needed, this sub_stream_id includes "10010001b" indicating the MPEG2 video stream of the Secondary Content, "10010010b" indicating the MPEG4/AVC stream of the Secondary Content, "10010011b" indicating the VC-1 stream of the Secondary Content, "11000***b" indicating the Dolby Digital+ (registered trademark) stream of the Secondary Content, "11001***b" indicating the DTSHD (registered name) stream of the Secondary Content, "11010***b" indicating the SDDS (registered trademark) audio stream of the Secondary Content, "11100***b" indicating the linear PCM audio stream of the Secondary Content, "11111111b" indicating the stream defined by the content provider, and others (for future presentation data). The sub_stream_id for the private_stream3 shown in FIG. 125 mainly includes the content pertaining to the "Secondary Content" in the sub_stream_id for the private_stream1 shown in FIG. 120.

FIG. 126 is a flowchart for explaining an example of the processing sequence when the primary object and/or secondary object is played back from the disc and/or external communication line (Web). In FIG. 126, a sequence for playing back the Secondary Content (or Secondary/2ndary Video Set) using a Markup document. That is, when no Markup document is present on the Disc (corresponding to information storage medium 1 shown in FIG. 50 and the like) (NO in step ST202), a player (e.g., a playback apparatus having the arrangement shown in FIG. 100) performs playback using a standard VTS (corresponding to normal DVD-video content, or HDVTS# shown in FIG. 1) (step ST204).

When the Markup document is present on the Disc (YES in step ST202), the player determines whether a NET (Web) connection destination is described in the Markup document. If no connection destination is described (NO in step ST206), it is checked whether the Secondary Video Set is present using the Markup document on the Disc (step ST208). If no Secondary Video Set is present (NO in step ST210), the Primary Video Set is played back (step ST212).

When the NET (Web) connection destination is described in the Markup document (YES in step ST206), the connection state is checked. If no connection is assured (NO in step ST214), the flow advances similar to the preceding step (NO in step ST206), and then the

Primary Video Set is played back (step ST212) or the Secondary Video Set is played back (step ST224), using the Markup document on the Disc (step ST208).

When the NET connection is assured (YES in step
5 ST214), it is determined whether the Secondary Video Set is stored on the NET. If no Secondary Video Set is stored (NO in step ST216), it is determined whether the Markup document is stored on the NET. If neither
10 Secondary Video Set nor the Markup document are present on the NET (NO in steps ST216 and ST218), the Secondary Video Set is played back (step ST224) or the Primary Video Set is played back (step ST212), using the Markup document on the Disc (step ST208).

If only the Secondary Video Set is present on the
15 NET (YES in step ST216), and no Markup document is present on the NET (NO in step ST226), the Secondary Video Set is loaded (step ST230), and updated attribute information and updated playback information in the TMAP and VTSI are loaded (step ST232). These loaded
20 pieces of information are added to current playback control information (navigation data) to start playback of the Secondary Video Set on the NET at the playback start timing of the Markup document on the Disc (step ST234).

25 Alternatively, when no Secondary Video Set is present on the NET (NO in step ST216), and only the Markup document is present on the NET (YES in step

ST218), the Markup document is updated (step ST220), and then the updated attribute information and the updated playback information in the TMAP and VTSI are loaded (step ST222). These loaded pieces of

5 information are added to the current playback control information (navigation data) to start playback of the Secondary Video Set on the Disc at the playback start timing of the updated Markup document (step ST224).

In this case, since the Secondary Video Set is not
10 updated, the TMAP and the like need not be updated. When both of the Markup document and the Secondary Video Set are present (YES in steps ST216 and ST226), the Markup document is updated (step ST228), and the required information is added (step ST232).

15 Accordingly, the Secondary Video Set on the NET is played back.

According to the process shown in FIG. 126, as needed, the Secondary Video Set can be played back from the Disc (step ST224) or the NET (Web) (step ST234).

20 At this time, an indicator (e.g., LED in different colors) (not shown) may be provided with the player, or the original from which the currently played back Secondary Video Set is obtained may be displayed by OSD (On Screen Display) on the screen of a monitor TV, such
25 that a user can easily recognize whether the Secondary Video Set obtained via the NET connection is currently displayed, or currently played back in accordance with

the information on the Disc.

Note that, in accordance with the load attribute (e.g., see <object load = "disc" data> and <object load = "net" data> in description example 3 shown in
5 FIG. 132 to be described later) in the Markup document, it can be determined whether the currently played back Secondary Video Set is obtained from the Disc or NET.

Note that in the process shown in FIG. 126, only the playback process of the Secondary Video Set is
10 described. However, of course, the Primary Video Set can be played back from the Disc simultaneously with the playback of the Secondary Video Set. In this case, the used Markup document (step ST208, ST220, or ST228) can designate the playback timing of the Secondary
15 Video Set (from the Disc or NET) with respect to the currently played back Primary Video Set. This description example of the Markup document will be described with reference to FIGS. 130 to 132.

FIG. 127 is a view for explaining the playback
20 path of the Primary Object/Primary Content (Primary Video Set) and Secondary Object/Secondary Content (Secondary Video Set) from the Disc. In this example, Secondary Content playback time or playback start enable time upon user's operation is described in the
25 Markup document recorded on the Disc. (This "playback start enable time upon user's operation" corresponds to the duration for holding the Secondary Content in

Buffers 114f, 116f, 118f, and the like shown in
FIGS. 114, 116, and 118.)

In FIG. 127, in interleaved block period T23
succeeding the period of VOB#1 (Primary Content)

5 recorded in contiguous block period T01 on the Disc, a
VOB#2 (Primary Content) and a VOB#3 (Secondary Content)
are interleaved and recorded using ILVUs. A VOB#4
(Primary Content) is recorded in contiguous block
period T04 succeeding the recorded period of VOB#2 and
10 VOB#3. At this time, when the VOB#2 is the Primary,
and the VOB#3 is the Secondary, the playback start time
and playback end time (or playback start available
period) of the Secondary Content (VOB#3) are set in the
Markup document on the Disc. The playback start and
15 end times are the times for starting and ending
playback of the VOB#3, literally. The playback start
available duration is a duration in which the Secondary
is stored in the Buffer, and playback can be started
upon user's operation. For example, period T23 shown
20 in FIG. 127 is the playback available duration, the
VOB#3 (Secondary Content) can be played back together
(simultaneously or synchronously) with the VOB#2
(Primary Content) at the timing defined by the TMAP of
the VOB#3, in period T23.

25 FIG. 128 is a view for explaining the playback
path of the Primary Object/Primary Content (Primary
Video Set) from the Disc, and the Secondary

Object/Secondary Content (Secondary Video Set) from the external communication line (NET/Web). In FIG. 128, in interleaved block period T27 succeeding the period of VOB#1 (Primary Content) recorded in contiguous block period T01 on the Disc, the VOB#2 (Primary Content) and the VOB#3 (Secondary Content) are interleaved and recorded using ILVUs. The VOB#4 (Primary Content) is recorded in contiguous block period T04 succeeding the period of VOB#2 and VOB#3. However, in this example, a VOB#7 (Secondary Content) from the NET/Web is played back together with the VOB#2 (Primary Content) in period T27, in place of the VOB#3 (Secondary Content) from the Disc.

In the example shown in FIG. 128, the new Secondary Video Set of the VOB#7, new Markup document, VTSI file, and TMAP file are obtained from the NET. In the new Markup document in this example, the VOB#3 is not described and displayed (even if the VOB#3 is recorded on the Disc). In addition, since the Markup document and TMAP are updated when the new Markup document is obtained from the NET, the playback duration of the VOB#3 defined in FIG. 127 and the playback duration of the VOB#7 shown in FIG. 128 need not be matched. (That is, if T23 in FIG. 127 = T27 in FIG. 128, the playback durations of the VOB#3 and VOB#7 can be individually and arbitrarily set if each of the playback durations falls within a time range

corresponding to period T27.)

FIG. 129 shows an example of the data structure of a time map information table including the time map type flag (TMAP_TYPE_FL). In FIG. 129, the flag
5 (TMAP_TYPE_FL) for determining whether the TMAP is the Primary or Secondary, is added to time map information search pointer 519b in time map information table 519 which is described above with reference to FIG. 58. Hence, when the player loads and extends the TMAP from
10 the Disc, the current TMAP of the player can be smoothly replaced with a new one.

Note that in FIG. 129, the TMAP_TYPE_FL includes only one bit since the TMAP_TYPE_FL is only used to determine "whether the TMAP is the Primary or
15 Secondary". However, the TMAP_TYPE_FL can be extended to include a plurality of bits. For example, when the TMAP_TYPE_FL includes two bits, "00b" can specify the Primary Object TMAP from the Disc, "01b" can specify the Secondary Object TMAP from the Disc, "10b" can
20 specify the Secondary Object TMAP from the NET/Web, and "11b" can specify the Secondary Object TMAP from others.

FIGS. 130 to 132 are views for explaining description examples 1 to 3 of the Markup document. In
25 these description examples, three types of objects are assumed. That is, the first type is the Primary Content (e.g., <object load = "disc" data = "main.mpg">

shown in FIG. 130), and the remaining two object types are the Secondary Content. One of these Secondary Content performs playback in a playback time defined by the TMAP (e.g., <object load = "disc" data = "sec.mpg"> shown in FIG. 131), and the other starts playback at the timing of user's operation in the playback time of the TMAP (e.g., <object load = "disc" data = "sec2.mpg"> shown in FIG. 132). These object types can be determined using the data attribute of an object tag, and a type tag. A server tag (e.g., <server url = "http://dvdrom/dvd_ihd"> shown in FIG. 130) indicates the connection destination of the NET connection, and an operation as shown in the flowchart in FIG. 126 is assumed. When the Markup document is included in this connection destination, the player replaces the Markup document of the NET connection destination with the Markup document on the Disc, and uses the replaced Markup document for playback.

In description example 1 of the Markup document shown in FIG. 130, the Secondary Object from the Disc is played back in the playback time "03:15" to "05:40" (TMAP in evobid = "2") of an EVOB#2 serving as the Secondary Object.

In description example 2 of the Markup document shown in FIG. 131, the Secondary Object from the Disc is started to be played back in the playback time "03:15" to "05:40" (TMAP in evobid = "2") (content for

total 2' 25'') of the EVOB#2 serving as the Secondary Object. Additionally, the Secondary Object from the Disc is started to be played back in the playback time "04:43" to "07:08" (TMAP in evobit = "2") (content for
5 total 2' 25''). Note that the TMAP (<type = "sec_sp"/>) in the playback time "04:43" to "07:08" is the example of the playback start timing designated by user's operation.

In description example 3 of the Markup document
10 shown in FIG. 132, the Primary Object from the Disc is played back in the playback time "00:00" to "07:10" (TMAP in evobid = "1") of the EVOB#1 serving as the Primary Object. Alternatively, the description is rewritten in accordance with the Markup document
15 obtained (<object load = "net" data = "sec2.mpg">) from the NET, to start playback of the Secondary Object from the NET in the playback time "02:55" to "03:58" (TMAP in evobid = "3"). (In this example, since the Markup document is obtained from the NET, for example, <object
20 load = "disc" data = "sec.mpg"> shown in FIG. 130 is rewritten to <object load = "net" data = "sec2.mpg"> shown in FIG. 132.)

FIG. 133 is a view showing another example of a case (case 1a) wherein one program stream (PS) obtained
25 by multiplexing the Primary Object (Movie Object) and the Secondary Object (Advanced Object) on the Disc is recorded, and the Advanced Object (Secondary Object) is

independently present as the program stream on the external communication line (NET/Web).

In the example shown in FIG. 133, the Secondary Content which have not multiplexed with the Primary Content are multiplexed with the Secondary Object (Secondary EVOB) in advance, and the multiplexed Secondary EVOB is multiplexed with the Primary Content to implement one PS arrangement in a multistage process (the multiplexed Secondary has been implemented when the Primary and Secondary are multiplexed). This means that the model of one PS shown in FIG. 133 is improved.

FIG. 134 is a view showing still another example of a case (case 1b) wherein one program stream (PS) obtained by multiplexing the Primary Object (Movie Object) and the Secondary Object (Advanced Object) on the Disc is recorded, and the Advanced Object (Secondary Object) is independently present as the program stream on the external communication line (NET/Web). FIG. 135 is a view for explaining the Decoding model in case 1a, and FIG. 136 is a view for explaining an example of a smoothing buffer operation of the decoding model in case 1a.

In the Decoding Model shown in FIG. 114, the overall system operates as a model having a constant bitrate. In this case, since the bitrate of the Primary is assumed to be higher than that of the Secondary, the bitrates of the system, Primary, and

Secondary are respectively set to be 30, 20, and 10 Mbps in the Decoding Model shown in FIG. 135. Even if the average bitrate of the Secondary is 10 Mbps, the data may be temporarily input at a system bitrate of 30 Mbps. Hence, each of Input Buffers 114g to 114m in the former stage of decoders 114n to 114s shown in FIG. 114 must have an appropriate size to avoid an overflow. Under this situation, in order to operate the Decoding Model shown in FIG. 135 without contradiction, the multiplexing process shown in FIG. 133 or 134 must be limited. The example of the limited Pack structure of the Secondary Video Set will be described below.

FIG. 136 is a view for explaining an example of the smoothing buffer operation in the decoding model in case 1a. The upper side in FIG. 136 shows an example of the Pack structure input to a model of 30 Mbps of the DeMUX1. The lower side in FIG. 136 schematically shows the Pack structure having a lowered bitrate of 10 Mbps. In order to absorb the difference with the model having a bitrate of 10 Mbps in smoothing buffer 135x shown in FIG. 135, an interval of at least two Packs is required as shown in FIG. 136. Because of the Buffer overflow, the next Pack cannot flow before the output of the stream in the lower side in FIG. 136 from smoothing Buffer 135x.

In addition, when one Pack is set to be 2 KB, the

values of a system clock reference SCR and a presentation time stamp PTS (decoding time stamp DTS) must have an interval of $(2 \text{ KB}/30 \times 10^6 \text{ bps}) \times 3 \text{ Pack} = 1.599 \text{ [ms]}$. In this period, i.e., period S_{NG} shown in FIG. 136, the Pack of the Primary Video Set can be multiplexed (MUX). The Secondary Pack may have an interval larger than that of the Primary Video Set. As shown in S4 and S5 in FIG. 136, an interval of three or more Packs is provided, and the Primary pack may be multiplexed in the interval.

FIG. 137 schematically shows the type and format of data which can be recorded on the Disc in the embodiment of the present invention. In FIG. 137, an "Advanced Navigation" indicates the data pertaining to playback control for playing back an advanced HD video title set and advanced content as shown in FIGS. 74 and 79, and also indicates a data file described in the Markup/Script language or the like.

Additionally, a "Primary Video Set" indicates DVD main picture stream data represented by an advanced VTS. In FIG. 137, for example, the "Primary Video Set" includes IFO data for storing management information of the main picture stream, TMAP data including a data table such as offset information of the time for each EVOB included in the main picture stream and the start position of the VOB serving as a unit of playback management, the EVOB included in one video sequence of

the main picture stream, and a P-EVOBS (Primary EVOBS) including the plurality of EVOBS.

The "Secondary Video Set" is a picture stream which is played back simultaneously with the main picture, and different from the main picture. The "Secondary Video Set" is different from a multi angle video implemented by the conventional DVD in that the "Secondary Video Set" can play back a picture stream while playing back the main picture, whereas one of the Multi Angle videos is selectively played back. The S-EVOB (Secondary EVOB) indicates the picture stream itself of the "Secondary Video Set". In this embodiment, the "Secondary Video Set" does not have a function of the Multi Angle and sub title in the conventional "Primary Video Set", and includes simple video and audio data. In this case, IFO information for managing playback sequence control or the like in detail is not always required. The TMAP information for specifying the simple playback stream position is prepared in correspondence with an "S-EVOB".

An "Advanced Element" indicates the playback data of the HDD player other than the "Primary Video Set" and the "Secondary Video Set". More specifically, the "Advanced Element" indicates a still picture data such as JPEG or PNG, the audio data used for an effect sound played back upon click of the button, text data which supplies character information to be described in the

text sub-title, and font data used to render the text data.

The data expressed in a "Multiplexed Data structure on the disc" is the data stored in the contiguous sectors on the disc. In this data, the "P-EVOBS", "S-EVOB", "Advanced Element", and the like are interleaved and arranged. This is a countermeasure for avoiding a problem that the "Secondary Video Set" is played back during playing back the "Primary Video Set", and the "P-EVOBS" data of the "Primary Video Set" is delayed to be supplied by reading out the data from the separated sector on the disc when the "Advanced Element" is stored in a data cache shown in FIG. 100.

In the embodiment of the present invention, since the "S-EVOB", "Advanced Element", and the like are interleaved in the overall sector data included in the "P-EVOBS" in a "Multiplexed Data structure", the "Multiplexed Data structure" is arranged at the position of the advanced HD video title set in the video data recording area in the data structure shown in FIGS. 74 and 79. Additionally, the IFO and TMAP of the "Primary Video Set" are also stored at the position of the advanced HD video title set in the video data recording area.

Alternatively, the TMAP of the "Secondary Video Set", the S-EVOB of the "Secondary Video Set" which is not interleaved in the "Multiplexed Data structure",

and the "Advanced Element" which is not interleaved in the "Multiplexed Data Structure" are stored in the advanced content recording area shown in FIGS. 74 and 79.

5 Furthermore, from the viewpoint of the file system shown in FIG. 2, these various data which are interleaved in the "Multiplexed Data structure" on the Disc cannot be discriminated in accordance with the file system. These data are managed as ".EVO" files of
10 the advanced VTS. The IFO and TMAP of the "Primary Video Set" can access to each other as the ".IFO" and ".MAP" files.

 The TMAP and S-EVOB data of the "Secondary Video Set which are not interleaved in the "Multiplexed Data
15 Structure" and the "Advanced Element" data which is not interleaved in the "Multiplexed Data Structure" are managed as the advanced content, and can access to each other as the file data in an ADV_OBJ directory.

 FIG. 138 is a view showing the playback system
20 model of the HD_DVD player as a functional module having a large unit in accordance with the embodiment of the invention. A "Data Source" indicates a data storage position to which the HD_DVD player can access for playback. In the "Data Source", a "Disc",
25 "Persistent Storage", "Network Server", and the like are included. The "Disc" corresponds to DVD disc 1 shown in FIG. 100.

The "Persistent Storage" corresponds to the persistent storage shown in FIG. 100. For example, an NAS (Network Attached Storage) on a home network is also included in the persistent storage.. The "Network
5 Server" indicates a server on the Internet. Generally, the "Network Server" is assumed to be managed by a filmmaker supplying the DVD disc.

An "Advanced Content Player" indicates the overall playback system model of the HD_DVD player. As a large
10 module, a "Data Access Manager", "Data Cache", "Navigation Manager", "Presentation Engine", "User Interface Controller", and "AV Renderer" are included.

The "Data Access Manager" manages the exchange of the data between the "Advanced Content Player" and the
15 "Data Source". The "Data Cache" is a data storage device for temporarily storing the data required for playback of the "Navigation Manager" and the "Presentation Engine".

The "Navigation Manager" loads and analyzes the
20 "Advanced Navigation", controls the "Presentation Engine" and "AV Renderer", and manages playback control of the disc of content type 2 or 3. When inserting the disc, the "Navigation Manager" loads a "Startup File", and sets the HD_DVD player required for playback
25 control.

In accordance with the playback control information of the "Advanced Navigation" on the basis

of the control command and the signal issued by the "Navigation Manager", the "Presentation Engine" loads the "Primary Video Set", "Secondary Video Set", and "Advanced Element" data from the "Data Source" using the "Data Access Manager". The "Presentation Engine" also loads the data from the "Data Cache", plays back the data, and sends the played back data to the "AV Renderer".

In accordance with the playback control information of the "Advanced Navigation", on the basis of the control command and signal issued from the "Navigation Manager", the "AV Renderer" performs α blending and mixing control of the video and audio data output from the "Presentation Engine", thereby outputting the signal from the last HD_DVD player to the external TV monitor and loudspeaker.

The "User Interface Controller" transmits as an event an input signal from a user interface such as a front panel, remote controller, or mouse, to the Navigation Manager. In addition to this, the "User Interface Controller" controls display of a mouse cursor.

FIG. 139 shows the detailed arrangement shown in FIG. 138 from the viewpoint of the data flow. As a result of playback control of the "Advanced Navigation", various data are stored in the "Persistent Storage" and the "Network Manager" as capacity allows.

The stored data can be loaded and written by the HD_DVD player. Generally, as the data loaded and used for playback by the "Advanced Content Player", the above-described "Advanced Navigation", "Advanced Element",
5 and "Secondary Video Set" are available. The "Primary Video Set" is stored only on the Disc, but in the "Persistent Storage" and the "Network Server".

As shown in FIG. 137, as the data stored on the "Disc", the "Advanced Navigation", "Advance Element",
10 "Primary Video Set", and "Secondary Video Set" are available. The "Disc" is a read-only medium so that no data is written by playback control of the "Advanced Navigation".

The "Data Access Manager" contains the "Persistent
15 Storage Manager", "Network Manager", and "Disc Manager". Generally, it can be assumed that these managers manage the data access from the "Persistent Storage", "Network Server", and "Disc". However, as for a "NAS (Network Attached Storage)", the "Persistent
20 Storage Manager" may be assumed to manage the data access using the function of the "Network Manager".

When inserting the disc, an arrow from the Disc Manager to the Navigation Manager indicates the flow of loading the "Startup File" included in the "Advanced
25 Navigation" by the "Navigation Manager" after a determination process of a predetermined disc type. An arrow from the Disc Manager to the Primary Video Player

indicates the data flow of the Primary Video Set. An arrow from the Disc Manager to the Secondary Video Player indicates the data flow of the Secondary Video Set interleaved in the Multiplexed Data Structure on the Disc.

An arrow from the Disc Manager to the File Cache Manager indicates the data flow of the Advanced Element interleaved in the Multiplexed Data Structure on the Disc. An arrow from the Disc Manager to the File Cache indicates the data flow of the Advanced Navigation, Advanced Element, and Secondary Video Set which are not included in the Multiplexed Data Structure on the Disc.

An arrow from the Persistent Storage and Network Server to the File Cache indicates the flow and its reverse flow of the Advanced Navigation, Advanced Element, and Secondary Video Set. An arrow from the Persistent Storage or Network Server to the Streaming Buffer indicates the flow of the Secondary Video Set.

An arrow from the File Cache to the Navigation Manager indicates the flow of mainly loading the Advanced Navigation using the Navigation Manager. An arrow from the File Cache Manager to the File Cache indicates the flow of writing the data of the Advanced Element in the File Cache using file units sent from the Disc Manager to the File Cache. An arrow from the File Cache to the Advanced Element and the Presentation Engine indicates the flow of the Advanced Element. An

arrow from the File Cache to the Secondary Video Player indicates the data flow when the TMAP or S-EVOB of the Secondary Video Set temporarily stored in the File Cache as the file data is played back.

5 An arrow from the Streaming Buffer to the Secondary Video Player indicates the flow of temporarily loading the large Secondary Video Set stored in the Persistent Storage and the Network Server little by little, and then supplying the loaded
10 Secondary Video Set to the Secondary Video Player. Generally, this is a countermeasure for avoiding interruption of playback of the Secondary Video Set by absorbing the fluctuation of the data loading rate when the data is supplied from the Data Source with unstable
15 data loading rate, like a network.

 An arrow from the Advanced Navigation to the Presentation Engine or the AV Renderer indicates a control signal. However, the arrow from the Advanced Navigation to the Presentation Engine can also indicate
20 that text subtitle data stored in the Advanced Navigation data including a Markup/Script is supplied.

 FIG. 140 shows the detailed arrangement shown in FIG. 139 from the viewpoint of data supply from the Disc. In FIG. 139, only the Disc Manager manages the
25 data from the Disc in the Data Access Manager. However, in FIG. 140, a Stream Dispatcher also manages the data.

The Stream Dispatcher receives the Multiplexed Data Structure shown in FIG. 137 from the Disc Manager, and supplies the data of the P-EVOBS, S-EVOB, and Advanced Element which are interleaved in the

5 Multiplexed Data Structure to the Demux device in the Primary Video Player, the Secondary Video Playback Engine of the Secondary Video Player, and the File Cache Manager in the Navigation Manager, respectively.

When the Disc is inserted to the player according

10 to the embodiment of the present invention, the Disc Manager supplies the Startup File on the Disc to the Navigation Manager. Each file of the Advanced Navigation, Advanced Element, and Secondary Video Set managed by a file system on the Disc is loaded to the

15 File Cache in accordance with the result obtained by interpreting the Startup File and the Advanced Navigation using the Advanced Navigation Engine in the Navigation Manager.

When the Primary Video Player plays back the

20 Primary Video Set, the IFO and TMAP data of the Primary Video Set is captured from the Disc Manager to the DVD Playback Engine, prior to playback. The Primary Video Player supplies an upper level control API (Application Interface) for playing back the Primary Video Set, to

25 the Navigation Manager. Note that for example, the upper level control API is the level API such as Play, FF, STOP, or PAUSE. The DVD Playback Engine performs

the detail playback control process of the Primary Video Set.

The DVD Playback Engine performs the playback control of the Primary Video Set in accordance with the upper level control API from the Advanced Navigation Engine based on the description of the "Advanced Navigation".

The Demux demultiplexes the P-EVOB data, and supplies a control pack (N_PCK), video pack (V_PCK), and sub-picture pack (SP_PCK), and audio pack (A_PCK) to the DVD Playback Engine, Video Decoder, SP Decoder, and Audio decoder. Each of these decoders decodes the obtained PCK data by appropriate units.

When the Secondary Video Player plays back the Secondary Video Set in which the S-EVOB is interleaved in the Multiplexed Data Structure on the Disc, the TMAP data of the Secondary Video Set is received from the Disc Manager to the Secondary Video Playback Engine, prior to playback. Additionally, the Secondary Video Set managed on the file system can be temporarily stored in the File Cache, and then loaded and played back by the Secondary Video Playback Engine.

The Secondary Video Player supplies the upper level control API for playing back the Secondary Video Set as the Primary Video Player, to the Navigation Manager.

The Secondary Video Playback Engine performs

playback control of the Secondary Video Set in accordance with the upper level control API from the Advanced Navigation Engine on the basis of the description of the "Advanced Navigation".

5 The Demux in the Secondary Video Player demultiplexes the S-EVOB data, and supplies the video pack (V_PCK) and audio pack (A_PCK) to the Video Decoder and Audio Decoder, respectively.

10 In this embodiment, the Secondary Video Set only includes the video and audio packs. However, the Secondary Video Set can include the sub-picture and the control pack.

 The File Cache Manager obtains the Advanced Element data pack output from the Stream Dispatcher.
15 After the pack data is supplied to form one file data, the data pack is written as one file belonging to the Advanced Element, to the File Cache.

 For example, when the file data as large as the font data is written in the File Cache, write of the
20 file data may be started in the File Cache before all data in the font file are collected to the File Cache Manager. The file data may be sequentially written, and the final font file may be arranged on the File Cache.

25 The Advanced Element stored in the Multiplexed Data Structure can also be compressed and then interleaved. In this case, the File Cache Manager

receives the compressed Advanced Element data to be extracted, and then extracts the input data. As a result, the generated Advanced Element file is written in the File Cache. The compressed Advanced Element data may be compressed using file units, or the plurality of archived Advanced Element files may be compressed.

An Advanced Element Presentation Engine loads the Advanced Element data from the File Cache, and decodes the Advanced Element in accordance with the control command/signal from the Advanced Navigation Engine based on the description of the Advanced Navigation.

FIG. 141 shows the more detailed arrangement shown in FIG. 139 from a viewpoint of the data supply from the Network Server and Persistent Storage. A device implementing the Persistent Storage can be divided into the Fixed Storage and the Additional Storage. The Fixed Storage is a recording medium which is fixed and connected to the HD_DVD player such as a FLASH memory.

The Additional Storage is a recording medium which can be connected to or separated from the HD_DVD player. The Additional Storage can use a memory card represented by an SD card, a memory device and HDD device via the connection interface such as a USB, and a NAS (Network Attached Storage) connected to the network.

As the supply model from the Disc shown in

FIG. 140, the data such as the Advanced Navigation, Advanced Element, and Secondary Video Set are supplied to the File Cache via the Network Manager and Persistent Storage Manager.

5 When the Secondary Video Set having the S-EVOB which is too large to be stored in the File Cache is played back, the data is directly and sequentially supplied to the Secondary Video Playback Engine to perform playback. At this time, in accordance with
10 described control in the Advance Navigation, the Secondary Video Playback Engine can perform playback while the data is temporarily stored in the Streaming Buffer. This is a countermeasure for reducing the possibility that playback of the Secondary Video Set is
15 interrupted when the data supply rate is unstable in the network or the like. Generally, the Streaming Buffer need not be used for playback of the Secondary Video Set captured in the File Cache.

FIG. 142 shows the detailed arrangement in
20 FIG. 139 from a viewpoint of the data storage flow of the Persistent Storage and the Network Server. An arrow from the Advanced Navigation to the Advanced Element indicates the flow of writing the Advanced Element such as a data file generated by the Advanced
25 Navigation Engine using a script language or the like, to the File Cache. In the Advanced Navigation, using, e.g., the Script language, the file for recording the

number of views of the Disc is generated. The file is stored in the Persistent Storage. Whenever the video on the Disc is viewed, the data in the file is updated. Accordingly, the number of updating processes can be
5 displayed on the screen, the score data of a game generated by the Script language can be generated, and the generated data can be sent to the Network Server to compete for high scores. These data generated by the Advanced Navigation Engine are temporarily stored in
10 the File Cache, and then copied/moved to the appropriate storage destinations.

An arrow from the Primary Video Player to the Advanced Element indicates the flow of interrupting the video currently played back by the Primary Video Set in
15 accordance with the description of the Advanced Navigation Engine and the interpretation of the user operation, and writing the Advanced Element such as an image file obtained by capturing the screen, to the File Cache. On the generated capture screen, an
20 original chapter group with an appropriate explanation may be generated, the data of the original chapter group may be stored in the Persistent Storage or the like, and a scene may be selected from the original chapters and viewed next time. As the destination of
25 the capture screen, a Secondary Video Set screen to which the Secondary Video Player outputs the data, a graphics screen to which the Advanced Element

Presentation Engine outputs the data, and an output image from the AV Renderer obtained by mixing the above data.

The data generated by the Navigation Manager and
5 Presentation Engine is temporarily stored in the File Cache, and then stored on the appropriate Data Source medium in accordance with the description of the Advanced Navigation. Similarly, when the content in the Persistent Storage, Network Server, and Disc are to
10 be uploaded or stored in the Persistent Storage and the Network Server, the generated data are temporarily loaded in the File Cache and then stored on the appropriate Data Source medium, in accordance with the description of the Advanced Navigation.

15 FIG. 143 shows the mixed model of the image output in detail. In FIG. 143, five image planes are assumed to be output. From the lower plane, there are a Primary Video Plane, Secondary Video Plane, Sub-Picture Plane, Graphicss Plane, and Cursor Plane.

20 The Primary Video Plane is a plane for video output from the Primary Video Set. In this model, the video is supplied to the AV Renderer through a scaling device. In this model, an α value (value for determining contrast ratio) is not assumed to be
25 applied to the Primary Video Plane. However, for example, when a background plane or the like is prepared on the lower layer of the Primary Video Plane,

the α value can be effectively applied for the Primary Video Plane to improve the expression.

The Secondary Video Plane is a plane for video output from the Secondary Video Set. In this model, the video is supplied to the AV Renderer through the scaling device. In this model, a Chroma Effect function is included in order to extract an object shape in the video to overlap the object on the output from the Primary Video. This process can be performed by filling the portion other than the object to be extracted in a specific color, and managing the colored portion as a transparent portion.

The Sub-Picture Plane is a plane for image output from the Sub-Picture of the Primary Video Set. In this model, the video is supplied to the AV Renderer via the scaling device. However, for example, when the Sub-Picture in the size of the SD card is prepared, or when the Sub-Picture for a Pan Scan output or Letter Box output in the size of the SD card, the scaling device performs no operation, and the Sub-Picture data corresponding to the output is output from the SP Decoder. Accordingly, the data is mixed with the overall image.

The Graphics Plane is a plane for image output from the Advanced Element Presentation Engine. In this model, the Advanced Graphics Decoder is assumed to process image data such as JPEG and PNG, and image data

such as cell animation and vector animation, and the Advanced Text Decoder is assumed to process the text image output using the font data. The decode result output for each object unit is sent to Layout/Alpha Control, and undergoes a layout and α blending process in accordance with the control information of the Navigation Manager which interprets the Advanced Navigation. This layout process also includes an object scaling process and the like.

10 The Cursor Plane is managed by and output from the Cursor Manager in the User Interface Controller. In this model, the α value is set for the Cursor object, and mixed with other planes.

15 These five image data are output from the respective decoders and the like in the format corresponding to the frame rate of the final video data output of the HD_DVD player. When capturing the data to the AV Renderer, all the plane data are supplied in the frame rate format.

20 A Graphics Composer is a module for managing the mixing process of the five image outputs in the AV Renderer, and includes α Blending Control, Position Control, Chroma Effect, and the like.

25 As described above, the Chroma Effect is a functional module for processing the color designated by the Navigation Manager as a transparent color, in order to extend a predetermined object shape from the

video output from the Secondary Video Player.

Actually, when the Secondary Video is output, a pixel color value can change as a Chroma Key since a Lossy codec such as MPEG2 is used. Hence, it is effective to
5 have a function of avoiding a monochrome Chroma Key, and accurately extracting the object shape by designating the chroma key with some variations, and performing the image process for the chroma key.

The Position Control supplies to the α Blending
10 Control the image controlled to determine the position of the input video data with respect to the overall image output size.

The α Blending Control mixes the above video data in accordance with the instruction of the Advanced
15 Navigation which is interpreted by the Navigation Manager, to generate the final video output image.

FIG. 144 shows an example of the actual image output from the image output mixed model shown in FIG. 143. Generally, the video output from the Primary
20 Video Set in the Primary Video Plane is a moving image data of the DVD main picture, and displayed on the entire screen. Generally, the video output from the Secondary Video in the Secondary Video Plane is arranged in the Primary Video Plane in the format of a
25 Picture In Picture, and undergoes the α blending process for the Primary Video image in accordance with the description of the Advanced Navigation. As

described above, the object shape can also be extracted to be mixed with the Primary Video Plane.

The output from the Sub-picture Plane is the image data of the Sub-Picture stored in the Primary Video Set, given the α value at a pixel level, and mixed with the mixed image of the Primary Video Plane or the Secondary Video Plane serving as the background.

The output from the Graphics Plane is controlled using the α value at the pixel level. Accordingly, the α value of the Graphics Plane is not controlled by the Navigation Manager. In comparison with the Graphics Plane, the Navigation Manager controls the α value using object units such as a button image and text arranged in the Graphics Plane. When the α value at the pixel level is to be controlled, the image object itself must use a format to describe the α value at the pixel level. As this format, PNG, JPEG 2000, and the like are available. As for the text, characters may be deformed by scaling an output character image, thereby decreasing the readability. Hence, the image data supplied to the Layout/ α Control is decoded in correspondence with the final output image size in advance to effectively avoid the deterioration of image quality.

The Cursor Plane is a pointer image which moves on the screen in accordance with the event of a direction key such as the mouse or remote controller. This

pointer image can be replaced with the arbitral
Advanced Element image in accordance with the
description of the Advanced Navigation. The α value
can be applied to the Cursor Plane at the object
5 (Plane) level.

FIG. 145 is a view showing the mixing model of an
audio output. In this model, three audio outputs are
mixed. That is, a Primary Audio output is an audio
output from the Primary Video Set. A Secondary Audio
10 output is an audio output from the Secondary Video Set.
Note that the Second Video Set need not always include
the Video output. The Secondary Video Set may include
only an Audio output.

Each of the Audio Decoders in the Primary Video
15 Player and the Secondary video player can interpret
Meta Data in the Audio Elementary Stream, and control a
change in mixing level at the frame level. In this
model, the Meta Data process is completed in each
decoder. However, the Meta Data information may be
20 sent to the Sound Mixer, and processed in the Sound
Mixer.

The Sound Decoder in the Advanced Element
Presentation Engine outputs an effect sound when the
button is clicked. The mixing process of the audio
25 output is performed by the Sampling Rate Converter and
the Sound Mixer in the AV Renderer.

In this model, the audio output of the Primary

Video is generally assumed to be supplied with highest sound quality, and the sampling rates of the Secondary Audio and the Effect Sound correspond to the Primary Audio. Hence, the Primary Audio output includes no
5 Sampling Rate Converter. When a function of deteriorating the audio output quality is implemented in order to realize the HD_DVD player at low cost, it is effective to include the Sampling Rate Converter prior to the Primary Audio output.

10 Each of the audio signals is supplied to the Sound Mixer in a state wherein the Sampling Rates are matched by the Sampling Rate Converter. The Sound Mixer mixes and outputs these three audio signals in accordance with the mixing level instructed on the basis of the
15 description of the Advanced Navigation. When the HD_DVD player outputs the analog audio signal, and the DA converter outputs the digital audio signal, these three audio signals are sent to an appropriate encoding processing device.

20 A Water Mark Detect is a module for examining the output audio signal from the Sound Mixer, and detecting the presence of copyright management information.

FIG. 146 is a view showing a User Interface process managed by the User Interface Controller. In
25 this model, as the User Input device, a Front Panel, Remote Controller, Keyboard, Mouse, and Game Pad are shown. As described above, the Cursor Manager controls

the display position of the cursor object on the screen in accordance with the direction key and moving event of the Remote Controller or the Mouse. The button pressure event of the Remote Controller or the Keyboard is notified to the Navigation Manager as the User Interface Event.

FIG. 147 is a flowchart showing the flow of the startup process after inserting the disc. When the disc is inserted to the HD_DVD player, first, the content type is detected. The content type can be detected under the condition of the presence of an Advanced VTS and specific Markup File. When the disc is a content type 2 or 3 disc (YES in step ST302), the Startup File is loaded from the disc (step ST304). As the content type 2 or 3 data structure, a disc including only the Advanced VTS shown in FIG. 74, or a disc including both the Advanced VTS and the Standard VTS shown in FIG. 79 is available.

After the Startup is interpreted, the setting of the player changes in accordance with the description (step ST306...player system setting: Configure Player System). The information to be changed includes the distribution of the File Cache of Data Cache and the Streaming Buffer, and network connection setting.

After that, the Advanced Navigation file including the Startup File for the initial operation is loaded from the Disc, Network Server, Persistent Storage, and the

like (step ST308). The Advanced Navigation process described in the Startup File then starts (step ST310).

Alternatively, when the disc is a content type 1 disc (NO in step ST302, and YES in step ST312), the content type 1 disc performs Standard VTS playback process to conform to the conventional DVD. The content type 1 disc includes only the Standard VTS as shown in FIG. 73. If the disc is a disc other than the content type 1 disc (NO in steps ST302 and ST312), each of playback processes is performed in accordance with the medium type supported by the individual HD_DVD player which plays back the disc (step ST316).

<Summary>

An information storage medium (high-definition video disc or the like) according to the embodiment of the present invention has a data area (12) that stores a video data recording area (20) including a management area (30) that records management information and an object area (40, 50) that records objects to be managed by this management information, and an advanced content recording area (21) including information (21A to 21E) different from the recording content (30 to 50) of this video recording area (20), and a file information area (11) storing file information corresponding to the recording content of this data area (12). In this information storage medium, the object area (40, 50) is configured to store expanded video objects (objects in

an HDVTS and abbreviated as an EVOBS or VOBS as needed) which undergo playback management using a logical unit called a program chain, and advanced objects (objects in an AHDVTS) recorded independently of the expanded video objects. The advanced objects are configured to store playback control information and the like, which give playback sequence information (playback control information implemented by a markup language and the like, as exemplified in FIGS. 95 to 98) that describes the playback order of expanded video objects, and the playback conditions (playback timings, picture output positions, display sizes, etc.) of other advanced objects.

The playback conditions (or playback control information, playback sequence information, etc.) can be described by a provider of the content recorded on the information storage medium using a predetermined language (markup language or the like). By supplying the markup language that gives the playback conditions to a player via a network (Internet or the like), management information which is recorded on the information storage medium and is uniquely determined so far can be updated.

Furthermore, for example, the playback control information that controls playback of video objects is distributed via the Internet or the like after the disc is produced, or the aforementioned playback control

information is added to a video disc which is produced once, thus producing a new disc without re-producing the disc. More specifically, video objects, which cannot be played back upon delivery of a DVD-Video disc, are allowed to be played back using playback control information distributed via the Internet under specific conditions, or problems can be corrected by controlling parts that include errors upon delivery of a DVD-Video disc.

Put differently, the embodiment of the present invention provides a scheme that allows the user to freely change and enjoy the playback sequence of advanced objects and/or expanded video objects using playback control information implemented by the markup language at the time of production or after sales of an information storage medium (ROM-based disc).

The data area (12) is a group of one or more primary objects (EVOB#1, #2, and the like) whose relationship with the playback time (TM_DIFF or the like) and the recording position (TM_EN_ADR or the like) is managed by one or more time maps (TMAP#1, #2, and the like; corresponding to TMAPIT). The data area (12) can store a primary object set (P-EVOBS) included in the main picture stream, and a secondary object (S-EVOB) included in another picture stream which serves as an object for managing the relationship between a playback time (TM_DIFF) and recording position

(TM_EN_ADR) by individual time map (TMAP) and is included in another picture stream played back simultaneously with the main picture stream.

Note that playback of one or more primary objects (EVOB#1, #2, and the like) can be managed on the basis of the playback time using one or more time maps (TMAP#1, #2, and the like; corresponding to TMAPIT). In addition, playback of the secondary object (S-EVOB) which can be played back simultaneously (synchronously) with the arbitral object of the primary objects (EVOB#1, #2, and the like) can be managed on the basis of the playback time using the individual time map (TMAP). In this case, the playback timing and/or playback duration of the secondary object played back simultaneously (or synchronously) with a given primary object can be freely set using the predetermined language (Markup language or the like).

In each of the aforementioned embodiments described above with reference to the accompanying drawings, information elements (e.g., 310 to 318 in the example of FIG. 3) are arranged in the illustrated order. This arrangement corresponds to the order indicating which information element is to be loaded first by the player upon playback of disc 1.

The present invention is not limited to the aforementioned specific embodiments, but can be embodied by variously modifying constituent elements

without departing from the scope of the invention when it is practiced. For example, the present invention can be applied not only to DVD-ROM Video that has currently spread worldwide but also to
5 recordable/reproducible DVD-VR (video recorder) whose demand is increasing in recent years. Furthermore, the present invention can be applied to a reproduction system or a recording/reproduction system of next-generation HD-DVD which will be spread in the near
10 future.

Furthermore, various inventions can be formed by appropriately combining a plurality of required constituent elements disclosed in the respective embodiments. For example, some required constituent
15 elements may be omitted from all required constituent elements disclosed in the respective embodiments. Furthermore, required constituent elements across different embodiments may be appropriately combined.

C L A I M S

1. An information storage medium which comprises
a data area (12) in which a video data recording area
(20) including a management area (30) for recording
management information (HDVMG) and an object area (40,
50) for recording object (HDVTS, AHDVTS) managed by the
management information (HDVMG), and an advanced content
recording area (21) including information (21A-21E)
different from recording content (30-50) in the video
data recording area (20) are stored, and a file
information area (11) in which file information
(FIG. 2) corresponding to the recording content in the
data area (12) is stored,

wherein the data area (12) is configured to store
a primary object set (P-EVOBS) which is a group of at
least one primary object (EVOB#1, #2, ...) for managing
a relationship between a playback time and a recording
position in accordance with at least one time map
(TMAP#), and includes a main picture stream, and a
secondary object (S-EVOB) in which a relationship
between the playback time and the recording position is
managed in accordance with an individual time map, and
includes another picture stream to be played back
simultaneously with the main picture stream.

2. A medium according to claim 1, wherein
the object area (40, 50) includes a time map
information table (TMAPIT) configured to be formed of

at least one time map information,

the time map information table (TMAPIT) includes
one or more time map information search pointers
(TMAPI_SRP#) respectively corresponding to one or more
5 items of said time map information,

the time map information search pointer
(TMAPI_SRP#) includes start address information
(TMAPI_SA) of the corresponding time map information,
and a time map type flag (TMAP_YTPE_FL) which specifies
10 whether an object managed by the time map is the
primary object (EVOB).

3. A playback method of playing back an
information storage medium which comprises a data area
in which a video data recording area including a
15 management area for recording management information
and an object area for recording object managed by the
management information, and an advanced content
recording area including information different from
recording content in the video data recording area are
20 stored, and a file information area in which file
information corresponding to the recording content in
the data area is stored, wherein the data area stores a
primary object set which is a group of at least one of
primary objects for managing a relationship between a
25 playback time and a recording position in accordance
with at least one of time maps, and includes a main
picture stream, and a secondary object which is an

object for managing the relationship between the playback time and the recording position in accordance with an individual time map, and includes another picture stream to be played back simultaneously with the main picture stream, said playback method comprising

playing back (ST212, etc.) the primary object, and playing back (ST224, etc.) the secondary object.

4. A playback apparatus having an information storage medium which comprises a data area in which a video data recording area including a management area for recording management information and an object area for recording object managed by the management information, and an advanced content recording area including information different from recording content in the video data recording area are stored, and a file information area in which file information corresponding to the recording content in the data area is stored, wherein the data area stores a primary object set which is a group of at least one of primary objects for managing a relationship between a playback time and a recording position in accordance with at least one of time maps, and includes a main picture stream, and a secondary object which is an object for managing the relationship between the playback time and the recording position in accordance with an individual time map, and includes another picture stream to be

played back simultaneously with the main picture stream, said playback apparatus comprising

a first playback unit (DVD playback engine) configured to play back the primary object, and

5 a second playback unit (interactive engine of FIG. 100/secondary video player in presentation engine of FIG. 139, etc.) configured to play back the secondary object.

10 5. A playback method of playing back an information storage medium as defined by claim 1, which comprises a data area in which a video data recording area including a management area for recording management information and an object area for recording object managed by the management information, and an
15 advanced content recording area including information different from recording content in the video data recording area are stored, and a file information area in which file information corresponding to the recording content in the data area is stored, wherein
20 the data area is configured to store a primary object set which is a group of at least one of primary objects for managing a relationship between a playback time and a recording position in accordance with at least one of time maps, and includes a main picture stream, and a
25 secondary object which is an object for managing the relationship between the playback time and the recording position in accordance with an individual

time map, and includes another picture stream to be played back simultaneously with the main picture stream, said playback method comprising

playing back the (ST212, etc.) primary object, and

5 obtaining (ST230) to play back (ST234) the secondary object corresponding to said another picture stream from an external communication line.

6. A playback apparatus using an information storage medium as defined by claim 1, which comprises a data area in which a video data recording area including a management area for recording management information and an object area for recording object managed by the management information, and an advanced content recording area including information different from recording content in the video data recording area are stored, and a file information area in which file information corresponding to the recording content in the data area is stored, wherein the data area is configured to store a primary object set which is a group of at least one of primary objects for managing a relationship between a playback time and a recording position in accordance with at least one of time maps, and includes a main picture stream, and a secondary object which is an object for managing the relationship between the playback time and the recording position in accordance with an individual time map, and includes another picture stream to be played back simultaneously

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with the main picture stream, said playback apparatus comprising

a first playback unit (DVD playback engine) configured to play back the primary object, and

5 a second playback unit (interactive engine of FIG. 100/secondary video player in presentation engine of FIG. 139, etc.) configured to play back the secondary object corresponding to said another picture stream from an external communication line.

10 7. A playback method of using an information storage medium as defined by claim 1, which comprises a data area in which a video data recording area including a management area for recording management information and an object area for recording object
15 managed by the management information, and an advanced content recording area including information different from recording content in the video data recording area are stored, and a file information area in which file information corresponding to the recording content in
20 the data area is stored, wherein the data area stores a primary object set which is a group of at least one of primary objects for managing a relationship between a playback time and a recording position in accordance with at least one of time maps, and includes a main
25 picture stream, and a secondary object which is an object for managing the relationship between the playback time and the recording position in accordance

with an individual time map, and includes another picture stream to be played back simultaneously with the main picture stream, said playback method comprising:

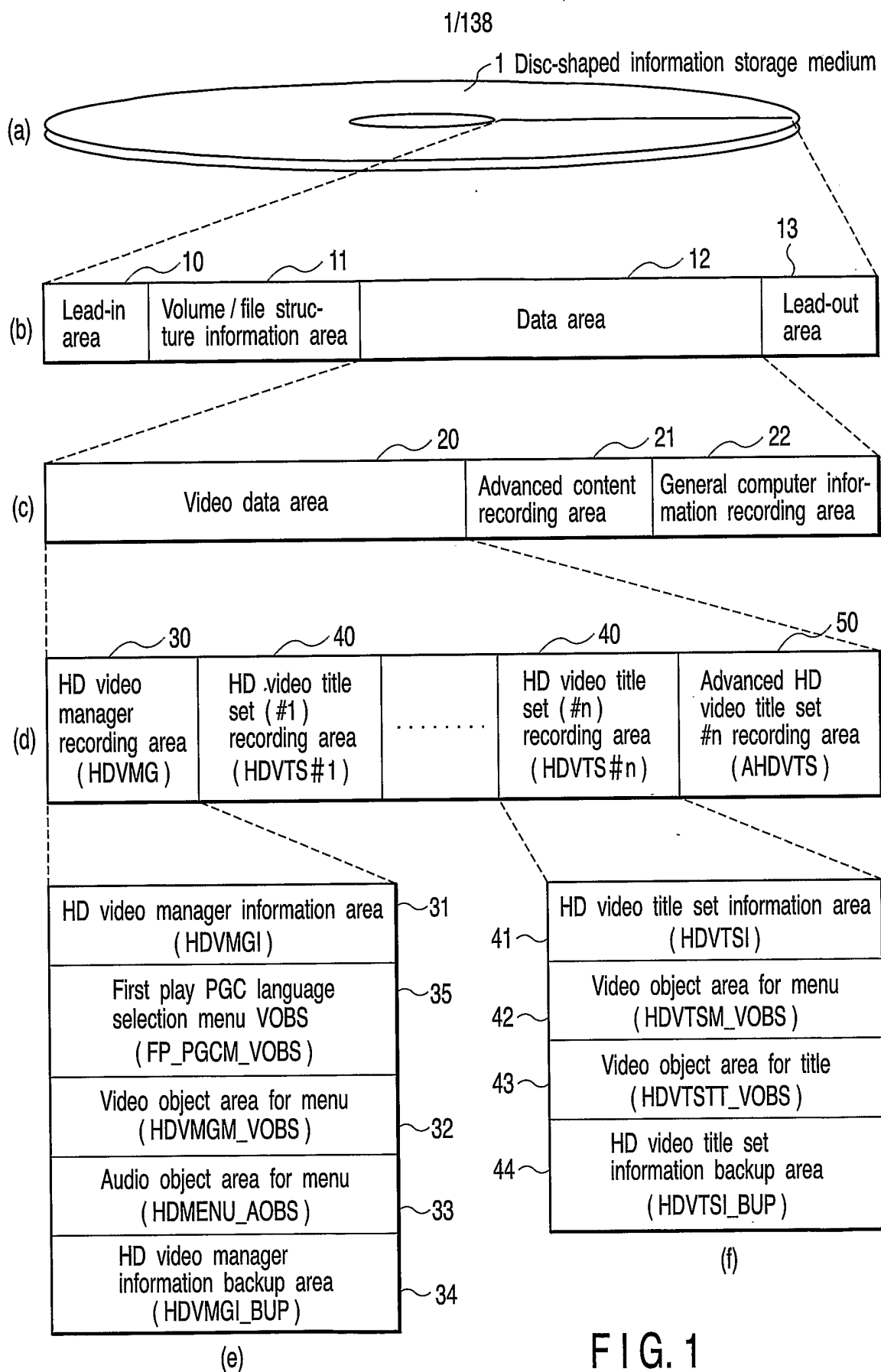
5 obtaining a first data stream (data stream <1> in FIG. 135) including the primary object set and the secondary object from the information storage medium at a first bit rate (e.g., 30 Mbps);

 obtaining a second data stream (data stream <2> in
10 FIG. 135) including another secondary object from a line (web) other than the information storage medium at a second bit rate (e.g., 10 Mbps) different from the first bit rate (30 Mbps);

 converting (by smoothing buffer in FIG. 135) the
15 bit rate of the secondary object included in the first data stream, to a rate (10 Mbps) corresponding to the second bit rate;

 decoding content of the primary object set included in the first data stream at the first bit rate
20 (30 Mbps); and

 decoding content of the secondary object included in the first stream converted to the rate corresponding to the second bit rate (10 Mbps), and content of said other secondary object included in the second data
25 stream at the rate (10 Mbps) corresponding to the second bit rate.



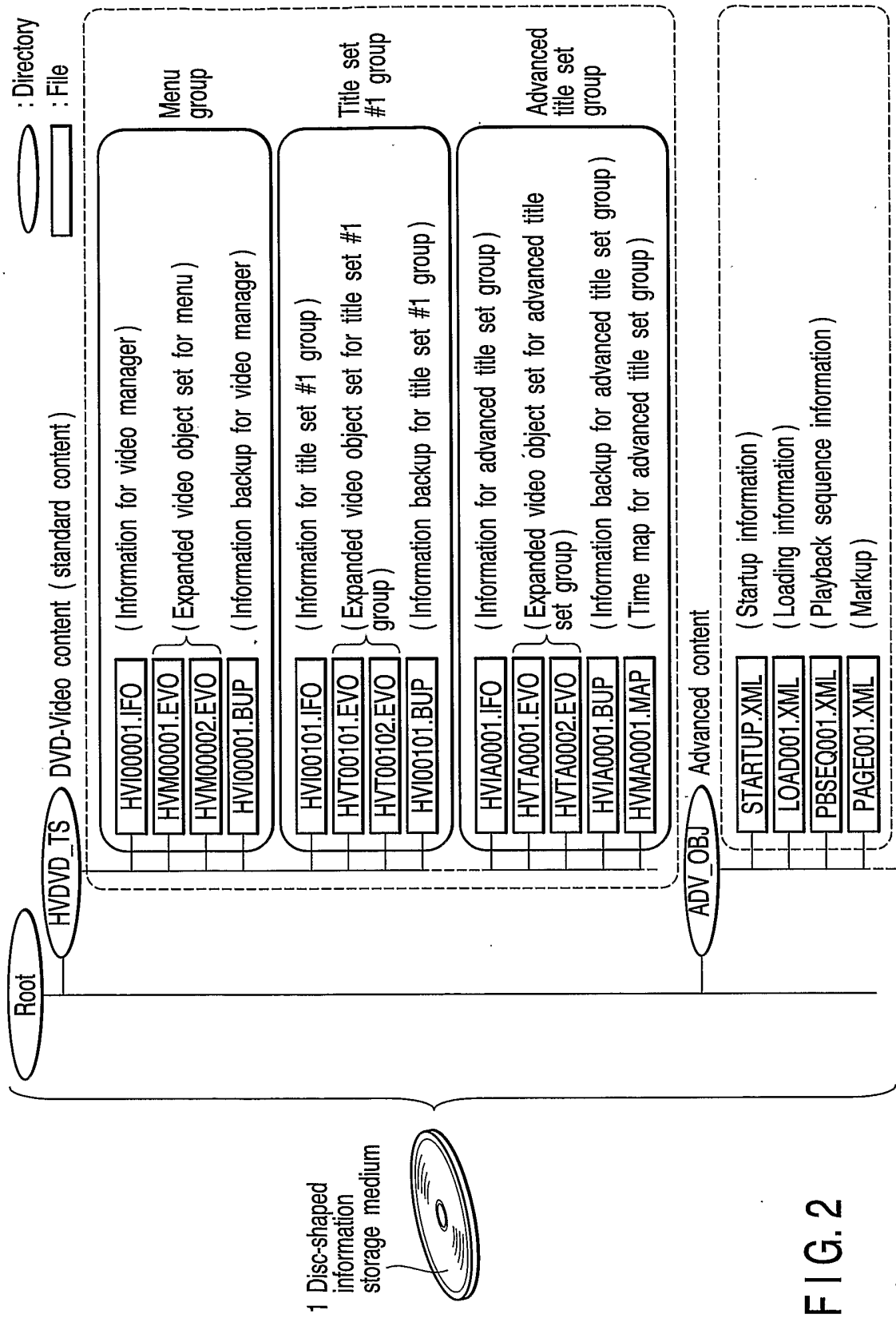


FIG. 2

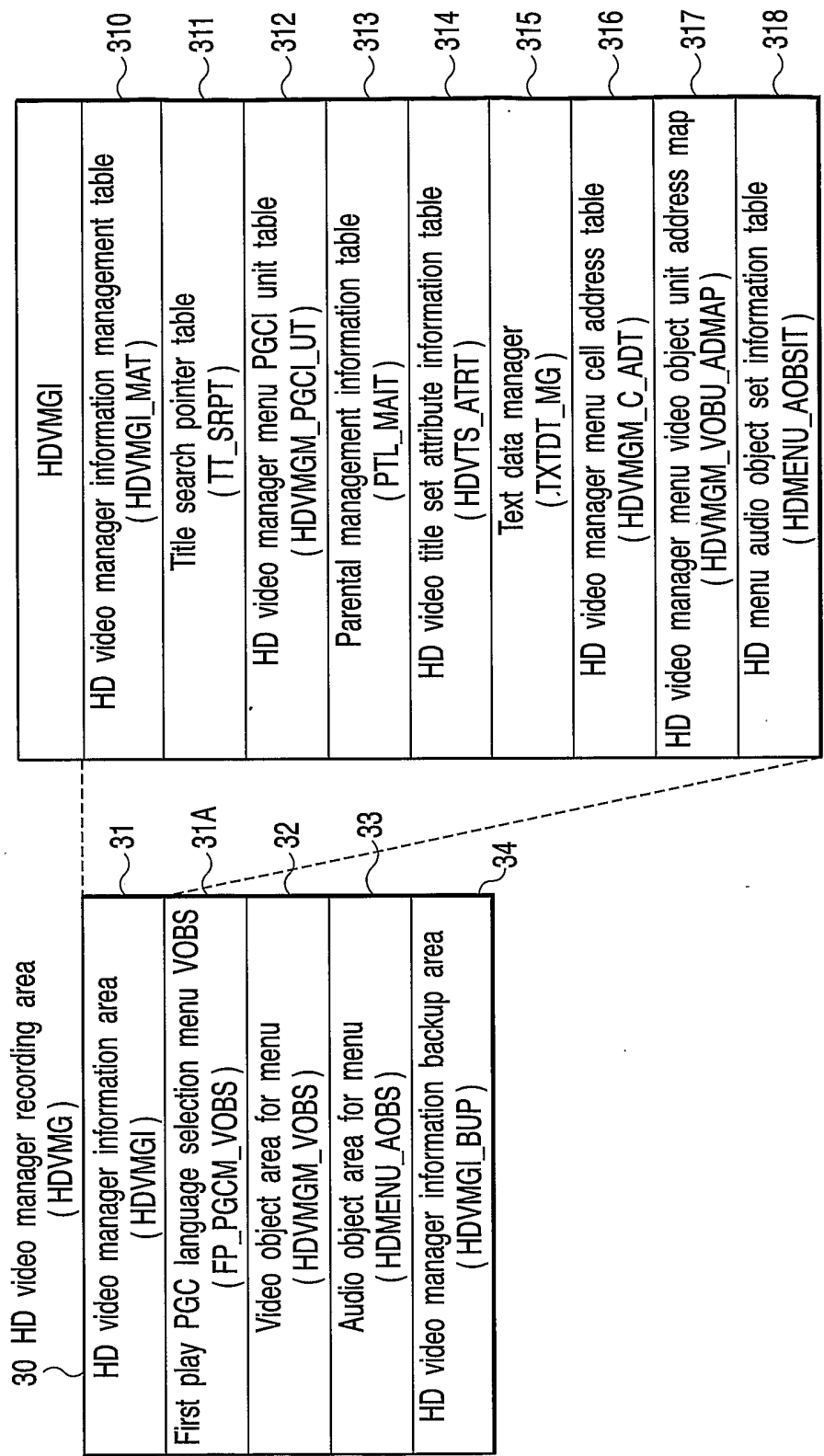
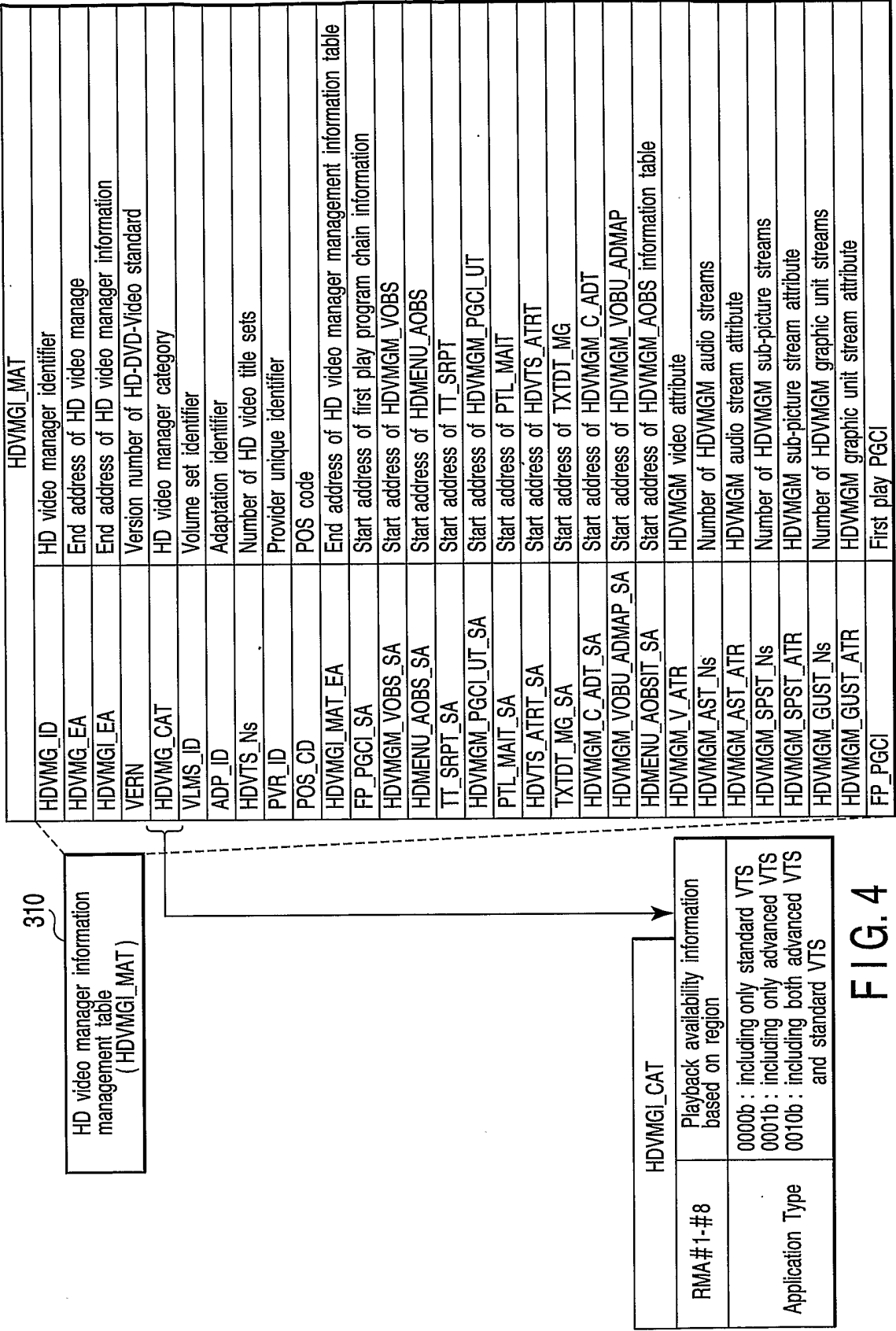


FIG.3



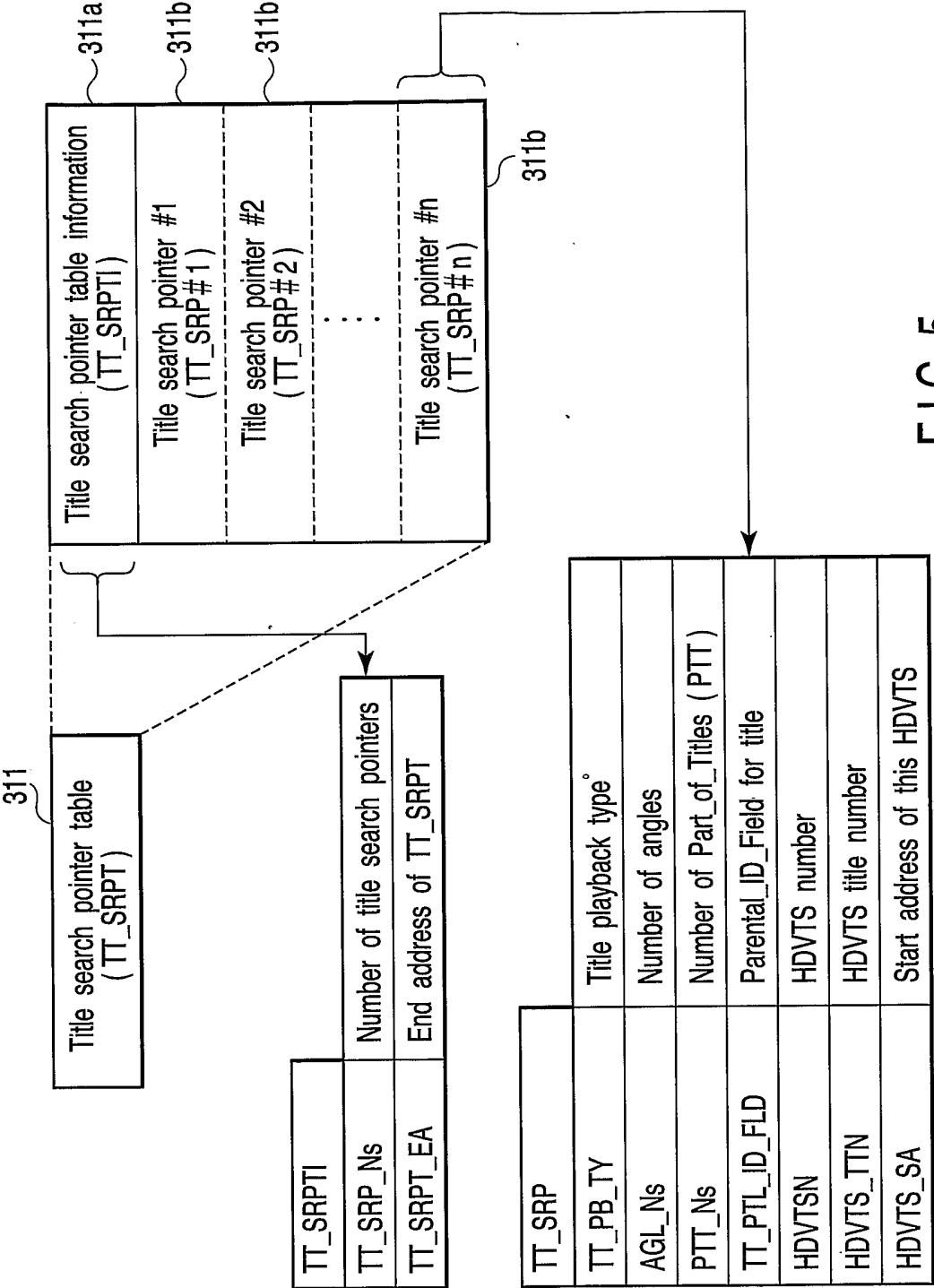


FIG. 5

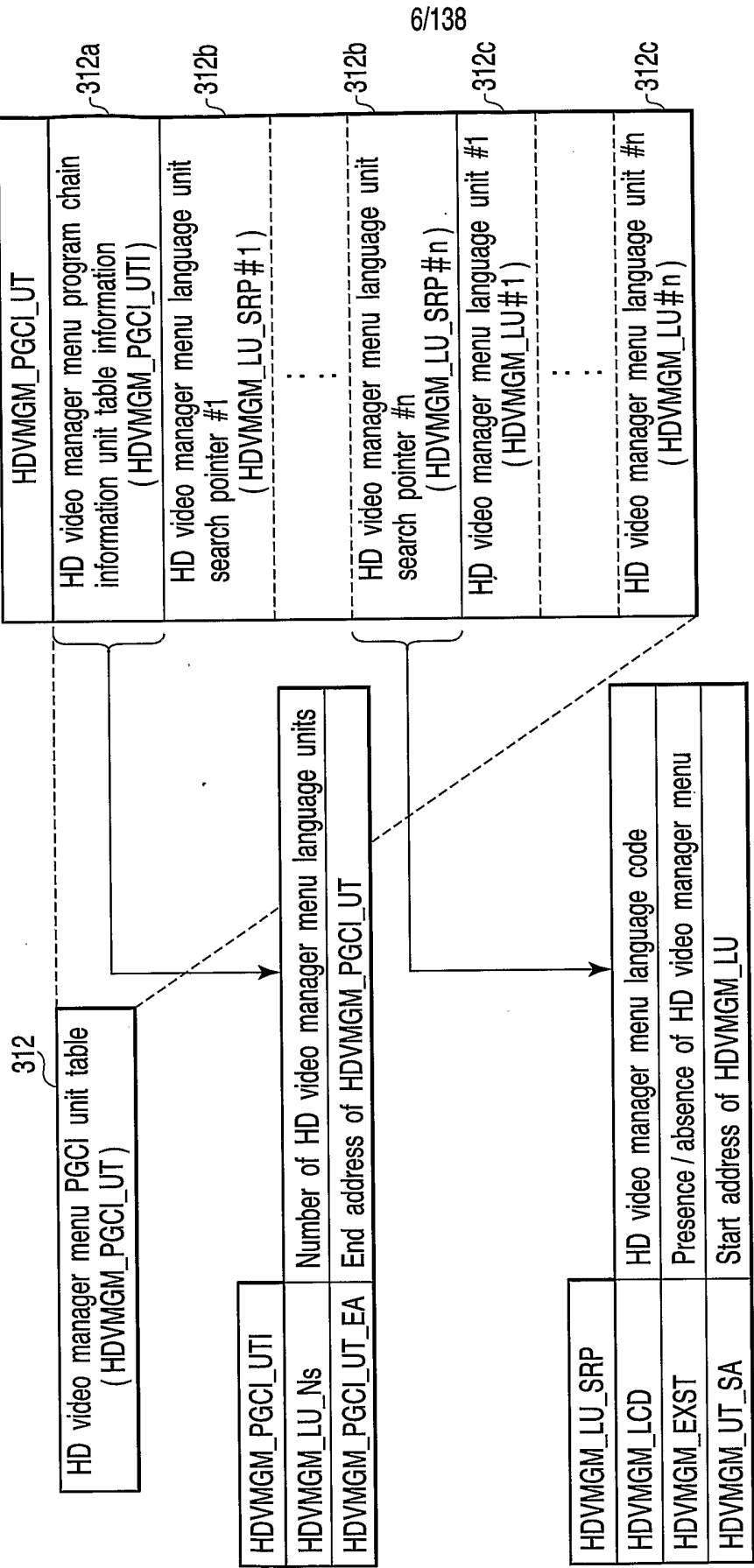


FIG. 6

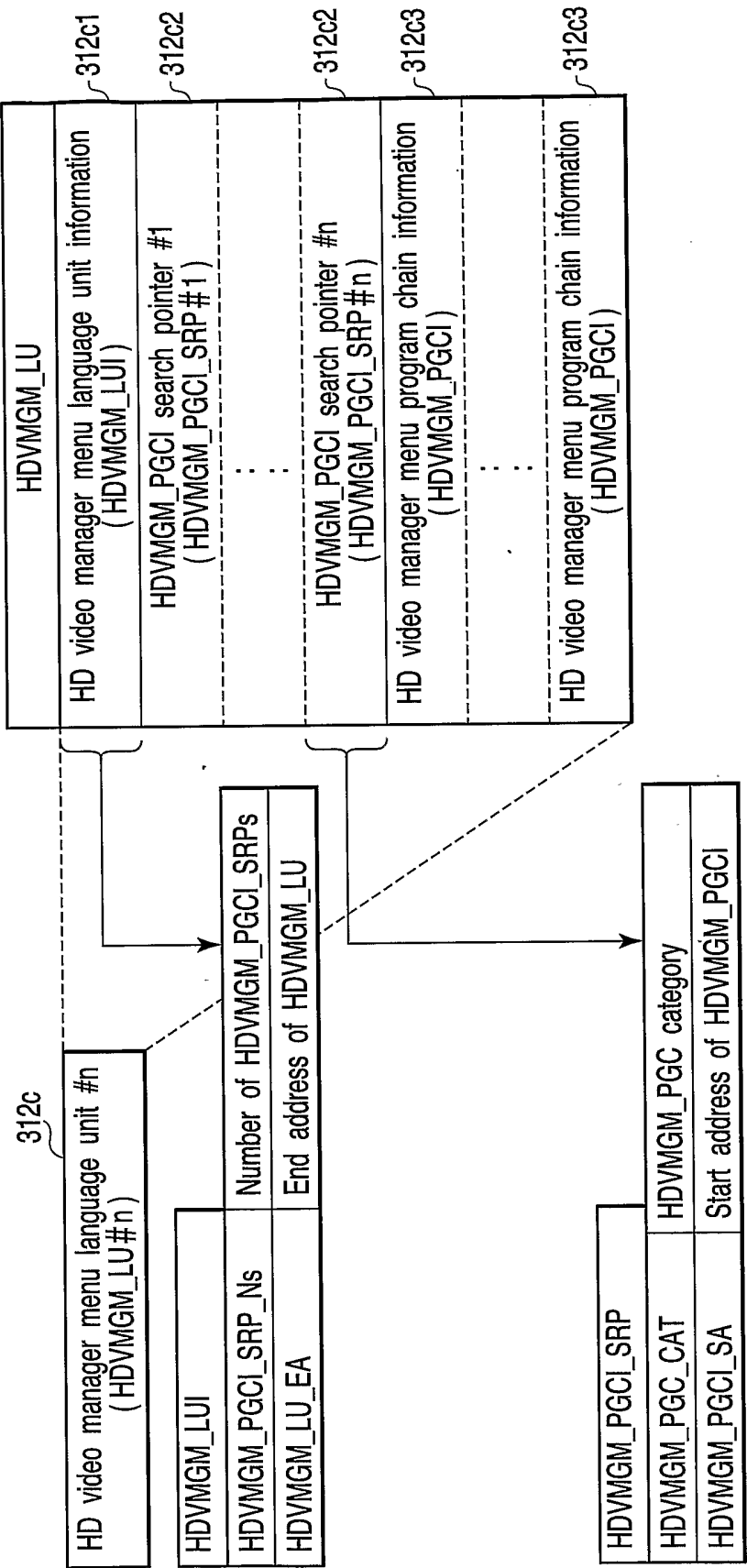


FIG. 7

HDMVMGM_PGC_CAT									
Byte \ Bit	b7	b6	b5	b4	b3	b2	b1	b0	
0	Entry type	Reserved	Audio information selection (Audio selection)			Menu			
1	Block mode		Block type		Audio information number (Audio number)				
2	PTL_ID_FLD (Upper bits)								
3	PTL_ID_FLD (Lower bits)								

Audio information selection (Audio selection)
Indicates selection of HDMENU_AOBS audio playback or HDMVMGM_VOBS audio playback, and start / end trigger of HDMENU_AOBS
...00b : audio in VOB designated by PGC is played back (stop HDMENU_AOBS)
...10b : playback of HDMENU_AOBS is continued (ignore audio in VOB)
...11b : playback of HDMENU_AOBS starts (ignore audio in VOB)

Audio information number (Audio number)
Designates AOB number #n to be played back in HDMENU_AOBS

FIG. 8

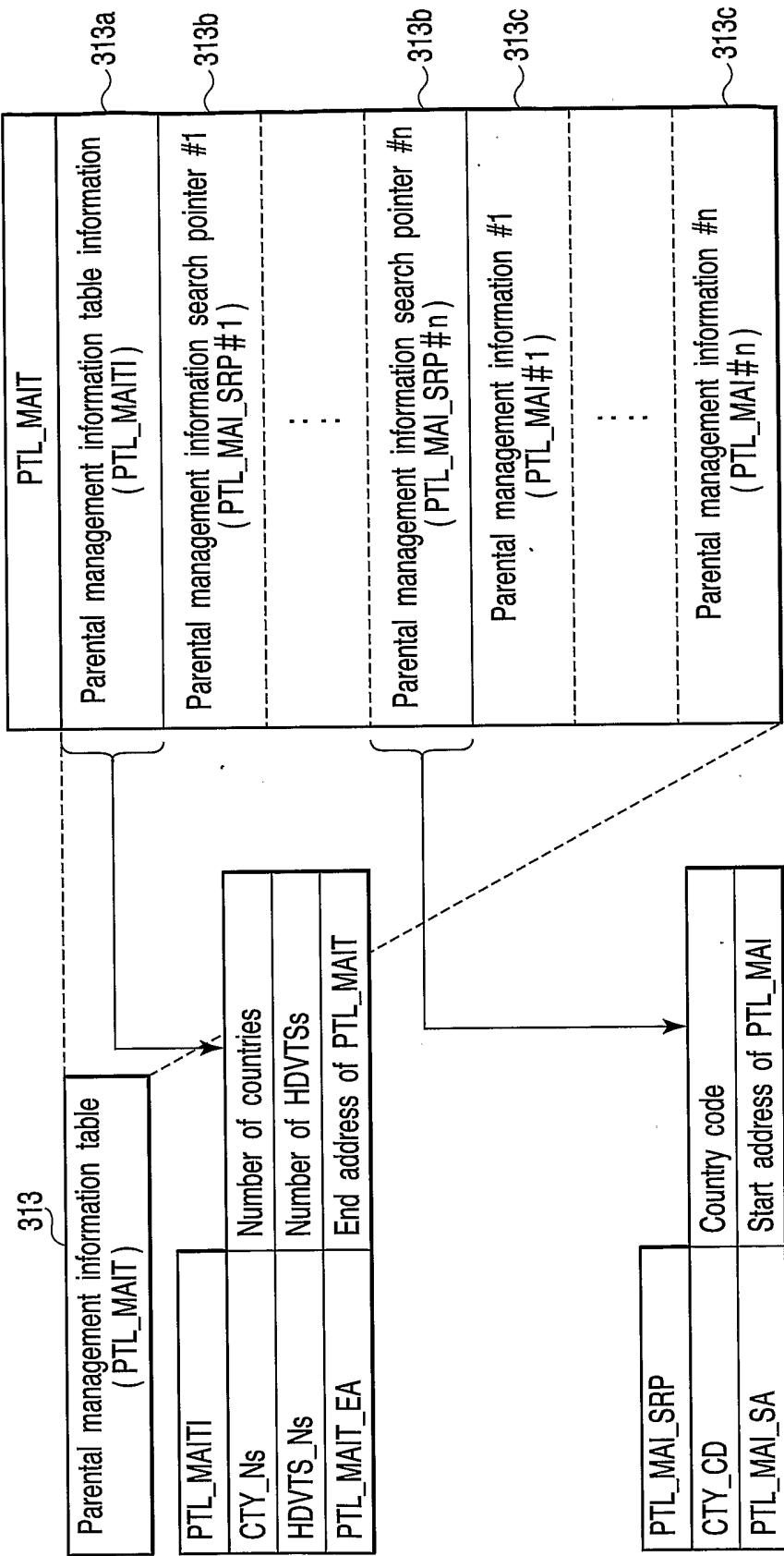


FIG. 9

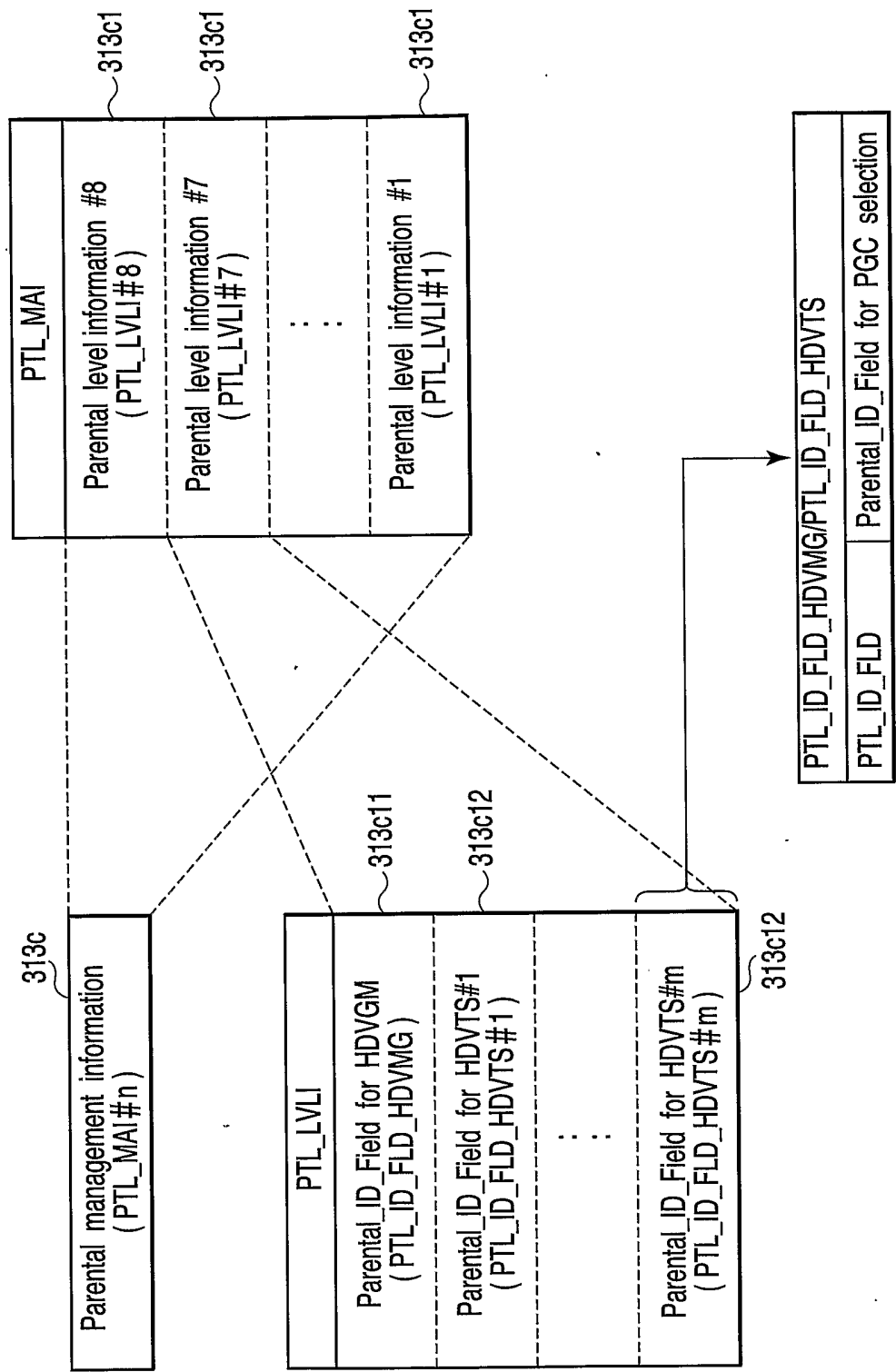


FIG. 10

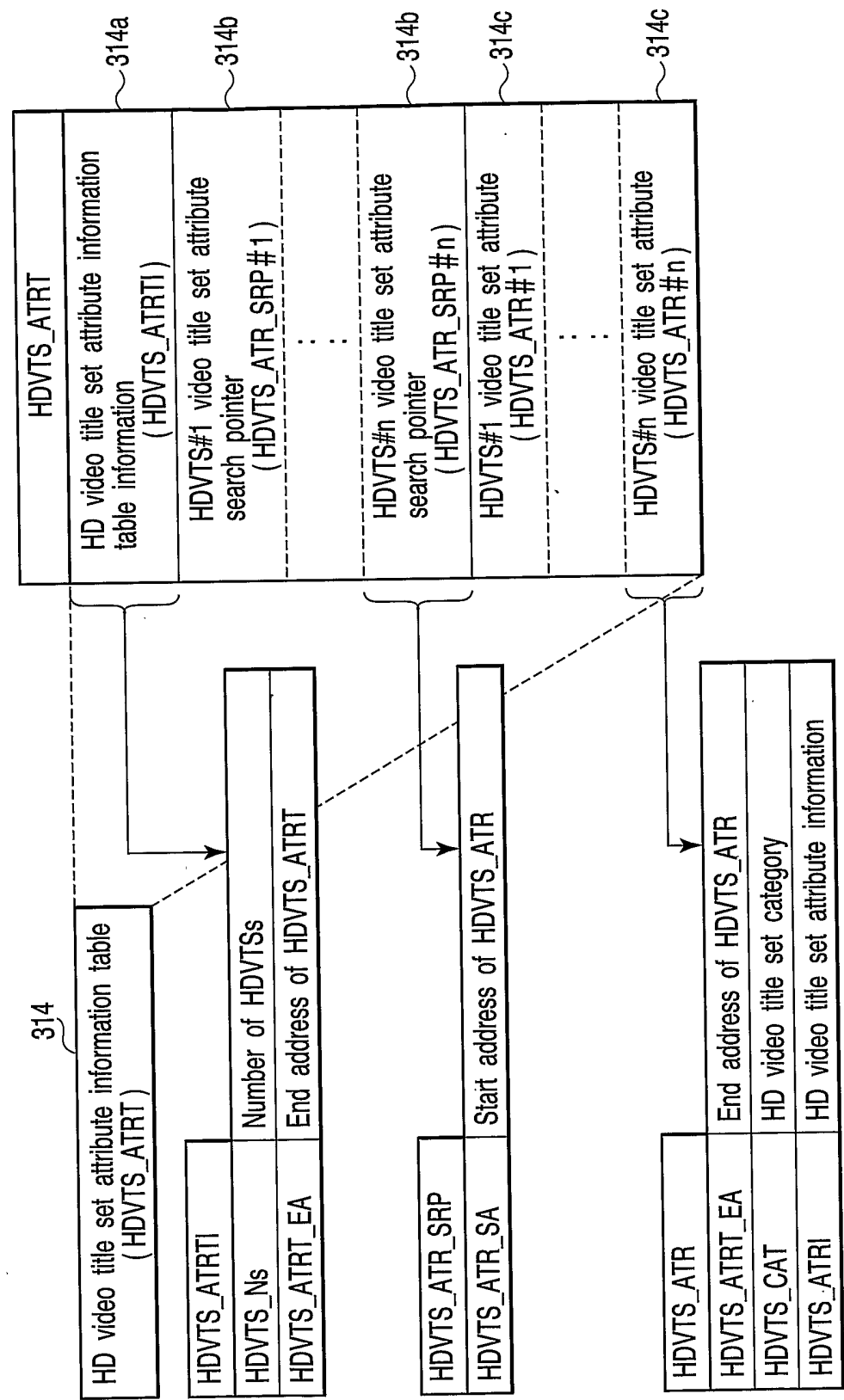


FIG. 11

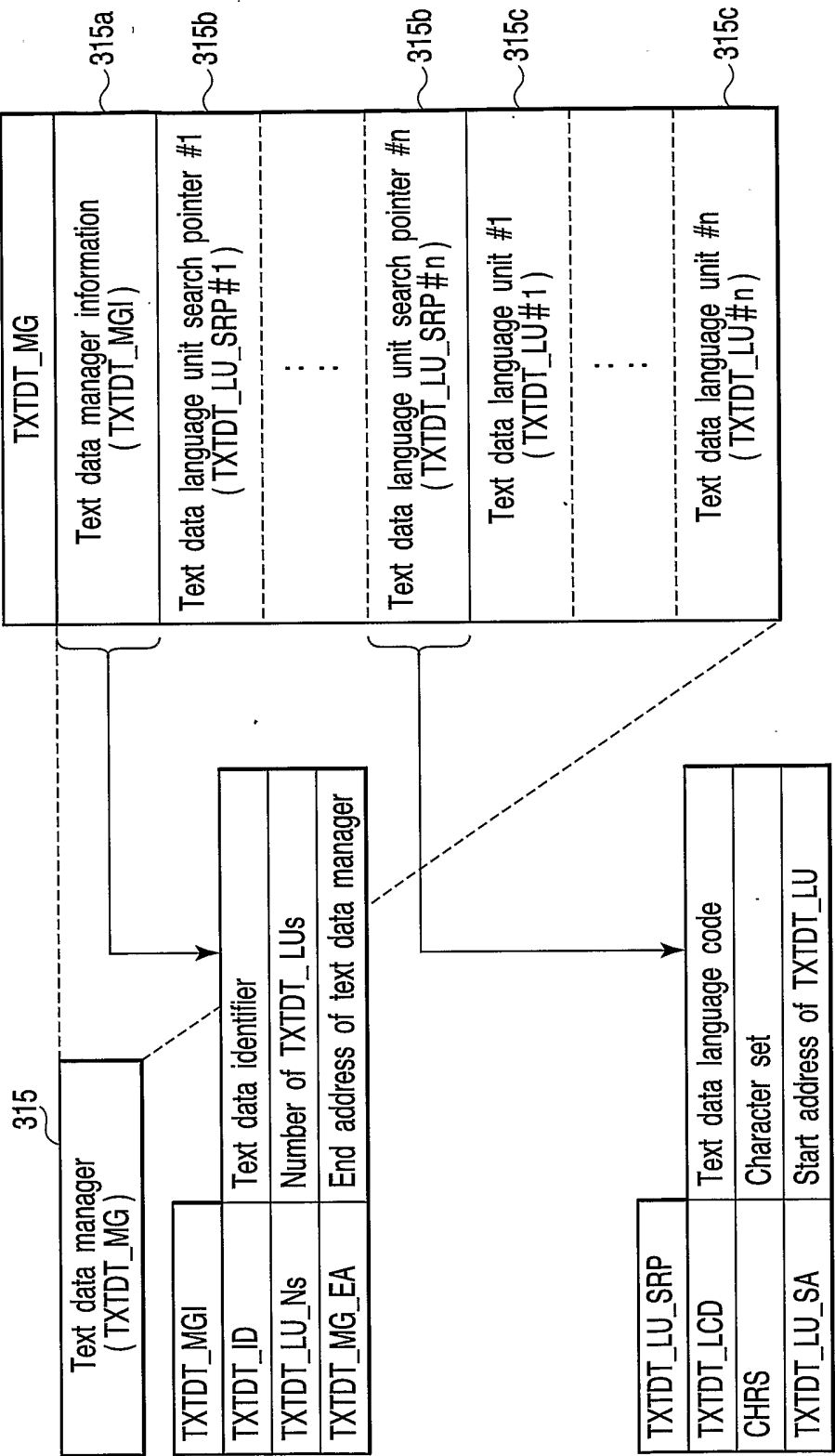


FIG.12

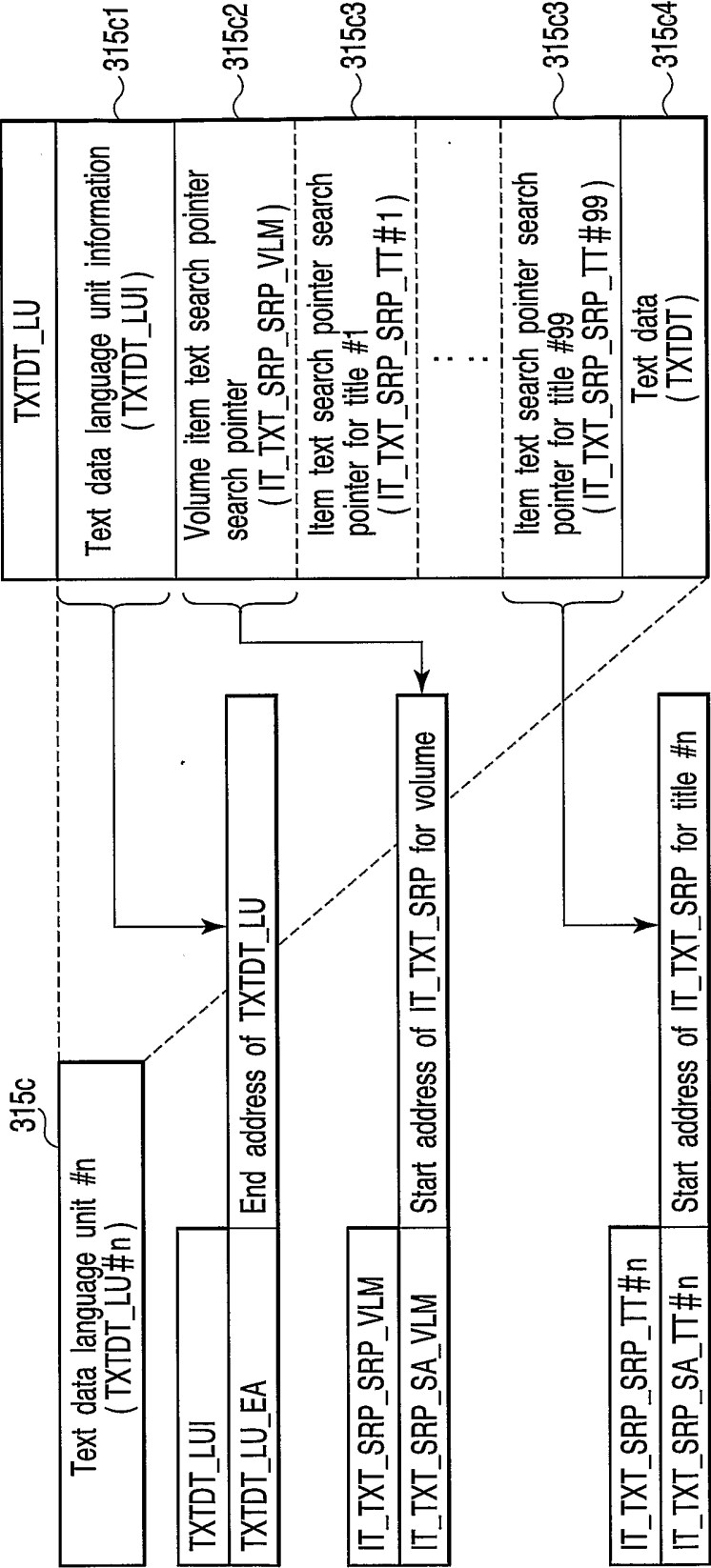


FIG. 13

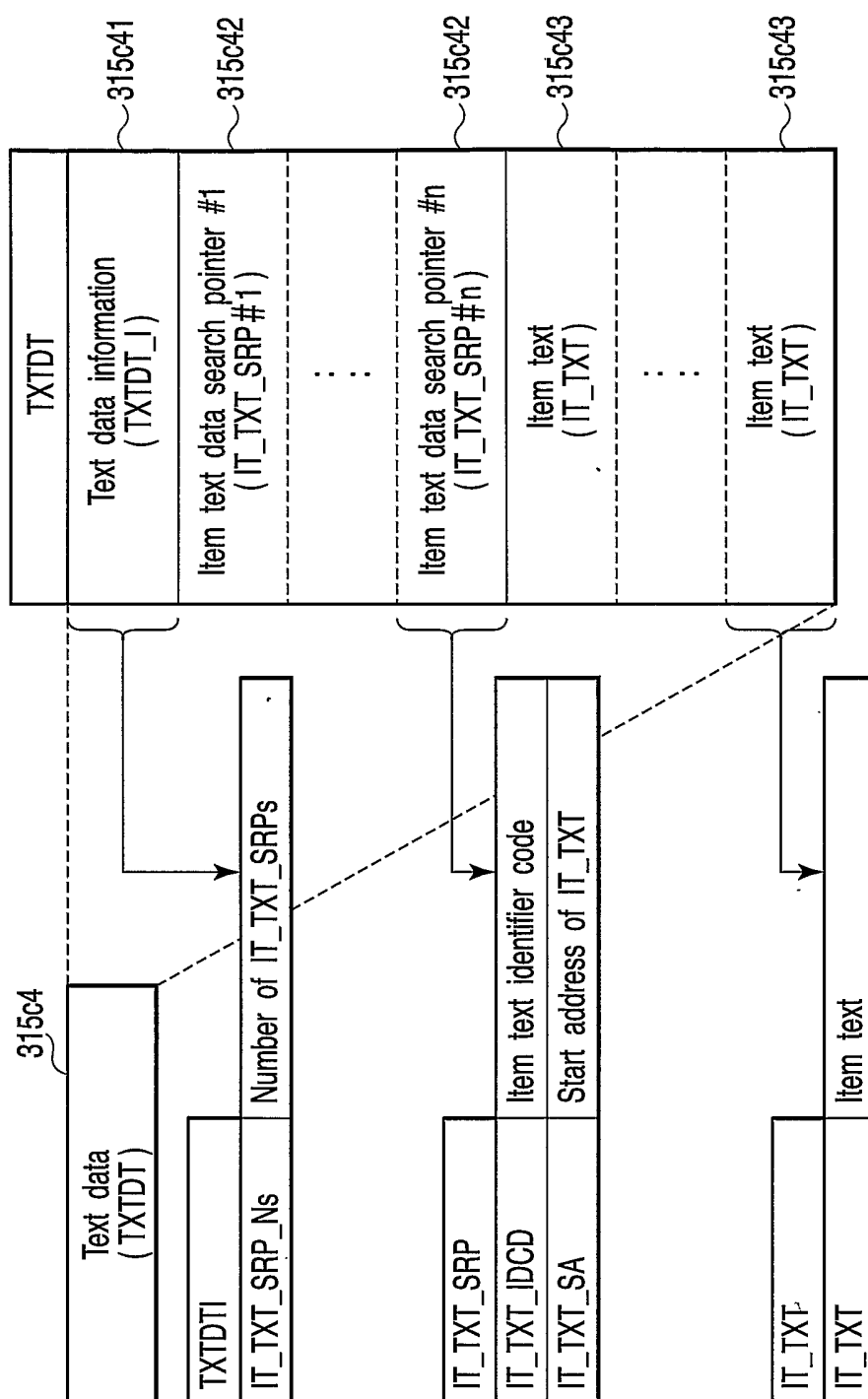


FIG. 14

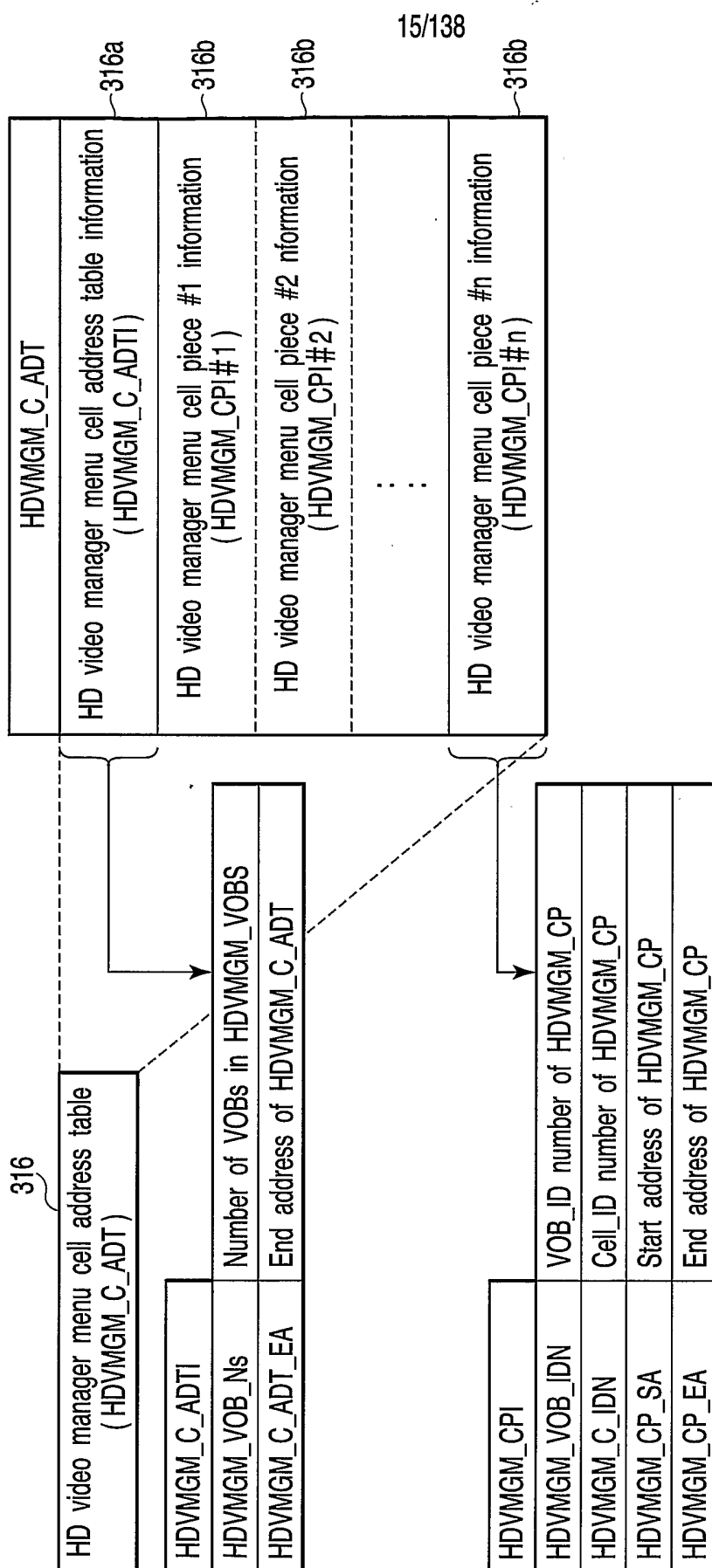


FIG. 15

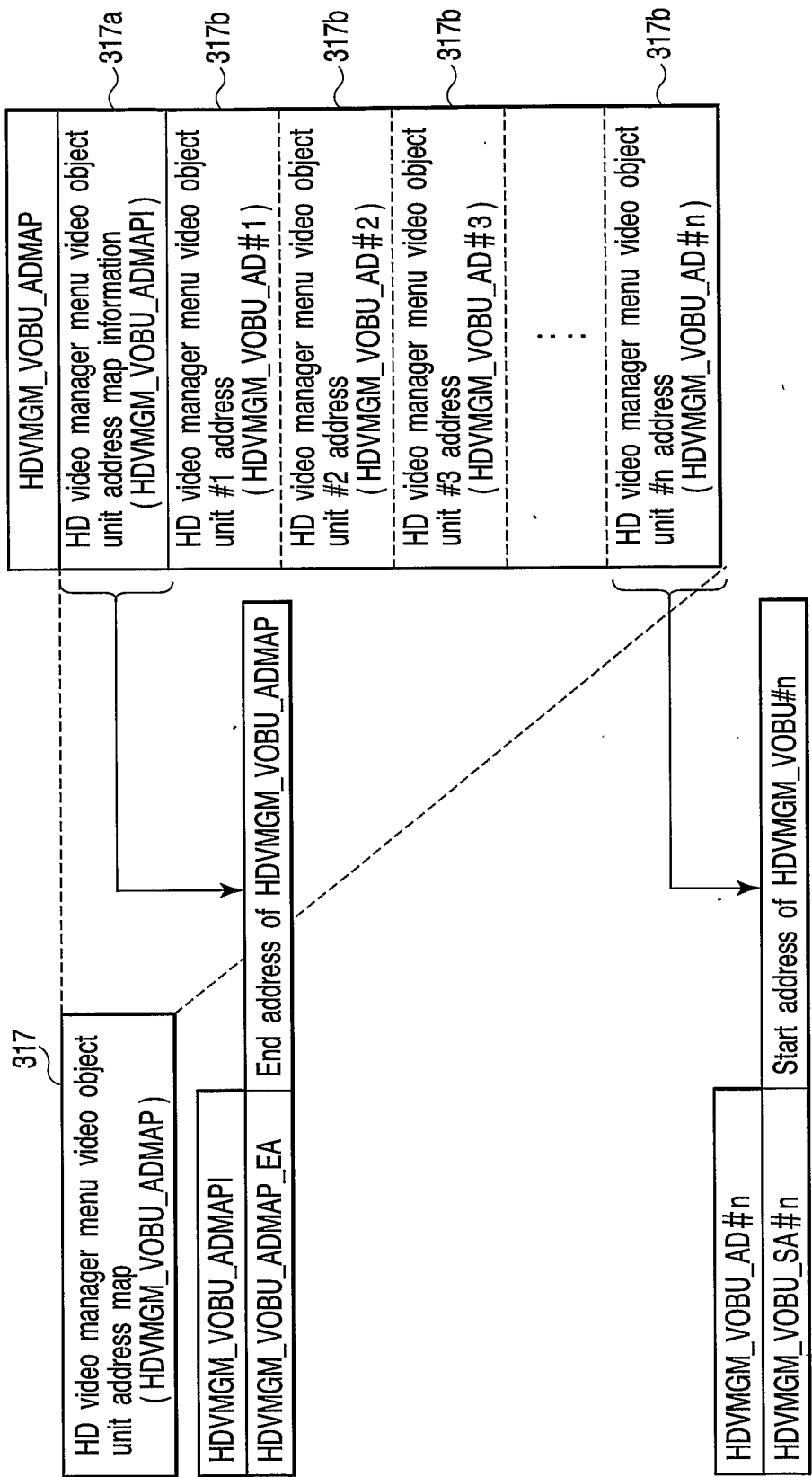


FIG. 16

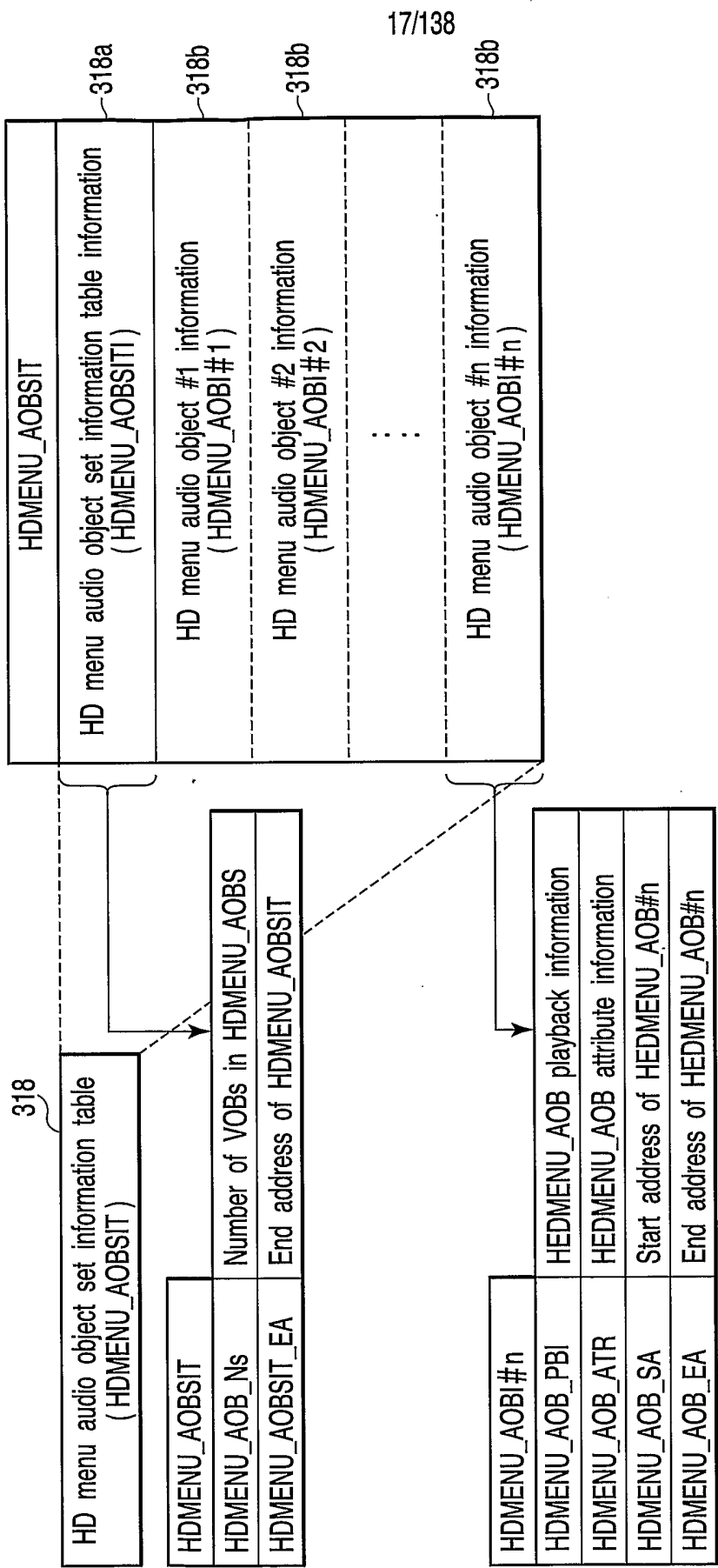


FIG.17

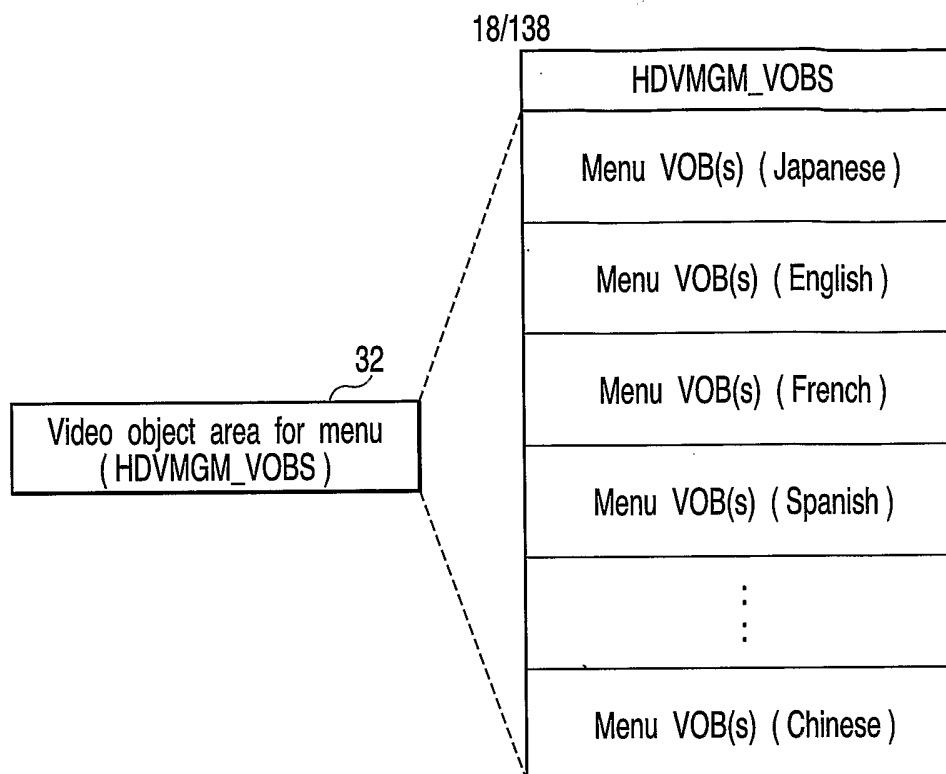


FIG. 18

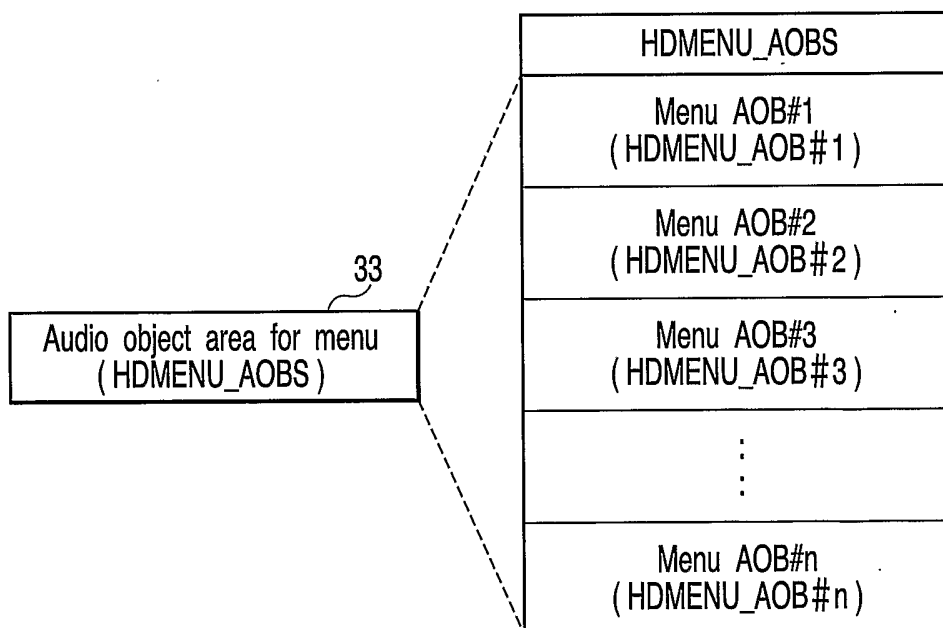


FIG. 19

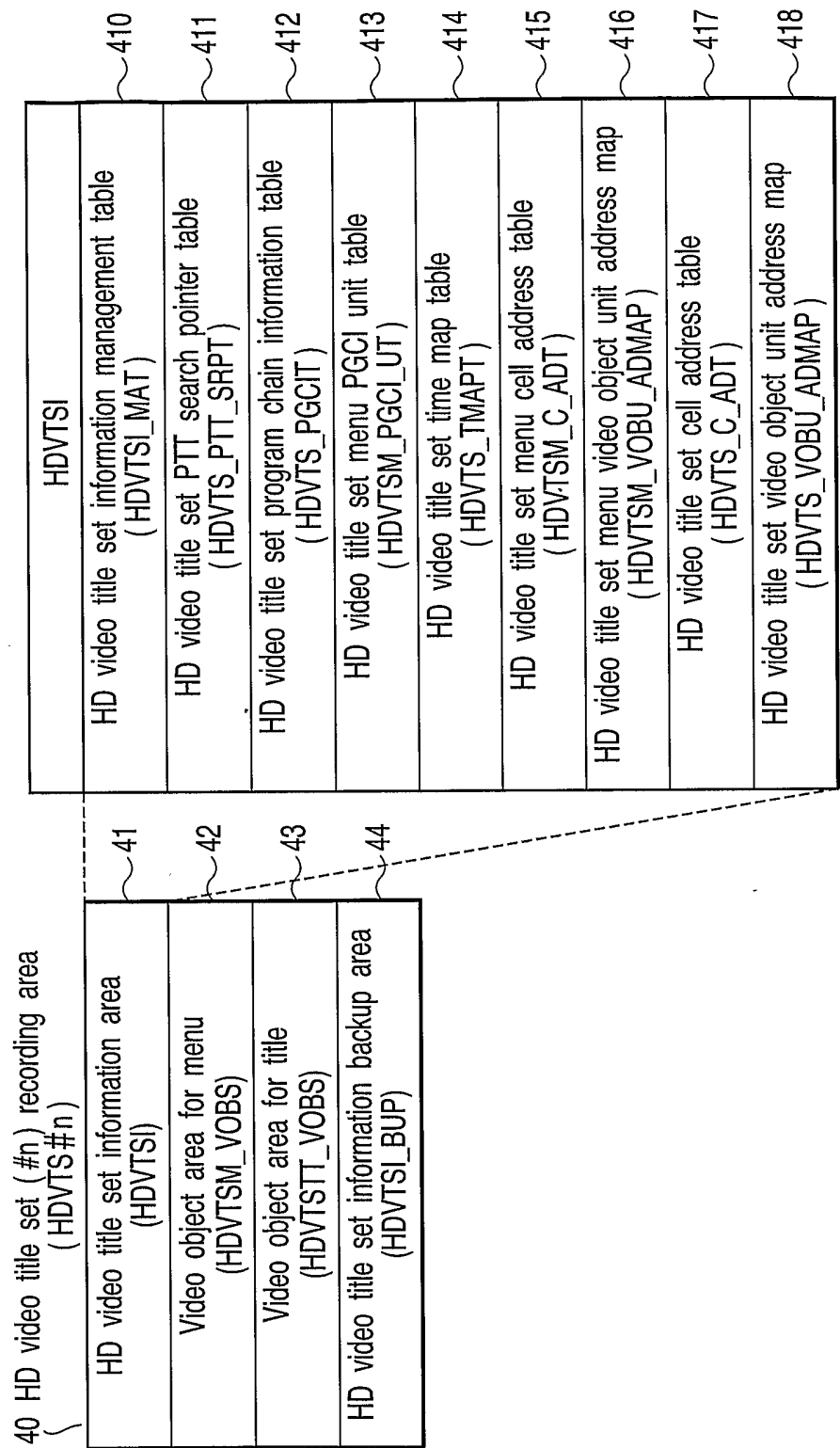


FIG. 20

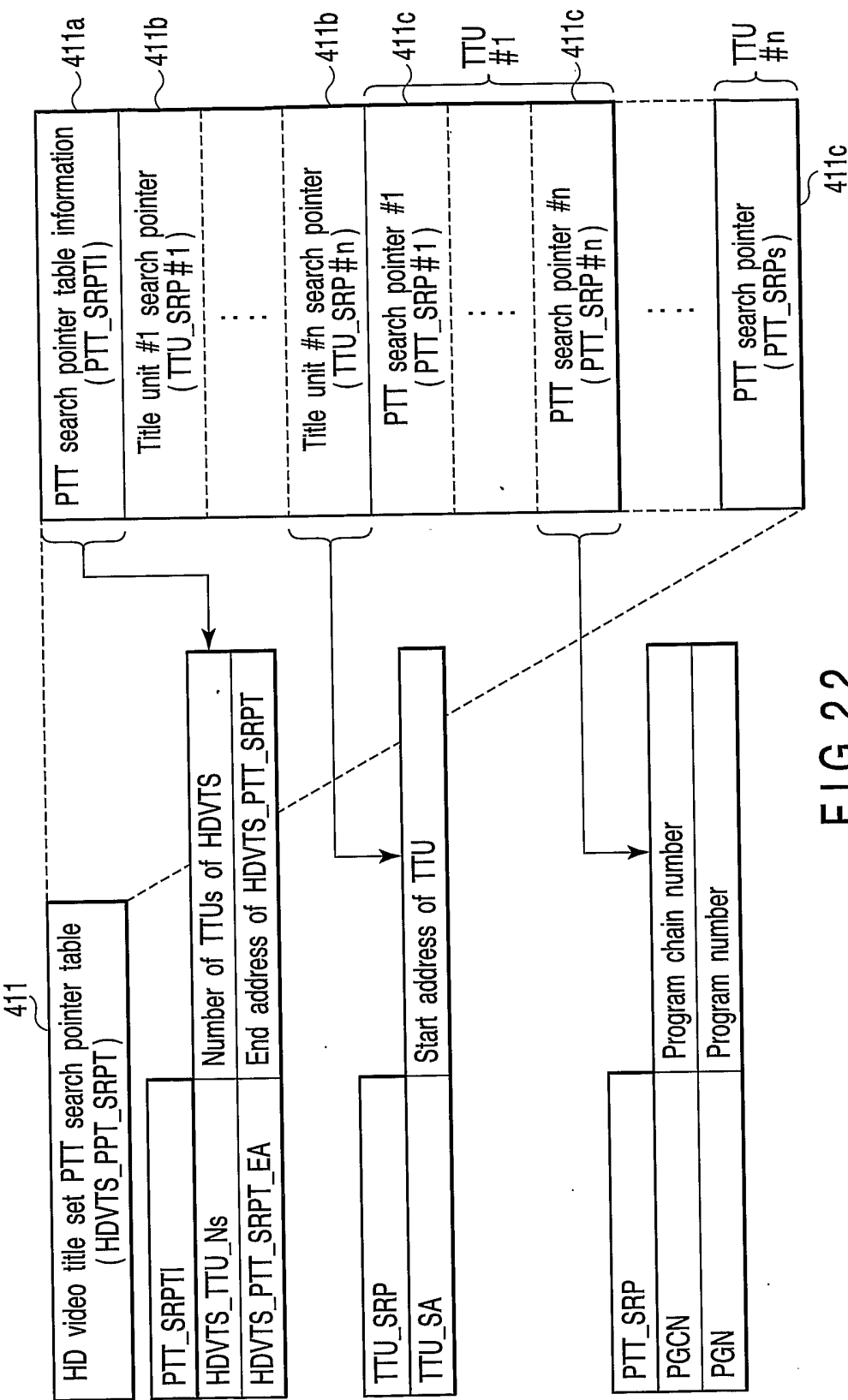
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HDVTSI_MAT	
HDVTS_ID	HD video title set identifier
HDVTS_EA	End address of HDVTS
HDVTSI_EA	End address of HDVTSI
VERN	Version number of HD-DVD-Video standard
HDVTS_CAT	HDVTS category
HDVTSI_MAT_EA	End address of HDVTSI_MAT
HDVTSM_VOBS_SA	Start address of HDVTSM_VOBS
HDVTSI_VOBS_SA	Start address of HDVTSI_VOBS
HDVTS_PTT_SRPT_SA	Start address of HDVTS_PTT_SRPT
HDVTS_PGCIT_SA	Start address of HDVTS_PGCIT
HDVTSM_PGCIT_SA	Start address of HDVTSM_PGCIT
HDVTS_TMAP_SA	Start address of HDVTS_TMAP
HDVTSM_C_ADT_SA	Start address of HDVTSM_C_ADT
HDVTSM_VOBU_ADMAP_SA	Start address of HDVTSM_VOBU_ADMAP
HDVTS_C_ADT_SA	Start address of HDVTS_C_ADT
HDVTS_VOBU_ADMAP_SA	Start address of HDVTS_VOBU_ADMAP
HDVTSM_V_ATR	HDVTSM video attribute
HDVTSM_AST_Ns	Number of HDVTSM audio streams
HDVTSM_AST_ATR	HDVTSM audio stream attribute
HDVTSM_SPST_Ns	Start address of number of HDVTSM sub-picture streams
HDVTSM_SPST_ATR	HDVTSM sub-picture stream attribute
HDVTS_V_ATR	HDVTS video attribute
HDVTS_AST_Ns	Number of HDVTS audio streams
HDVTS_AST_ATR	HDVTS audio stream attribute table
HDVTS_SPST_Ns	Number of HDVTS sub-picture streams
HDVTS_SPST_ATR	HDVTS sub-picture stream attribute table
HDVTS_MU_AST_ATR	HDVTS multi-channel audio stream attribute table
HDVTSM_GUST_Ns	Number of HDVTSM graphics unit streams
HDVTSM_GUST_ATR	HDVTSM graphics unit stream attribute
HDVTS_GUST_Ns	Number of HDVTS graphics unit streams
HDVTS_GUST_ATR	HDVTS graphics unit stream attribute table

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HD video title set information management table
(HDVTSI_MAT)

FIG. 21



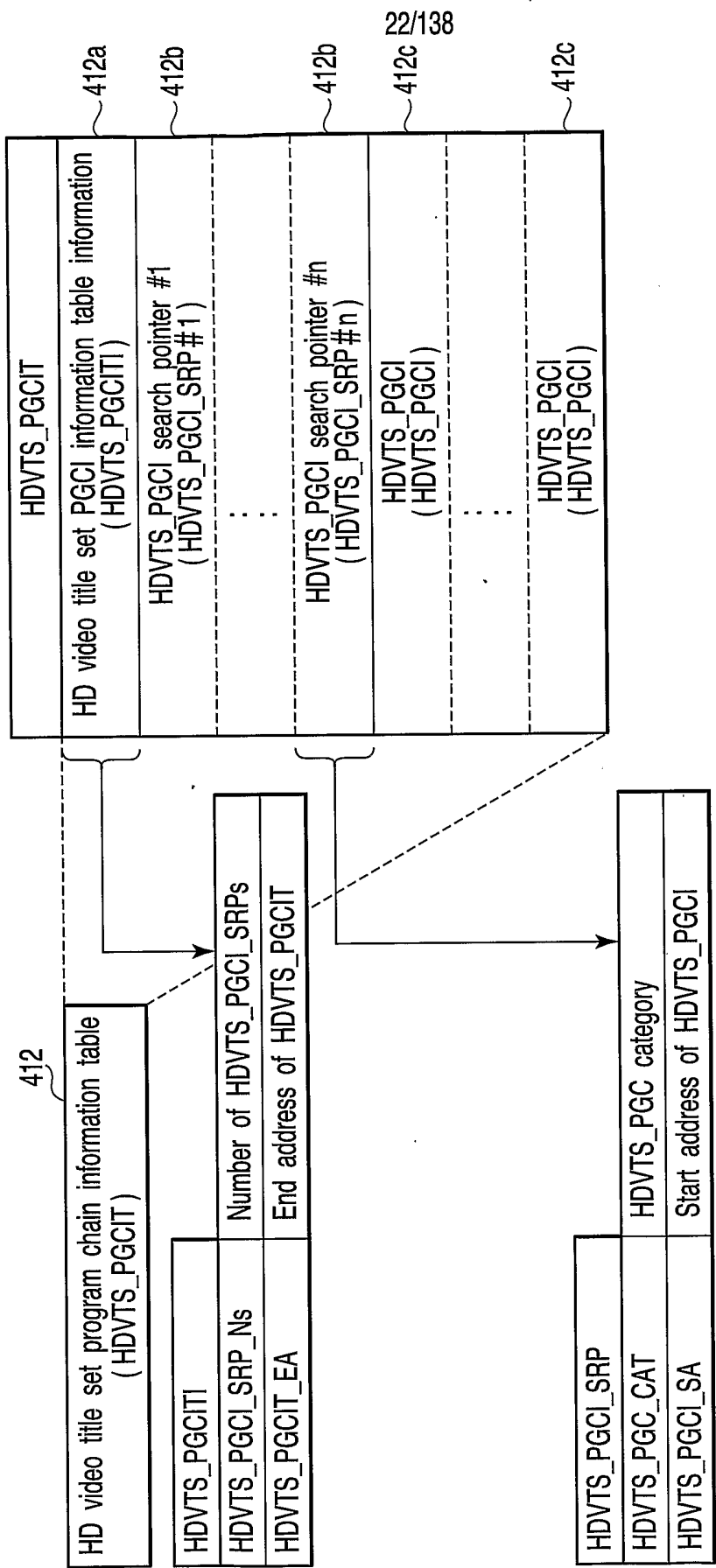


FIG. 23

HDVTS_PGC_CAT										
Byte \ Bit	b7	b6	b5	b4	b3	b2	b1	b0		
0	Entry type		VTS_TTN							
1	Block mode		Block type		Reserved			RSM permission		
2	PTL_ID_FLD (Upper bits) (RSM permission)									
3	PTL_ID_FLD (Lower bits)									

RSM permission
Indicates whether or not playback resume process based on RSM command or Resume () function is permitted in this PGC
...0b : permitted (update RSM information)
...1b : inhibited (do not update RSM information)

FIG. 24

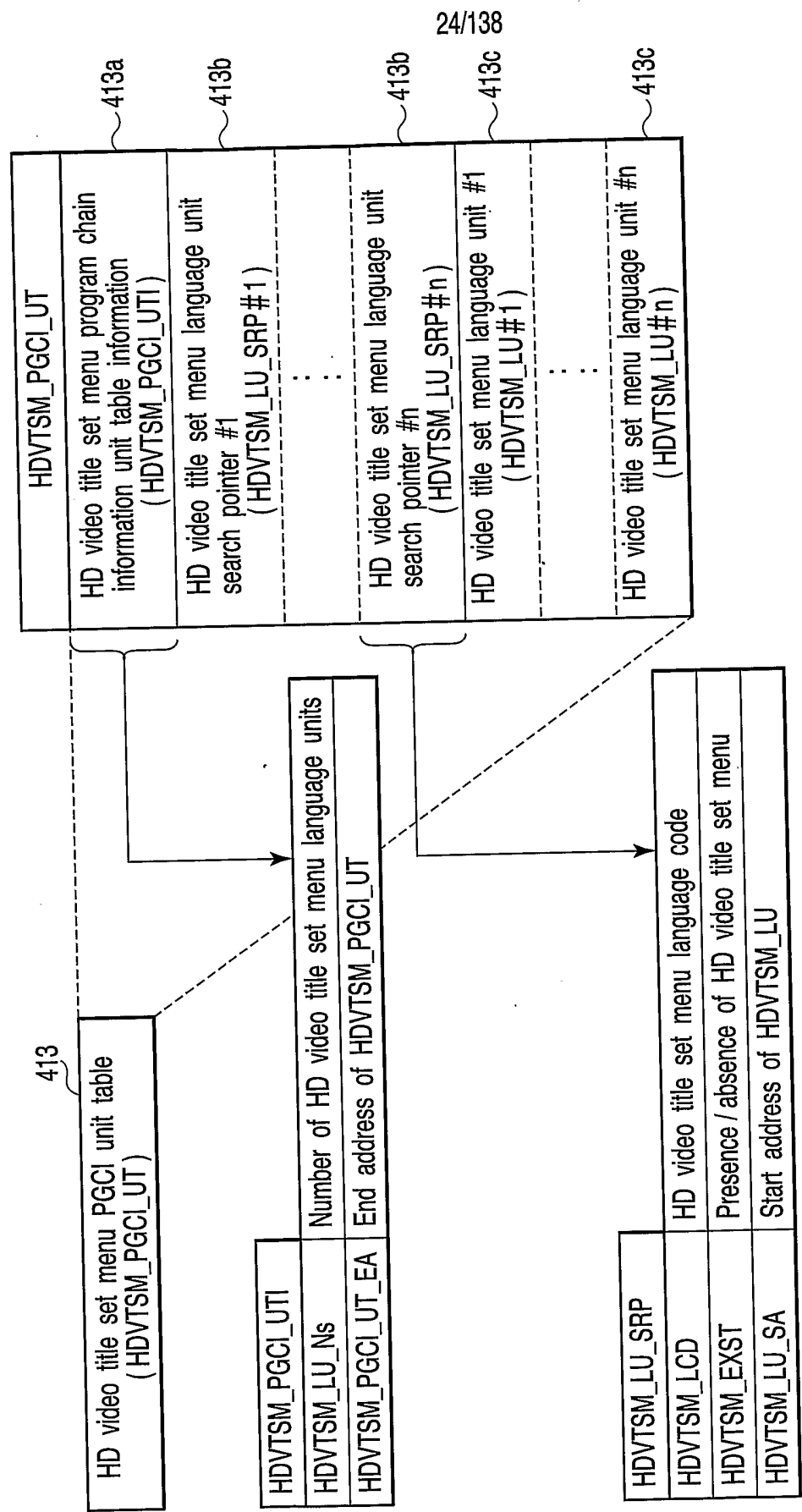


FIG. 25

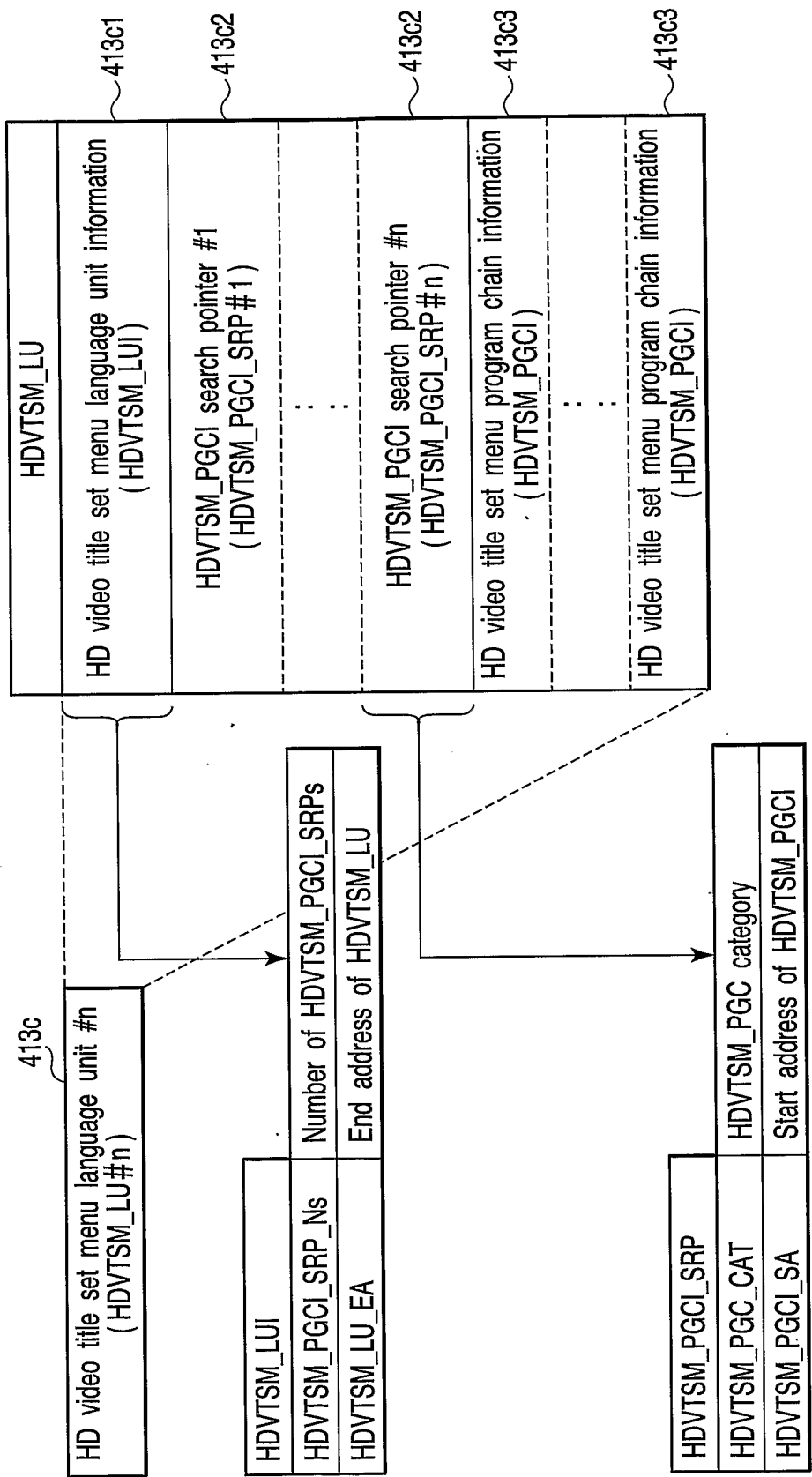


FIG. 26

HDTVSM_PGC_CAT									
Byte \ Bit	b7	b6	b5	b4	b3	b2	b1	b0	
0	Entry type	Reserved	Audio information selection (Audio selection)				Menu ID		
1	Block mode		Block type		Audio	information number (Audio number)			
2	PTL_ID_FLD (Upper bits)								
3	PTL_ID_FLD (Lower bits)								

Audio information selection (Audio selection)
Indicates selection of HDMENU_AOBS audio playback or HDTVSM_VOBS audio playback, and start/end trigger of HDMENU_AOBS
...00b : audio in VOB designated by PGC is played back (stop HDMENU_AOBS)
...10b : playback of HDMENU_AOBS is continued (ignore audio in VOB)
...11b : playback of HDMENU_AOBS starts (ignore audio in VOB)

Audio information number (Audio number)
Designates AOB number #n to be played back in HDMENU_AOBS

FIG. 27

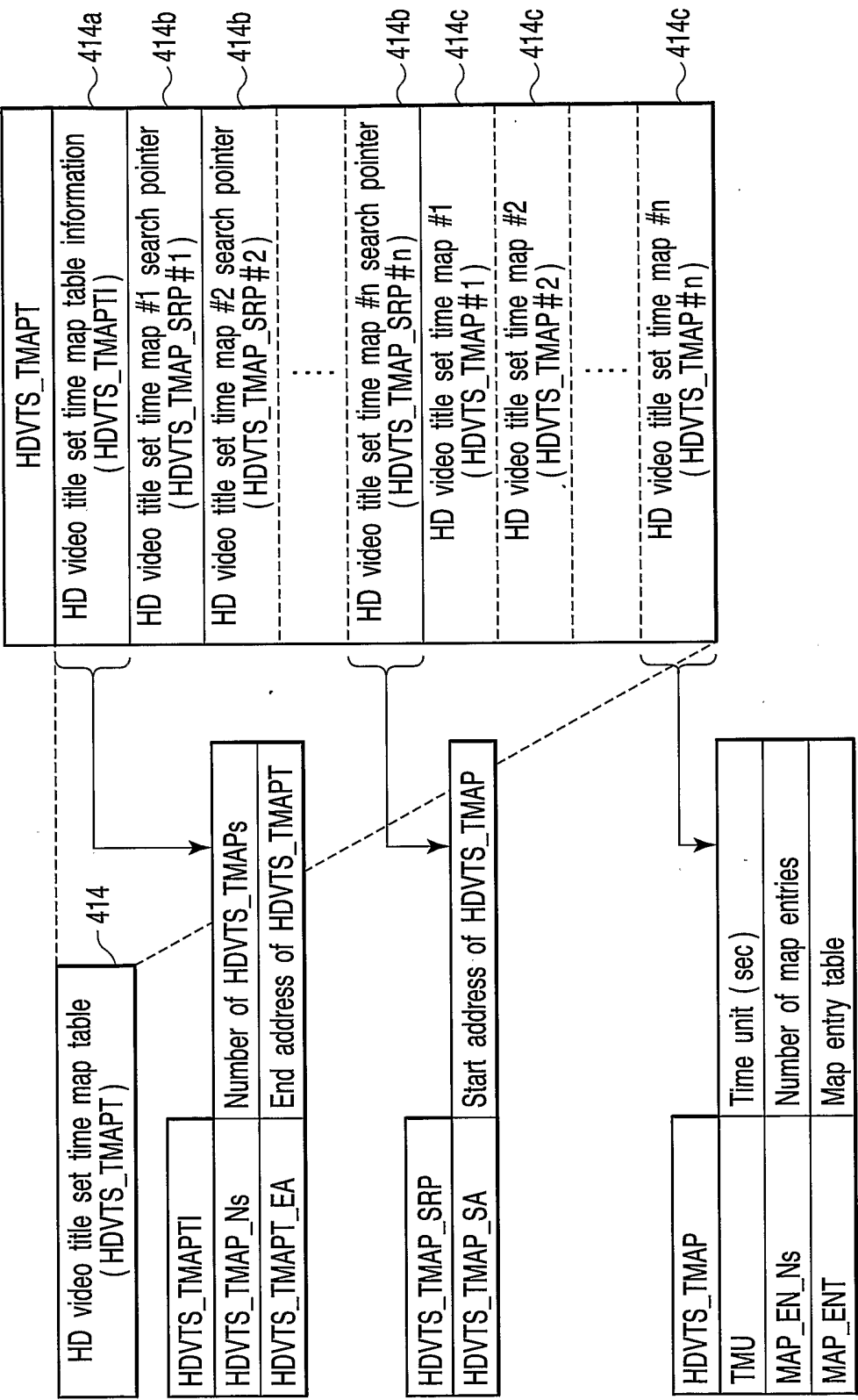


FIG. 28

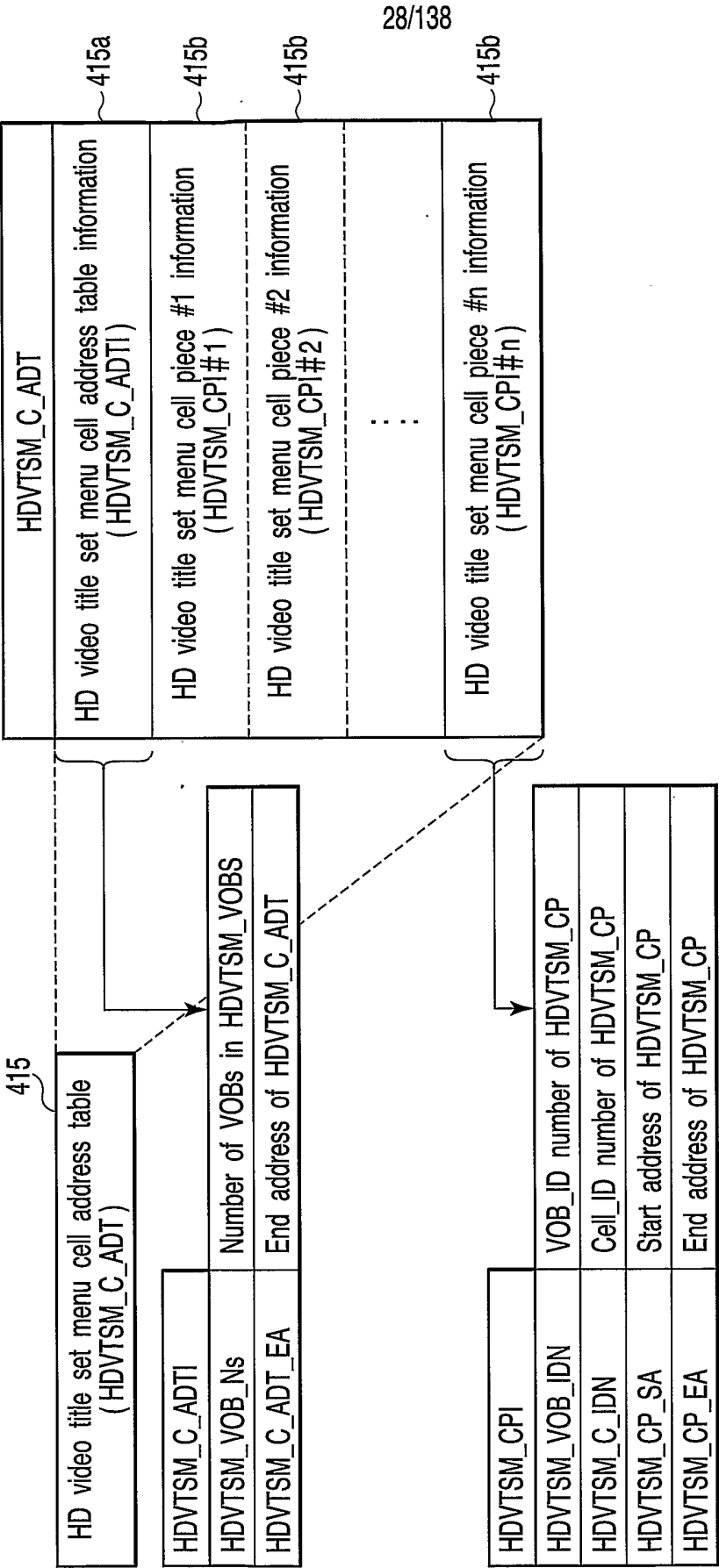


FIG. 29

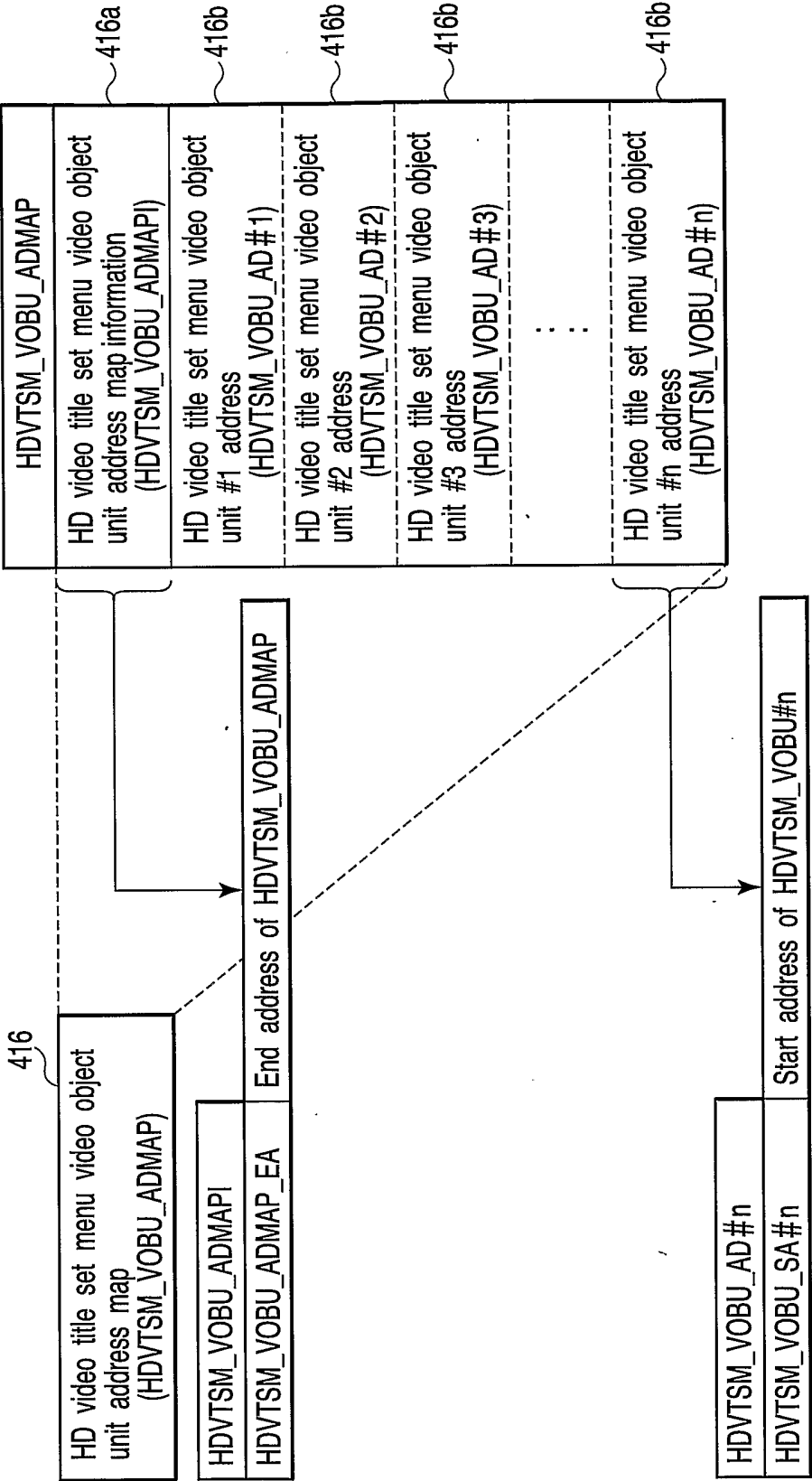


FIG. 30

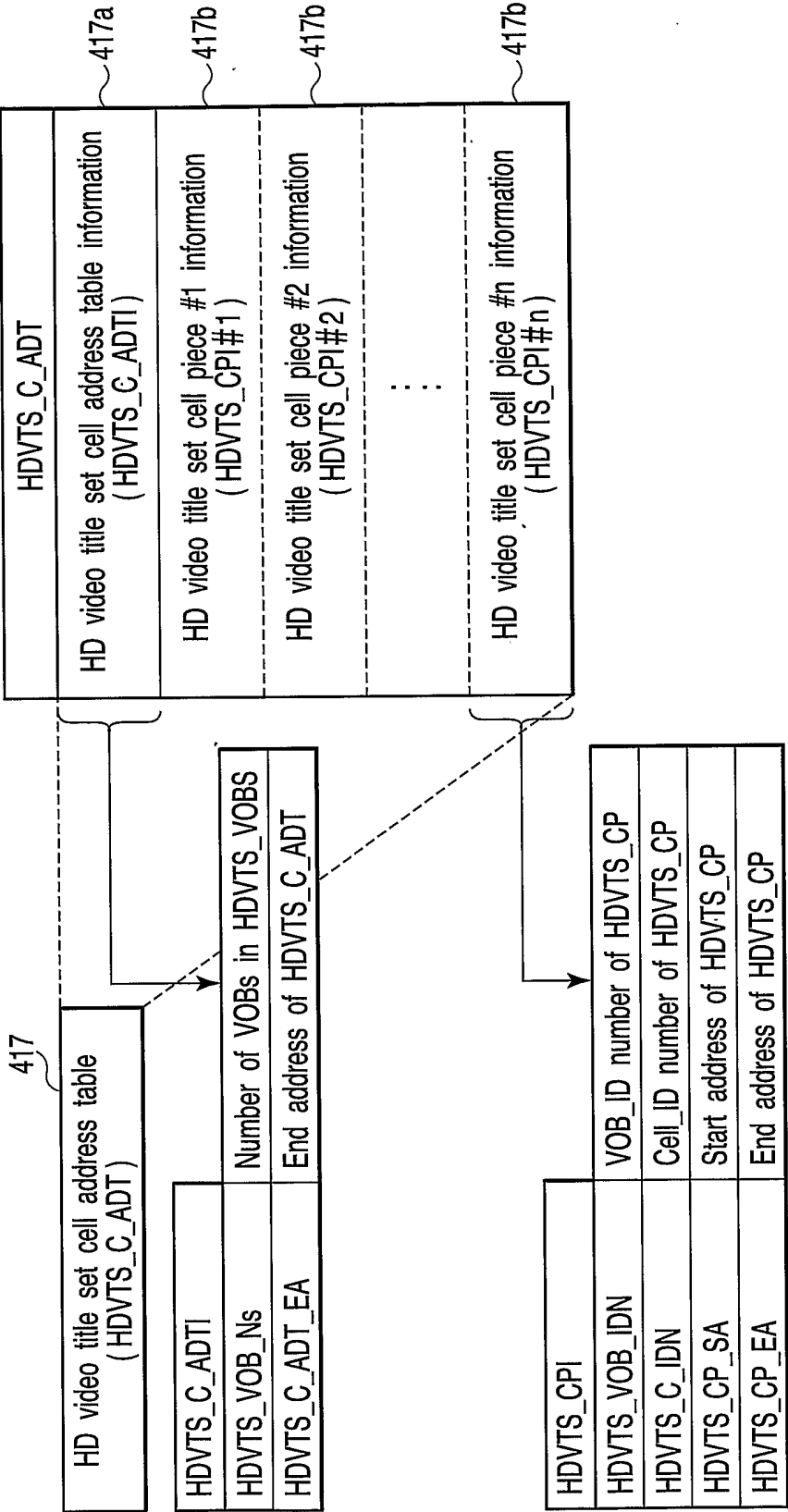


FIG. 31

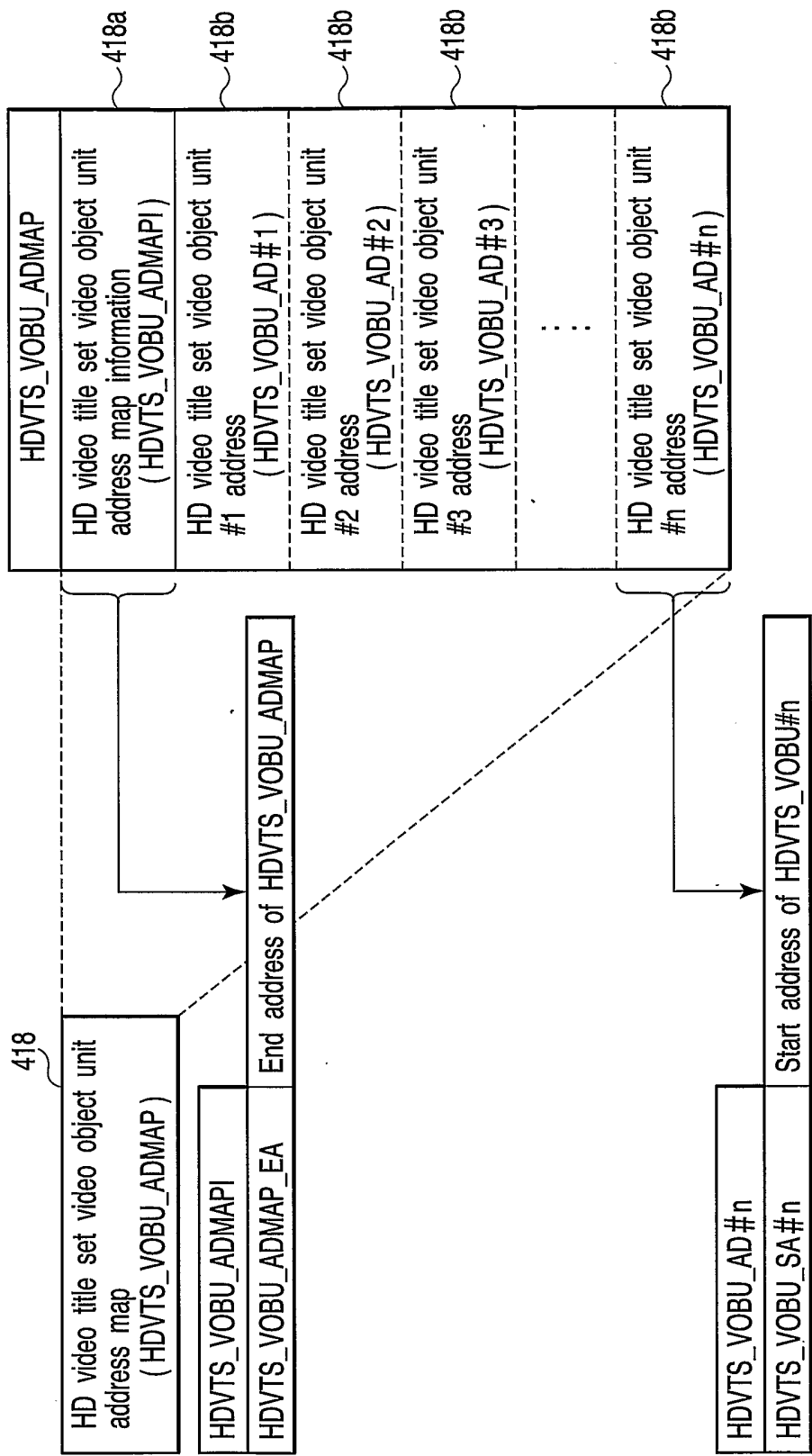


FIG. 32

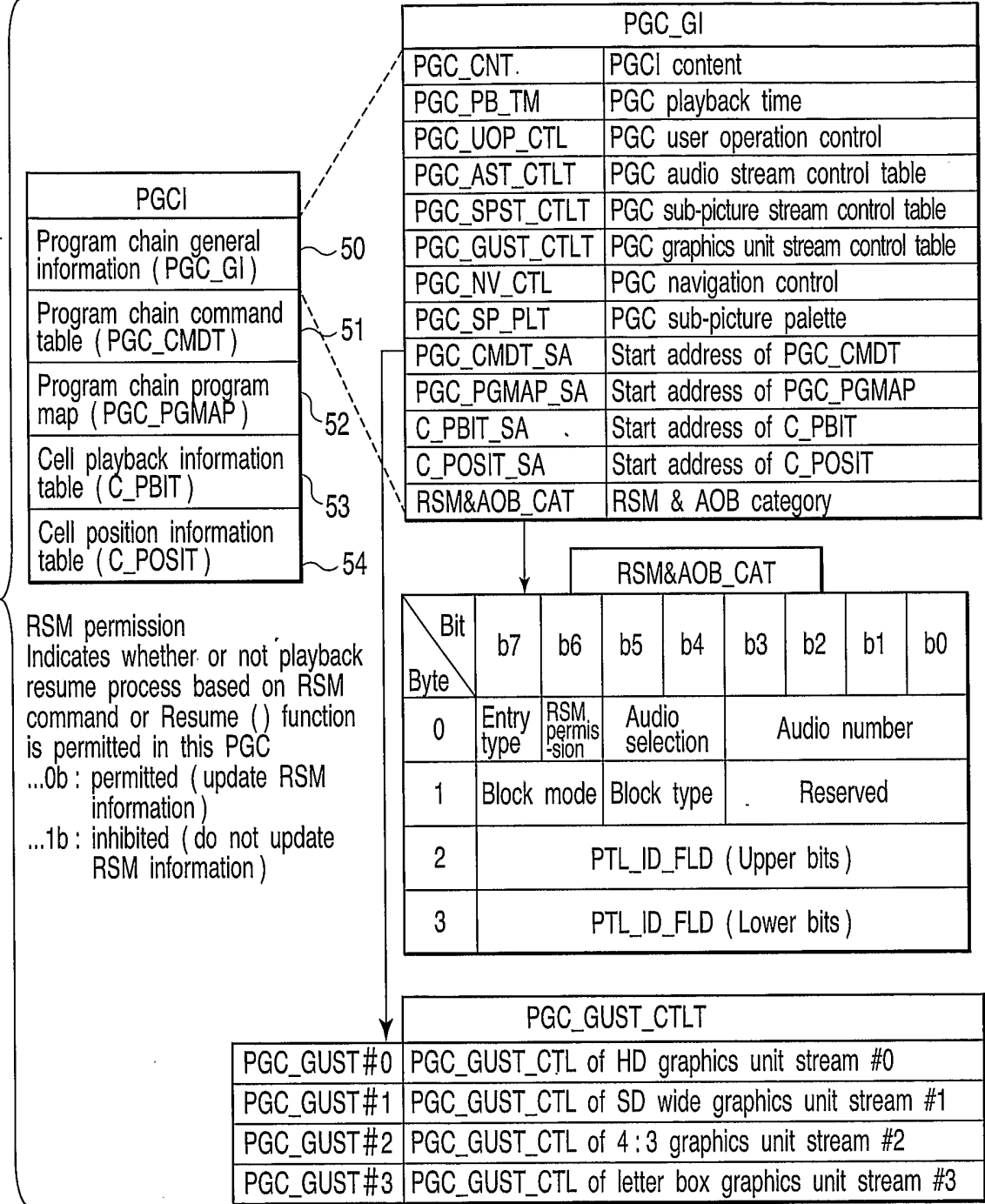


FIG. 33

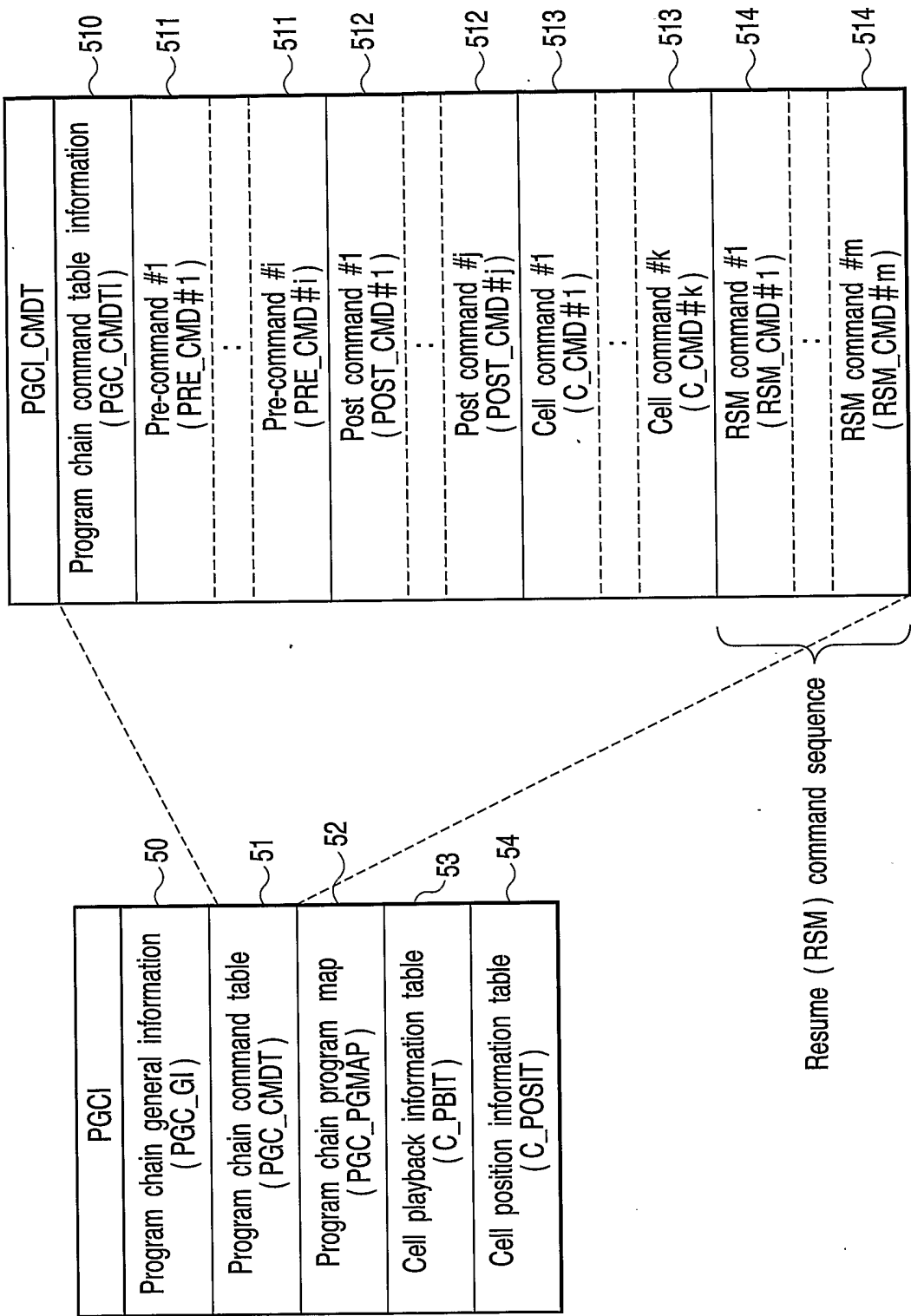


FIG. 34

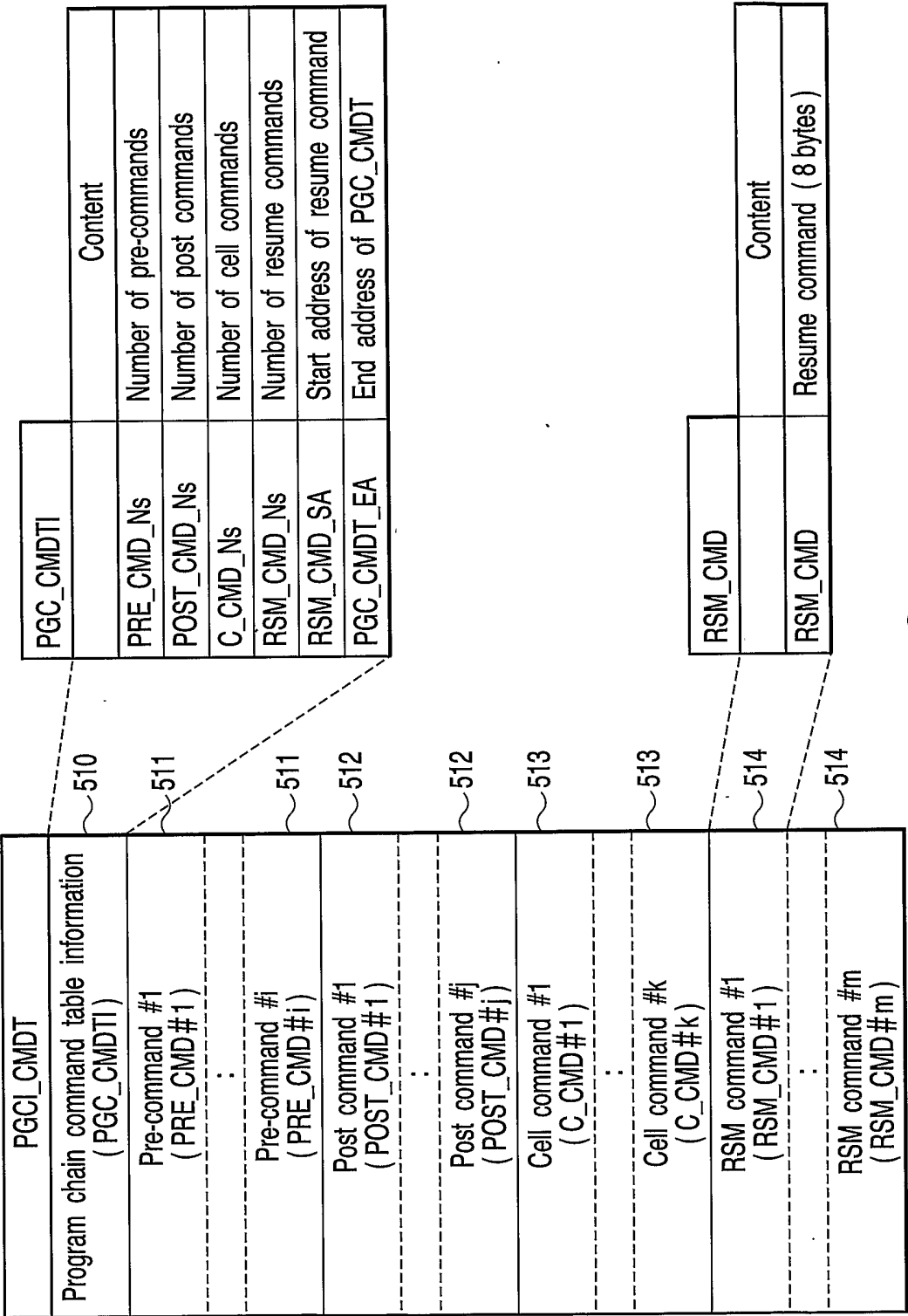


FIG. 35

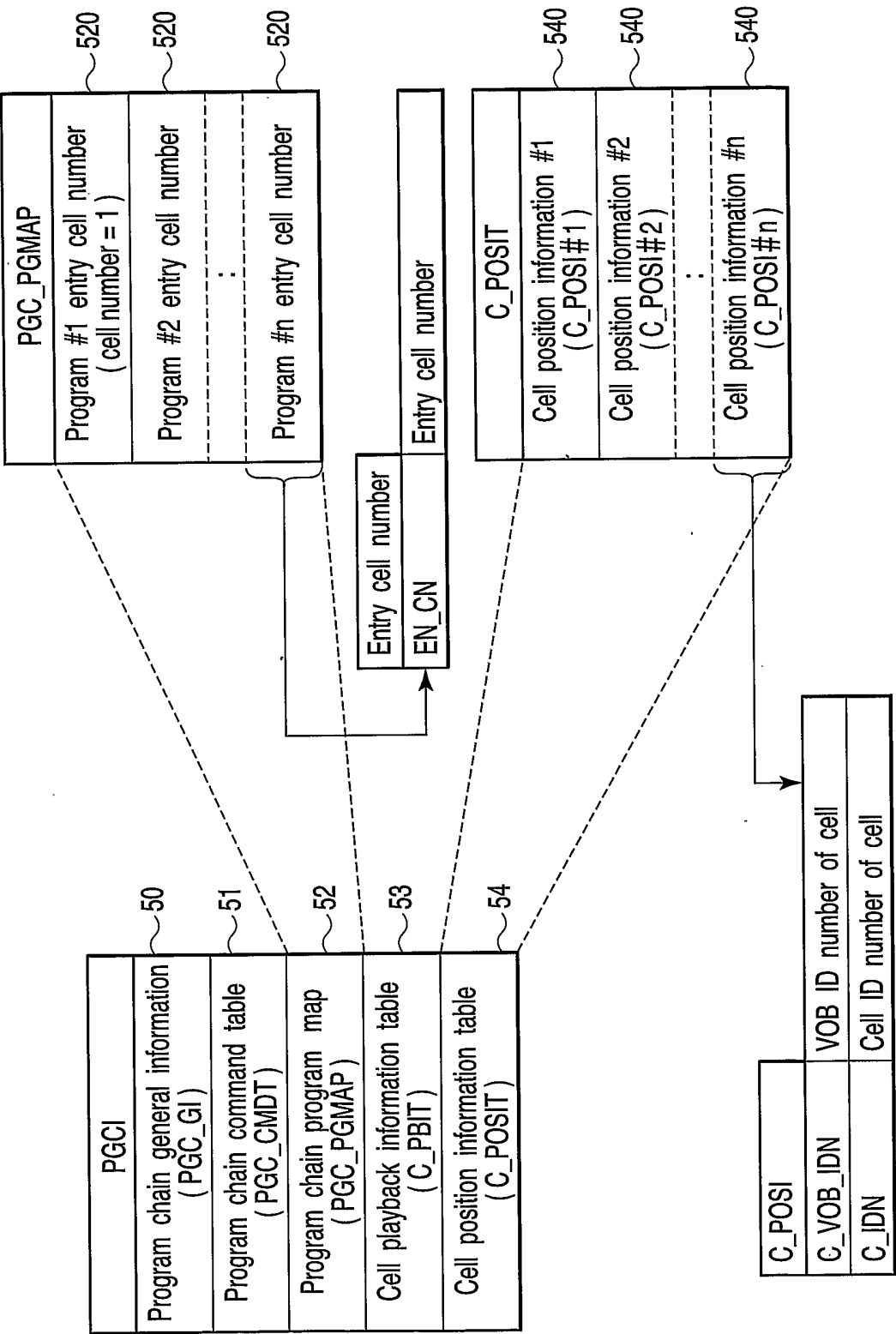


FIG. 36

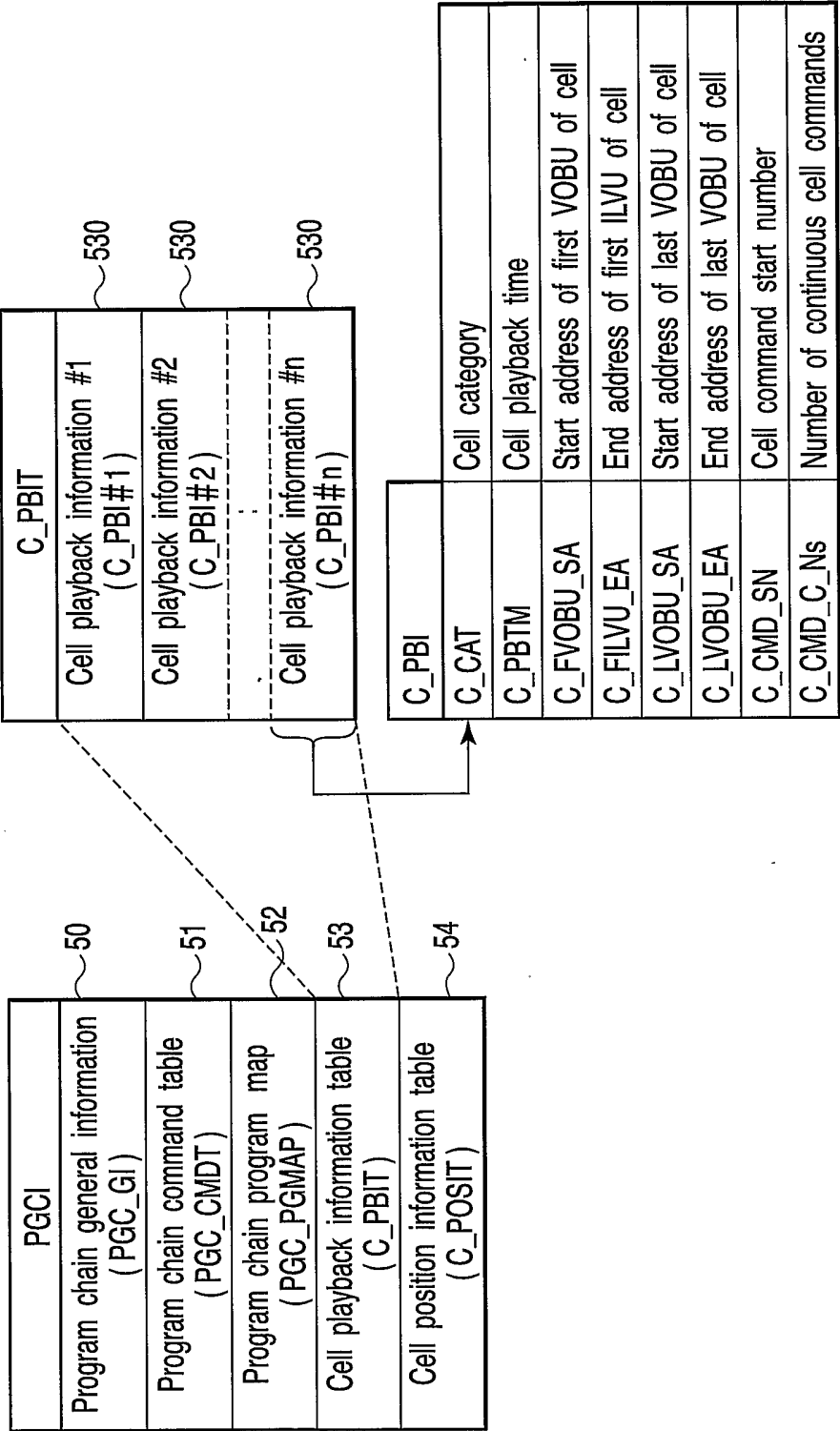


FIG. 37

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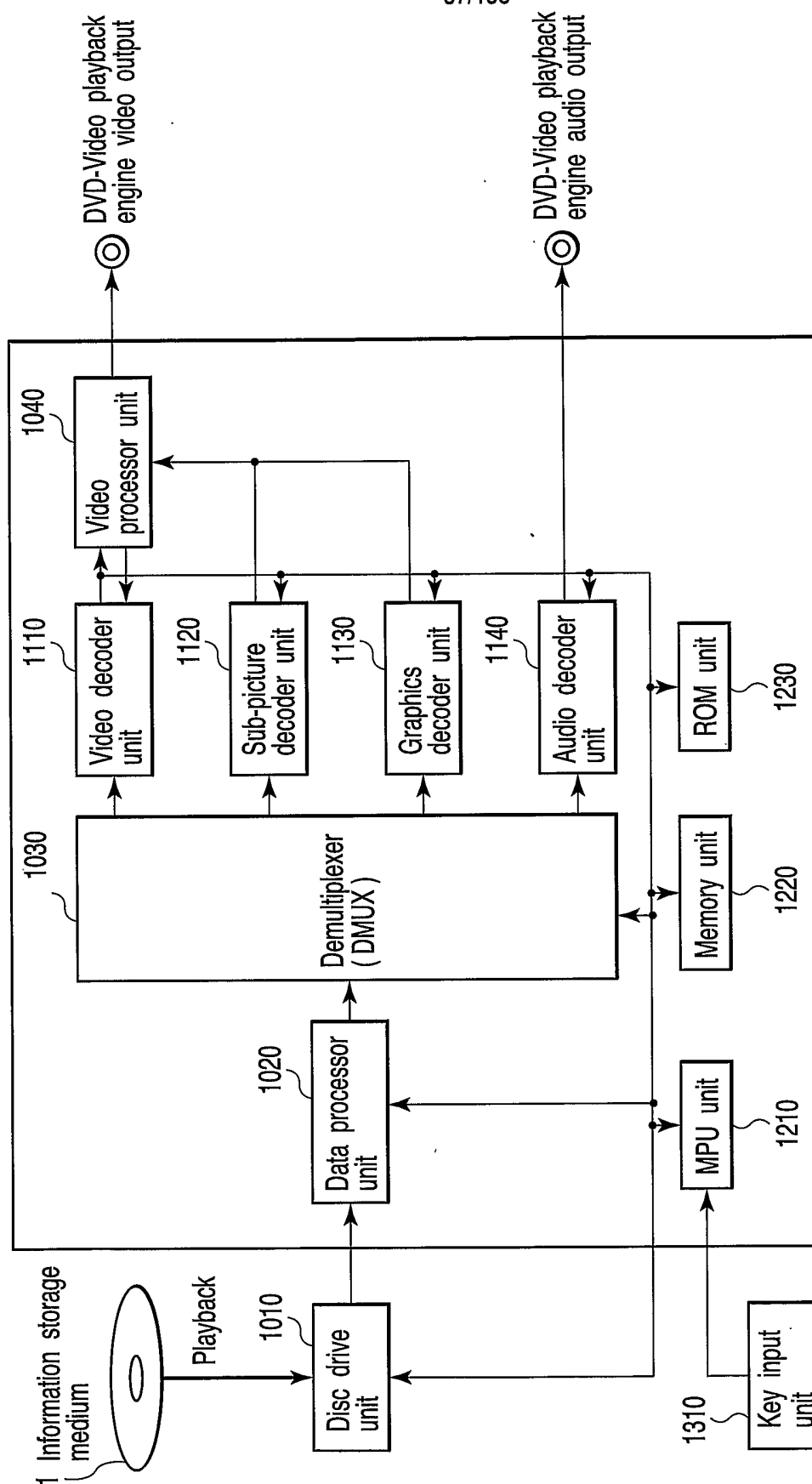


FIG. 38

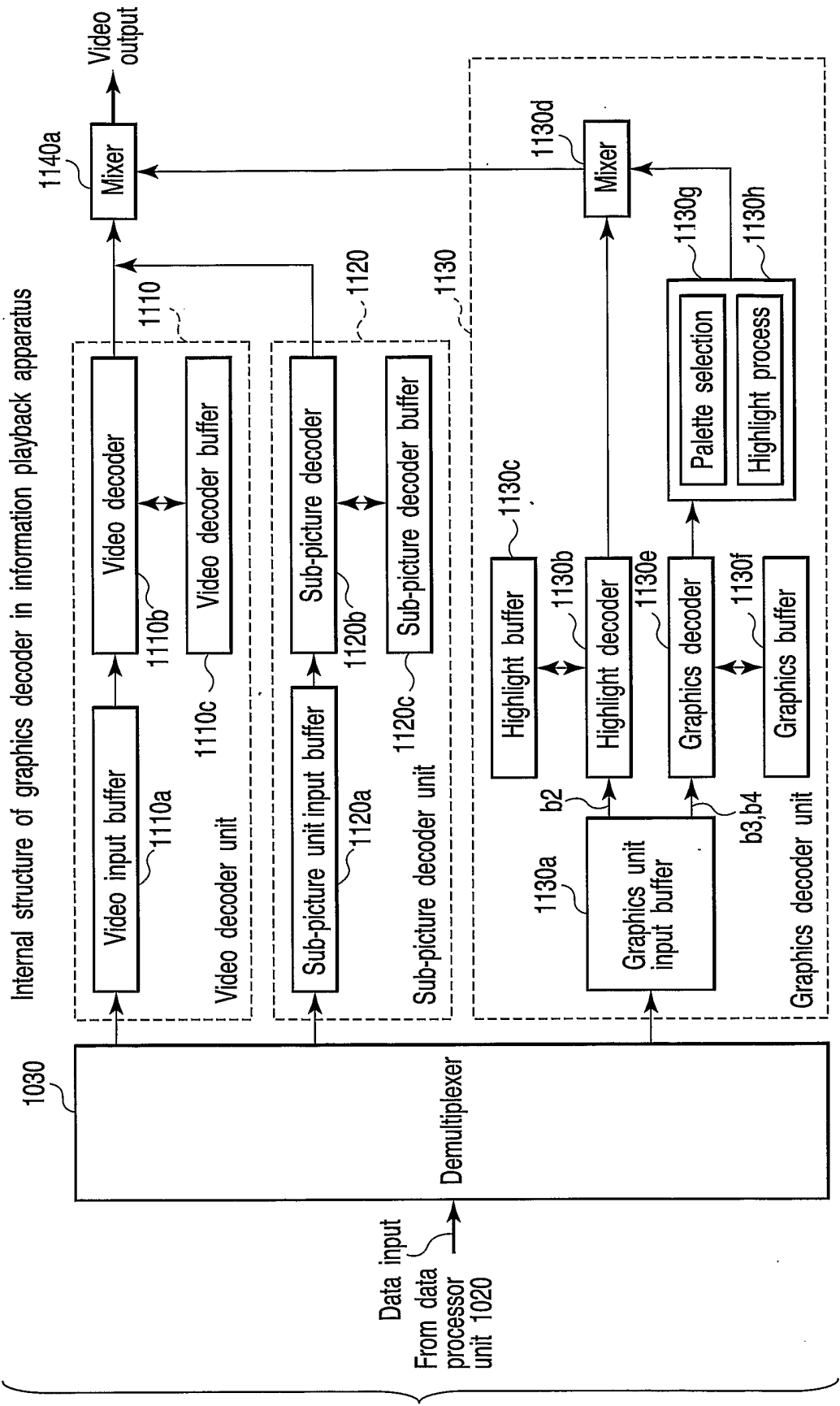


FIG. 39

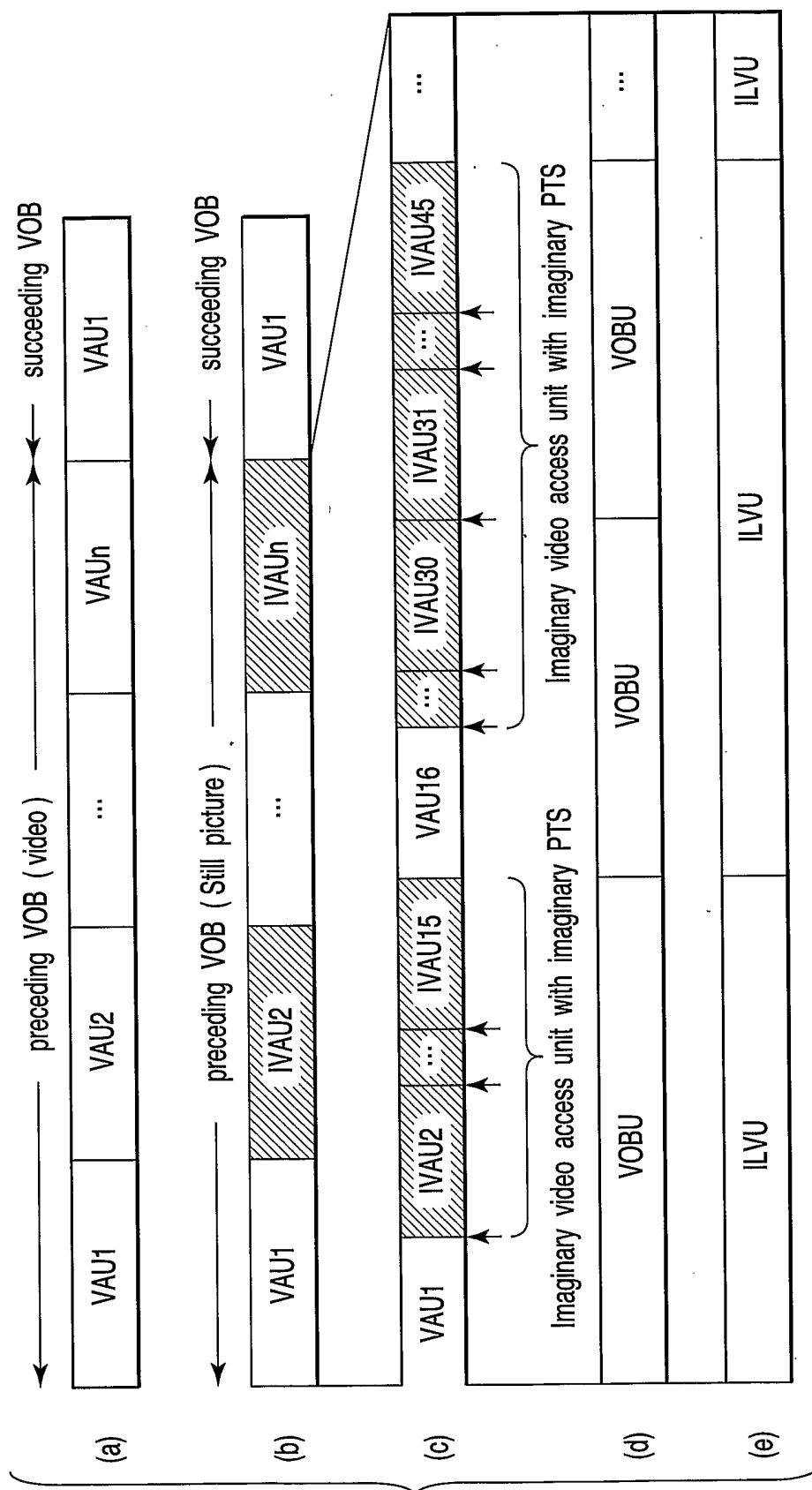


FIG. 40

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	SPRM	Meaning
(a)	0	Current Menu Language Code during Playback (CM_LCD) Changed / set by both user and command
(b)	1	Audio stream number (ASTN) for TT_DOM
(c)	2	Sub-picture stream number (SPTN) and On / Off flag for TT_DOM
(d)	3	Angle number (AGLN) for TT_DOM
(e)	4	Title number (TTN) for TT_DOM
(f)	5	VTS Title number (VTS_TTN) for TT_DOM
(g)	6	Title PGC number (PTTN) for TT_DOM
(h)	7	Part_of_Title number (PTTN) for One_Sequential_PGC_Title
(i)	8	Highlighted Button number (HL_BTN) for Selection state
(j)	9	Navigation Timer (NV_TMR)
(k)	10	TT_PGCN for NV_TMR
(l)	11	Player Audio Mixing Mode (P_AMXMD) for Karaoke
(m)	12	Country Code (CTY_CD) for Parental Management
(n)	13	Parental Level (PTL_LVL)
(o)	14	Player Configuration (P_CFG) for Video
(p)	15	P_CFG for Audio
(q)	16	Initial Language Code (INI_LCD) for AST
(r)	17	Initial Language Code extension (INI_LCD_EXT) for AST
(s)	18	INI_LCD for SPST
(t)	19	INI_LCD_EXT for SPST
(u)	20	Player Region Code
(v)	21	Initial Menu Language Code as Player Parameter (IM_LCD) Player setting language that can be changed / set by user only
(w)	22	reserved
(x)	23	reserved for extended playback mode

FIG. 41

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Command type	Command name	Command D-1	Command ID-2	Setting flag	Area type	Comparison I-flag	Compare Field	Branch Field	E-flag	Extended Field
1	Nop	000	0	/	0000	0	000	0000	0	0000000
1	GoTo	000	0	/	0000	0	000	GoTo Option	0	0000000
1	Compare GoTo	000	0	/	0000	0/1	Compare Option	GoTo Option	0	0000000
1	Link	001	0	/	0000	0	000	Link Option	0	0000000
1	Compare Link	001	0	/	0000	0/1	Compare Option	Link Option	0	0000000
1	Jump	001	1	/	0000	0	000	Jump Option	0	0000000
1	Compare Jump	001	1	/	0000	0	Compare Option	Jump Option	0	0000000
2	SetSystem	010	/	0/1	Set System	0	000	0000	0	0000000
2	Compare SetSystem	010	/	0/1	Set System	0	Compare Option	0000	0	0000000
2	SetSystem Link	010	/	0/1	Set System	0	000	Link Option	0	0000000
2	Set	011	/	0/1	Set	0	000	0000	0	0000000
2	Compare Set	011	/	0/1	Set	0/1	Compare Option	0000	0	0000000
2	Set Link	011	/	0/1	Set	0	000	Link Option	0	0000000
2	Set GoTo	011	/	0/1	Set	0	000	GoTo Option	1	Extra Set Option
2	Set Jump	011	/	0/1	Set	0	000	Jump Option	1	Extra Set Option
3	Set Compare LinkSIns	100	/	0/1	Set	0/1	Compare Option	0000	0	0000000
3	Compare&Set-LinkSIns	101	/	0/1	Set	0/1	Compare Option	0000	0	0000000
3	Compare&Set&LinkSIns	110	/	0/1	Set	0/1	Compare Option	0000	0	0000000
4	Call INTENG	111	—	—	—	—	—	0000	0	0000000

FIG. 42

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(a) Compare Field (Compare Option)

Value	Compare Condition	Meaning
001b	BC	Bitwise Compare
010b	EQ	Equal
011b	NE	Not Equal
100b	GE	Greater than or equal to
101b	GT	Greater than
110b	LE	Less than or equal to
111b	LT	Less than

(c) Branch Field (Link Option)

Value	Instruction
0001b	LinkSIns
0100b	LinkPGCN
0101b	LinkPTTN
0110b	LinkPGN
0111b	LinkCN
1100b	LinkPGCNR(*1)
1101b	LinkPTTNR(*1)
1110b	LinkPGNR(*1)
1111b	LinkCNR(*1)

(e) SetSystem Field
(SetSystem Option)

Value	Instruction
0001b	SetSTN
0010b	SetNVTMR
0011b	SetGPRMMD
0100b	SetAMXMD
0110b	SetHL_BTNN
1000b	SetM_LCD(*3)
1001b	SetRSMI(*4)

(b) Branch Field (GoTo Option)

Value	Instruction
001b	GoTo
010b	Break
011b	SetTmpPML

(d) Branch Field (Jump Option)

Value	Instruction
0001b	Exit
0010b	JumpTT
0011b	JumpVTS_TT
0101b	JumpVTS_PTT
0110b	JumpSS
1000b	CallSS
1001b	ReturnSS(*2)
1010b	ChangeTT(*2)
1011b	ChangeVTS_TT(*2)
1101b	ChangeVTS_PTT(*2)
1110b	Call INTENG

(f) Set Field (Set Option)

Value	Instruction
0001b	Mov
0010b	Swp
0011b	Add
0100b	Sub
0101b	Mul
0110b	Div
0111b	Mod
1000b	Rnd
1001b	And
1010b	Or
1011b	Xor
1100b	Exp(*5)

FIG. 43

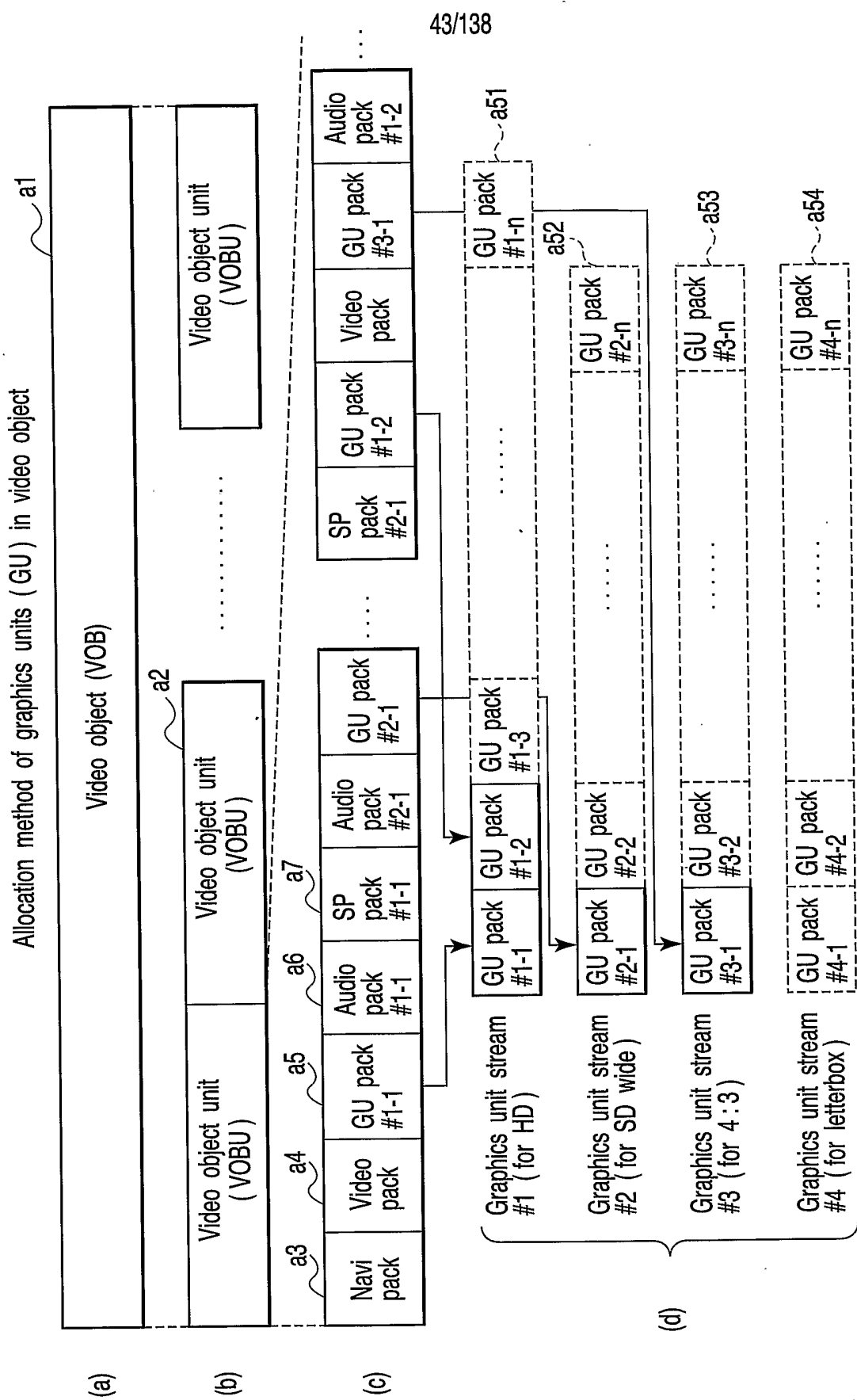


FIG. 44

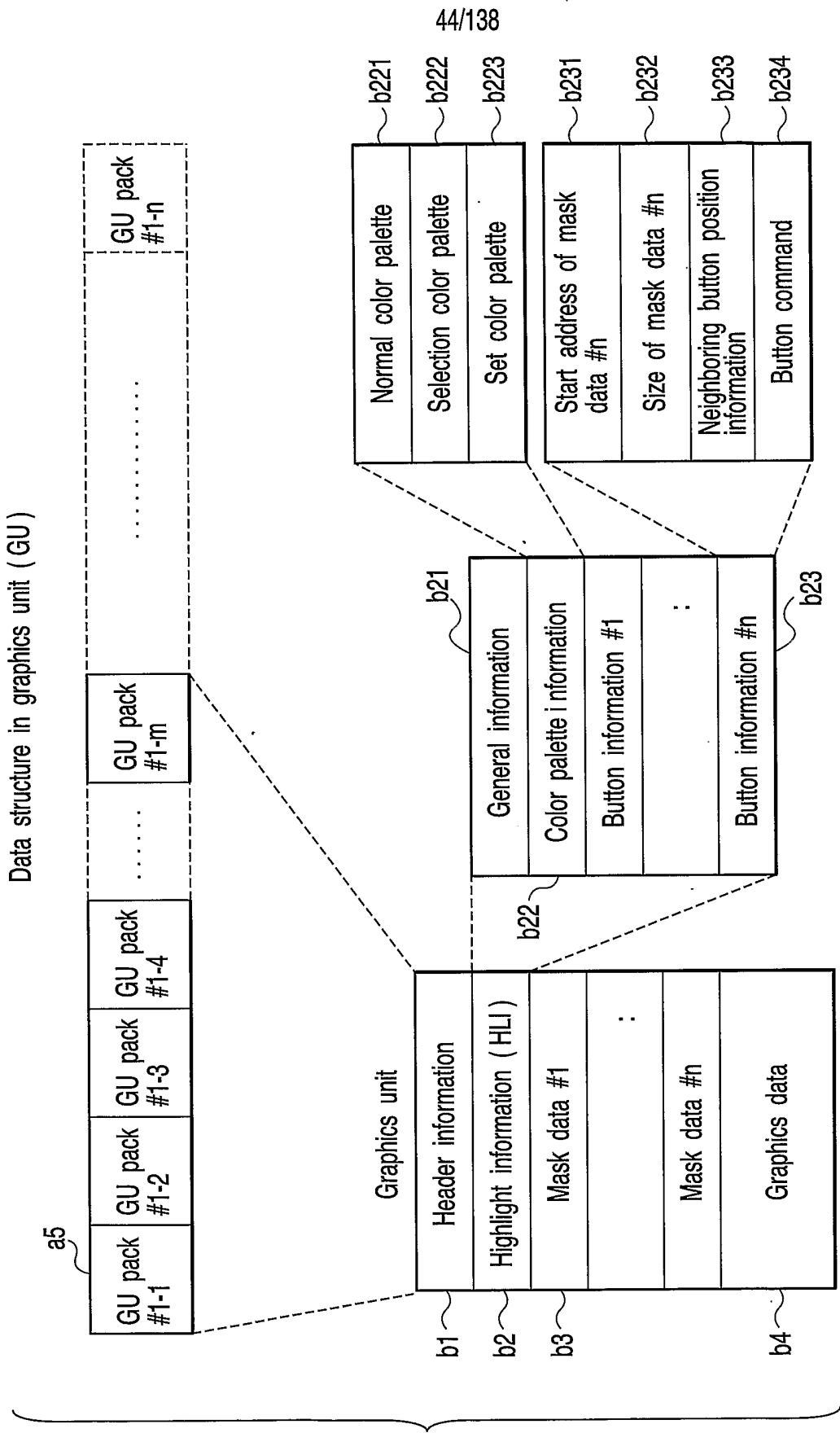


FIG. 45

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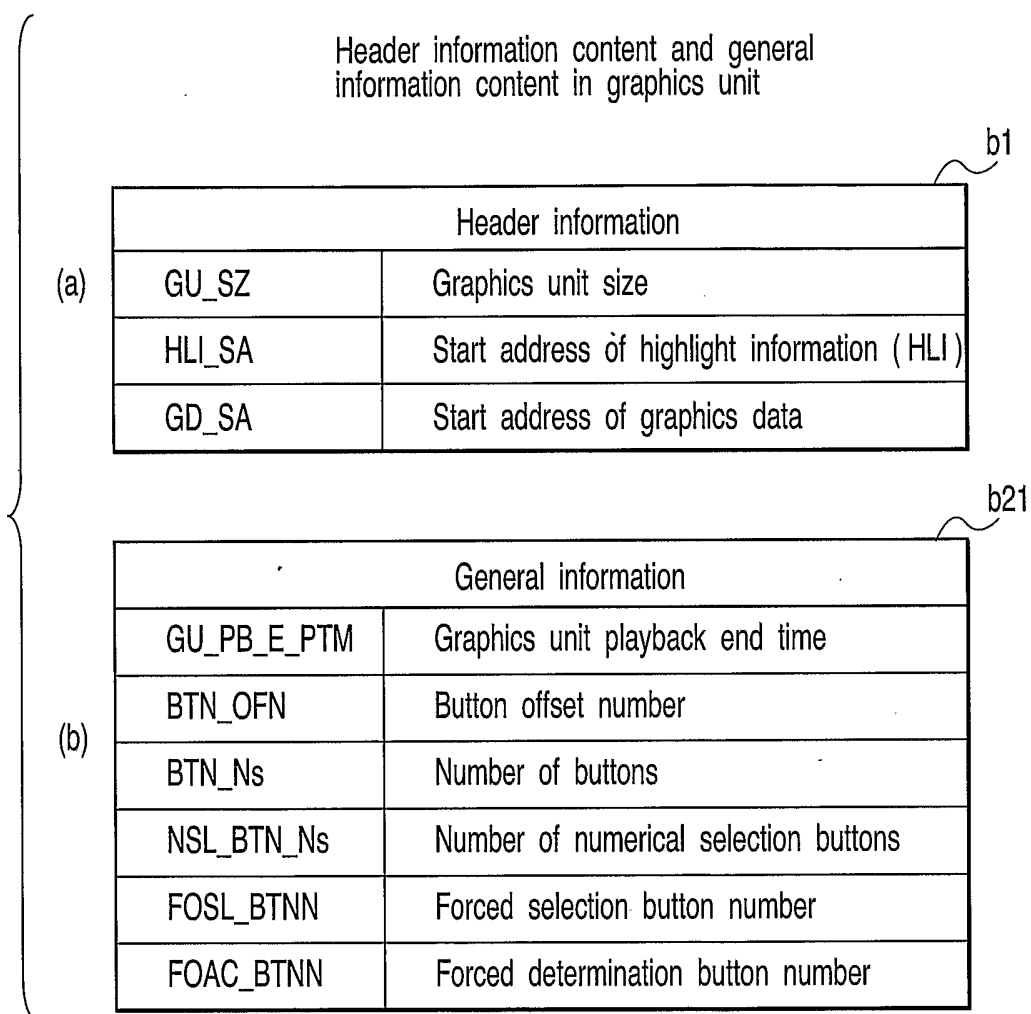


FIG. 46

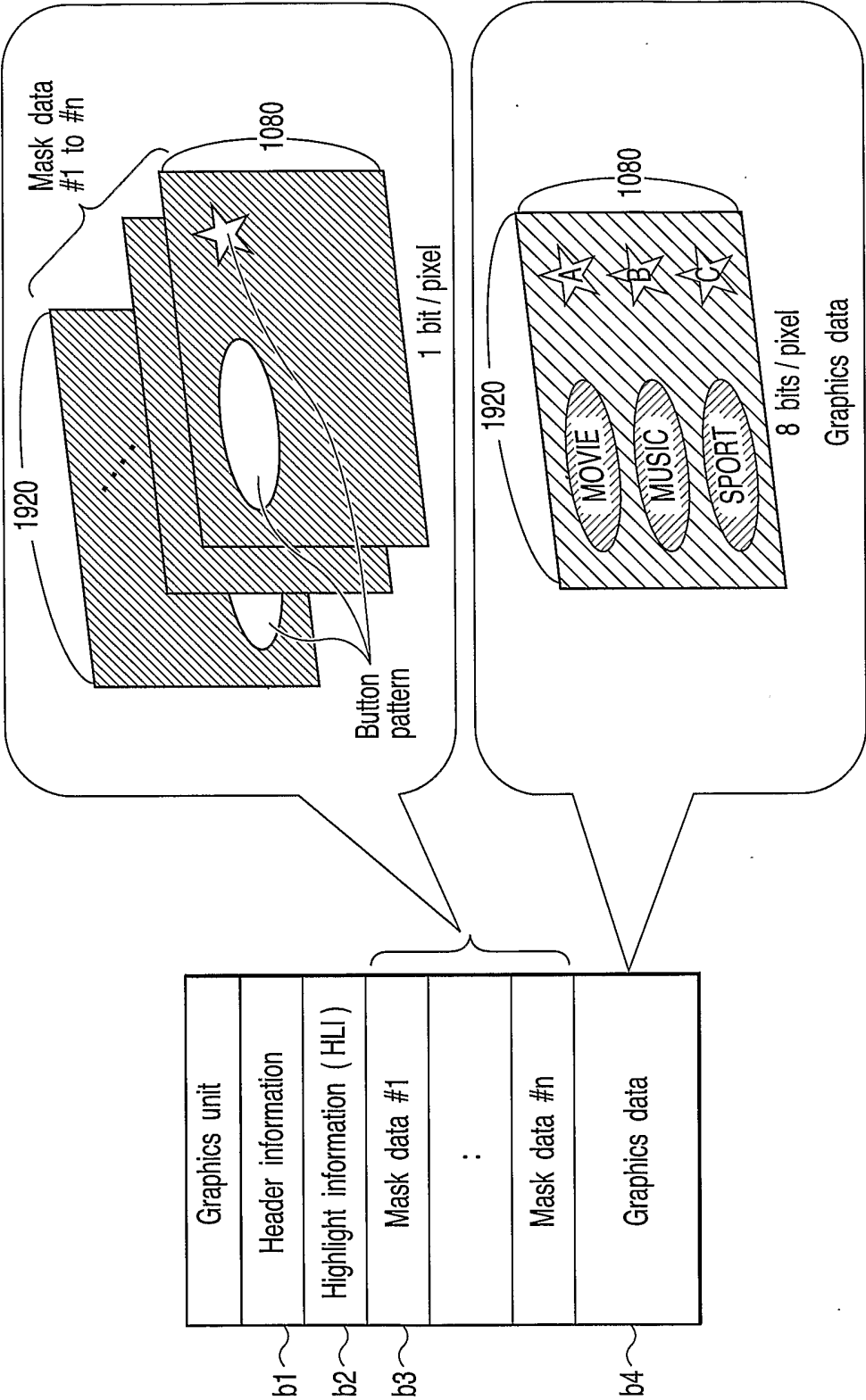
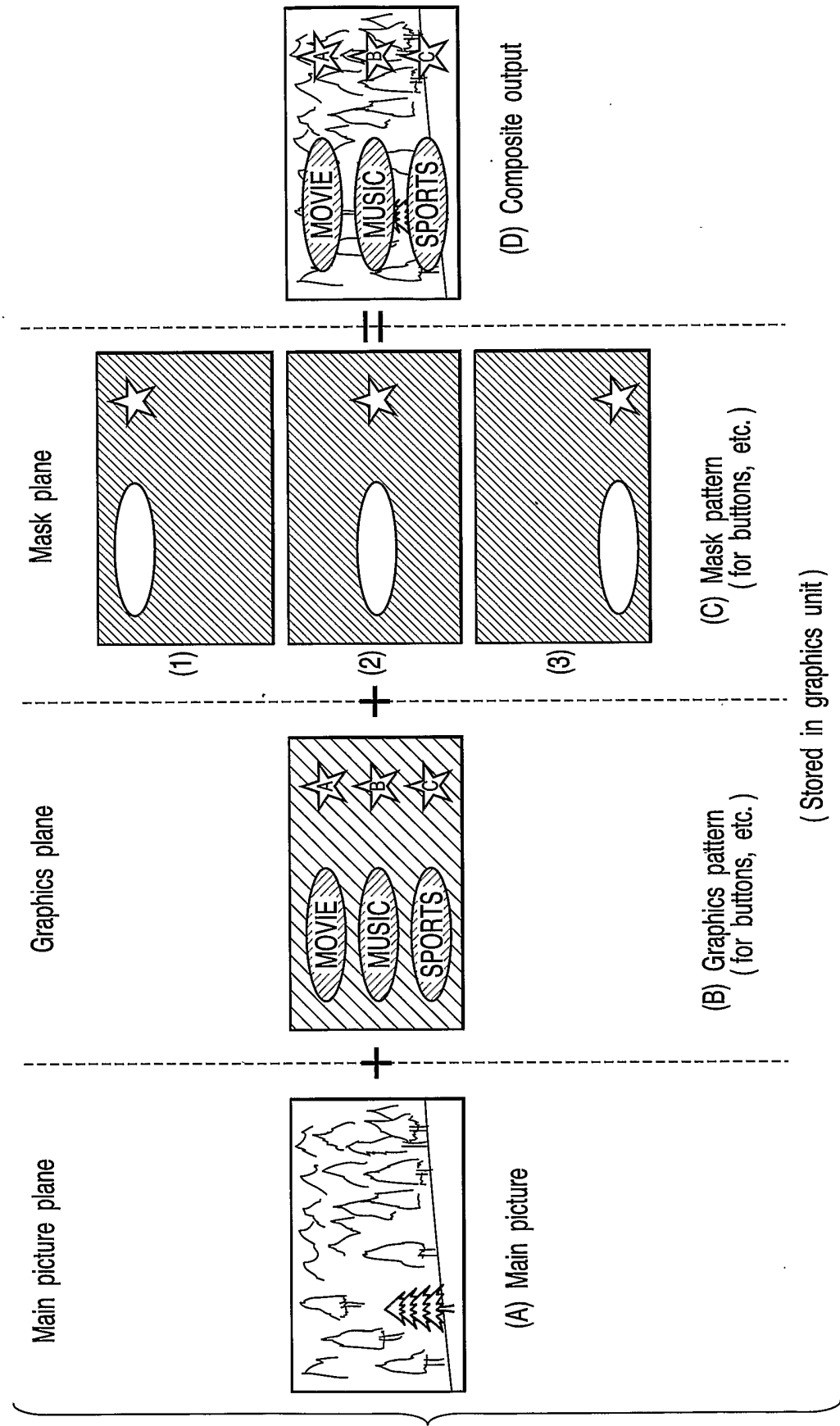


FIG. 47

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(Stored in graphics unit)

FIG. 48

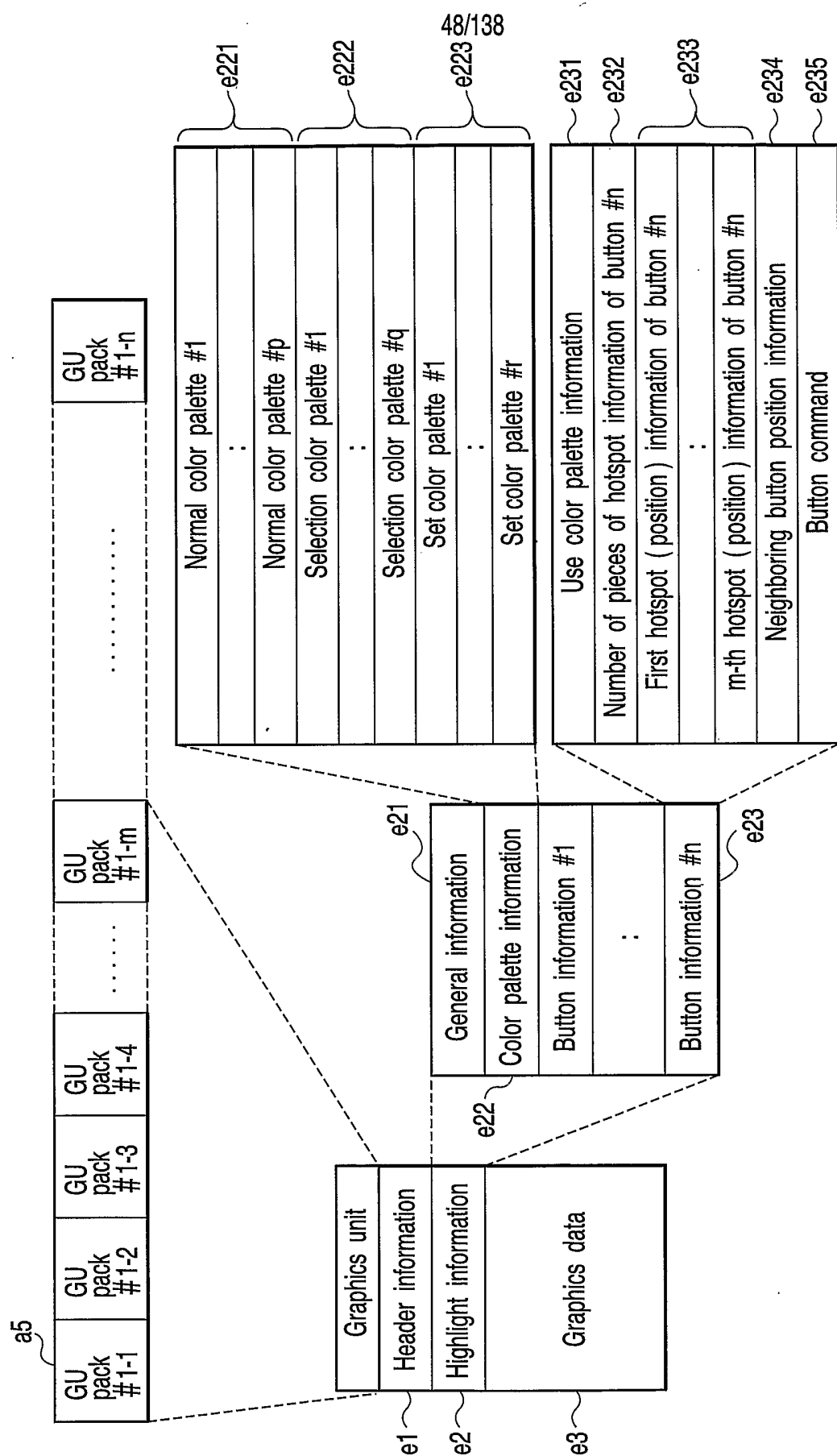


FIG. 49

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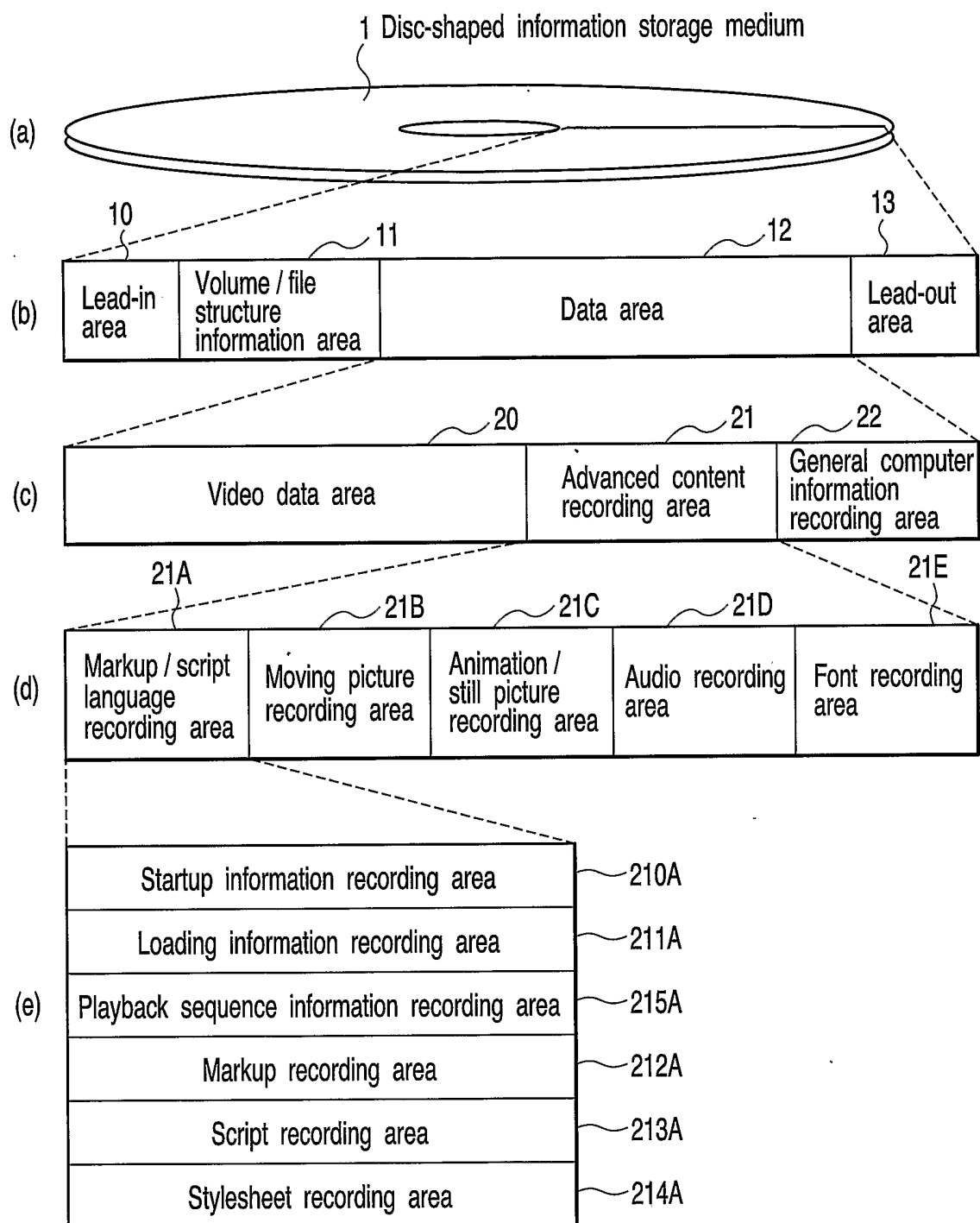


FIG. 50

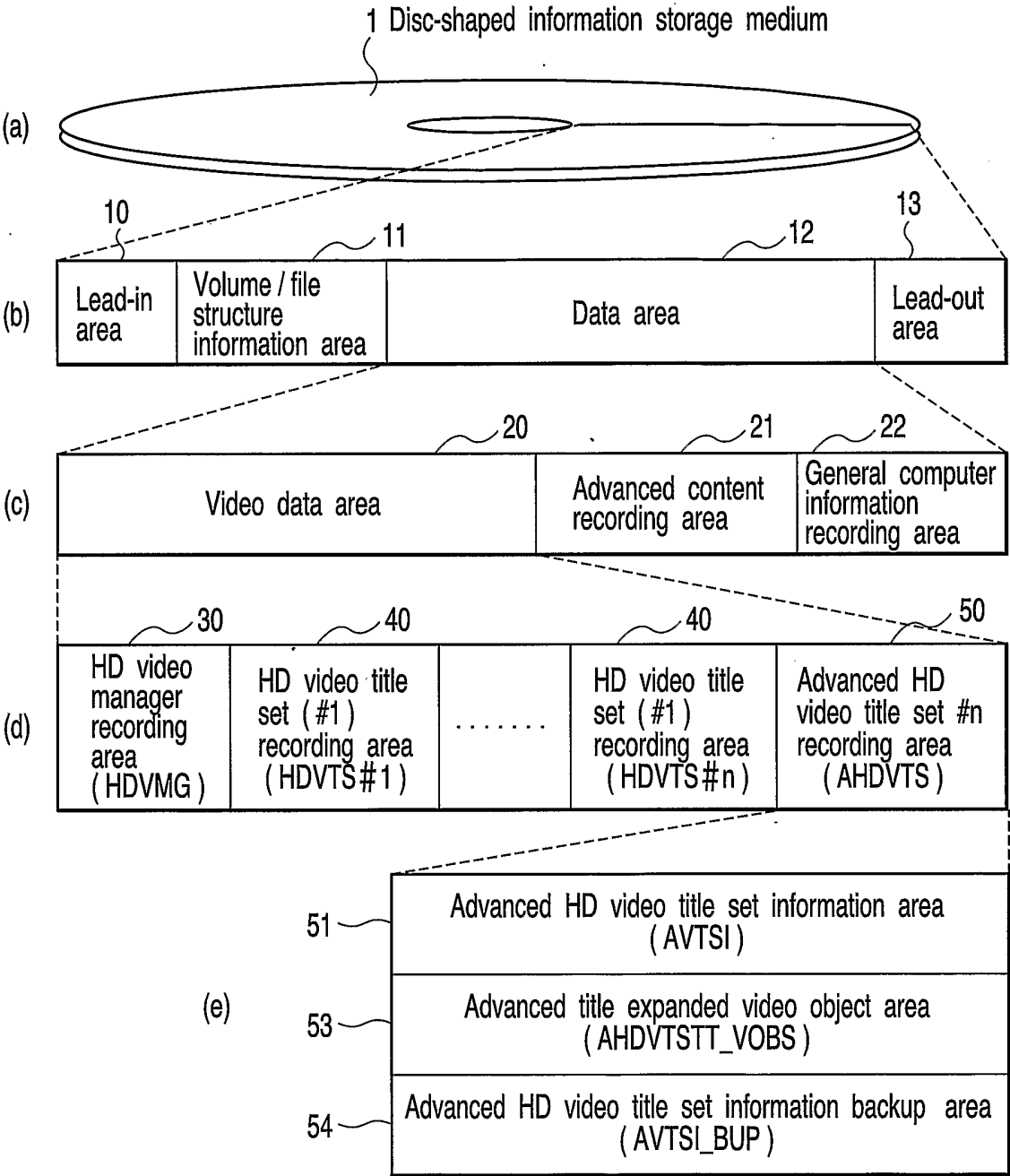


FIG. 51

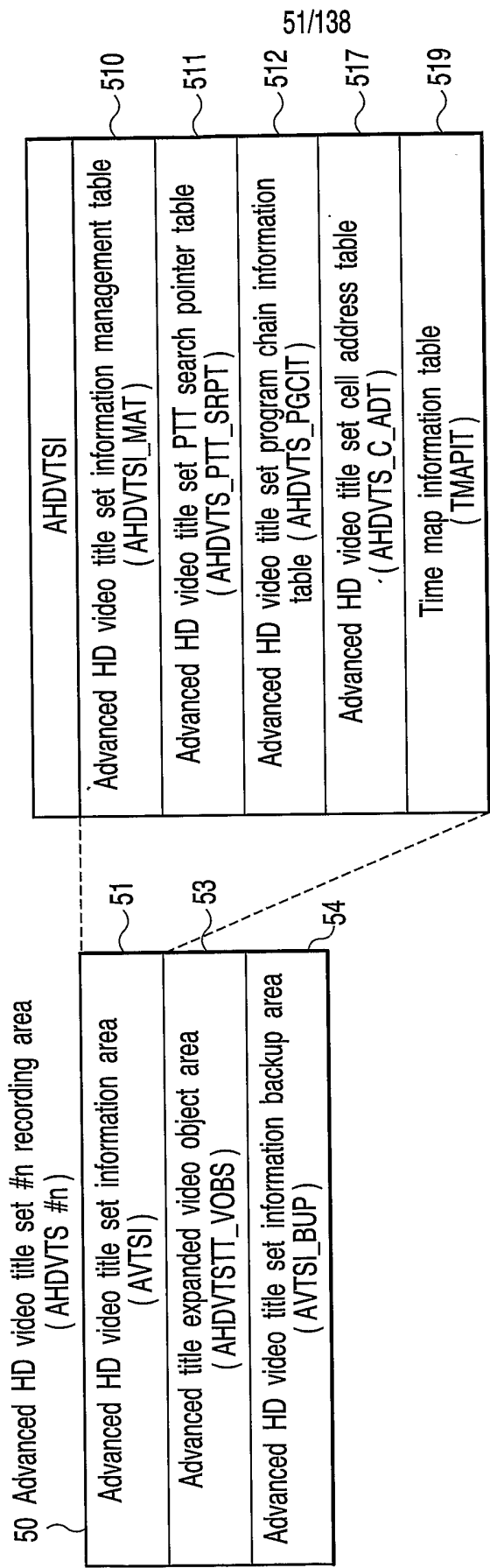


FIG. 52

AHDVTSI_MAT	
AHDVTS_ID	Advanced HD video title set identifier
AHDVTS_EA	End address of AHDVTS
AHDVTSI_EA	End address of AHDVTSI
VERN	Version number of HD-DVD-Video standard
AHDVTS_CAT	AHDVTS category
AHDVTSI_MAT_EA	End address of AHDVTSI_MAT
AHDVTSTT_VOBS_SA	Start address of AHDVTSTT_VOBS
AHDVTS_PTT_SRPT_SA	Start address of AHDVTS_PTT_SRPT
AHDVTS_PGCIT_SA	Start address of AHDVTS_PGCIT
AHDVTS_C_ADT_SA	Start address of AHDVTS_C_ADT
ATR1_AGL_Ns	Number of angles of ATR1
ATR1_V_ATR	Video attribute of ATR1
ATR1_AST_Ns	Number of audio streams of ATR1
ATR1_AST_ATRT	Audio stream attribute table of ATR1
ATR1_SPST_Ns	Number of sub-picture streams of ATR1
ATR1_SPST_ATRT	Sub-picture stream attribute table of ATR1
ATR1_MU_AST_ATRT	Multi-channel audio stream attribute table of ATR1
ATR2_AGL_Ns	Number of audio streams of ATR2
ATR2_V_ATR	Video attribute of ATR2
...	...

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Advanced HD video title set information management table
(AHDVTSI_MAT)

AHDVTSI_MAT	
Application Type	0000b : not specified 0001b : reserved 0010b : advanced VTS with advanced content 0011b : advanced VTS without advanced content Others : reserved

FIG. 53

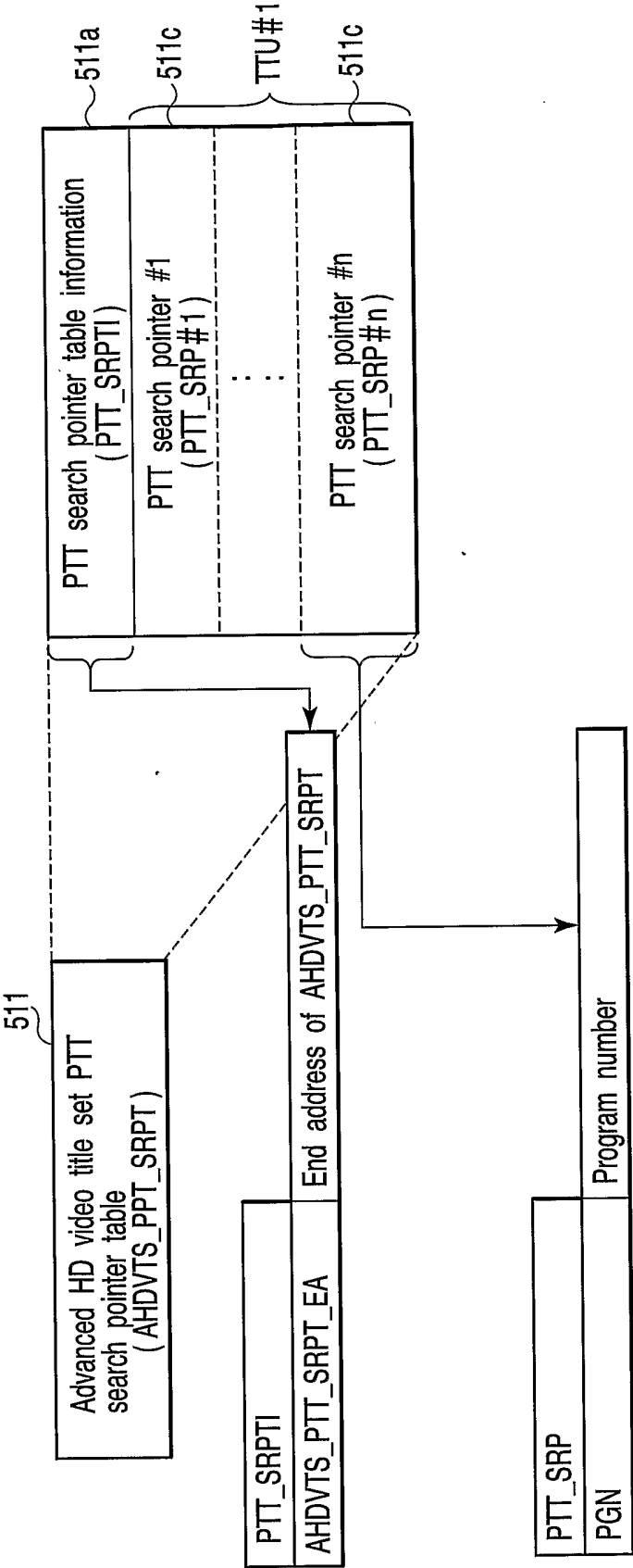


FIG. 54

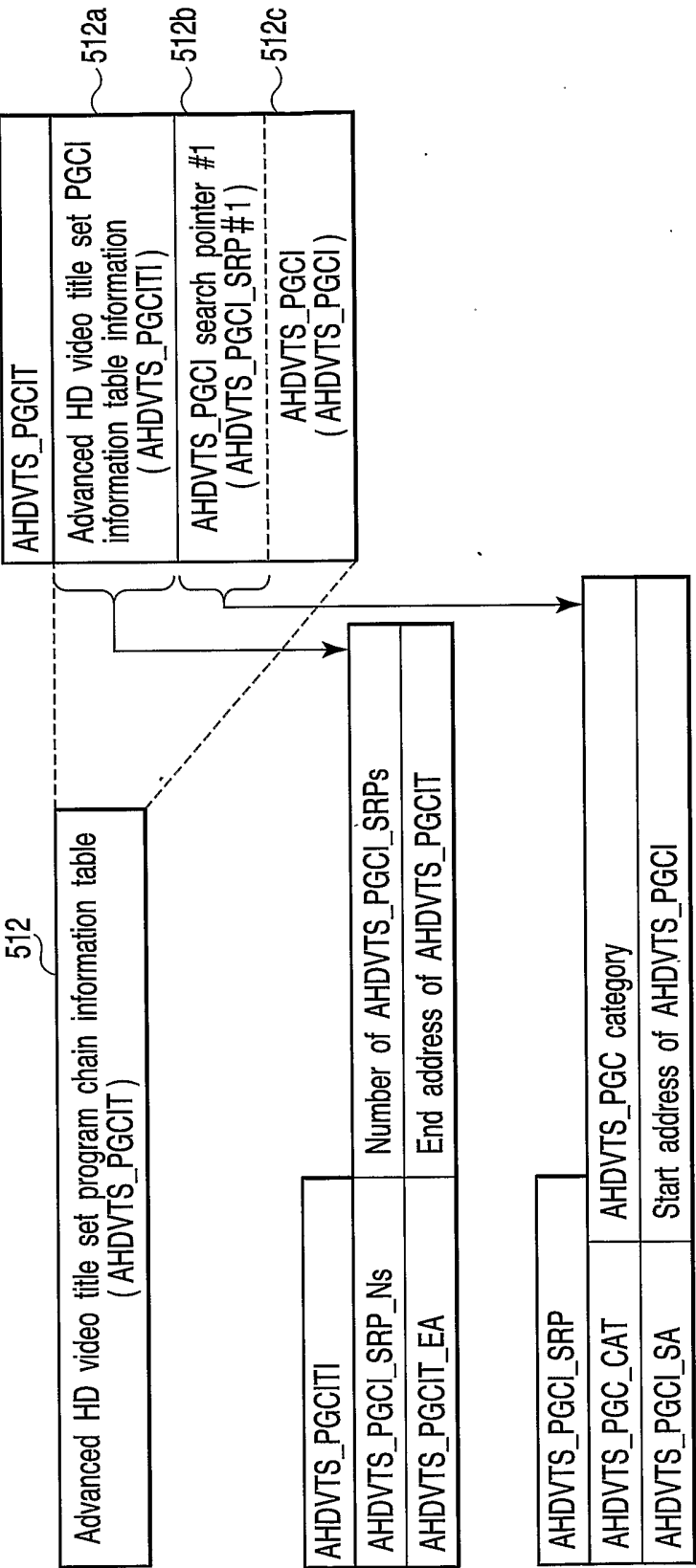


FIG. 55

PGCI		PGC_GI	
Program chain general information (PGC_GI)	Program chain general information (PGC_GI)	PGC_CNT	PGCI content
	Program chain program map (PGC_PGMAP)	PGC_PB_TM	PGC playback time
	Cell playback information table (C_PBIT)	PGC_AST_CTLT	PGC audio stream control table
	Cell position information table (C_POSIT)	PGC_SPST_CTLT	PGC sub-picture stream control table
		PGC_NV_CTL	PGC navigation control
		PGC_SP_PLT	PGC sub-picture palette
		PGC_PGMAP_SA	Start address of PGC_PGMAP
		C_PBIT_SA	Start address of C_PBIT
		C_POSIT_SA	Start address of C_POSIT
		RSM&AOB_CAT	RSM & AOB category

50
52
53
54

FIG. 56

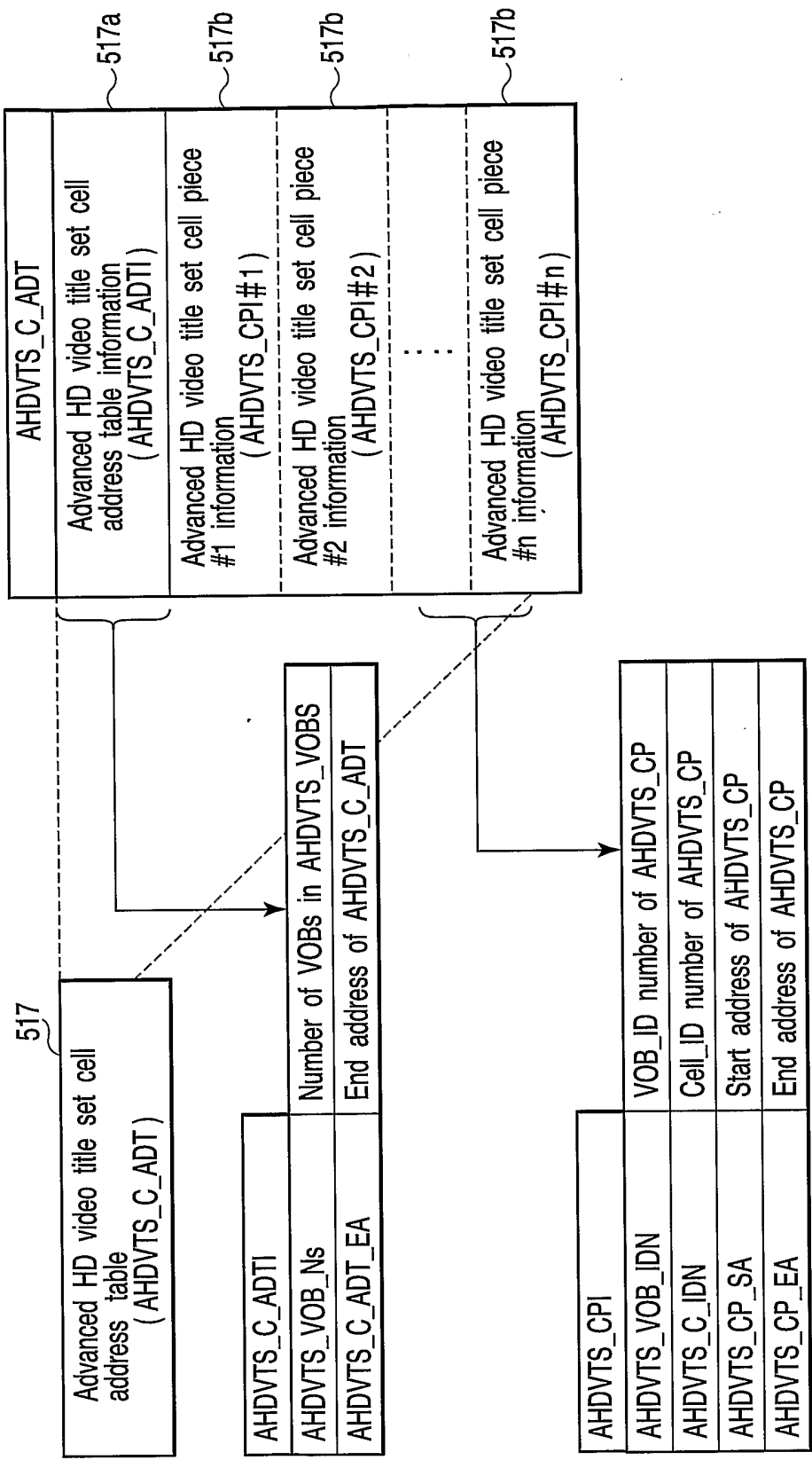


FIG. 57

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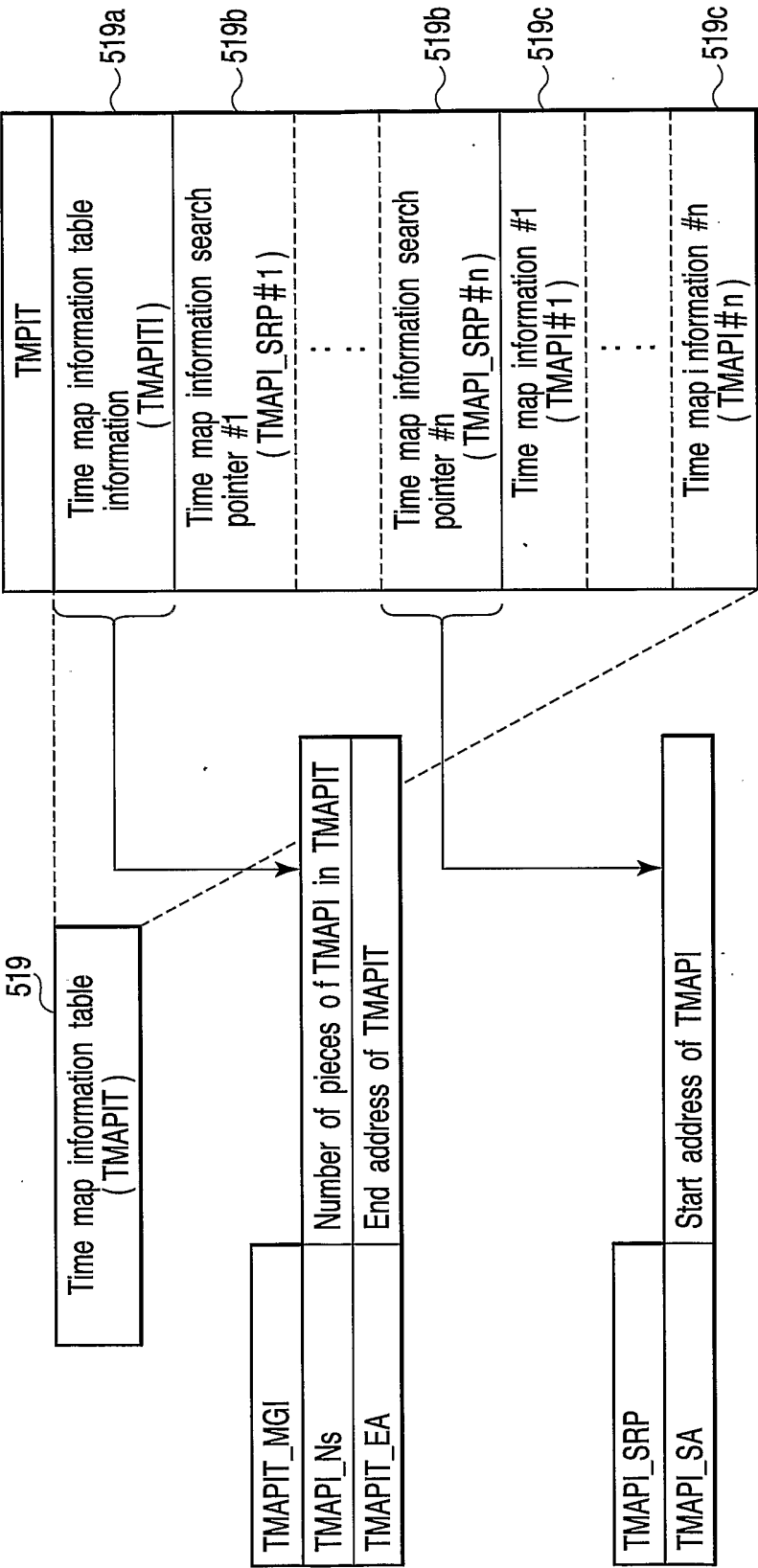


FIG. 58

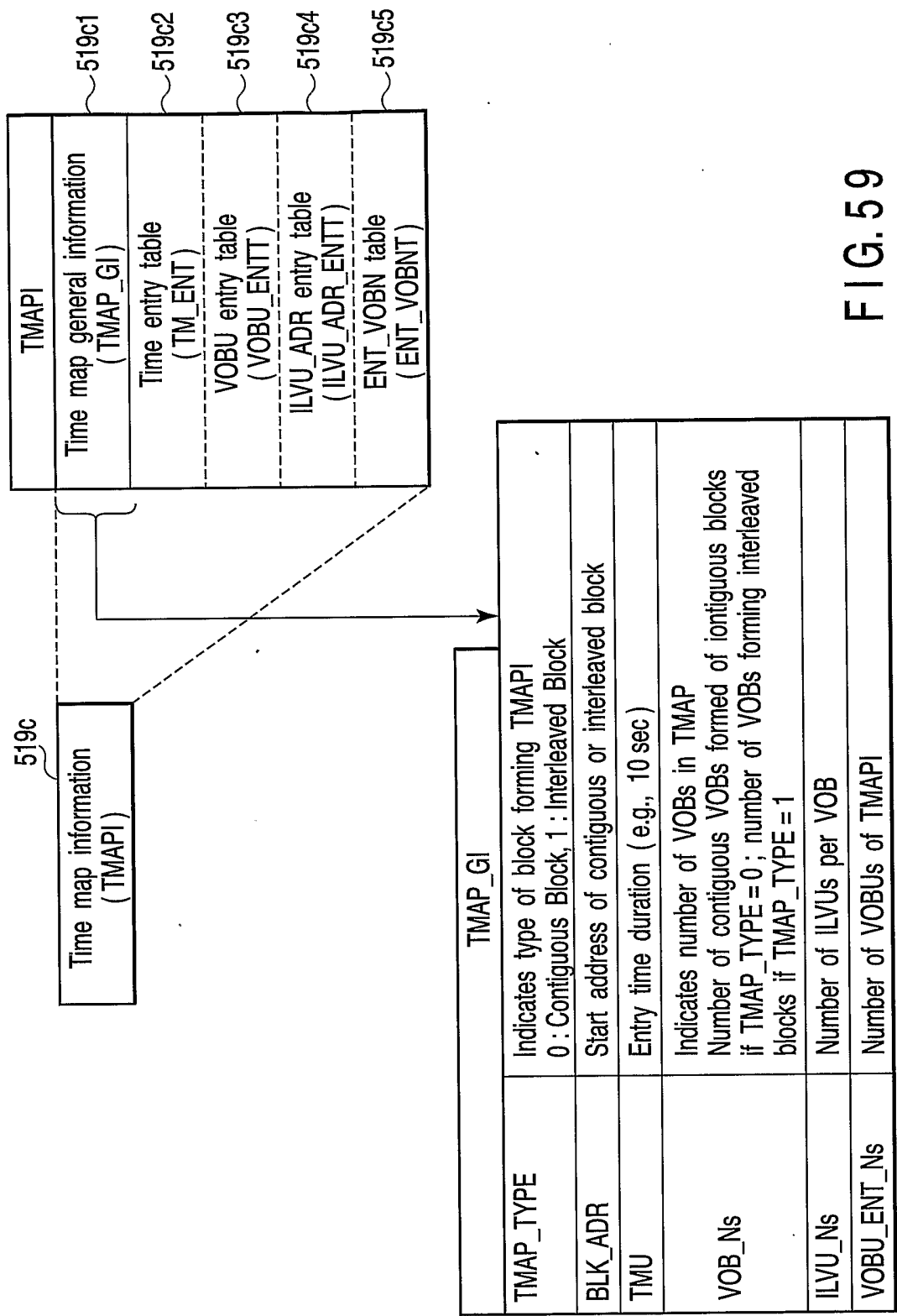


FIG. 59

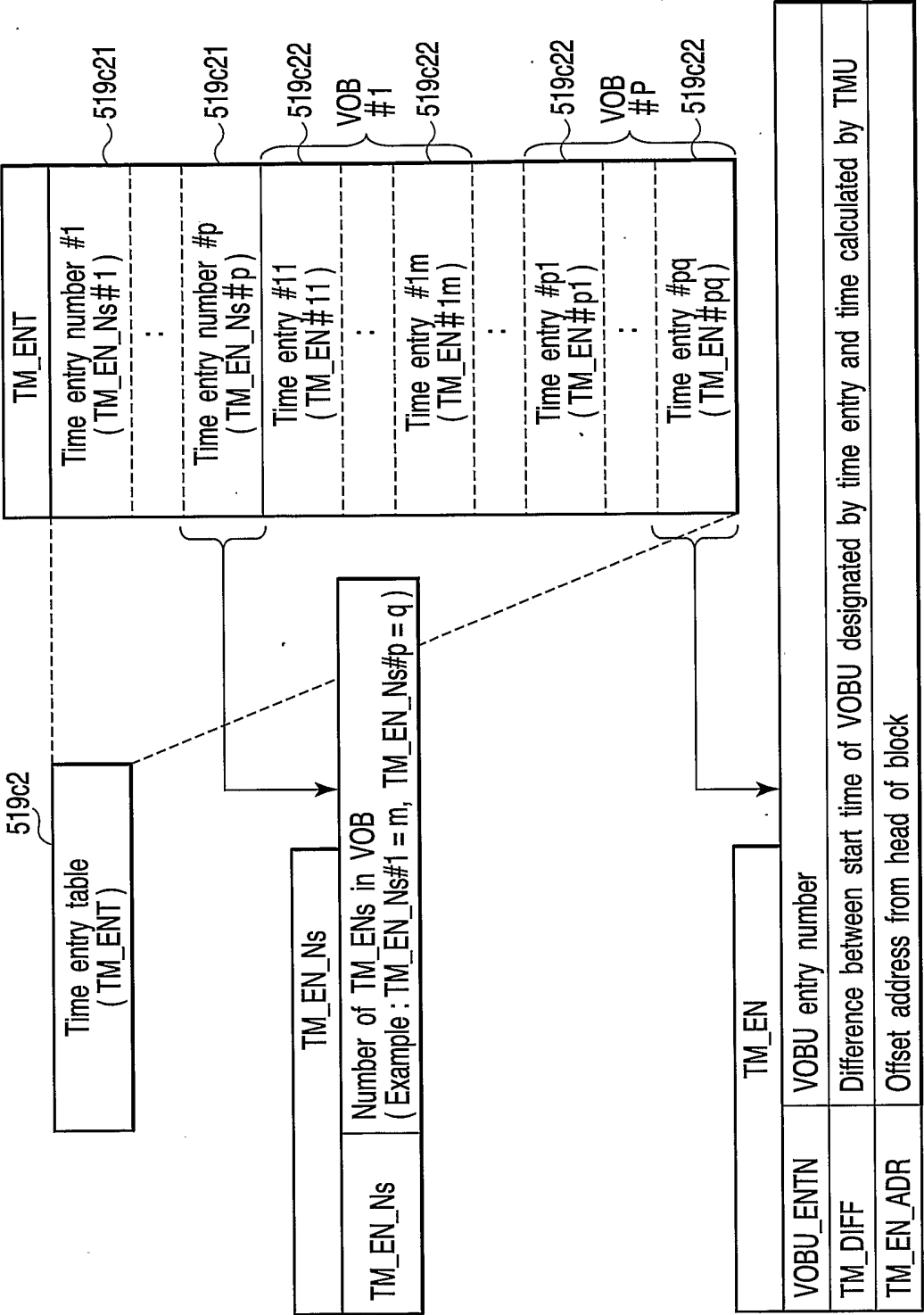


FIG. 60

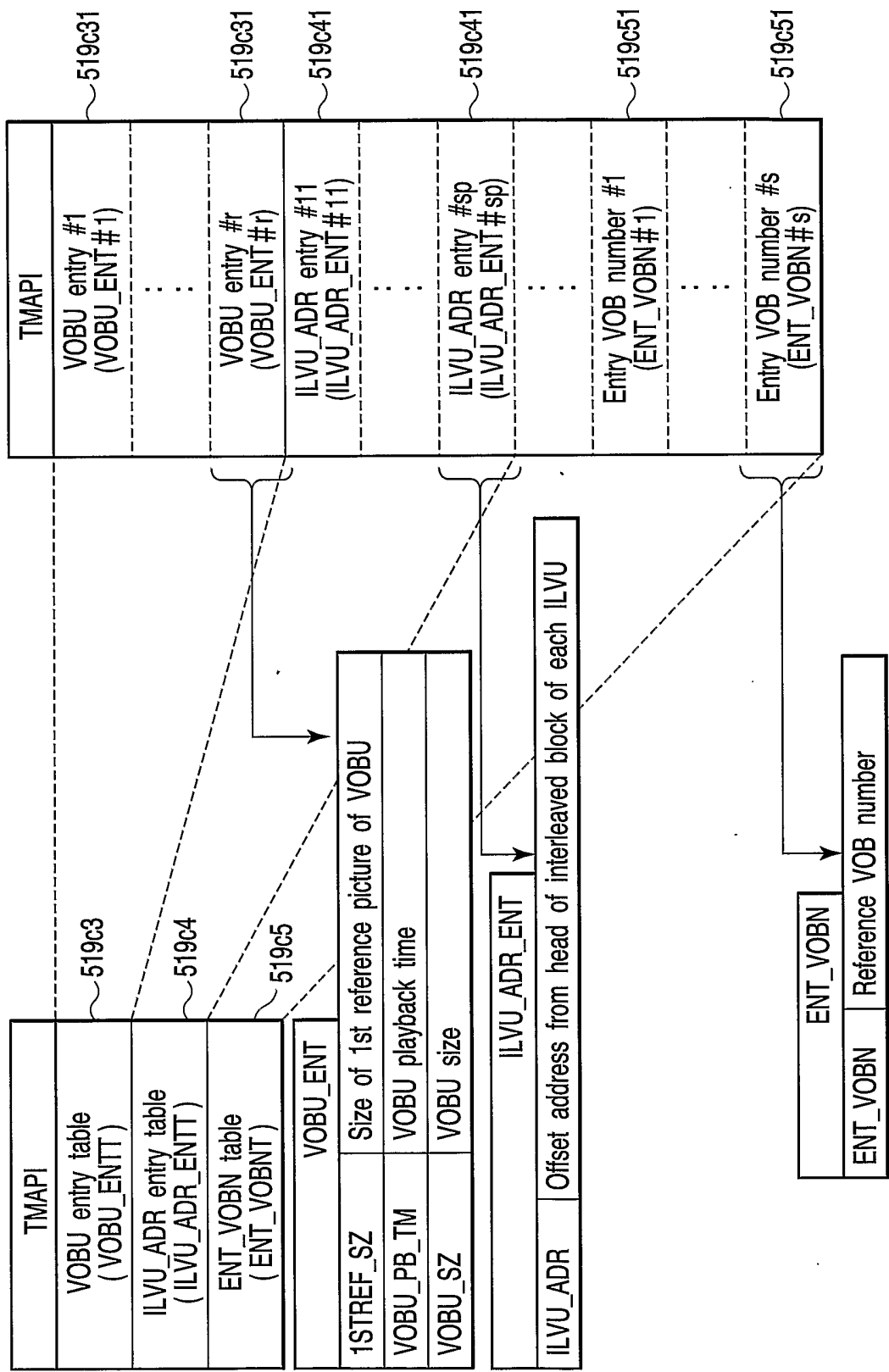


FIG. 61

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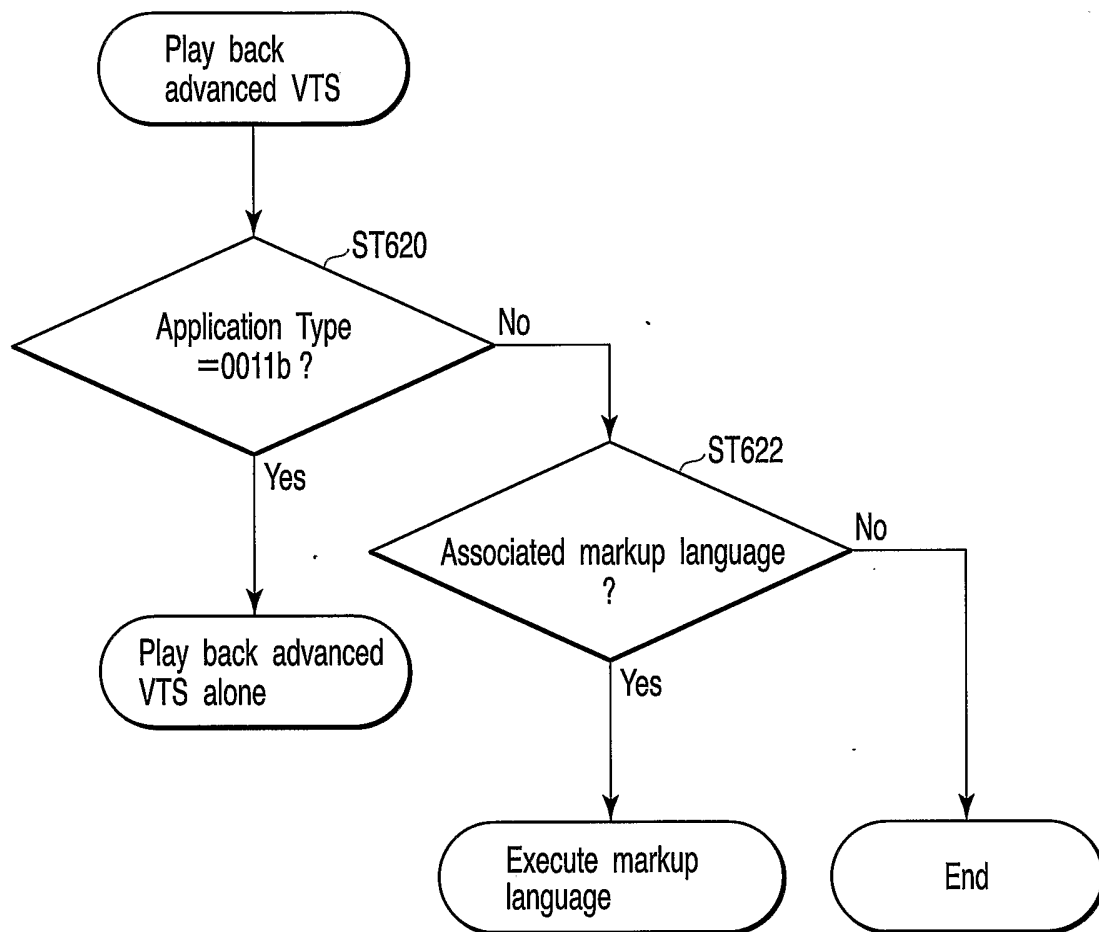


FIG. 62

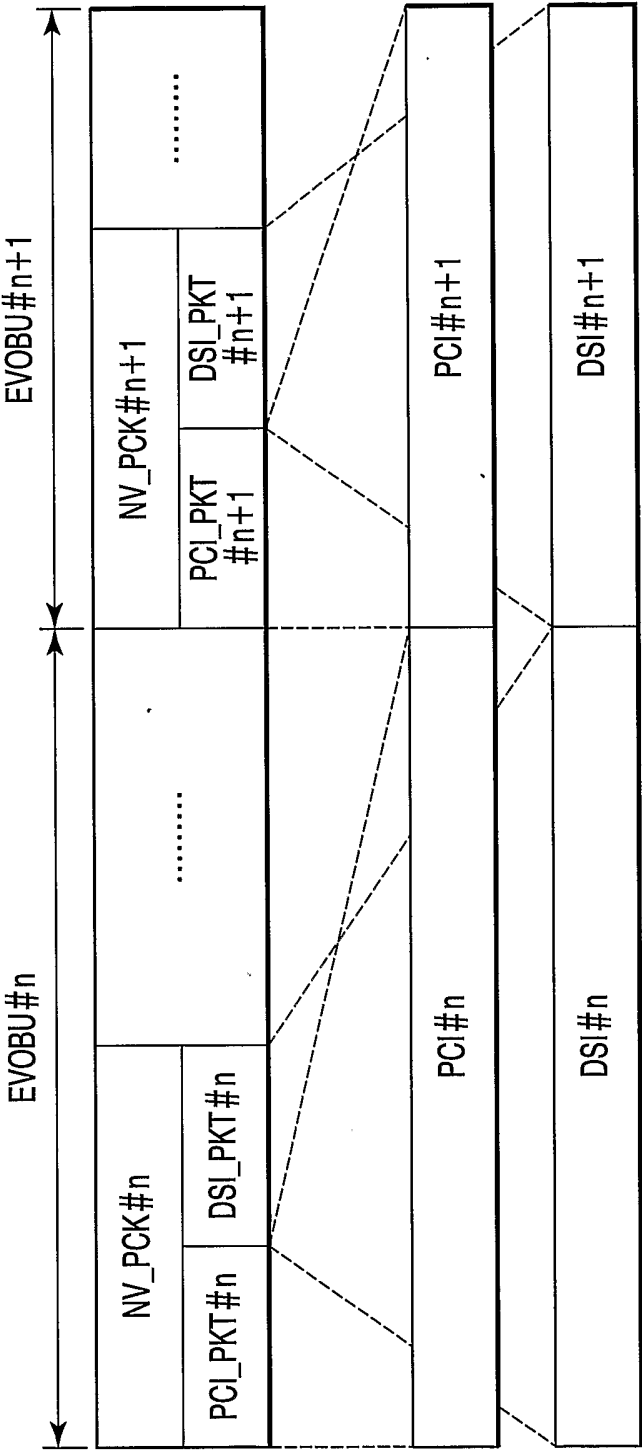


FIG. 63

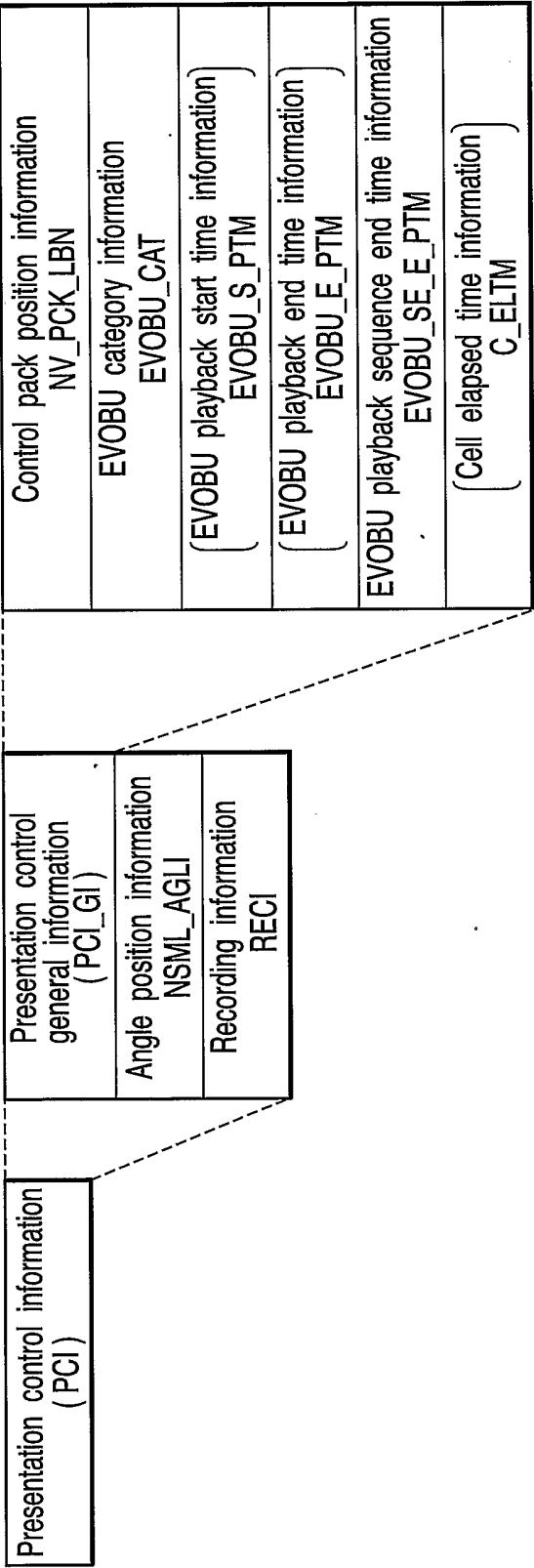


FIG. 64

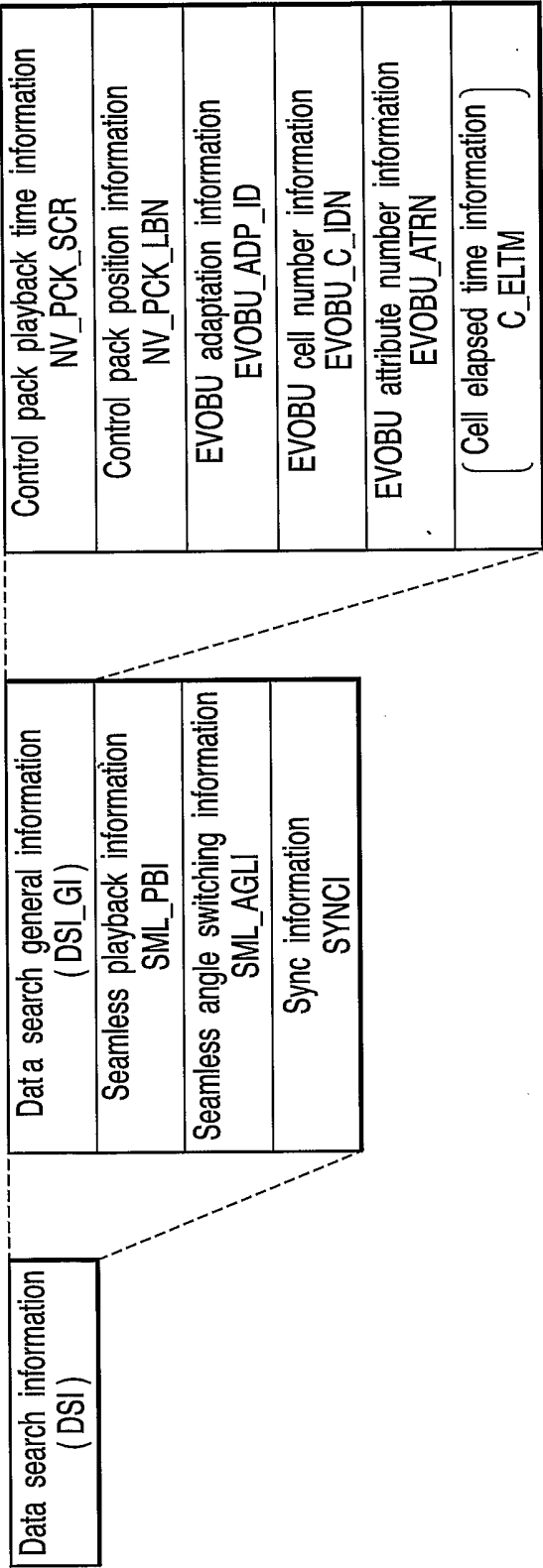


FIG. 65

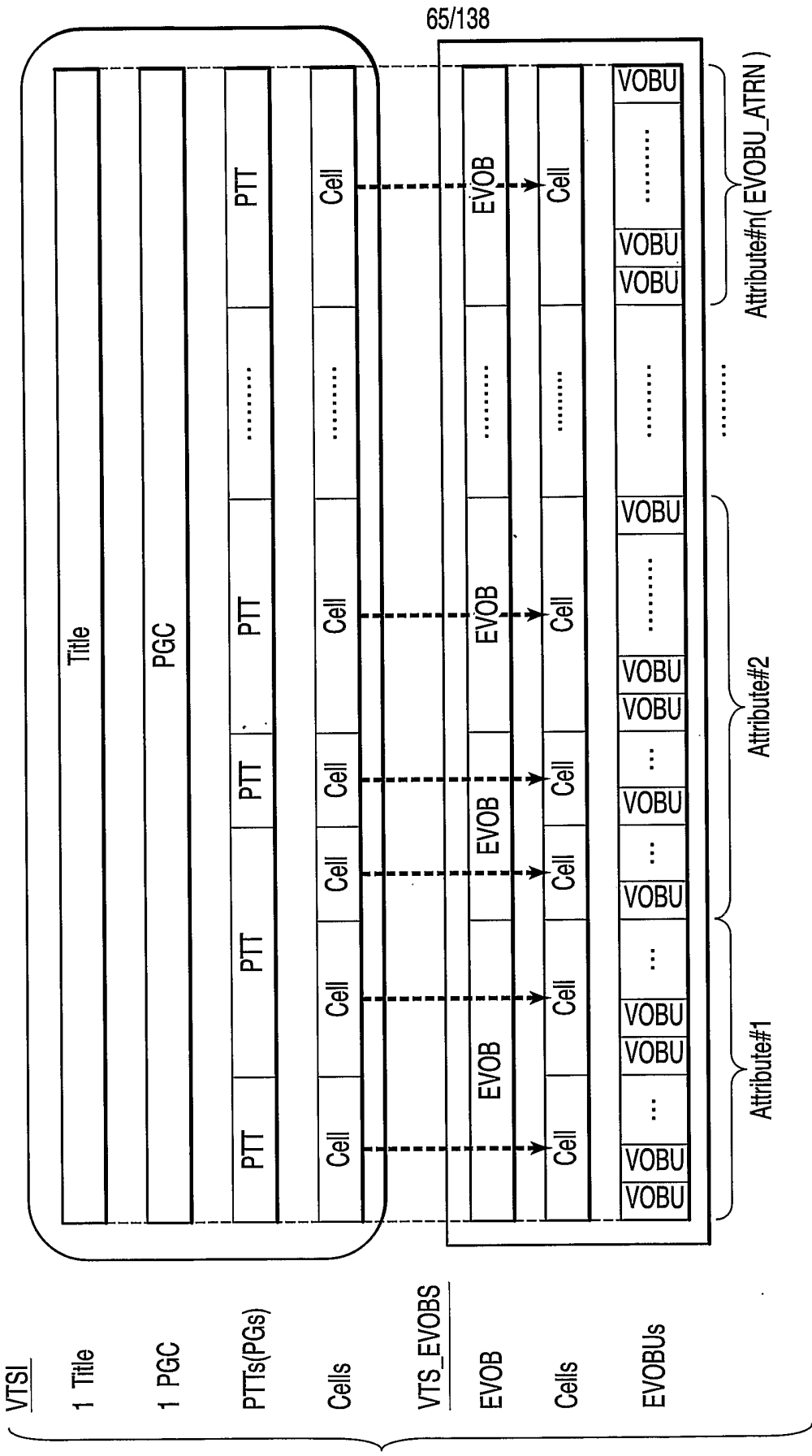


FIG. 66

TMAP element	
• Time — Starting point of description (time map unit) • PGC → head of PGC (existing standard) • VOB → head of VOB (VR standard) — Time interval • fixed → fixed at 600 video fields (10 sec) (VR standard) • variable → TMU can be set (range from 1 to 255 sec in increments of 1 sec) (existing standard) — Description upon forming ILVU • described in only path of first ILVU → describe only angle number 1 • described in all ILVUs	
• Address offset — Start address of each VOB • RLBN from first LB of VTSTT_VOBS • F_RLBN from first LB of file (current file is divided into a plurality of files)	
• VOB number — VOB entry • acquisition information, time information of I-picture	

FIG. 67

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Practical element of TMAP	
• BLK_ADR (Block Address)	— Designate start address of contiguous block or interleaved block using offset address from head of VTSTT_VOBS
• TM_EN_ADR (Time Entry Address)	— In contiguous block (single VOB) • Designate using offset address from head of block — Interleaved block (a plurality of VOBs) • Designate using offset address from head of block • Describe time entry tables as many as number of VOBs
• TMU (Time Unit)	— Fixed in VTSTT_VOBS (e.g., 10 sec).
• ILVU_ADR (Interleaved Unit Address)	— Designate address of each ILVU using offset address from head of interleaved block
• VOBU_SZ (VOBU size)	— Describe each VOBU size using number of packs in VOBU
• 1STREF_SZ (1 st Reference picture size)	— Describe I-picture size in each VOBU using number of packs

FIG. 68

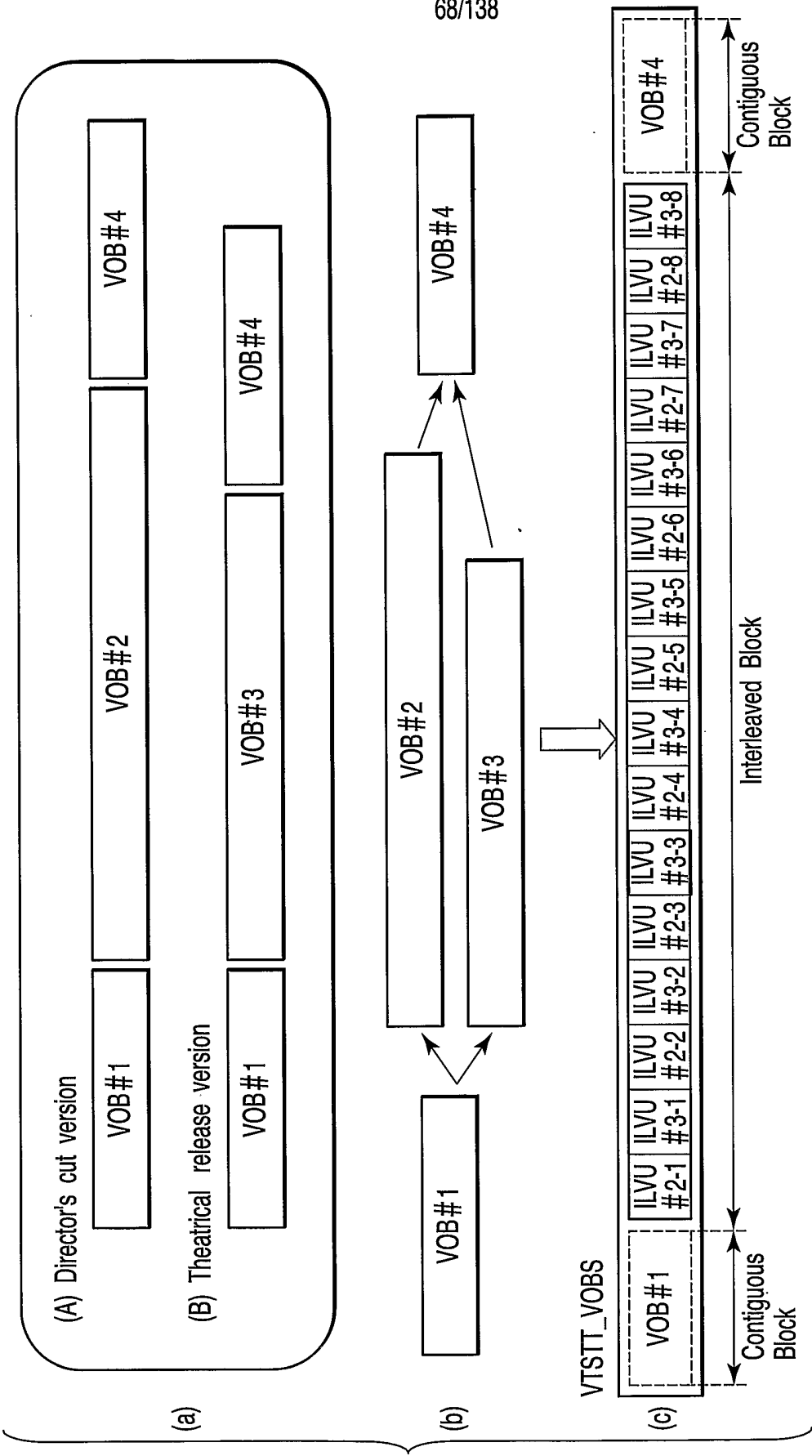
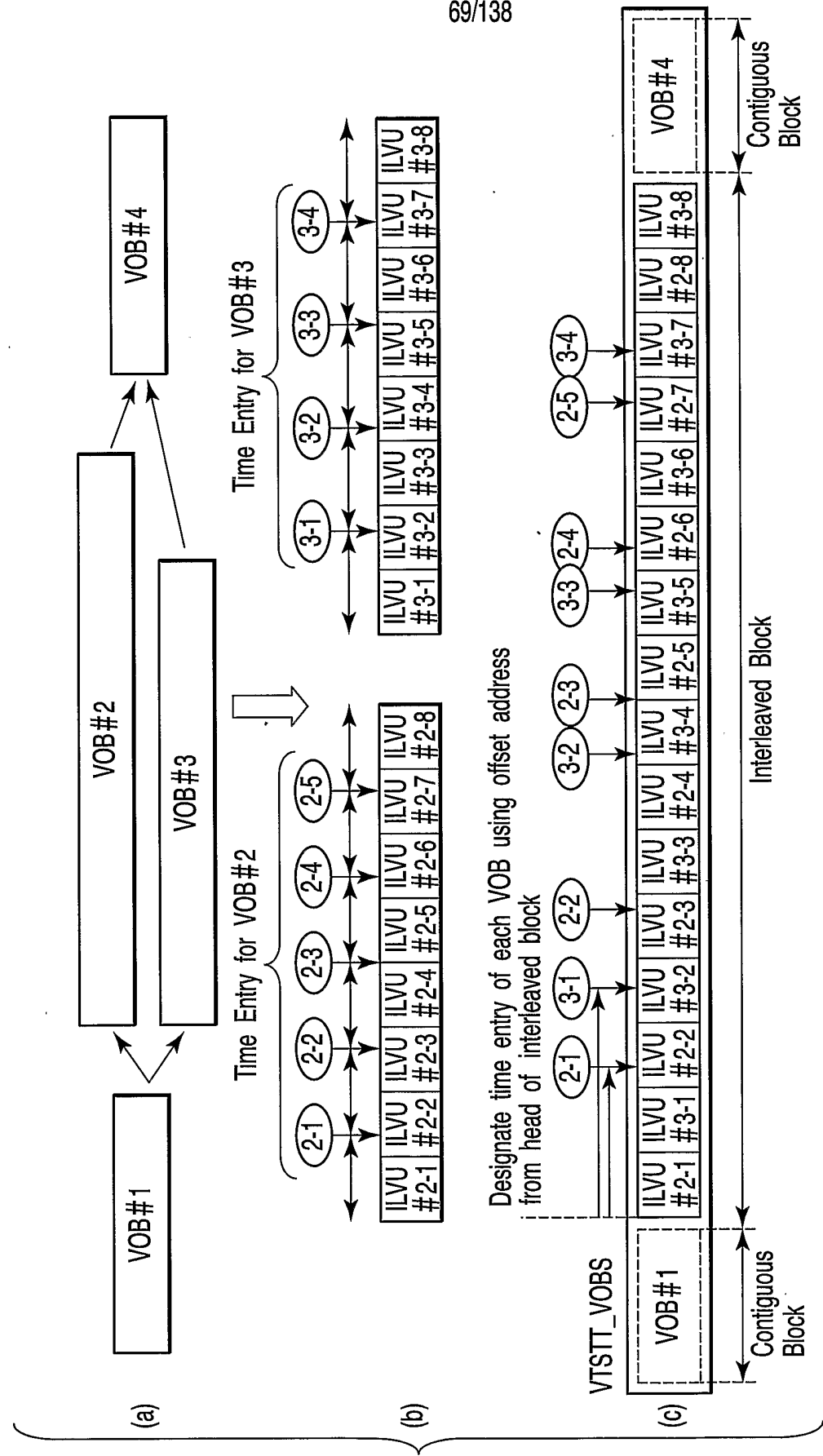
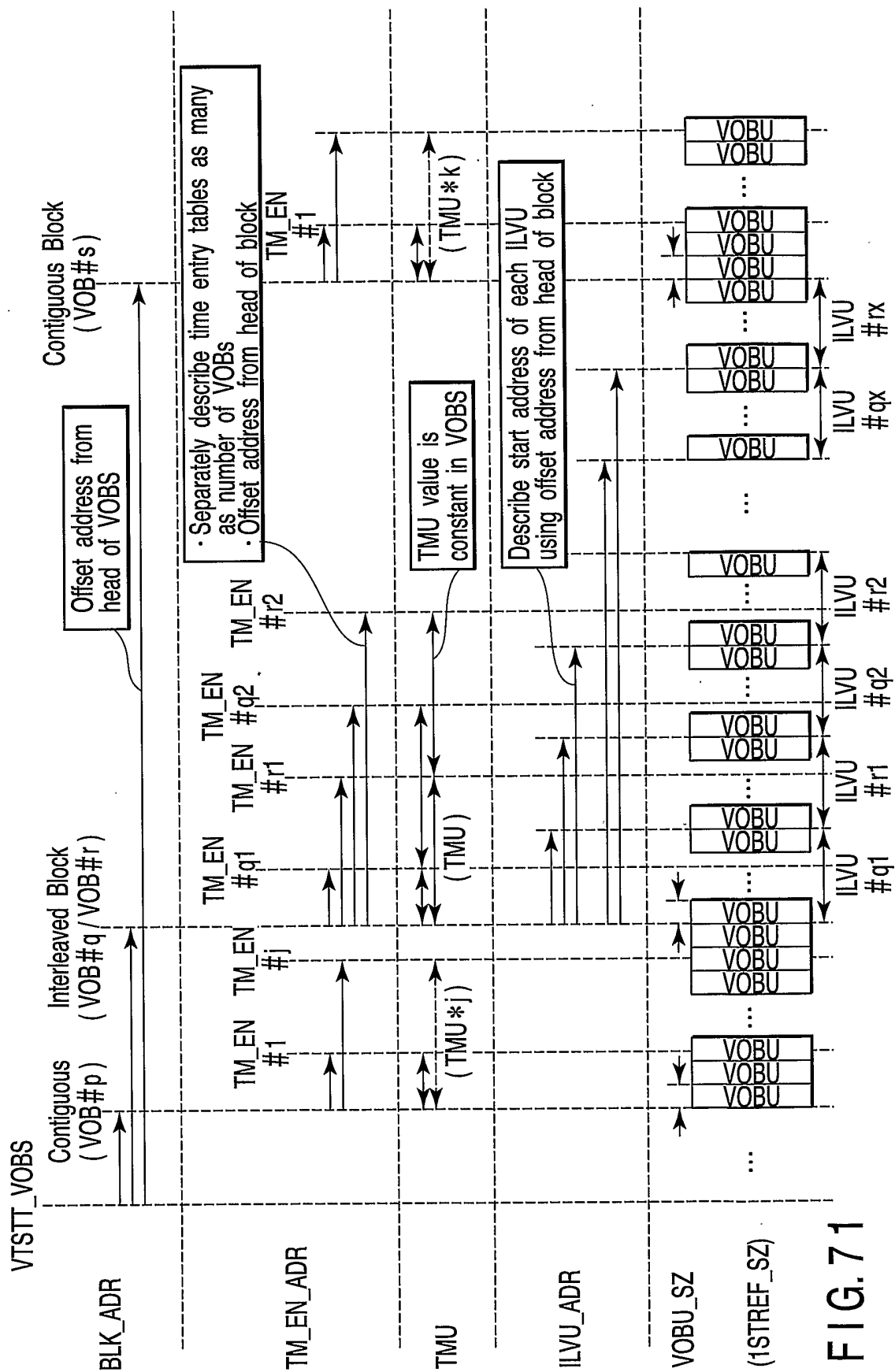


FIG. 69





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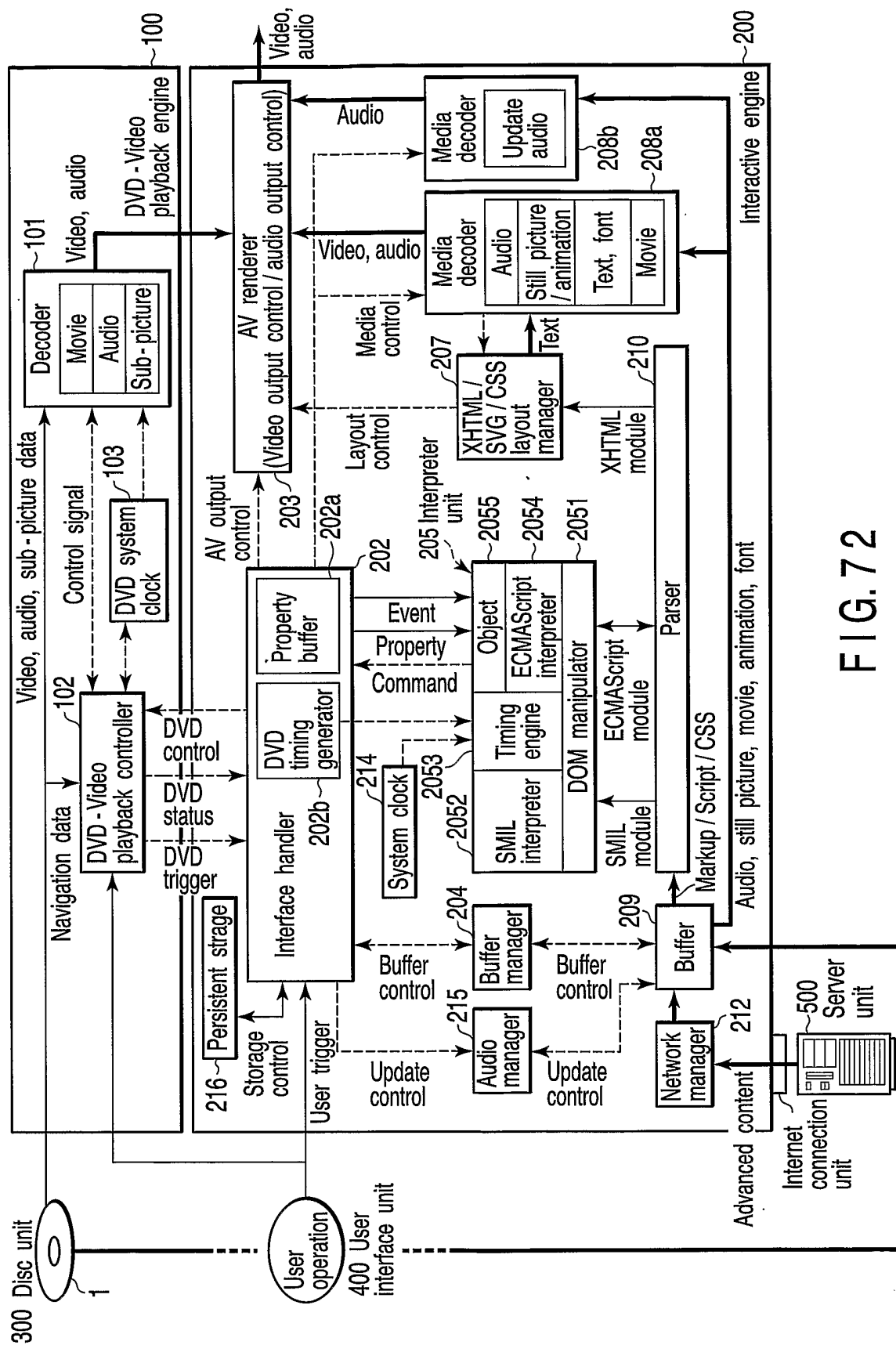


FIG. 72

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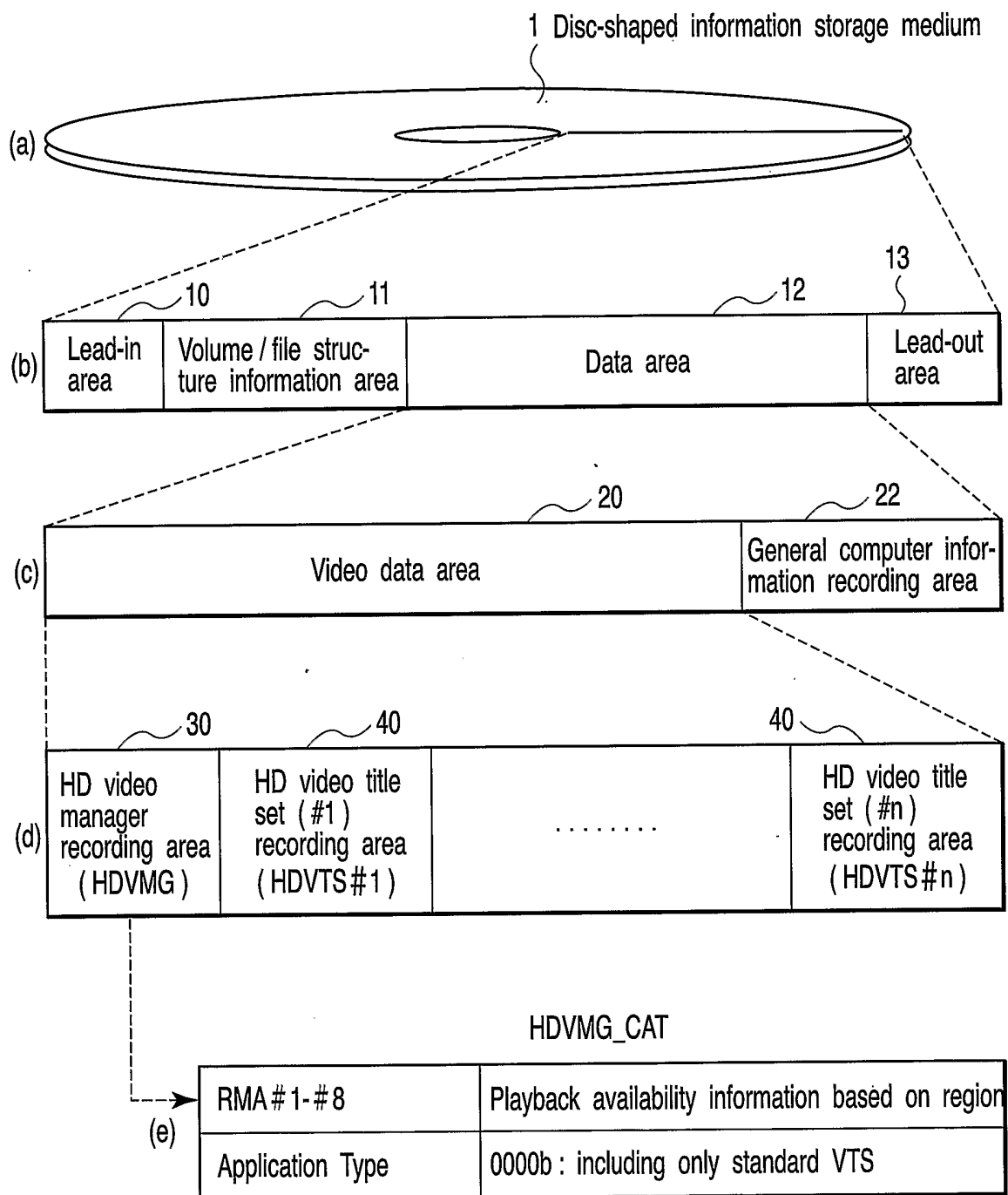


FIG. 73

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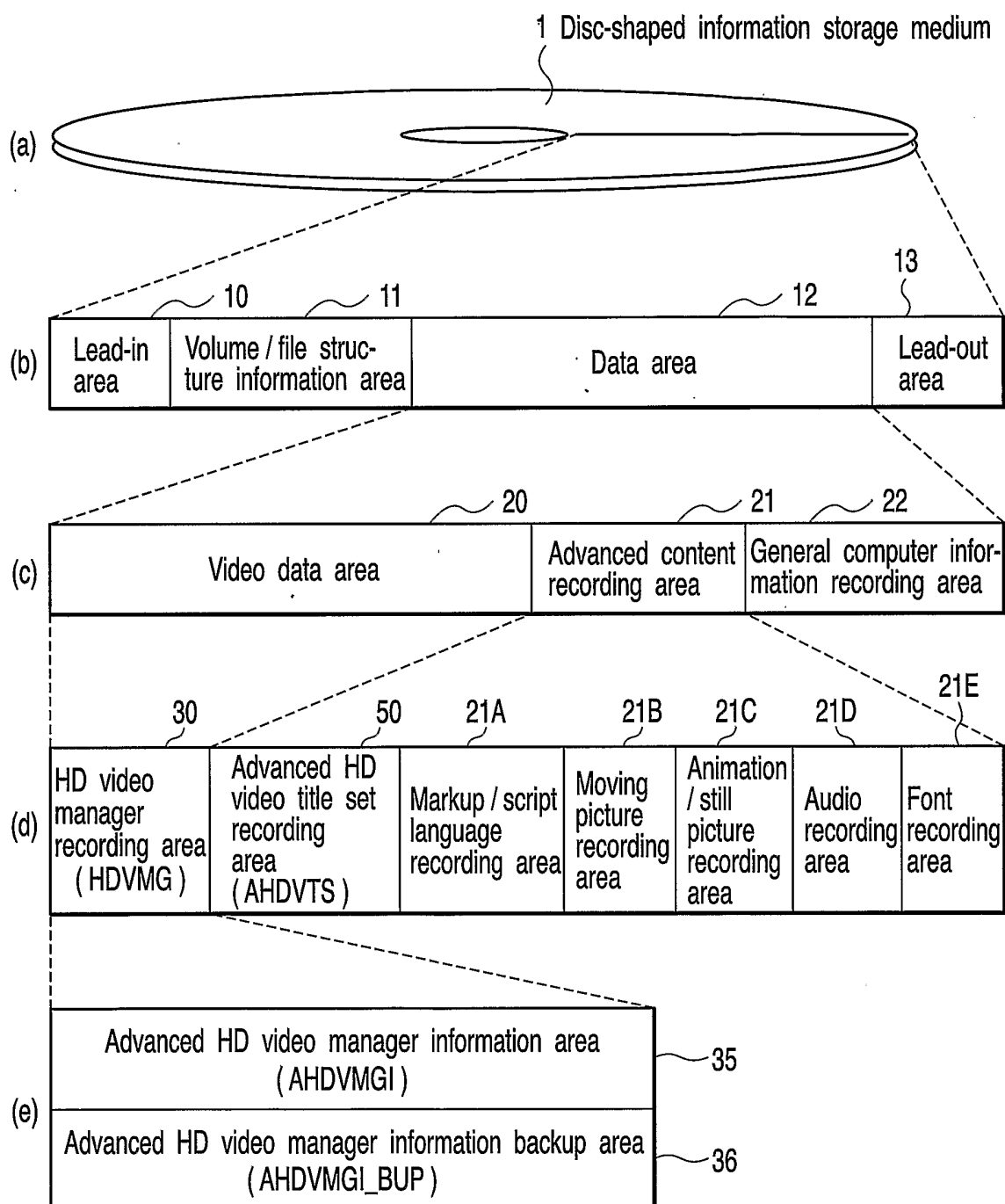


FIG. 74

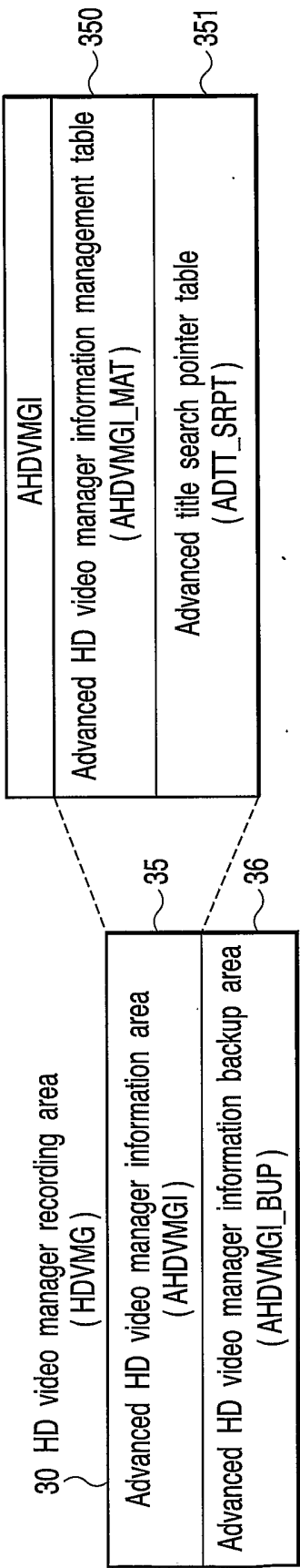


FIG. 75

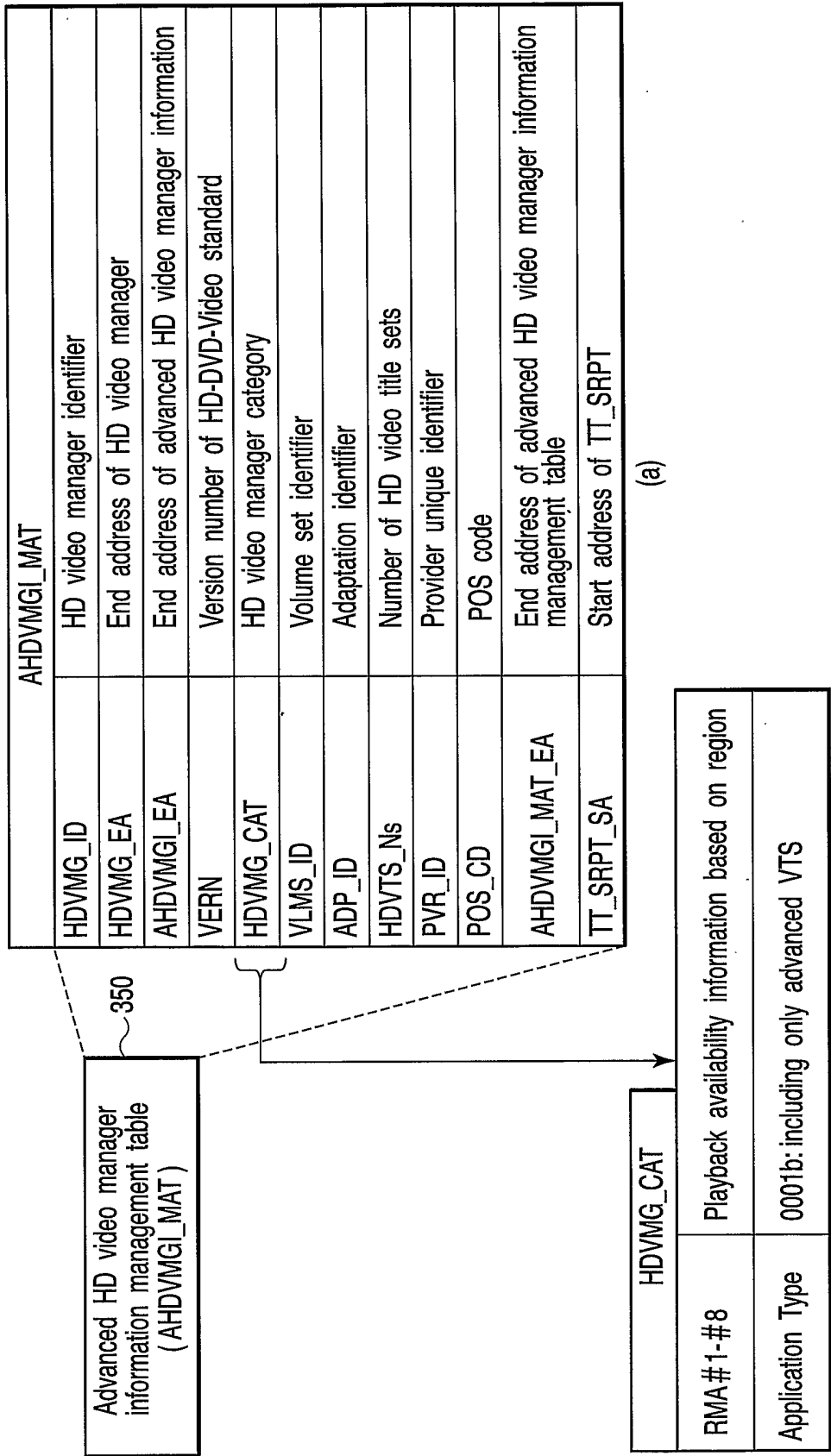


FIG.76

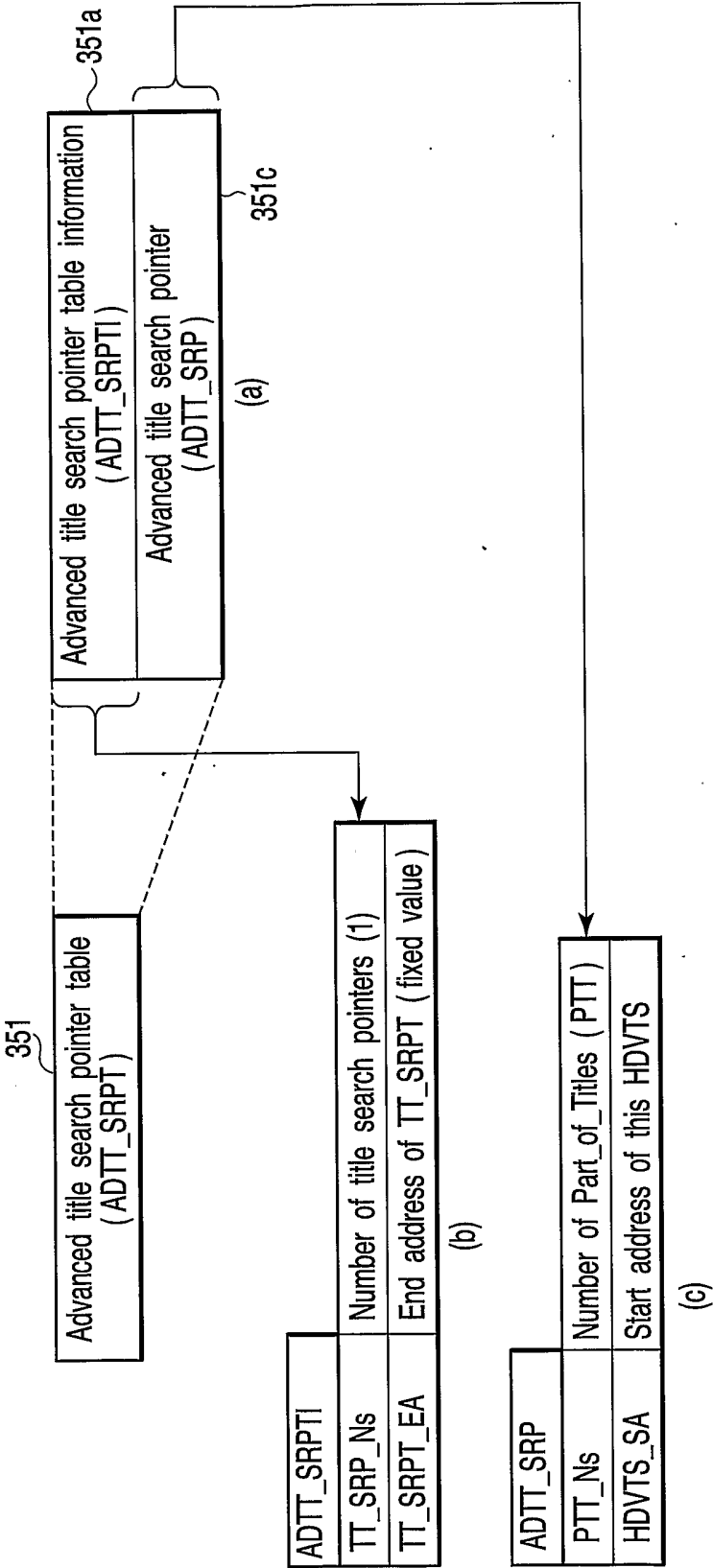
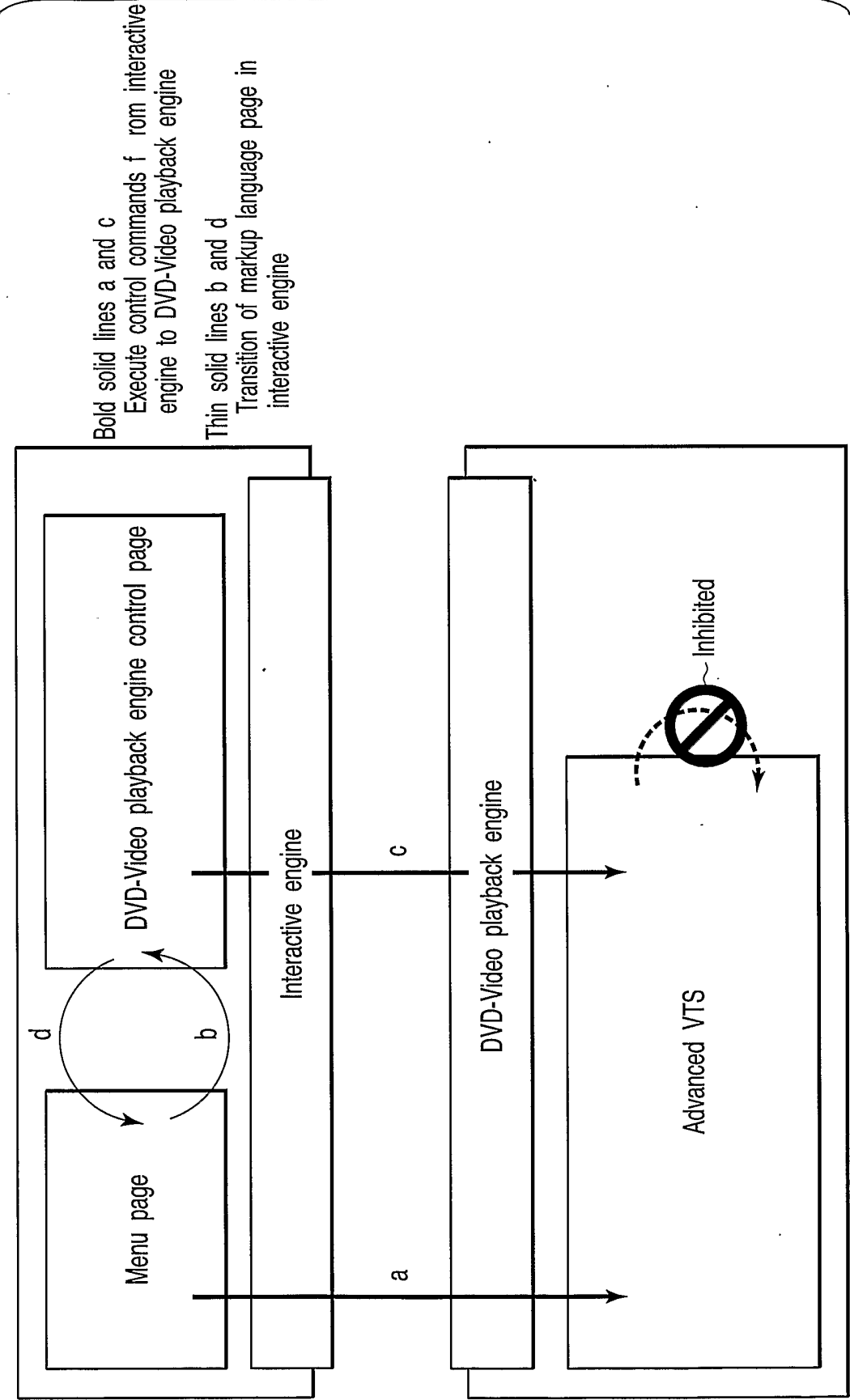


FIG. 77

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Bold solid lines a and c
Execute control commands from interactive engine to DVD-Video playback engine

Thin solid lines b and d
Transition of markup language page in interactive engine

FIG. 78

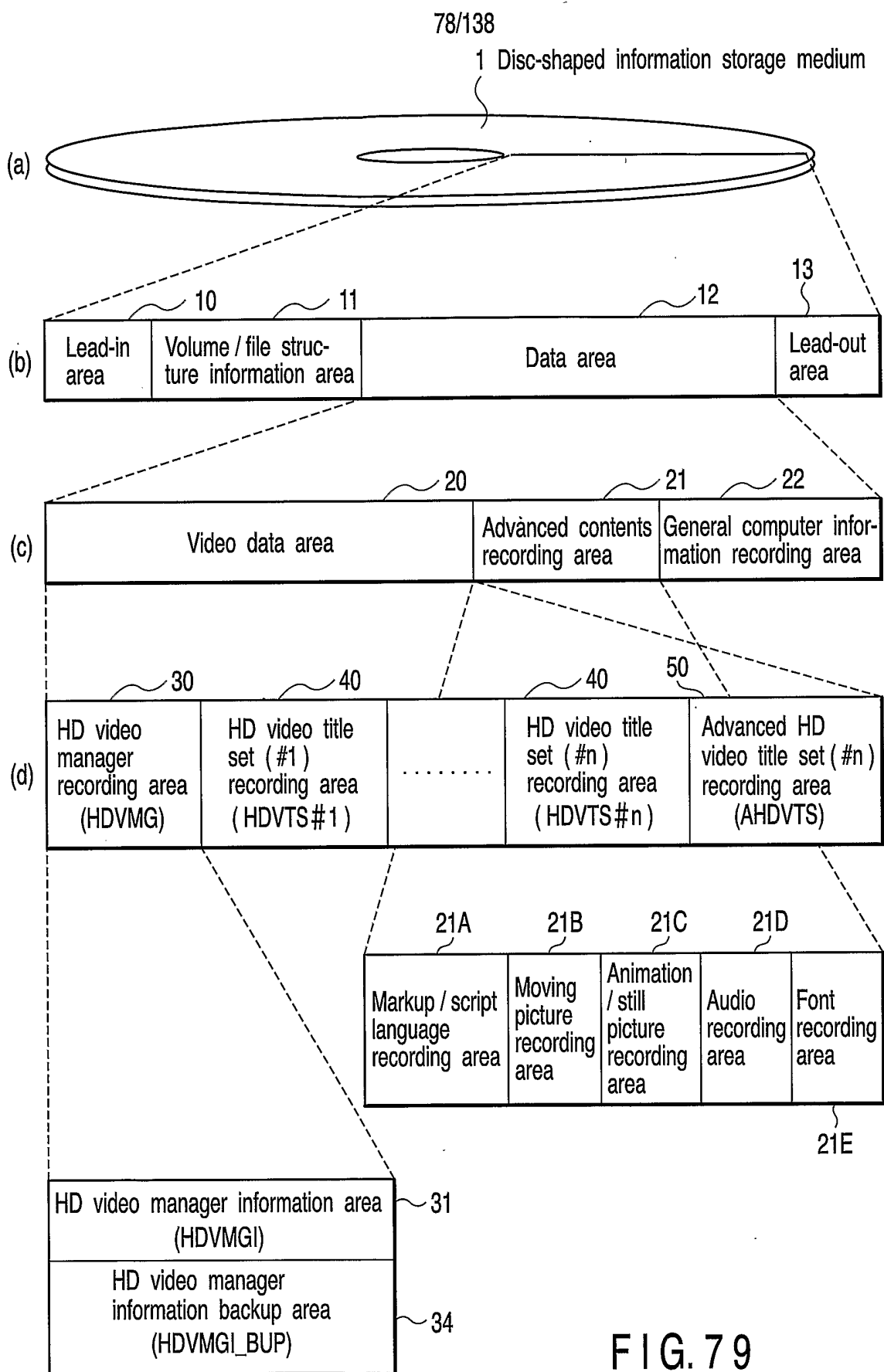


FIG. 79

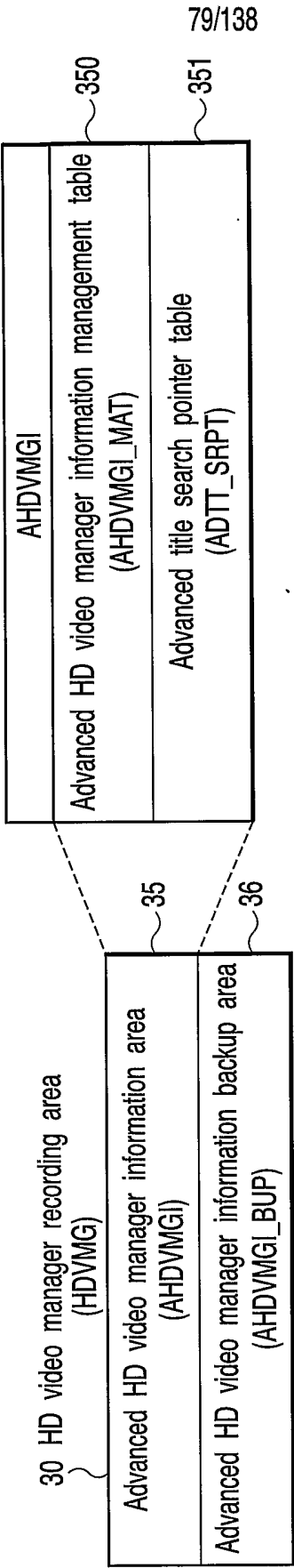


FIG. 80

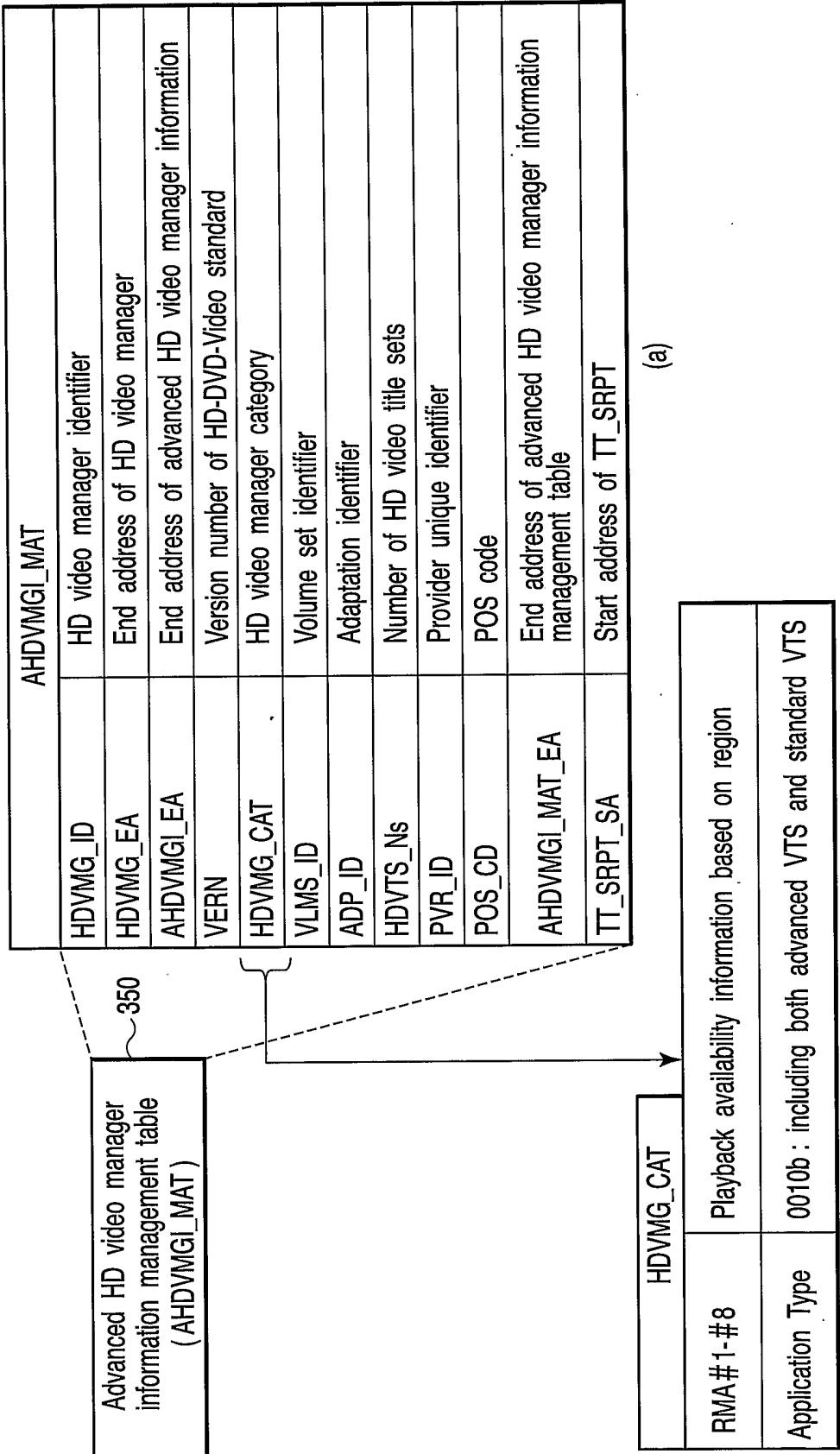
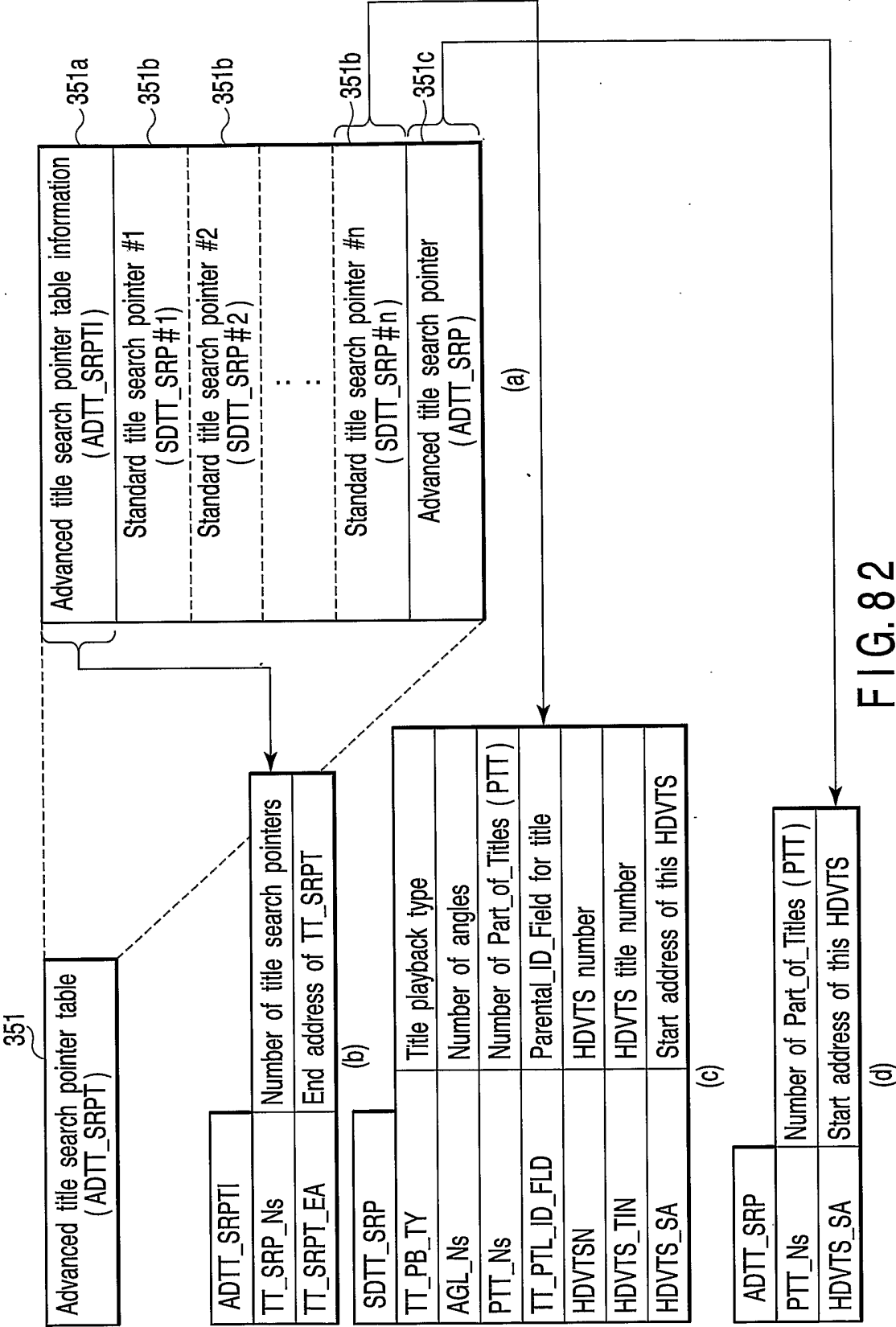


FIG. 81



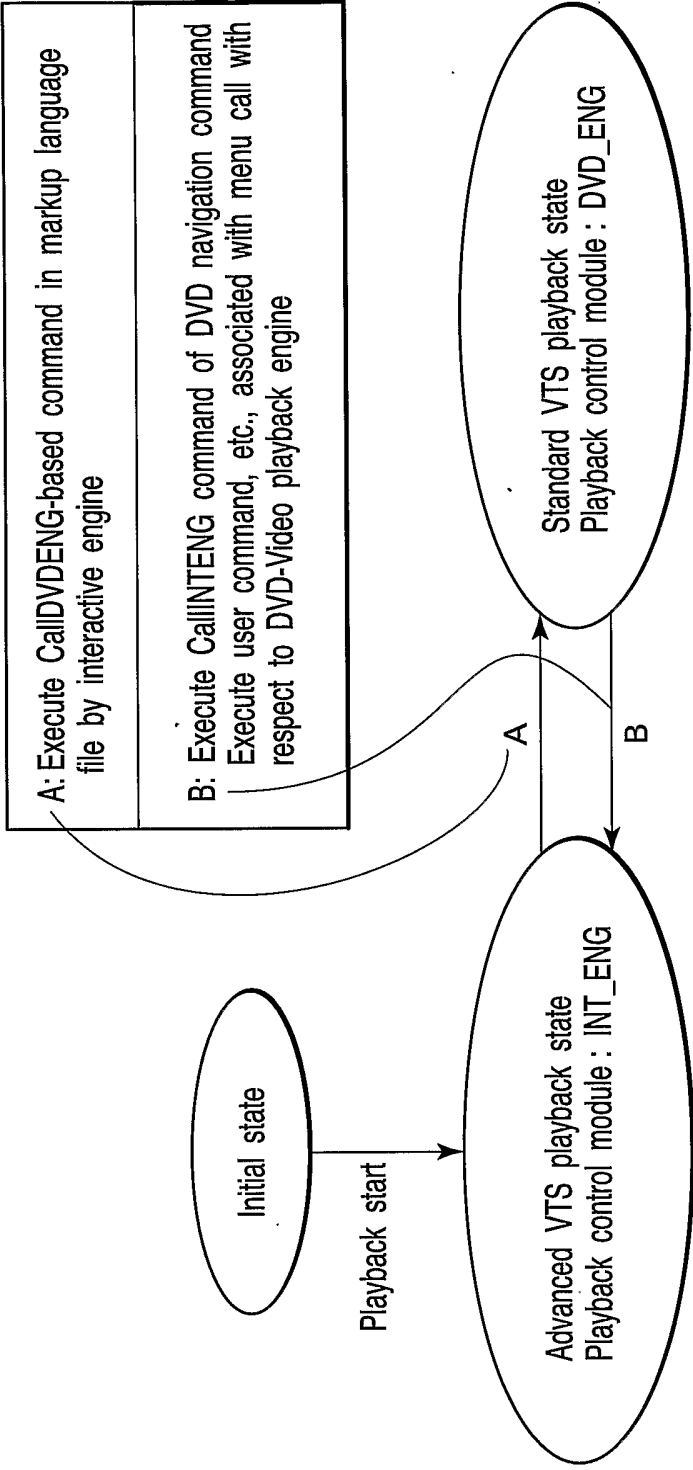


FIG. 83

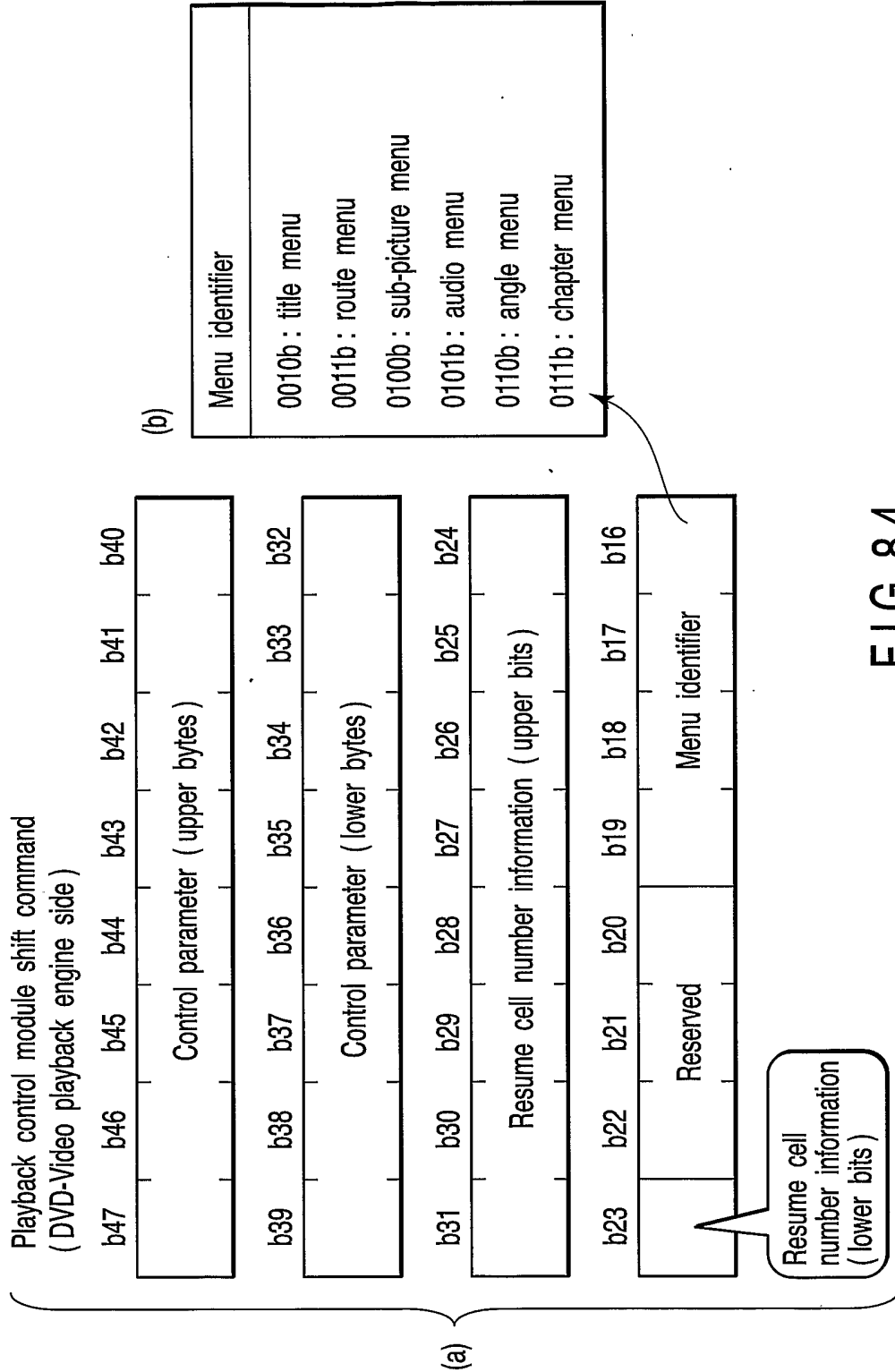


FIG. 84

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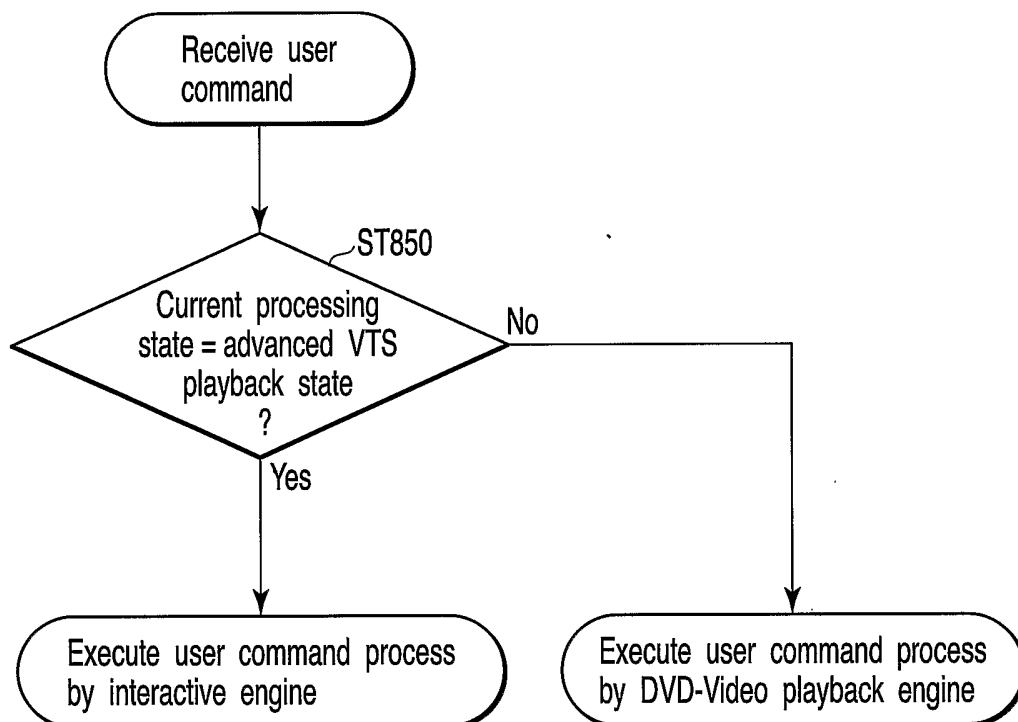


FIG. 85

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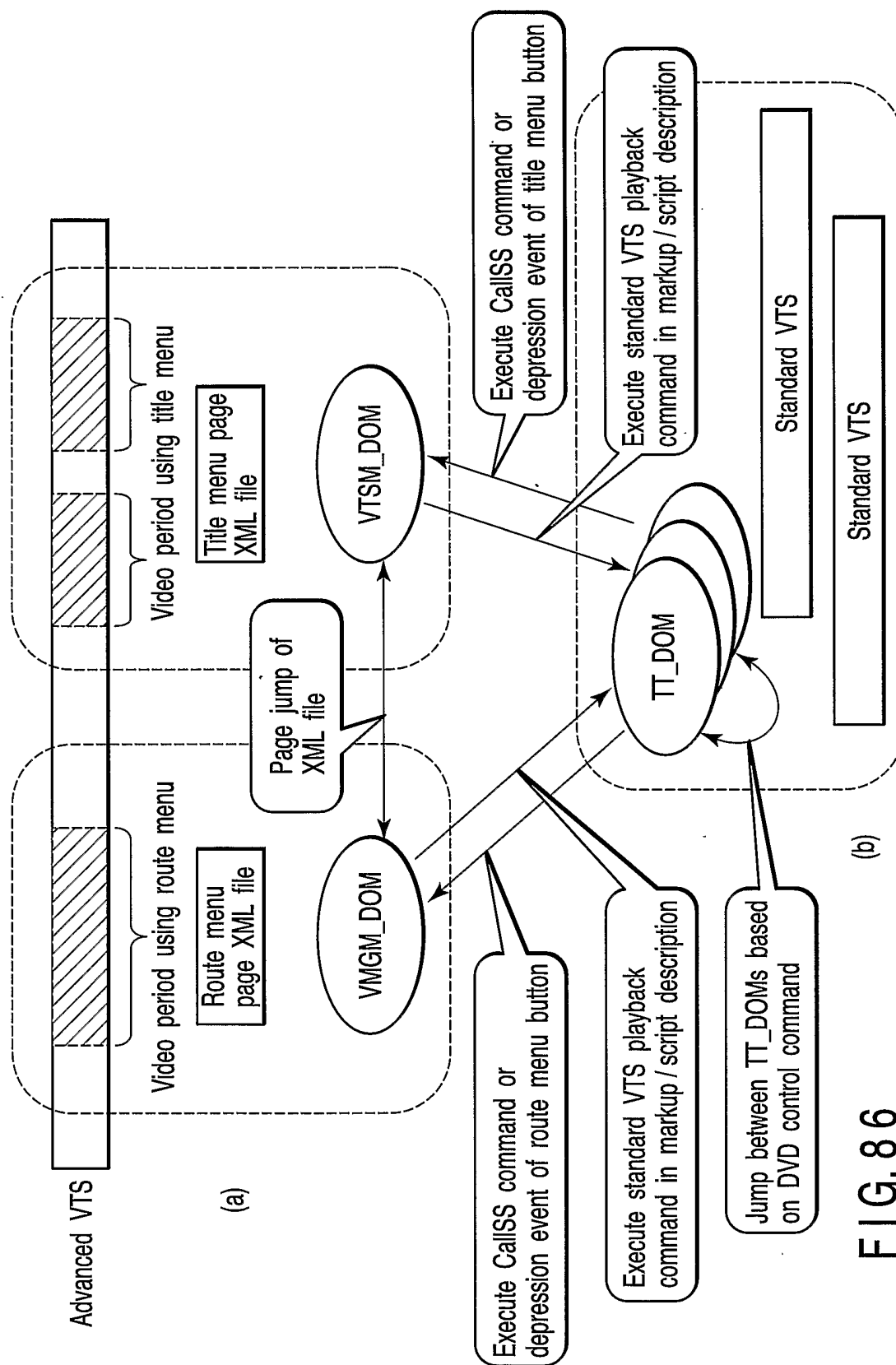


FIG. 86

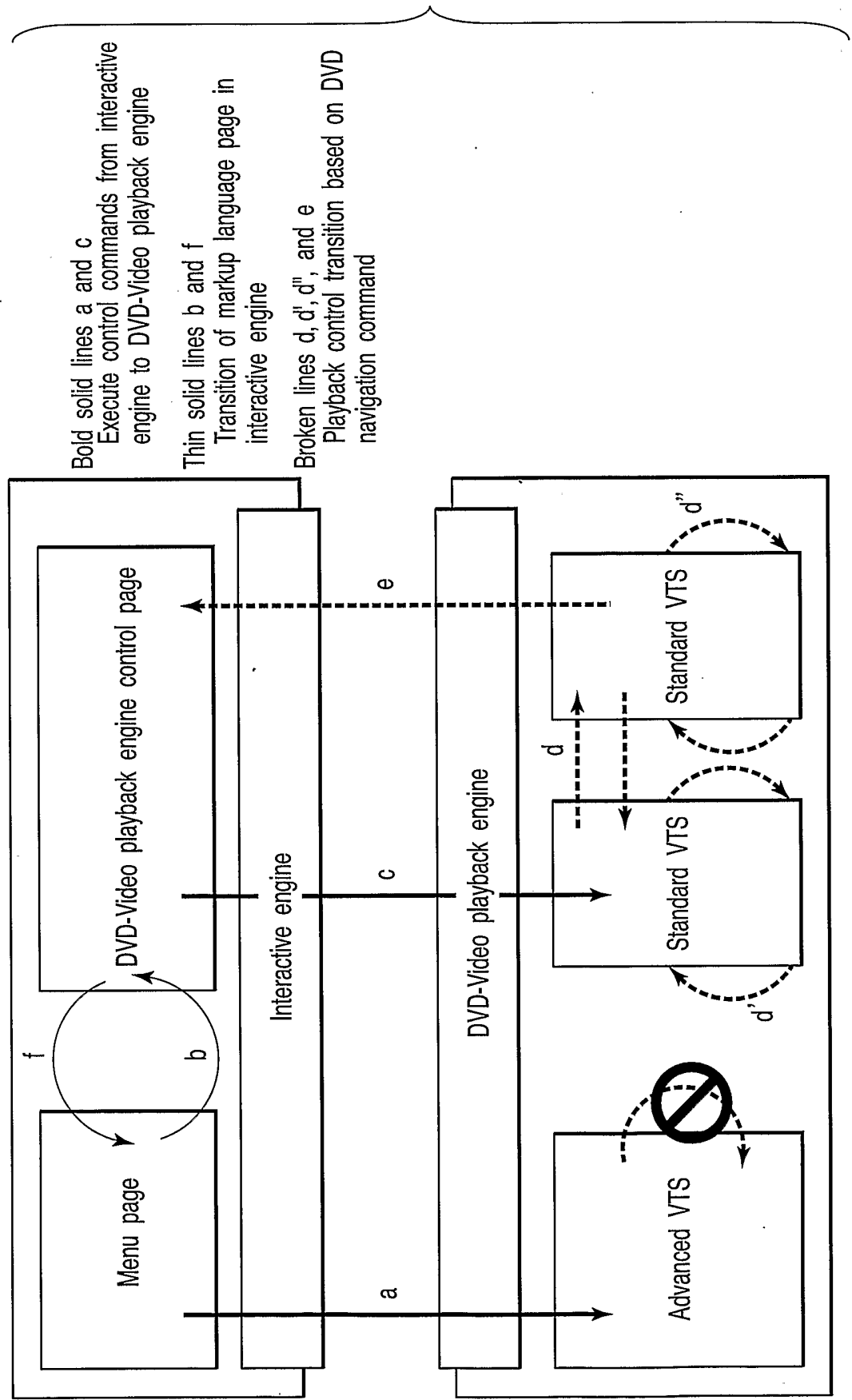


FIG. 87

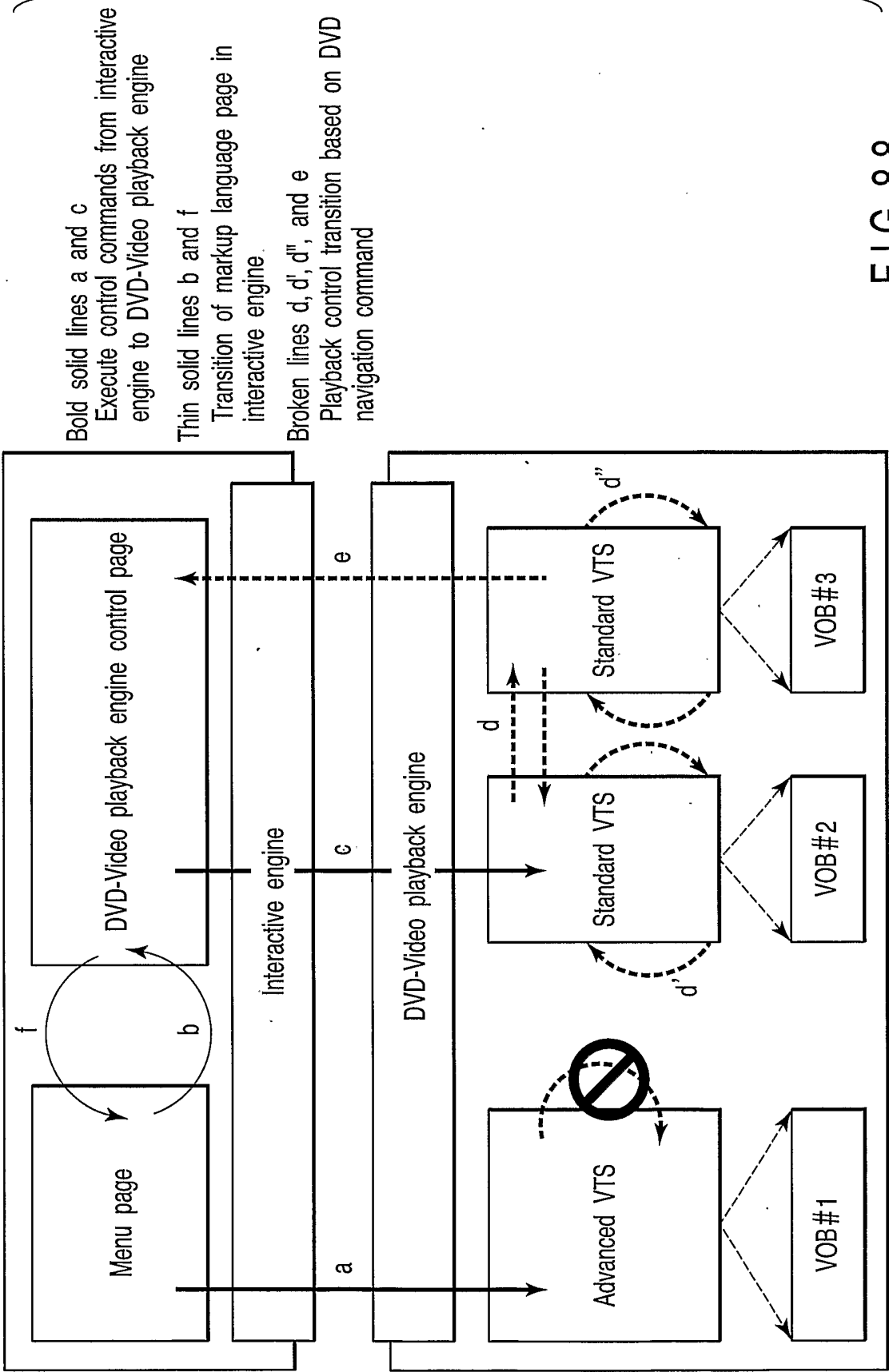


FIG. 88

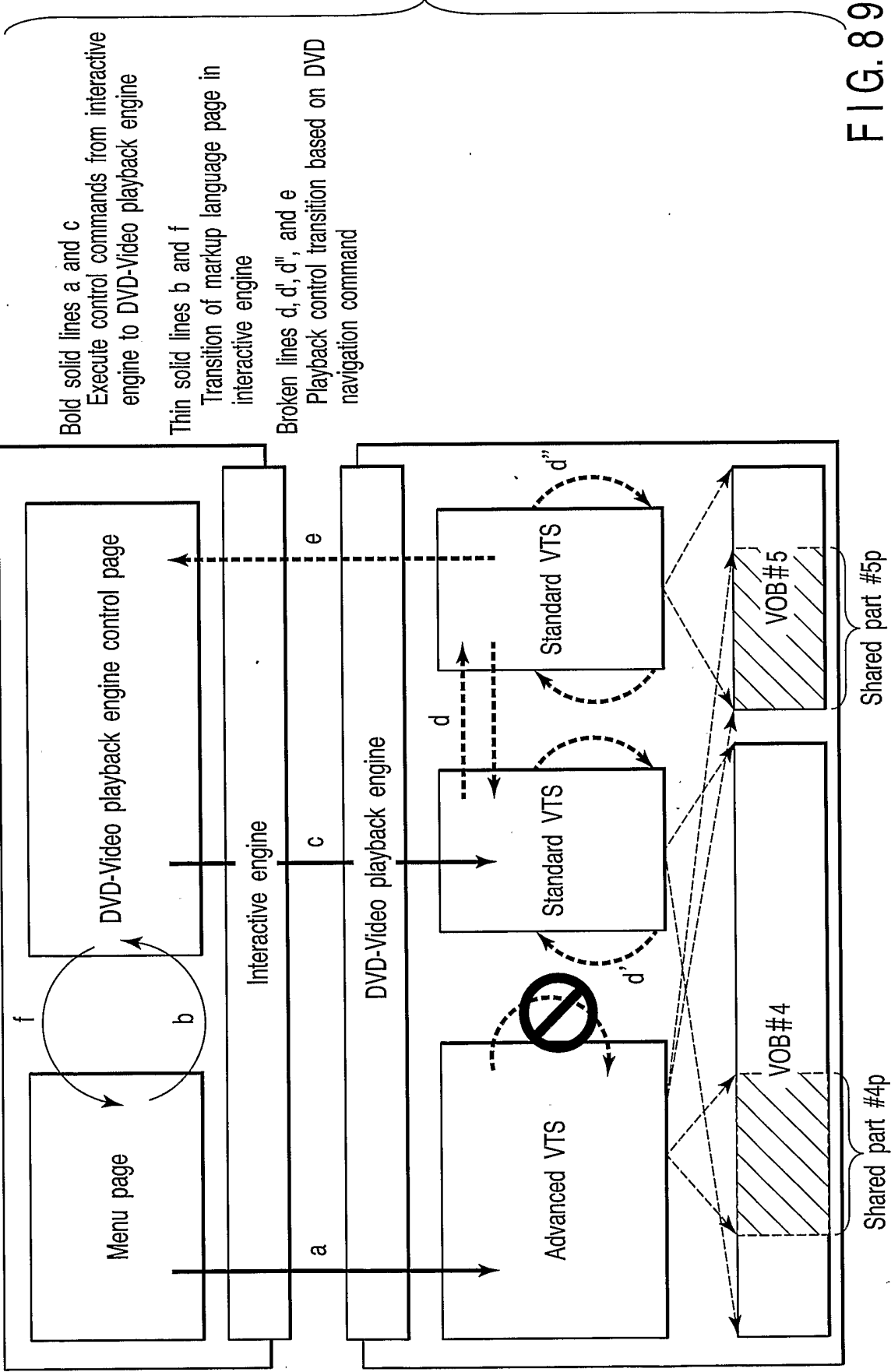


FIG. 89

(File name & location)	(File size)	(Content type)	(Reference start time)	(Reference end time)
dvd://DVD_ENAV/chapter1/index.htm	983,040	text/html	00:00:00	00:10:00
http://www.XXXXXX.XX.XX/DVD_ENAV/chapter1/bg.jpg	1,966,080	image/jpeg	00:00:00	01:00:00
http://www.XXXXXX.XX.XX/DVD_ENAV/chapter2/index2.htm	983,040	text/html	00:10:00	00:20:00
http://www.XXXXXX.XX.XX/DVD_ENAV/chapter3/anim.mng	2,949,120	video/x-mng	00:20:00	00:30:00
http://www.XXXXXX.XX.XX/DVD_ENAV/chapter3/pic2.png	1,966,080	image/png	00:20:00	01:00:00

FIG. 90

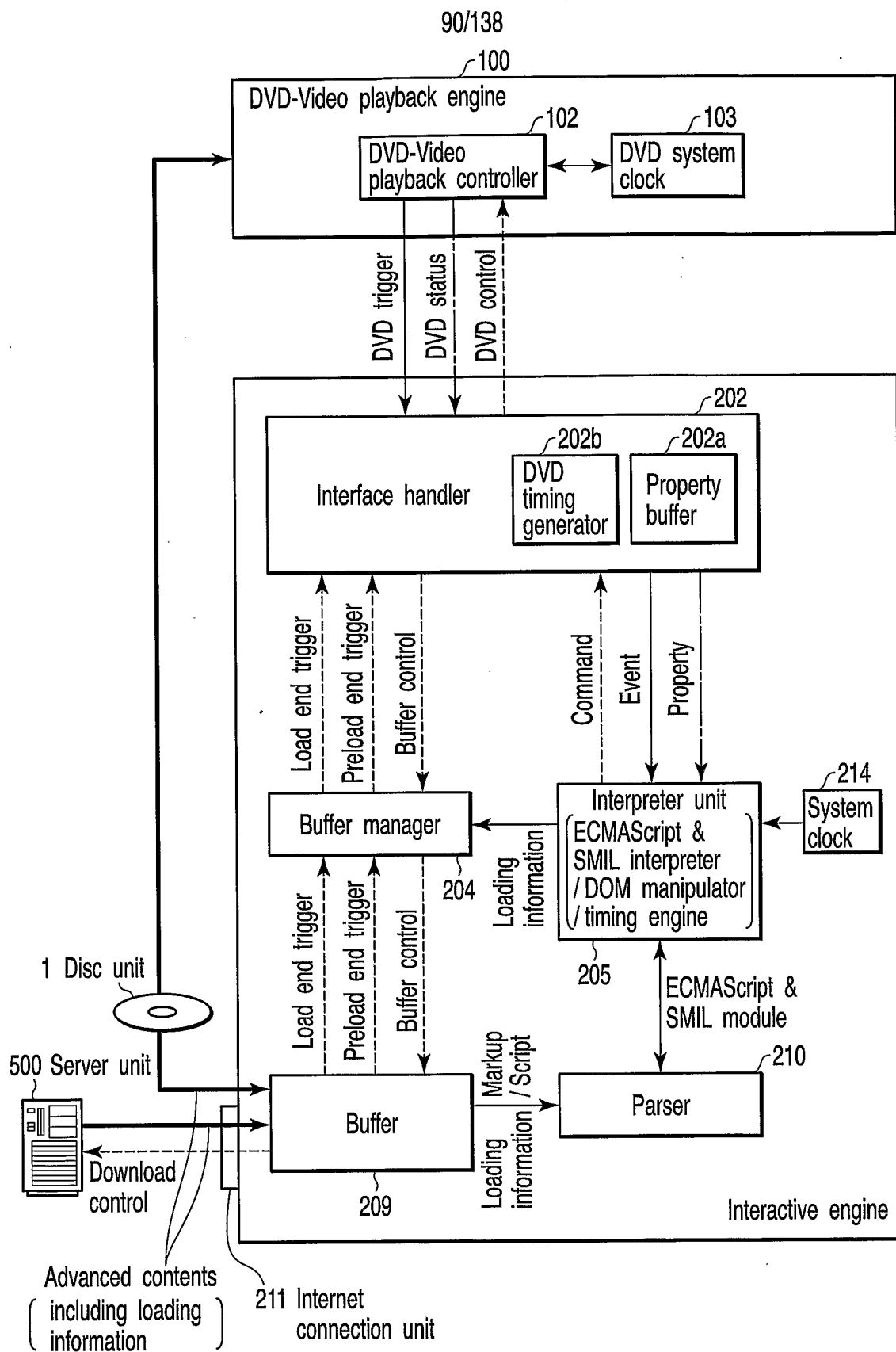


FIG. 91

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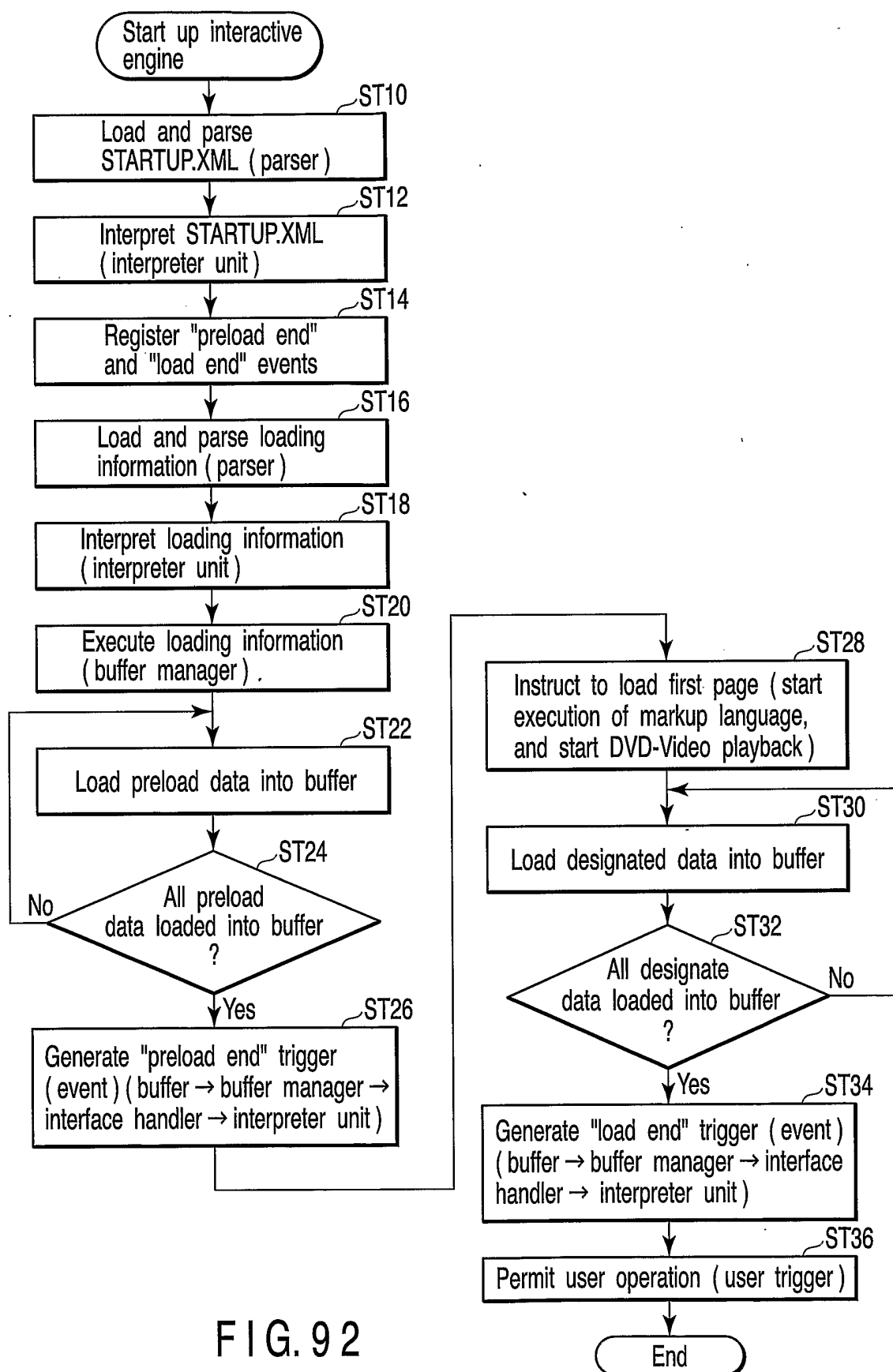


FIG. 92

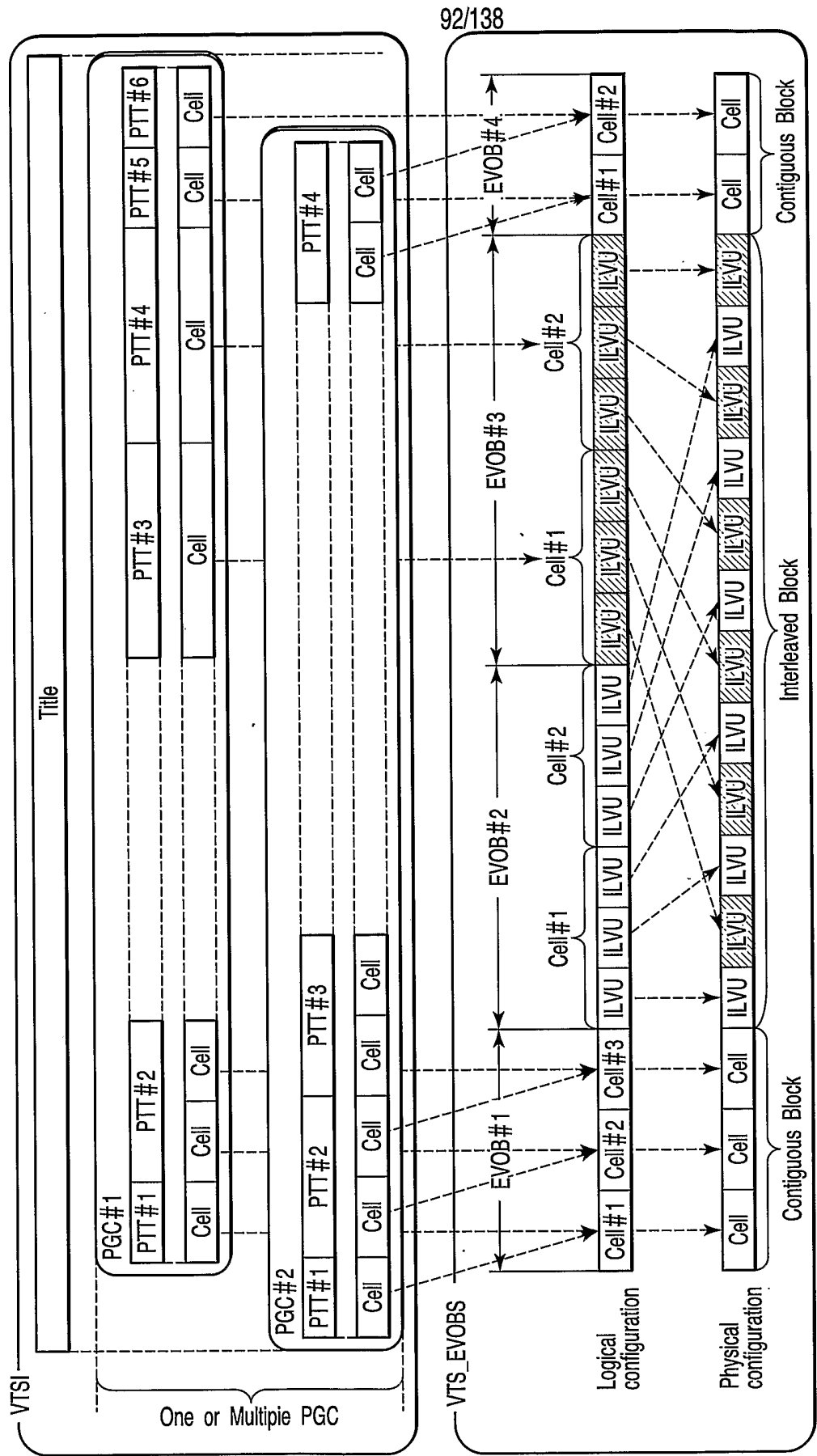


FIG. 93

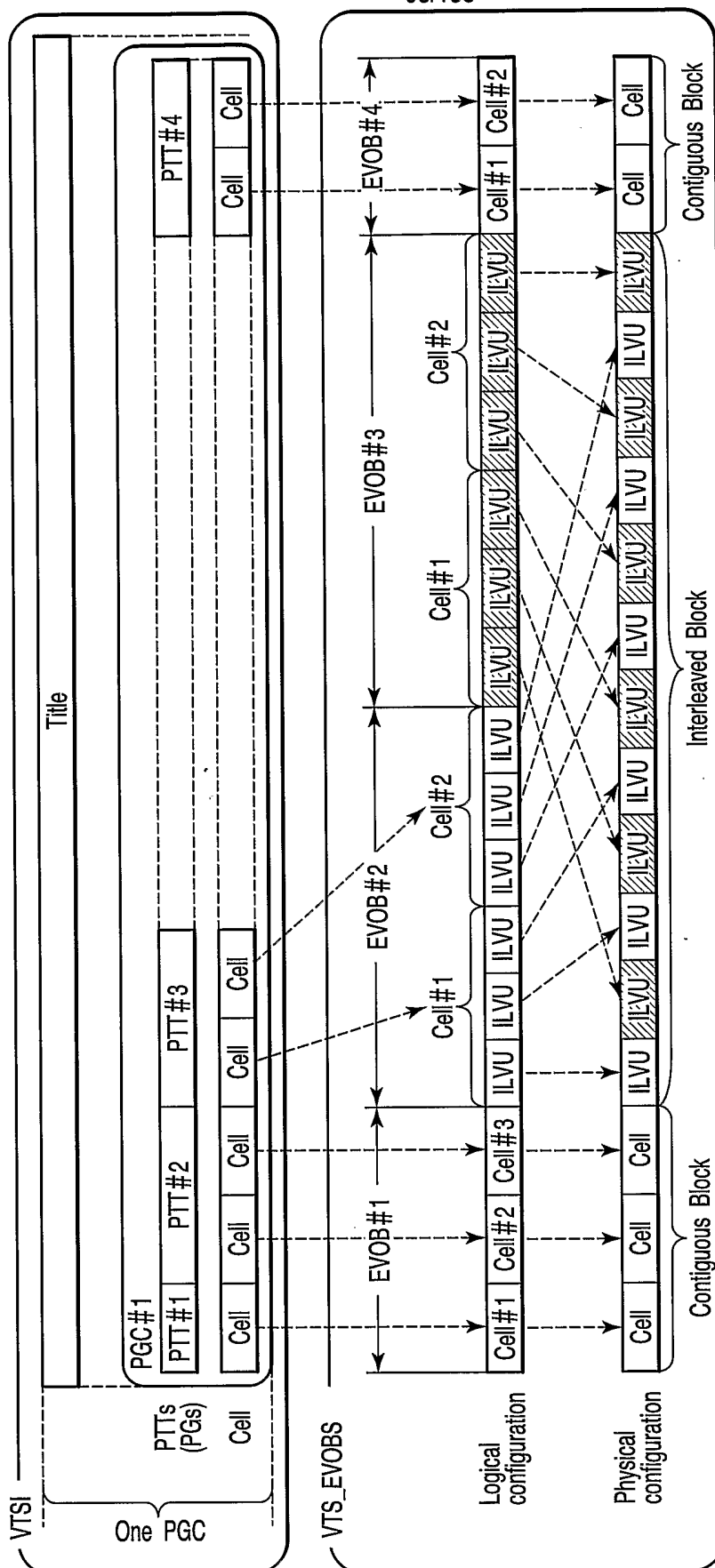


FIG. 94

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```

<chapter_seq "id"="directors_cut" "title"="1" "pgc"="1">
  <chapter "order"="1" "id"="start" "ptt"="1"/>
  <chapter "order"="2" "id"="middle" "ptt"="4"/>
  <chapter "order"="3" "id"="end" "ptt"="6"/>
</chapter_seq>

```

FIG. 95

```

<chapter_seq "id"="directors_cut" "title"="1" "pgc"="1">
  <chapter "order"="1" "id"="start" "cell"="1"/>
  <chapter "order"="2" "id"="middle" "cell"="5"/>
  <chapter "order"="3" "id"="end" "cell"="7"/>
</chapter_seq>

```

FIG. 96

```

<chapter_seq "id"="Bonus_comparison" "title"="1">
  <chapter "order"="1" "id"="directors_cut1" "pgc"="1" "ptt"="3"/>
  <chapter "order"="2" "id"="directors_cut2" "pgc"="1" "ptt"="4"/>
  <chapter "order"="3" "id"="theater1" "pgc"="2" "ptt"="3"/>
</chapter_seq>

```

FIG. 97

```

<chapter_seq "id"="Bonus_comparison" "title"="1">
  <chapter "order"="1" "id"="directors_cut1" "pgc"="1" "cell"="4"/>
  <chapter "order"="2" "id"="directors_cut2" "pgc"="1" "cell"="5"/>
  <chapter "order"="3" "id"="theater1" "pgc"="2" "cell"="4"/>
  <chapter "order"="4" "id"="theater2" "pgc"="2" "cell"="5"/>
</chapter_seq>

```

FIG. 98

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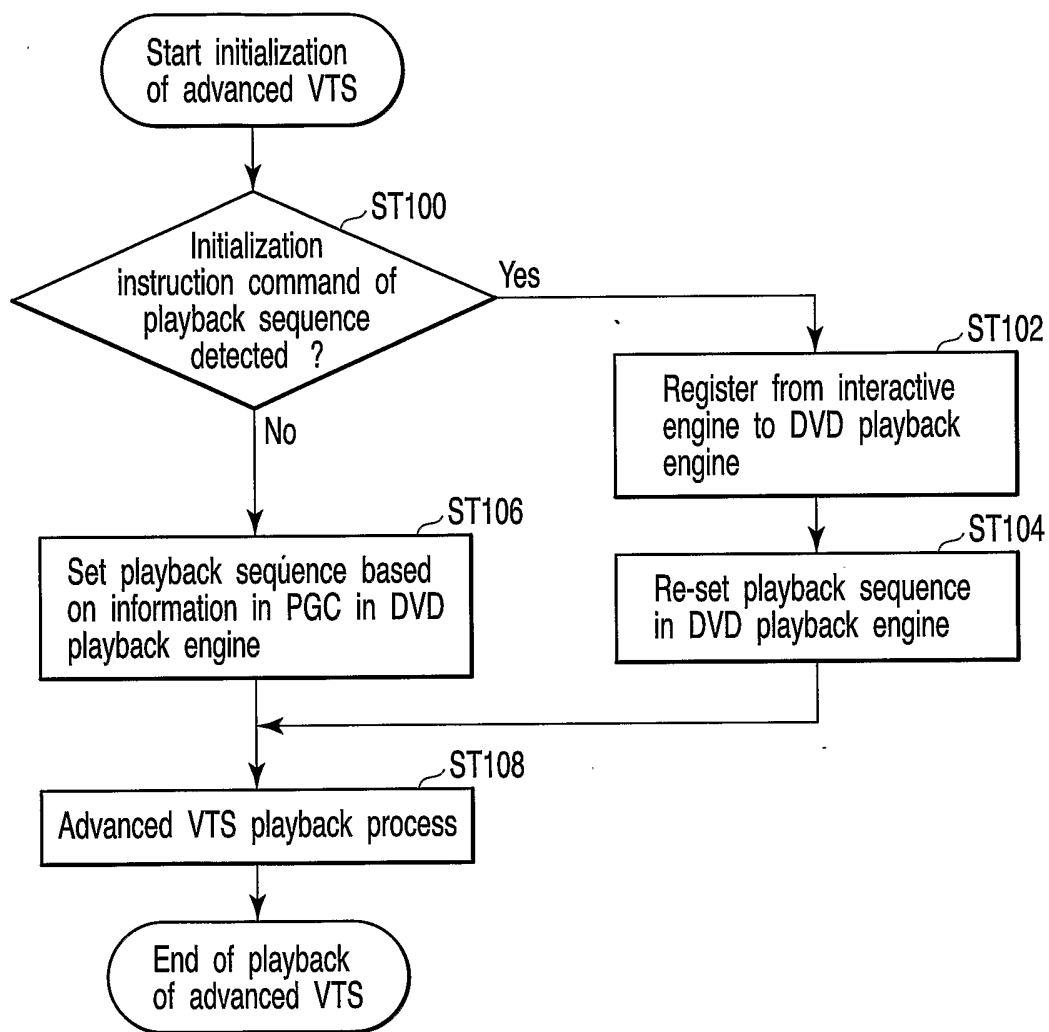


FIG. 99

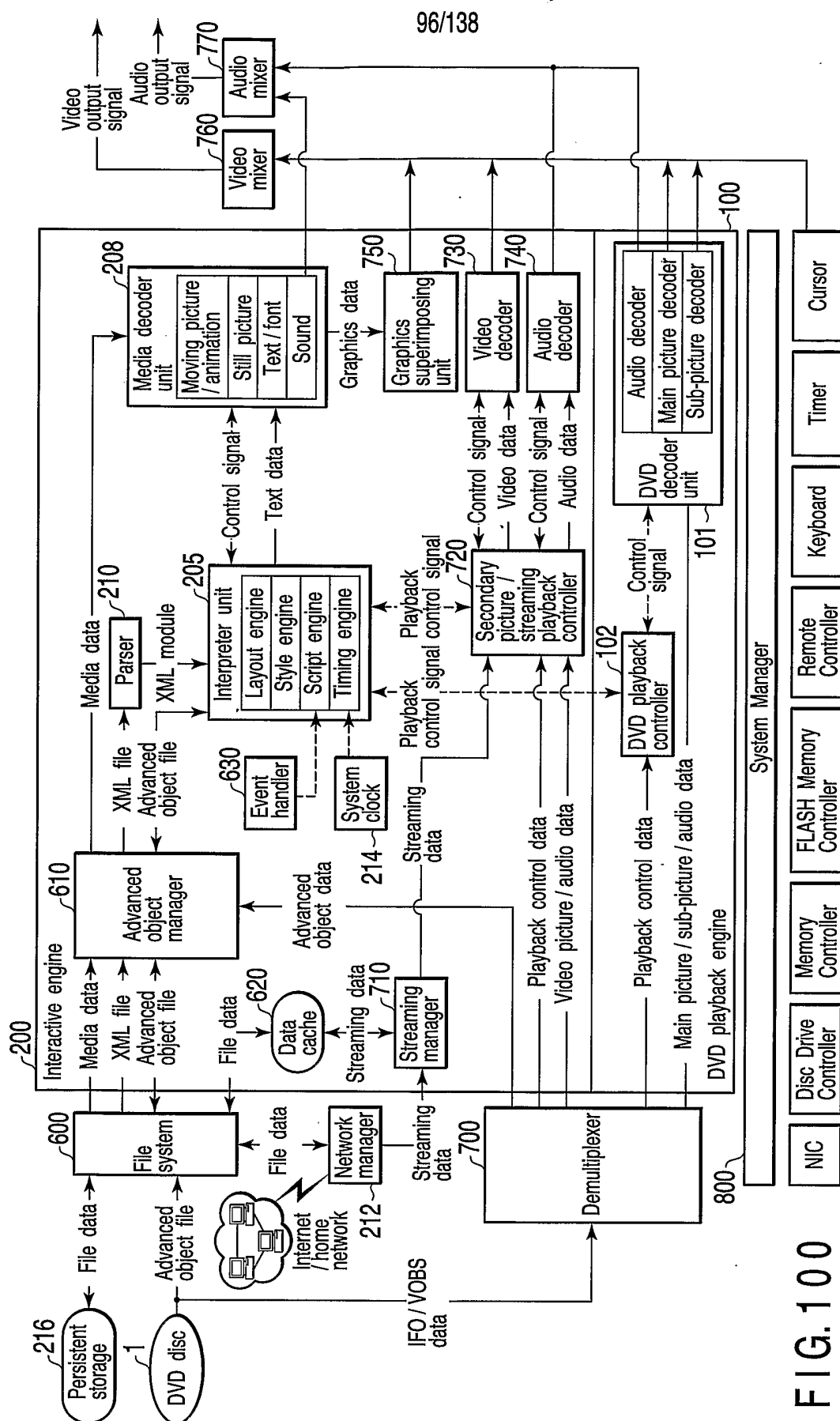


FIG. 100

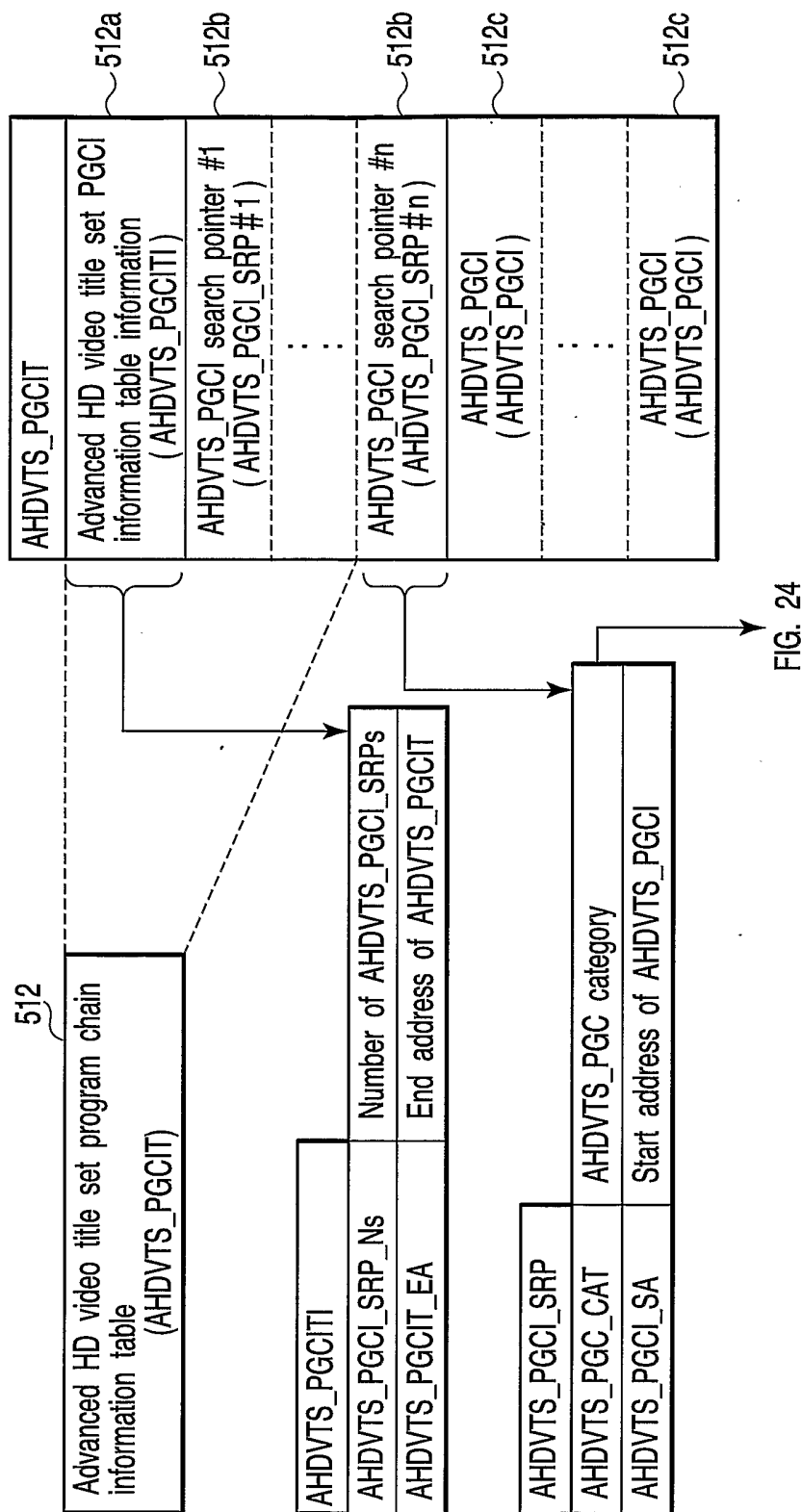


FIG. 101

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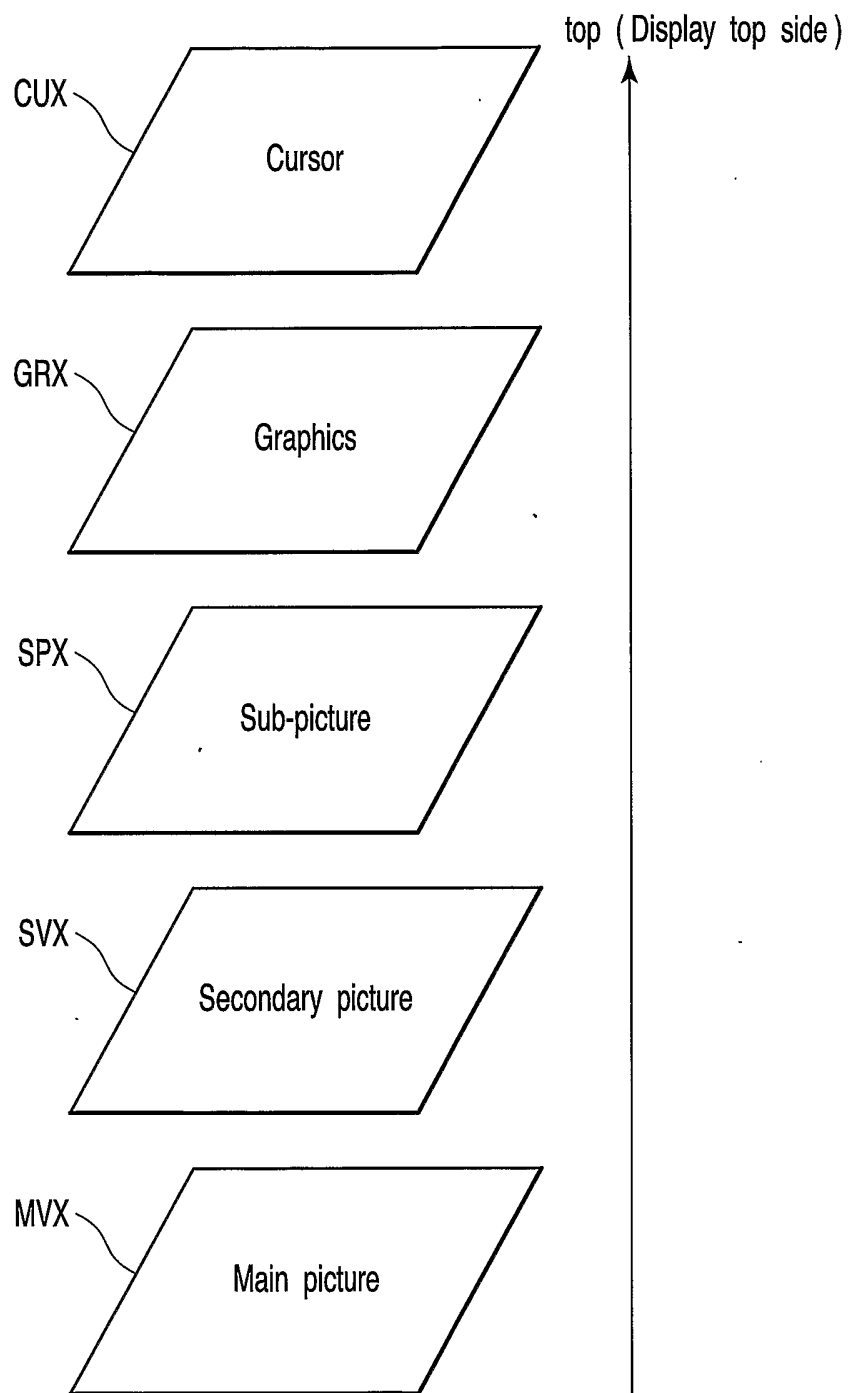


FIG. 102

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Example 1 : configuration of one
TMAP file that stores one TMAP

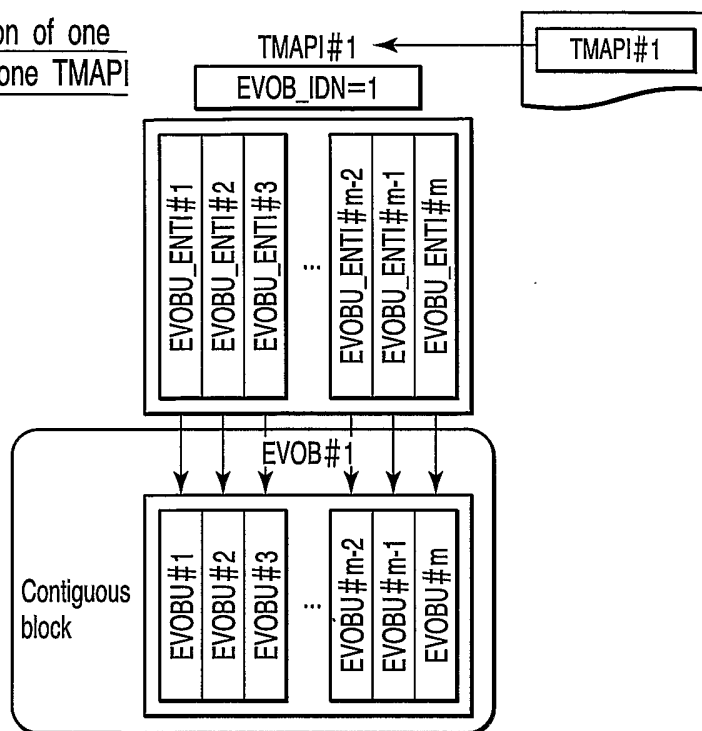


FIG. 103

Example 2 : configuration of one
TMAP file that stores a plurality
of pieces of TMAP

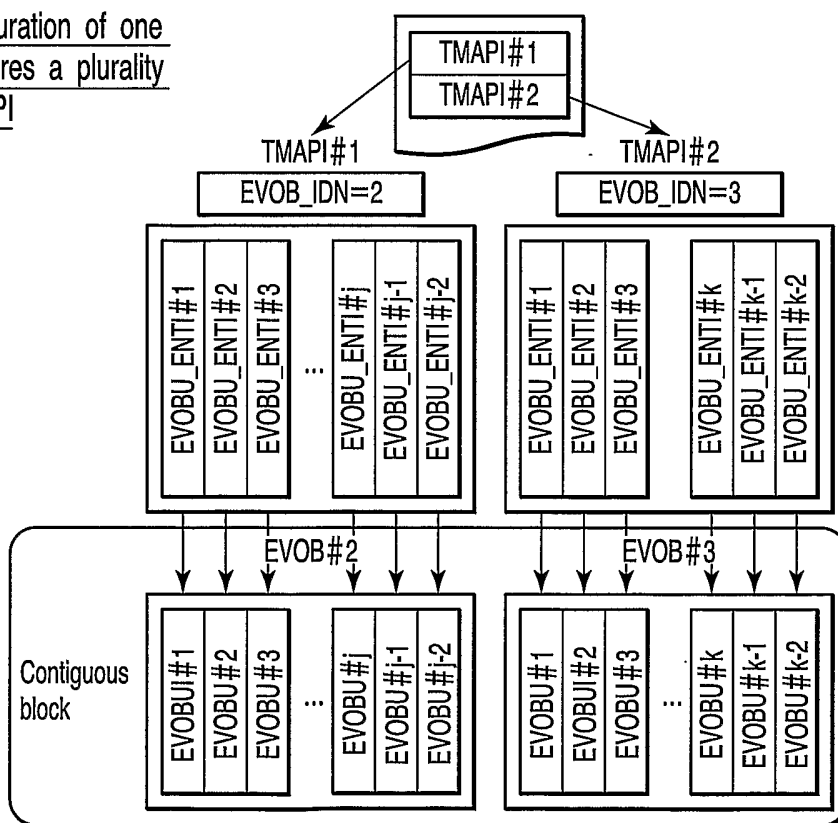


FIG. 104

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Example 3 : configuration of one TMAP file
that stores a plurality of pieces of TMAPi

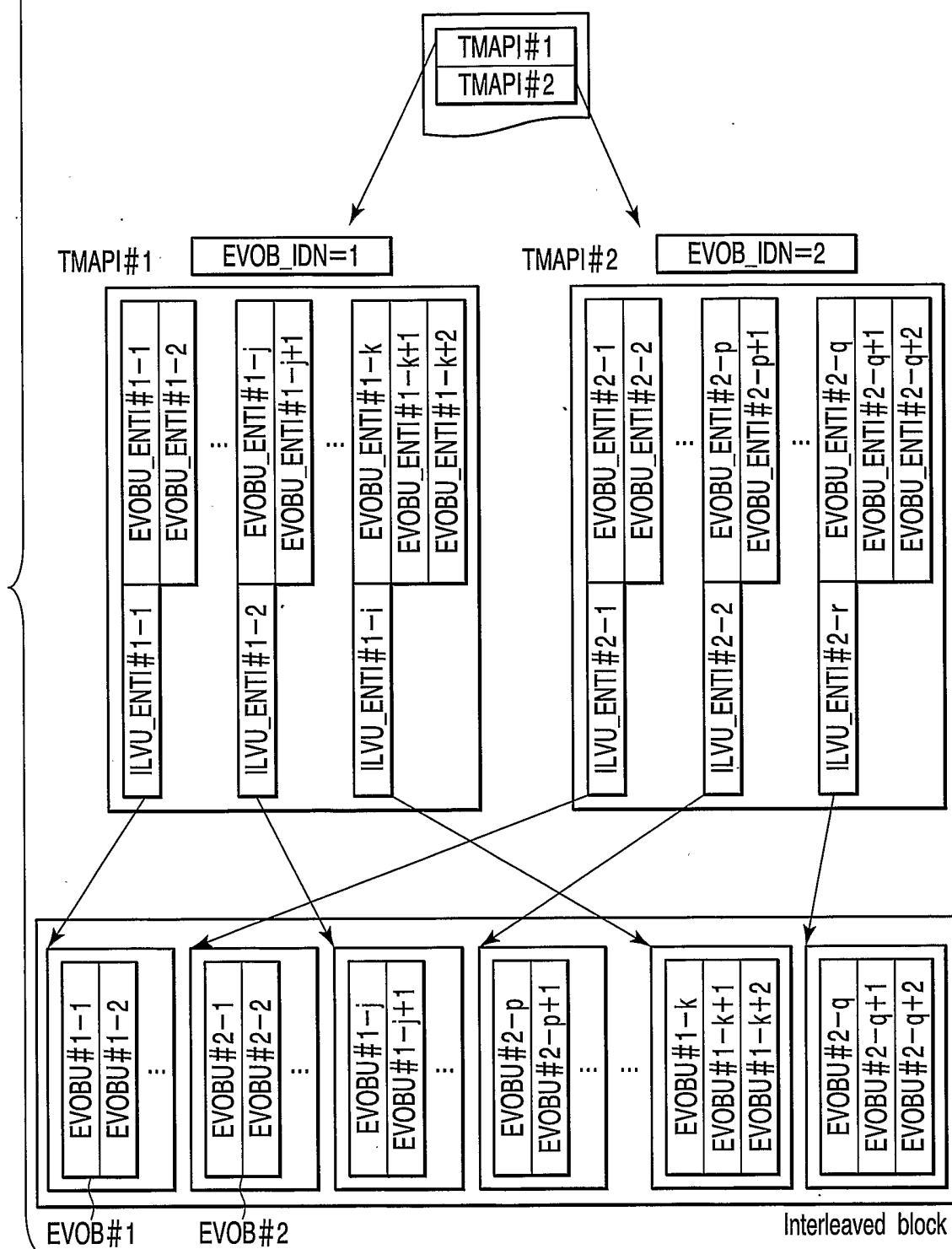


FIG. 105

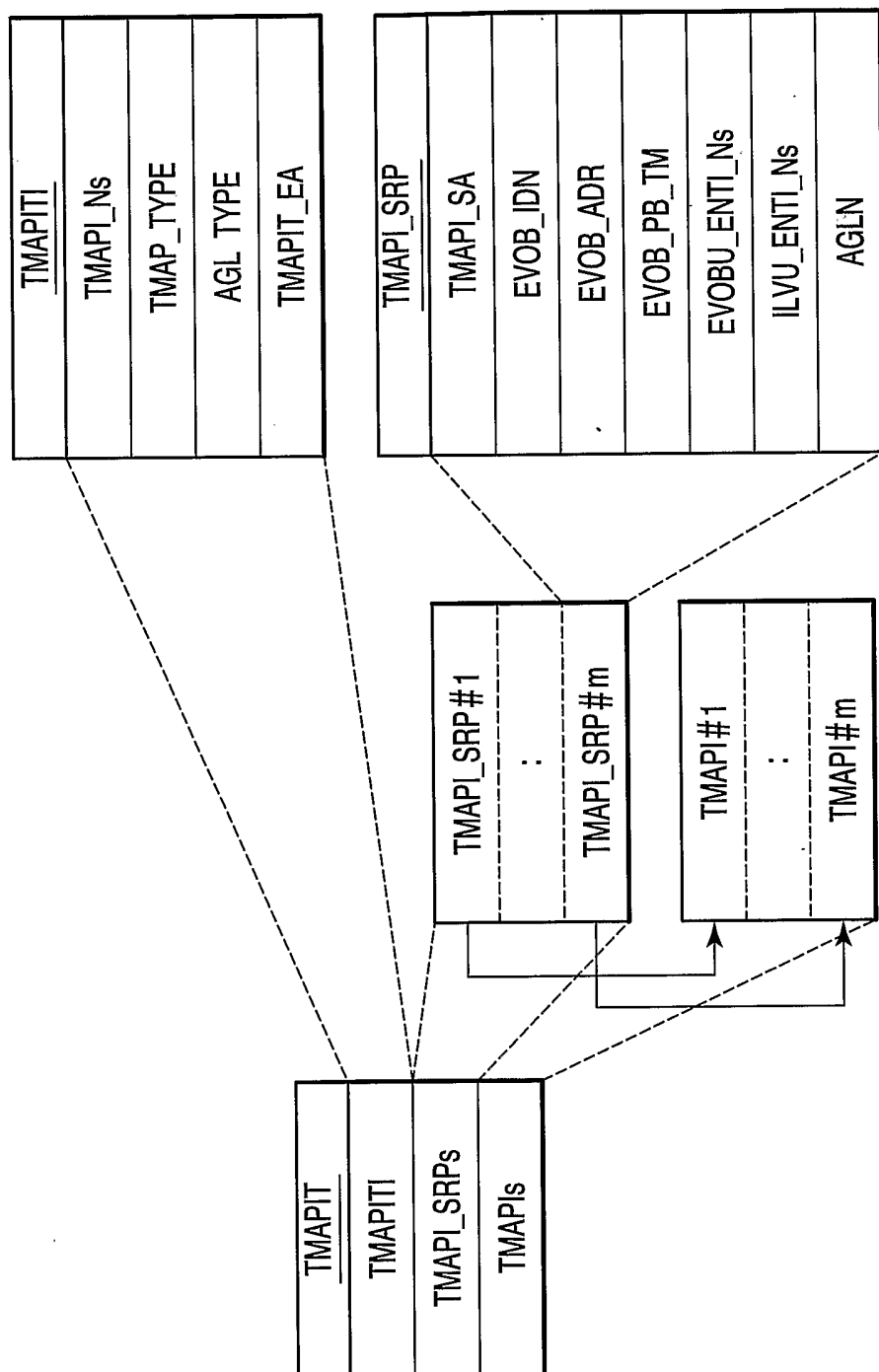


FIG. 106

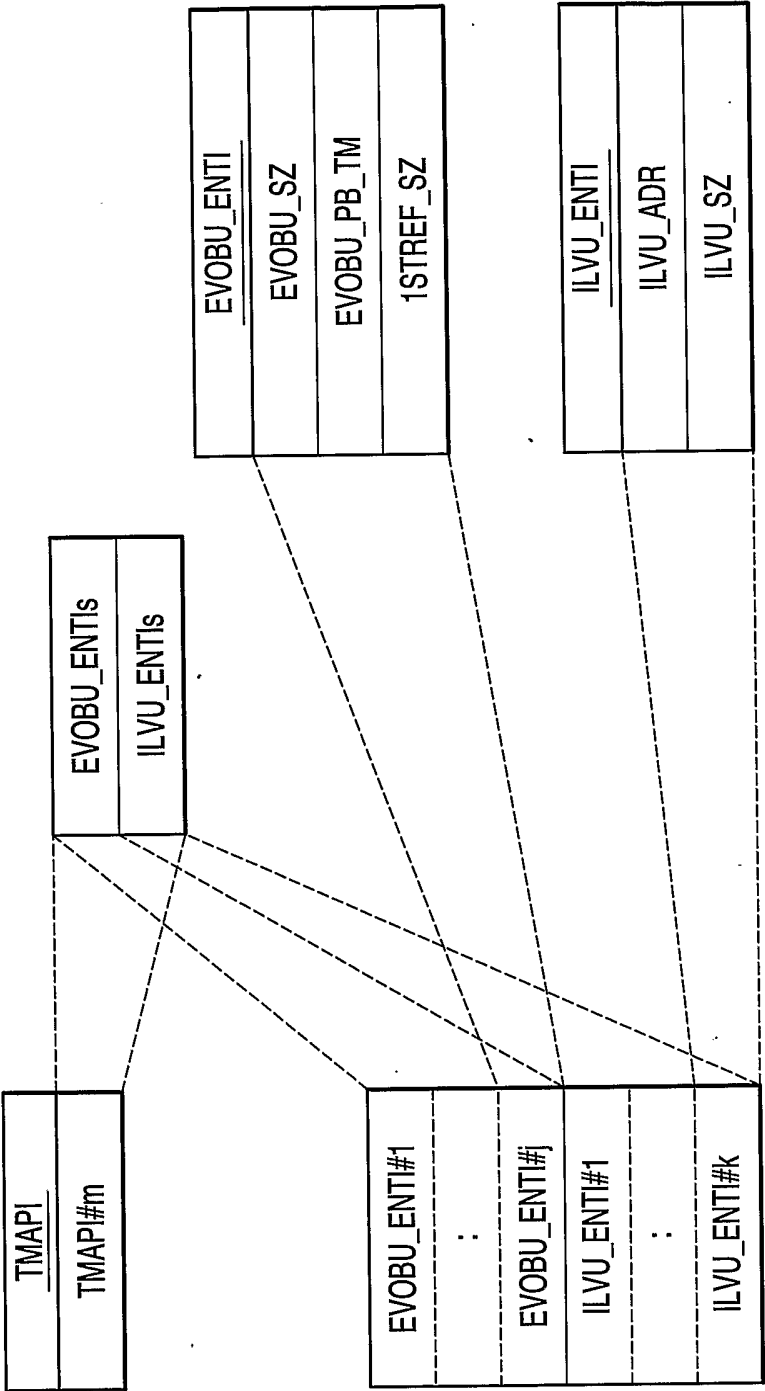


FIG.107

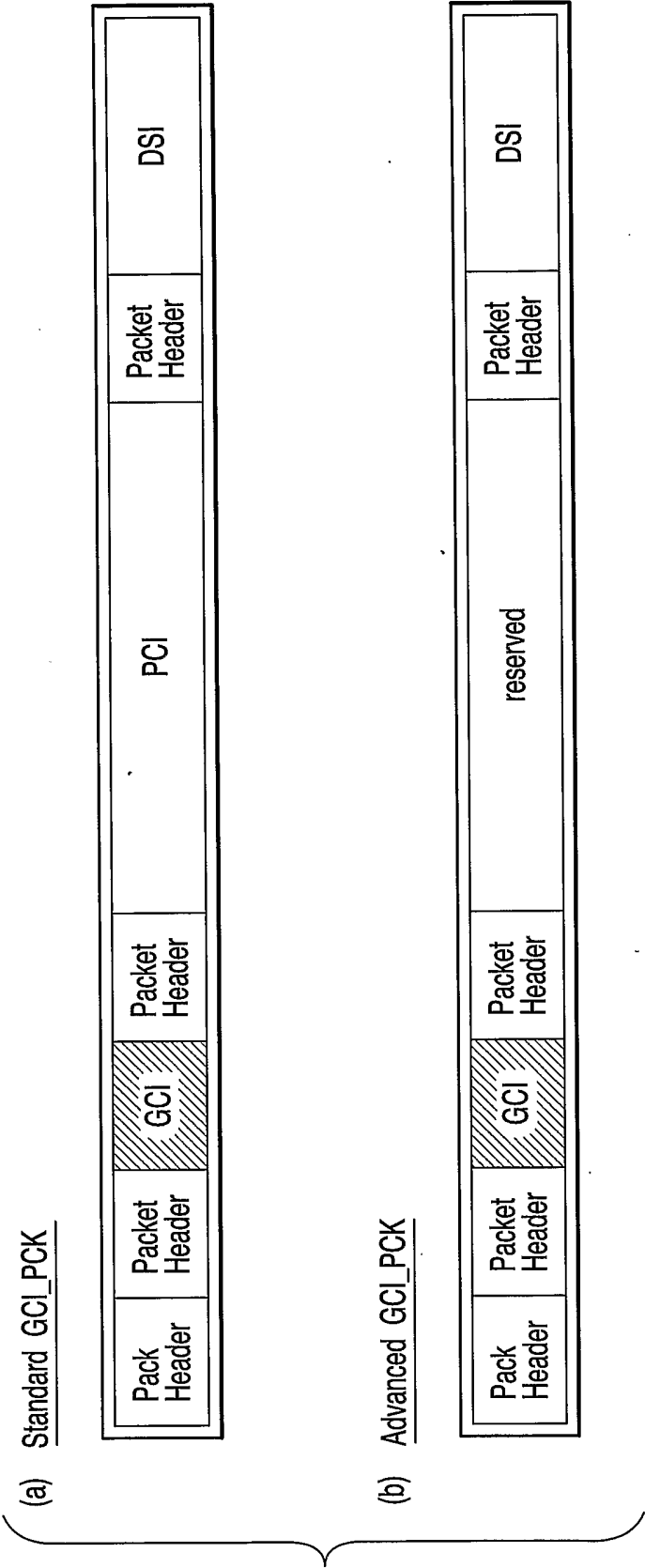


FIG. 108

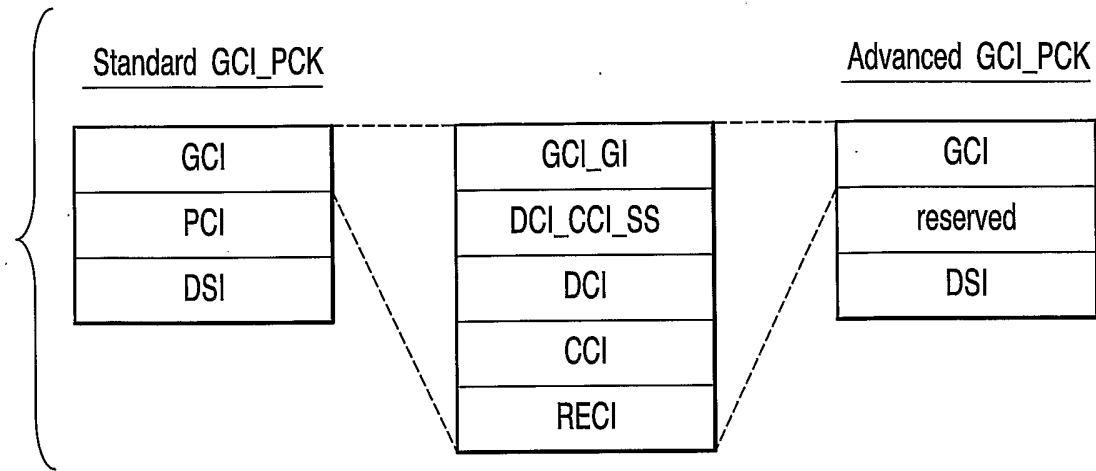


FIG. 109

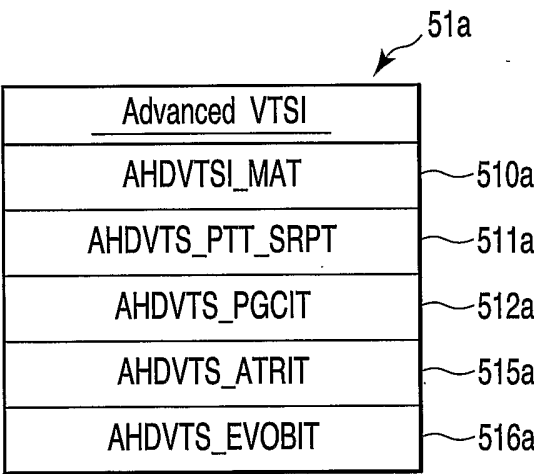


FIG. 110

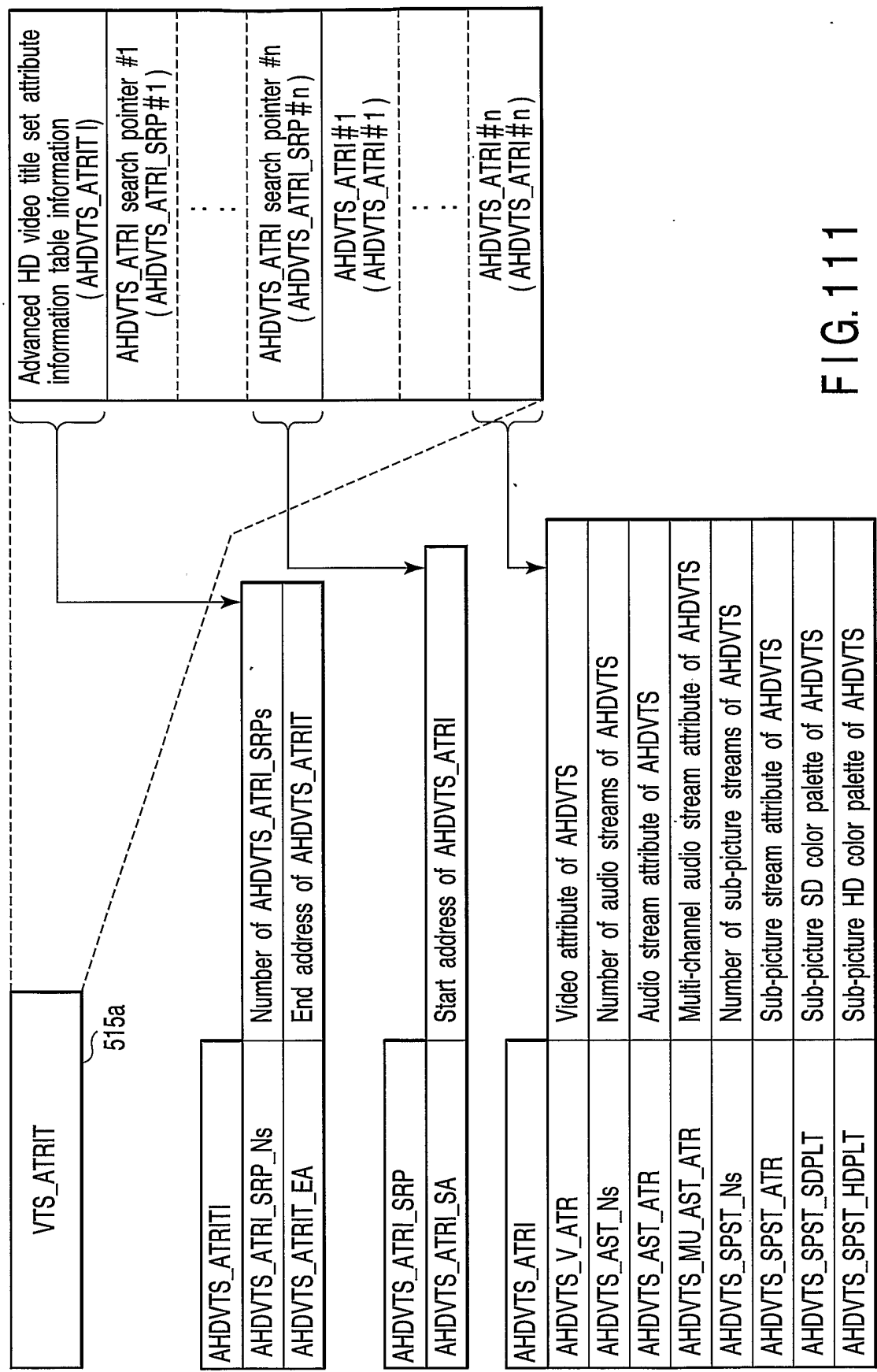


FIG. 111

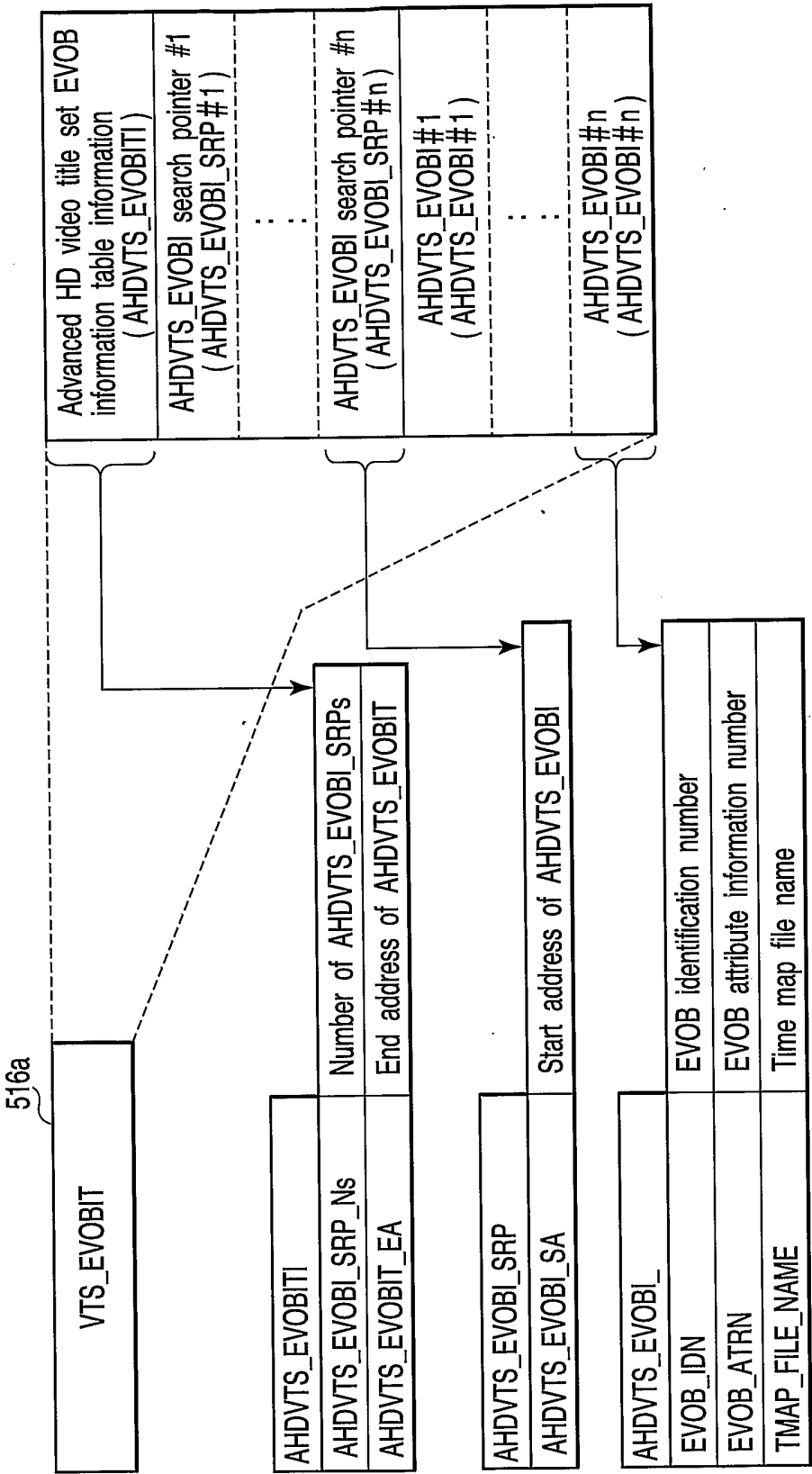


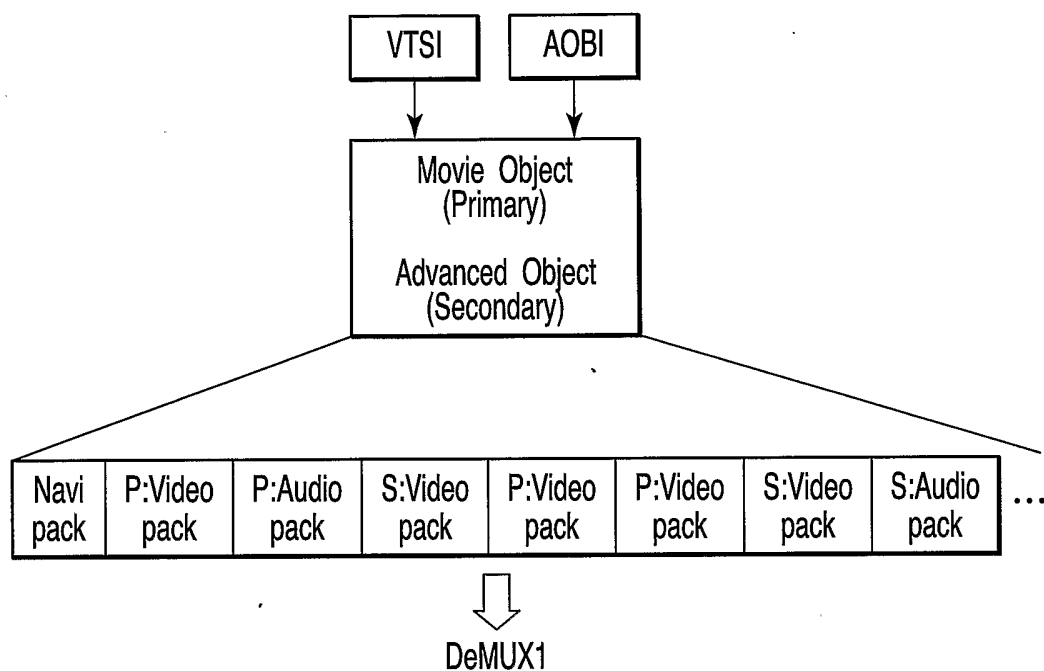
FIG.112

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Case1

(On Disc: one program stream (PS) multiplexed (MUX) with Movie)
(Web: Independent other PS)

(a) On Disc



(b) On Web

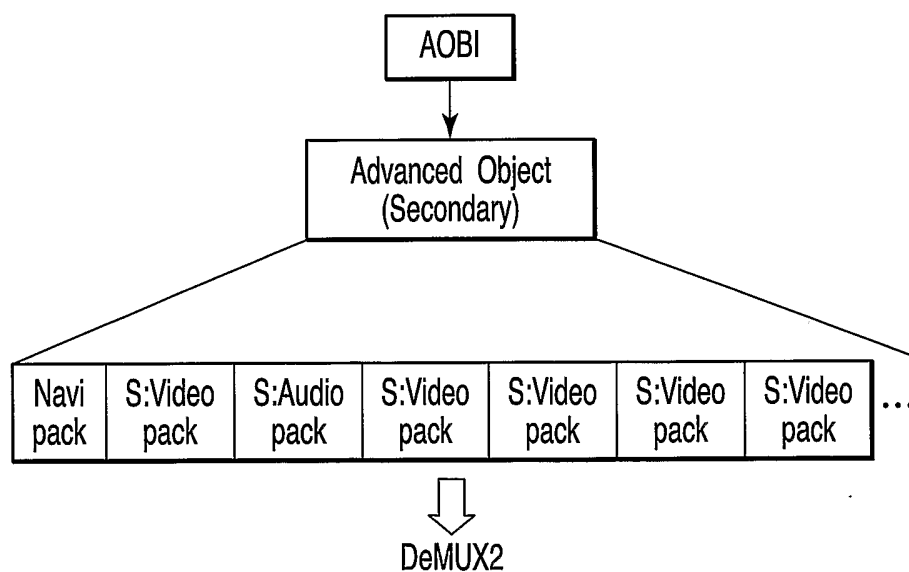
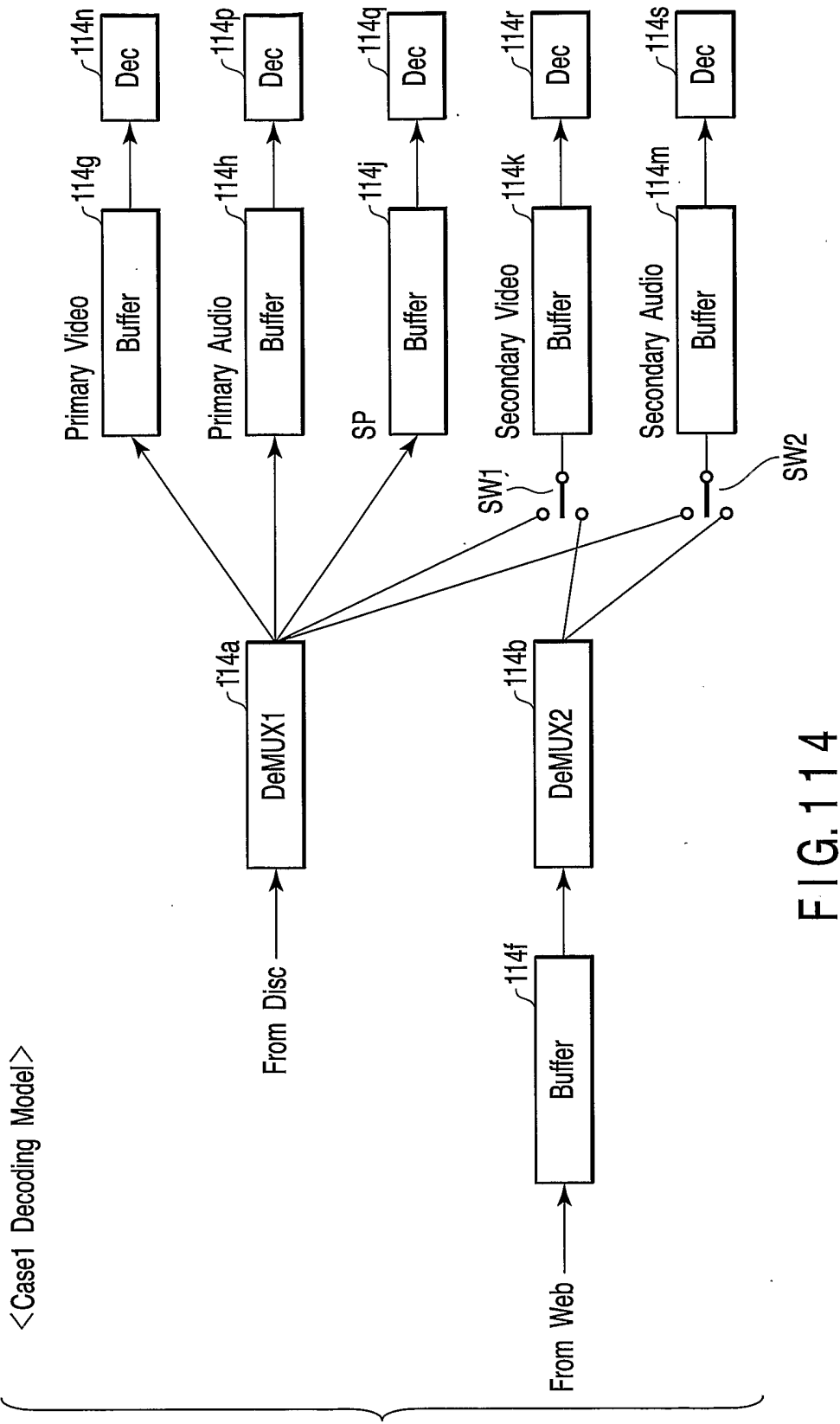


FIG. 113



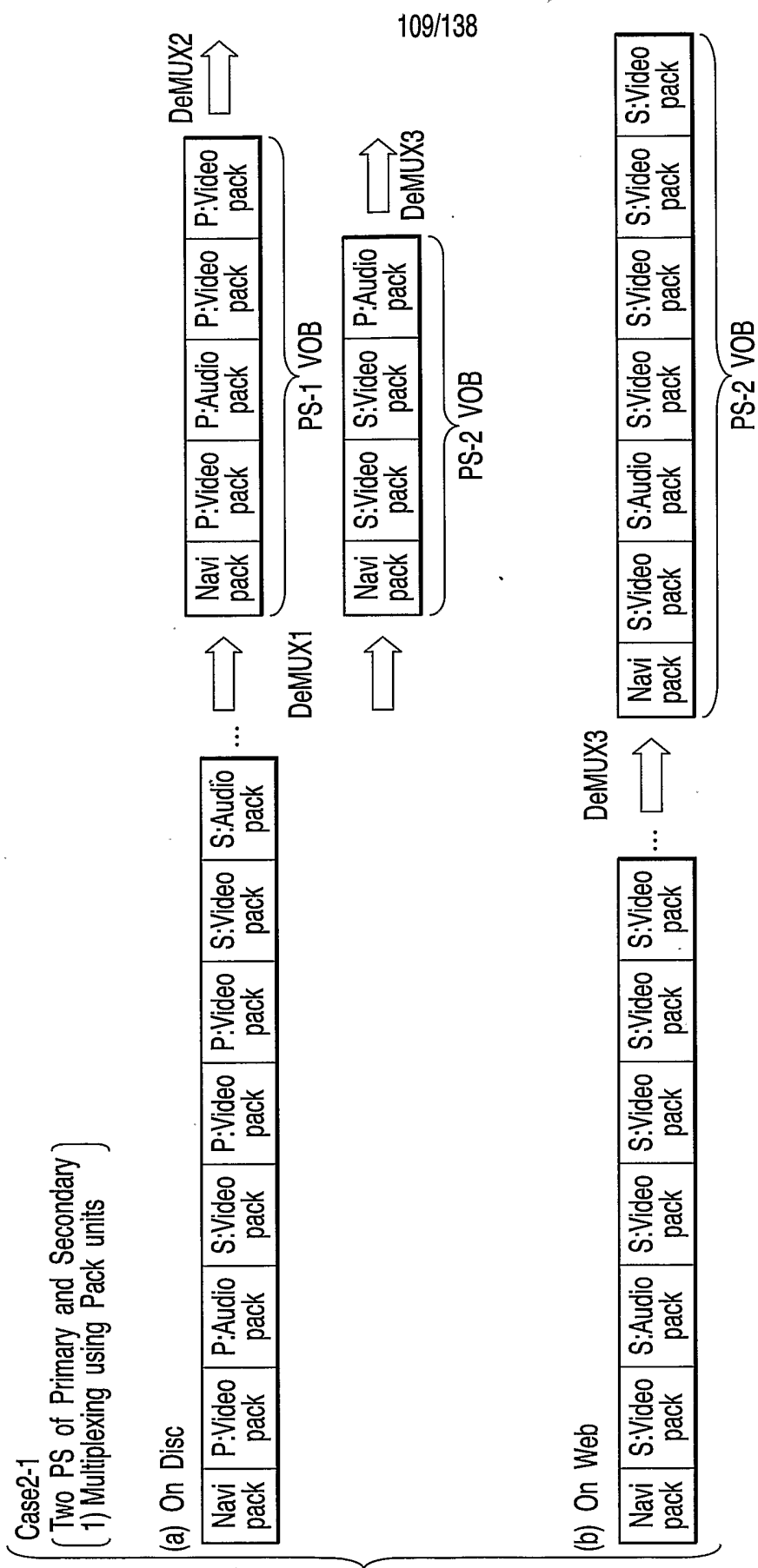


FIG. 115

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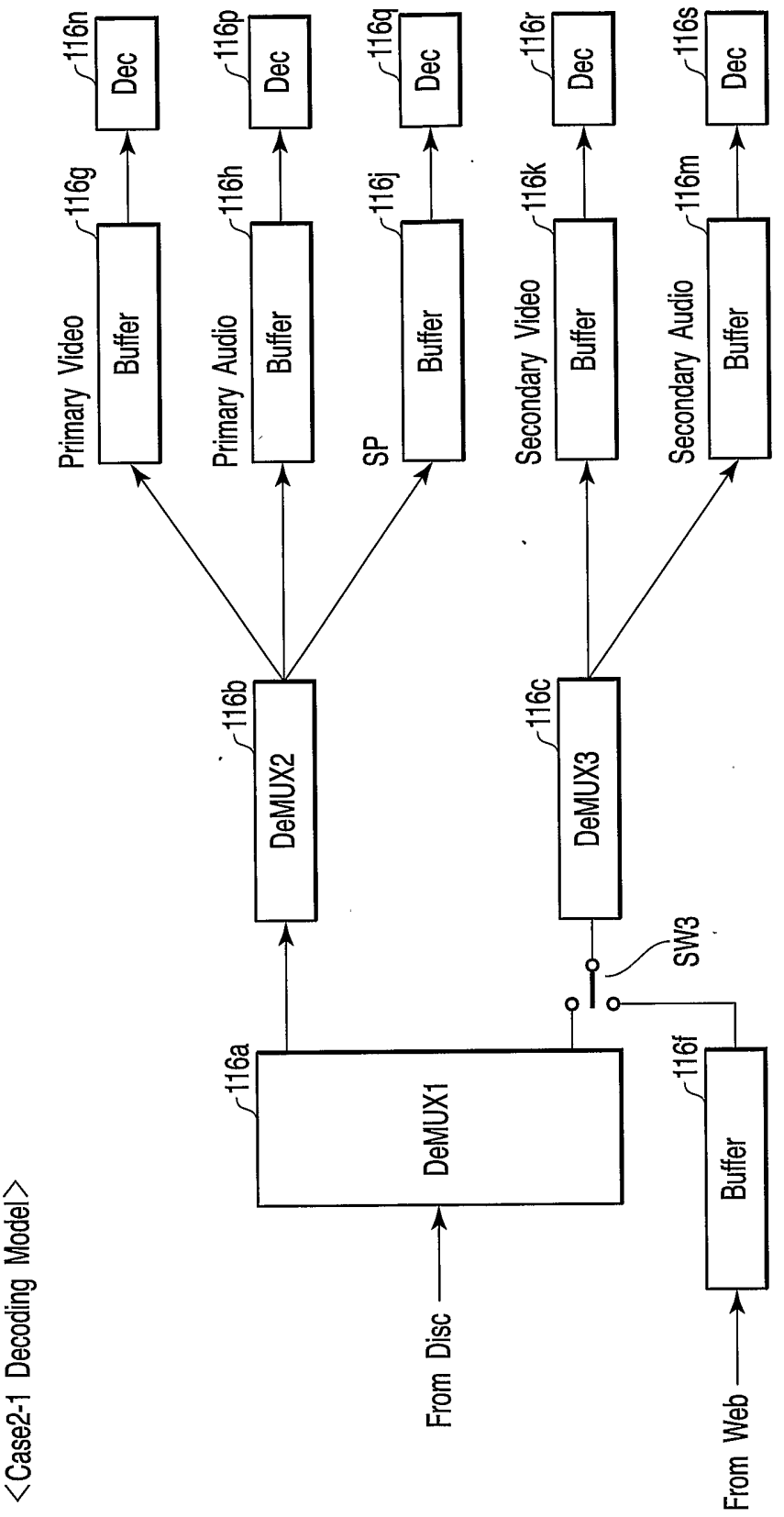


FIG. 116

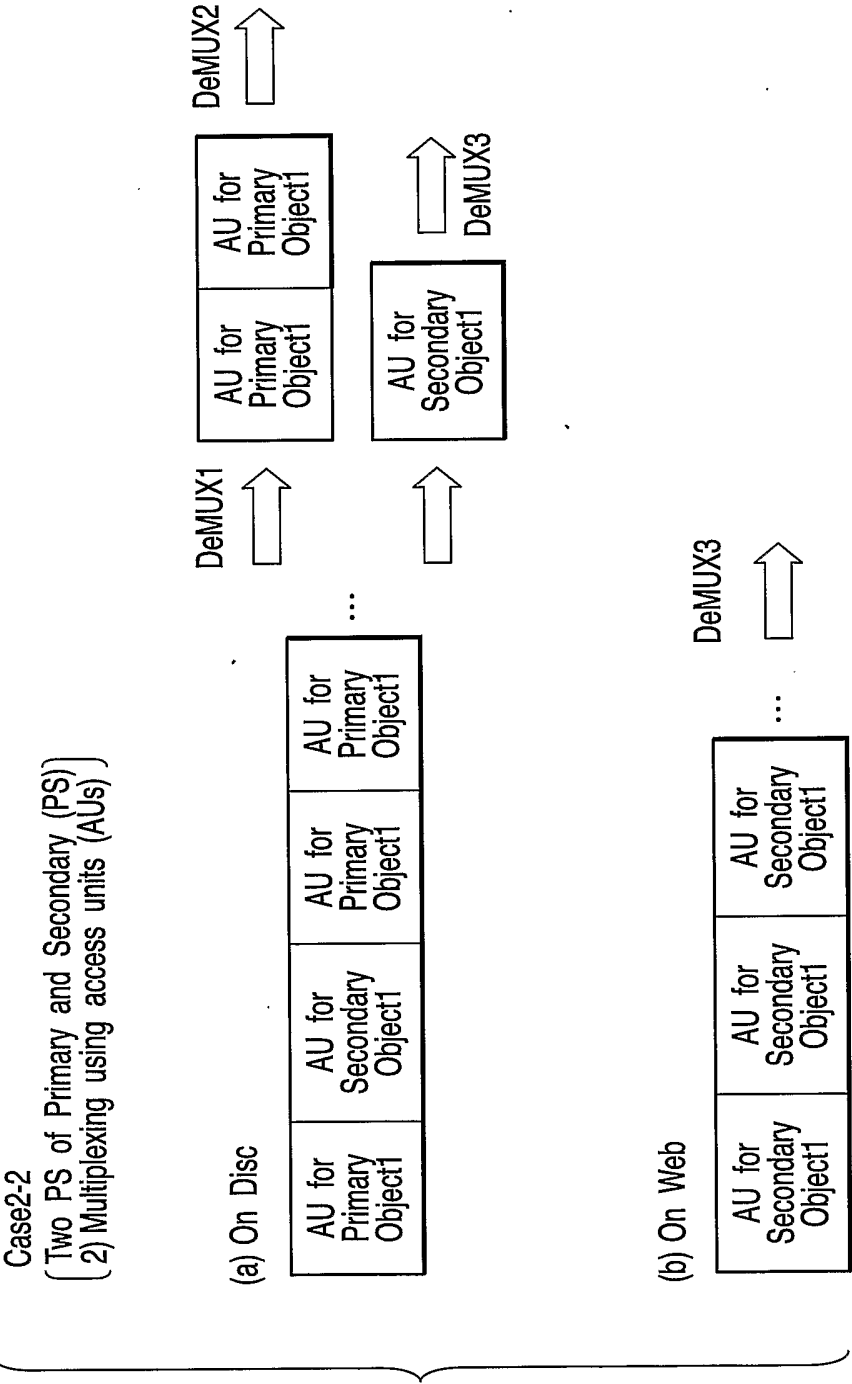


FIG.117

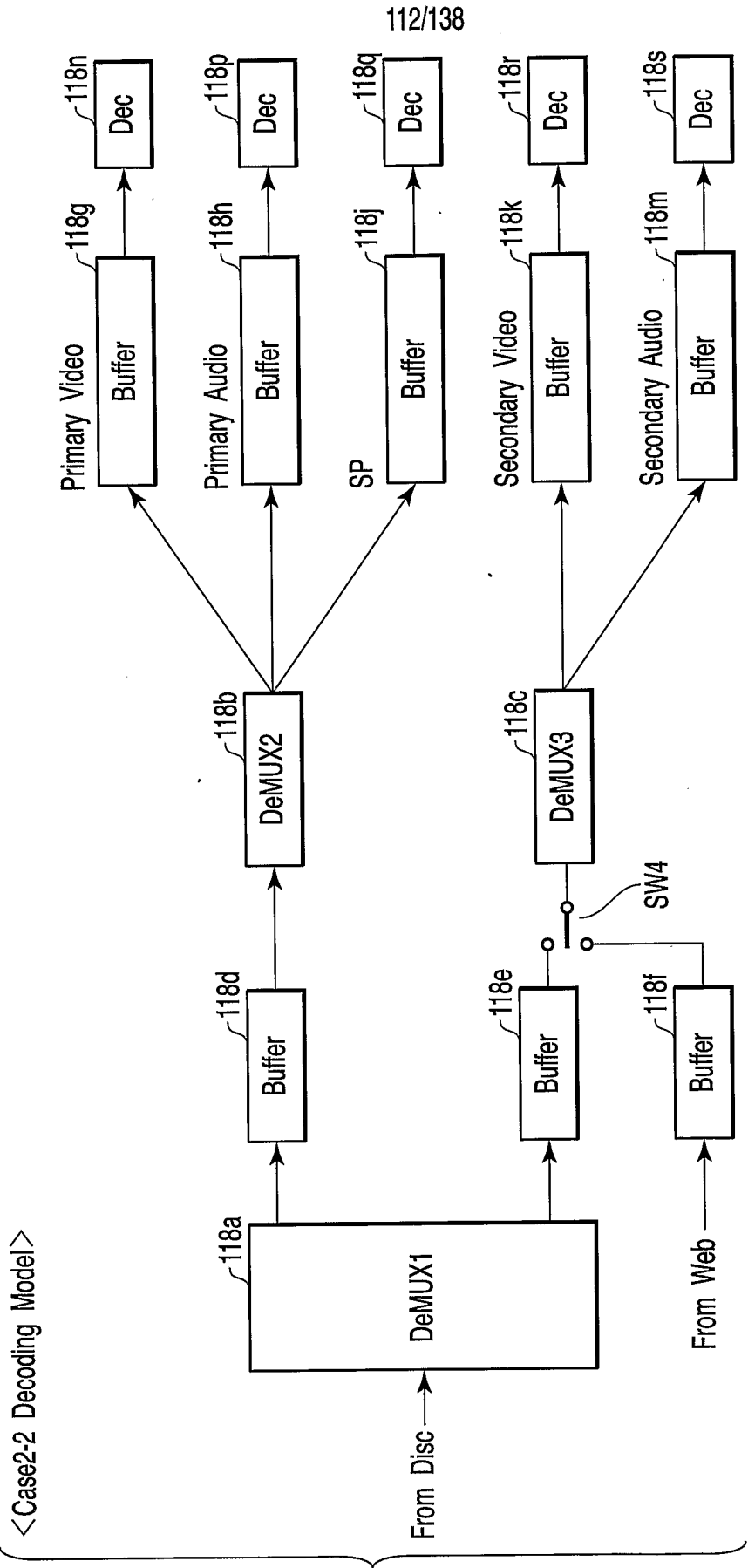


FIG.118

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stream_id for identifying Primary Content and Secondary Content (when using private_stream1)	
stream_id	Stream coding
110x0***b	MPEG audio stream*** = decoding audio stream number
1110 0000b	Video stream
1011 1101b	private_stream1
1011 1111b	private_stream2
Others	Unused

FIG. 119

sub_stream_id for private_stream1	
sub_stream_id	Stream coding
001* ****b	Sub-picture stream
0100 1000b	Reserved
011* ****b	Reserved (for expanded sub-picture)
1000 0***b	Dolby AC-3 audio stream
1000 1***b	DTS audio stream (option)
1001 0***b	SDDS audio stream
1010 0***b	Linear PCM audio stream
1111 1111b	Stream defined by provider
1001 0001b	Secondary Content MPEG2 video stream
1001 0010b	Secondary Content MPEG4 / AVC stream
1001 0011b	Secondary Content VC-1 stream
1100 0***b	Secondary Content Dolby digital+ stream
1100 1***b	Secondary Content DTS HD stream
1101 0***b	Secondary Content SDDS audio stream
1110 0***b	Secondary Content linear PCM audio stream
Others	Reserved (for future presentation data)

FIG. 120

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sub_stream_id for private stream2	
sub_stream_id	Stream coding
0000 0000b	PCI stream
0000 0001b	DSI stream
1111 1111b	Stream defined by provider
Others	Reserved (for future navigation data)

FIG. 121

stream_id for identifying Primary Content and Secondary Content (when using private_stream3)	
stream_id	Stream coding
110x0***b	MPEG audio stream*** = decoding audio stream number
1110 0000b	Video stream
1011 1101b	private_stream1
1011 1111b	private_stream2
1011 0000b	private_stream3
Others	Unused

FIG. 122

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sub_stream_id for private_stream1	
sub_stream_id	Stream coding
001* ****b	Sub-picture stream
0100 1000b	Reserved
011* ****b	Reserved (for expanded sub-picture)
1000 0***b	Dolby AC-3 audio stream
1000 1***b	DTS audio stream (option)
1001 0***b	SDDS audio stream
1010 0***b	Linear PCM audio stream
1111 1111b	Stream defined by provider
Others	Reserved (for future presentation data)

FIG. 123

sub_stream_id for private stream2	
sub_stream_id	Stream coding
0000 0000b	PCI stream
0000 0001b	DSI stream
1111 1111b	Stream defined by provider
Others	Reserved (for future navigation data)

FIG. 124

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sub_stream_id for private_stream3	
sub_stream_id	Stream coding
1001 0001b	Secondary Content MPEG2-video stream
1001 0010b	Secondary Content MPEG4 / AVC stream
1001 0011b	Secondary Content VC-1 stream
1100 0***b	Secondary Content Dolby digital+ stream
1100 1***b	Secondary Content DTSHD stream
1101 0***b	Secondary Content SDDS audio stream
1110 0***b	Secondary Content linear PCM audio stream
1111 1111b	Stream defined by provider
Others	Reserved (for future presentation data)

FIG. 125

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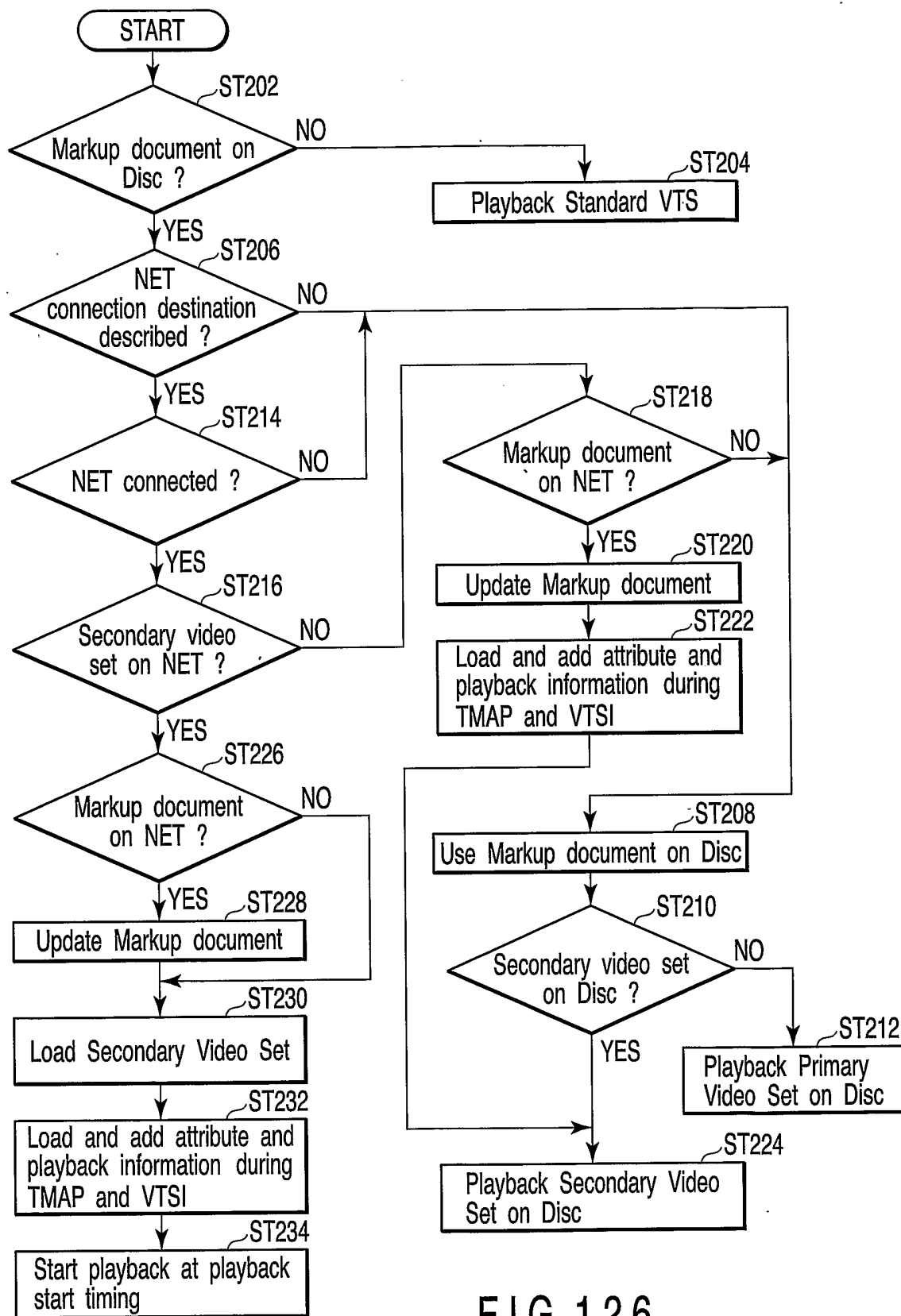


FIG. 126

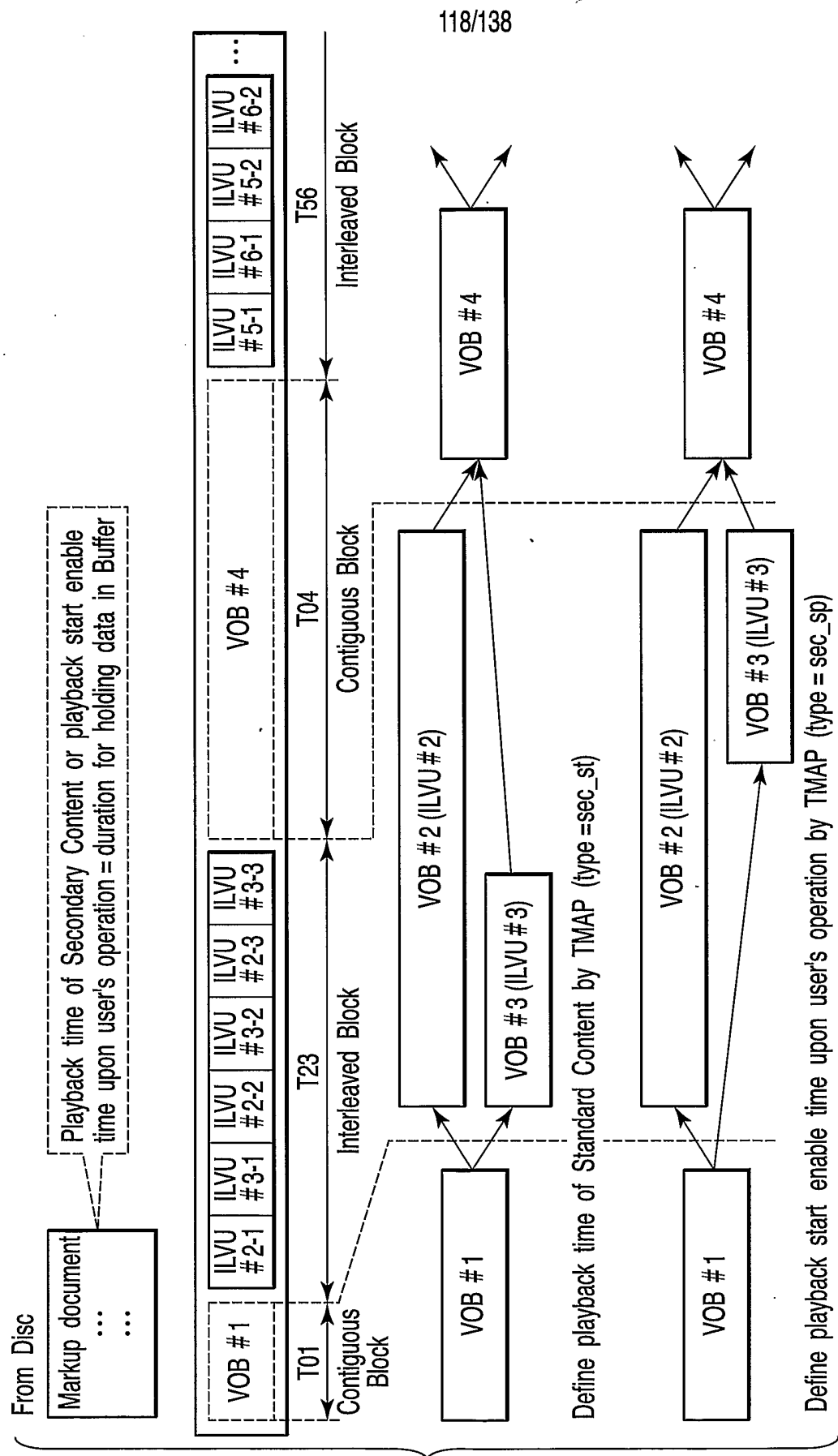


FIG.127

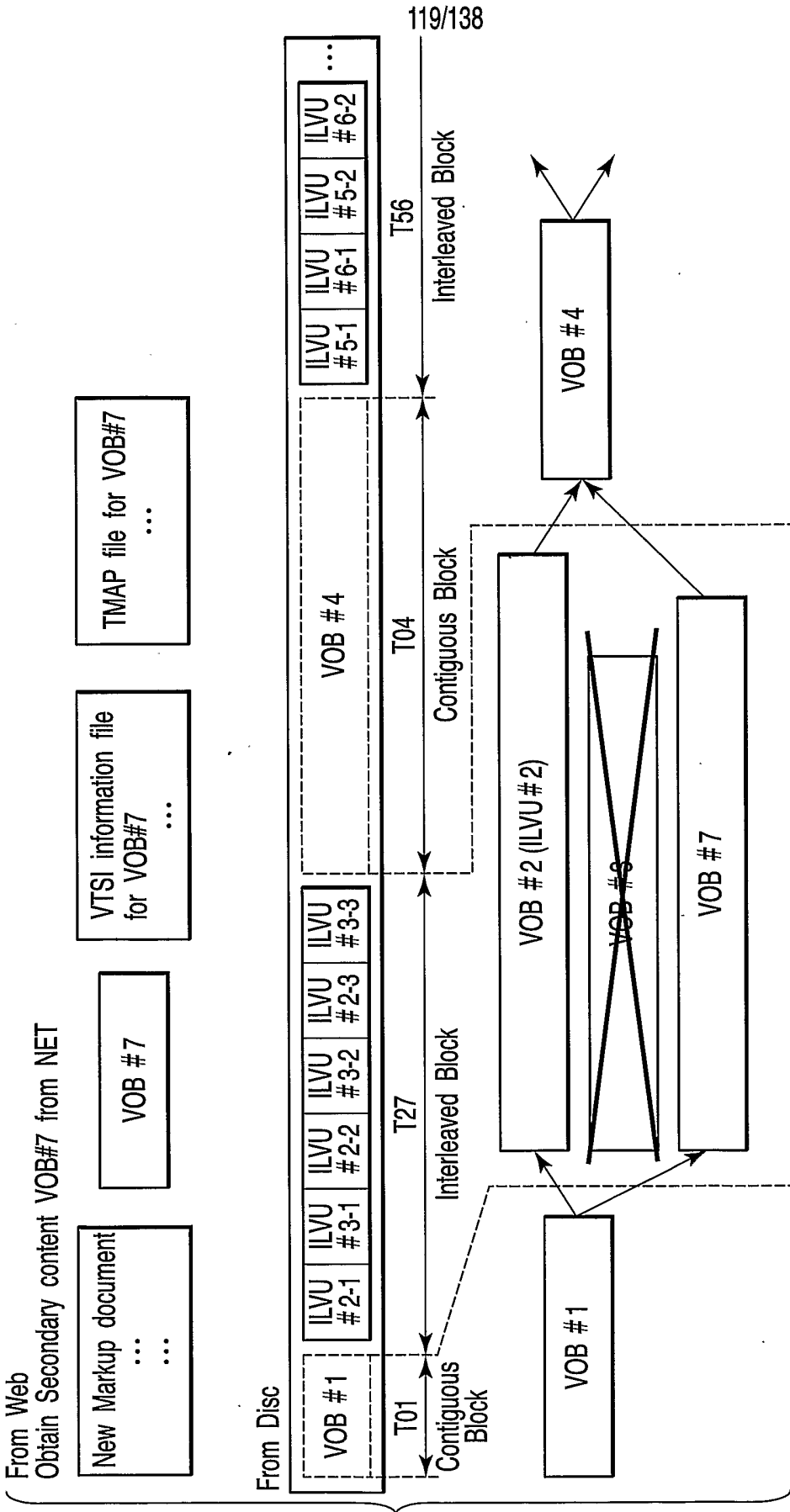


FIG.128

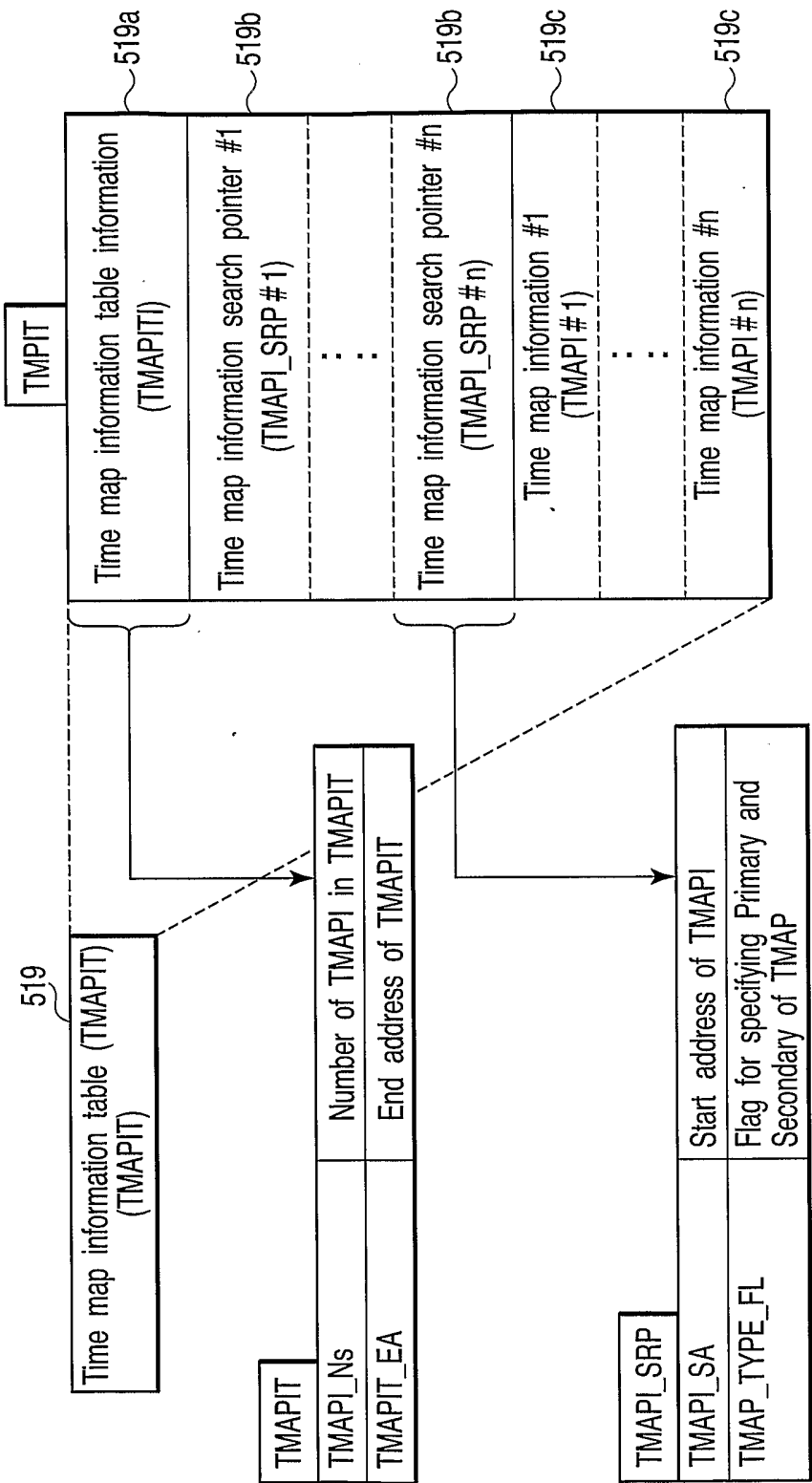


FIG.129

Markup description example 1
<pg num="1"> <server url="http://dvdrom/dvd_ihd"/> <object load="disc" data="main.mpg"> <type="pri"/> <start="00:00"/> <end="07:10"/> <evobid="1"/> </object> <object load="disc" data="sec.mpg"> <type="sec_st"/> <start="03:15"/> <end="05:40"/> <evobid="2"/> </object> </pg> • Use evobid for managing relationship with TMAP • Playback sec_st during playing back TMAP • Start playback sec_sp upon user's operation for playback time of TMAP • Replace when Markup document is obtained from Net • Determine by load attribute whether markup document is obtained from disc or Net

FIG.130

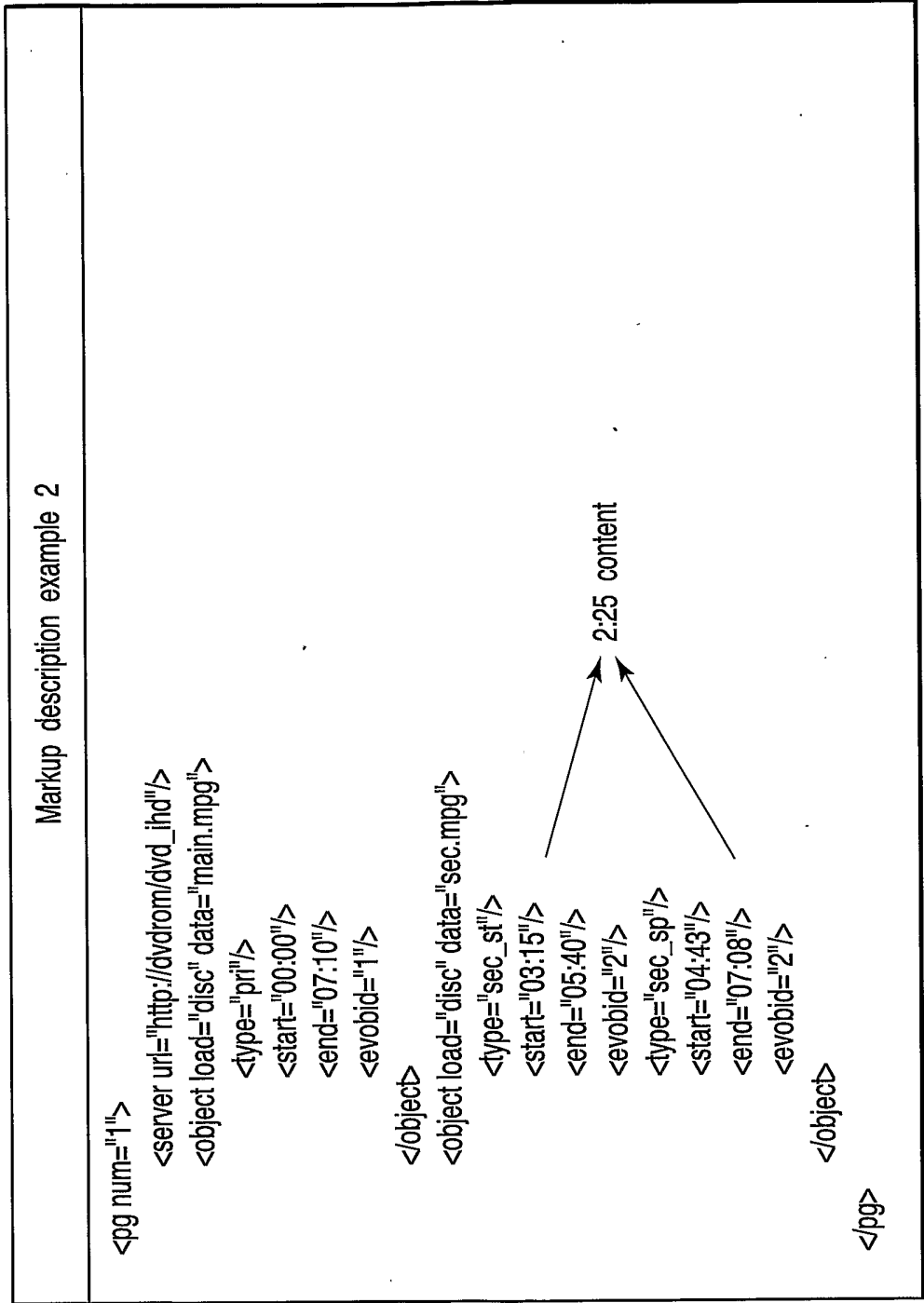


FIG.131

Markup description example 3
<pg num="1"> <server url="http://dvdrom/dvd_ihd"/> <object load="disc" data="main.mpg"> <type="pri"> <start="00:00"/> <end="07:10"/> <evobid="1"/> </object> <object load="net" data="sec2.mpg"/> <type="sec_st"> <start="2:55"/> <end="3:58"/> <evobid="3"/> </object> </pg> Example of obtaining and rewriting data from Net • id is unique since TMAP is rewritten to VTSI

FIG.132

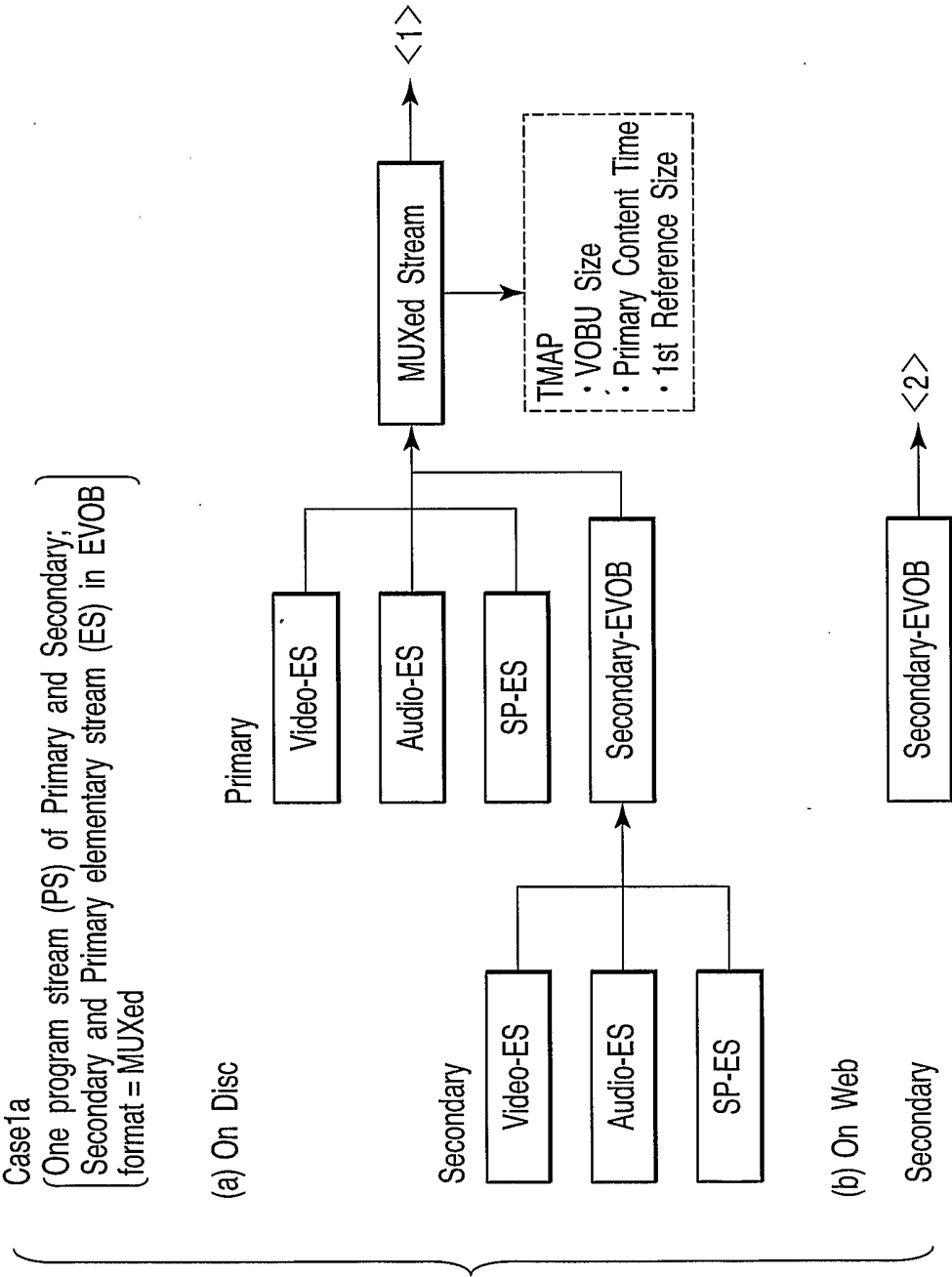


FIG. 133

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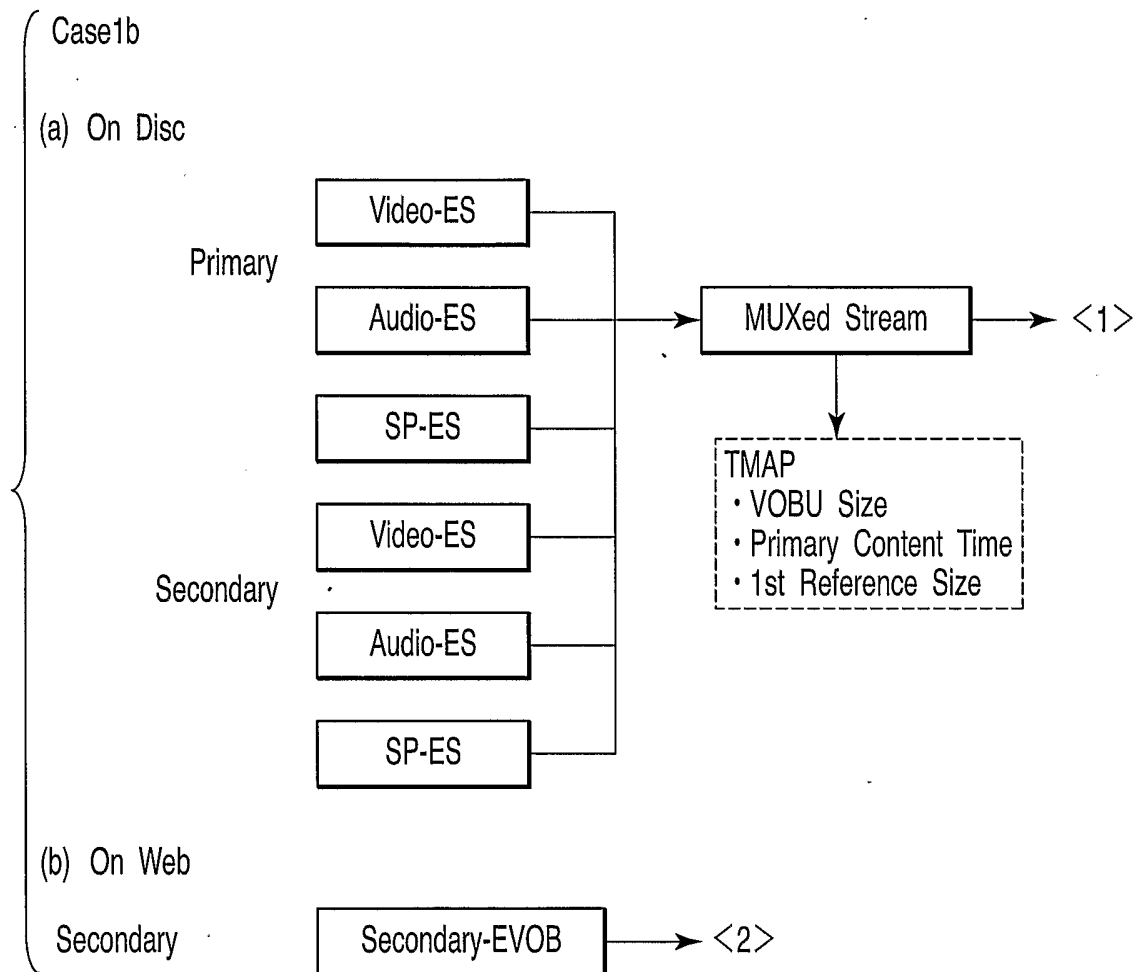


FIG. 134

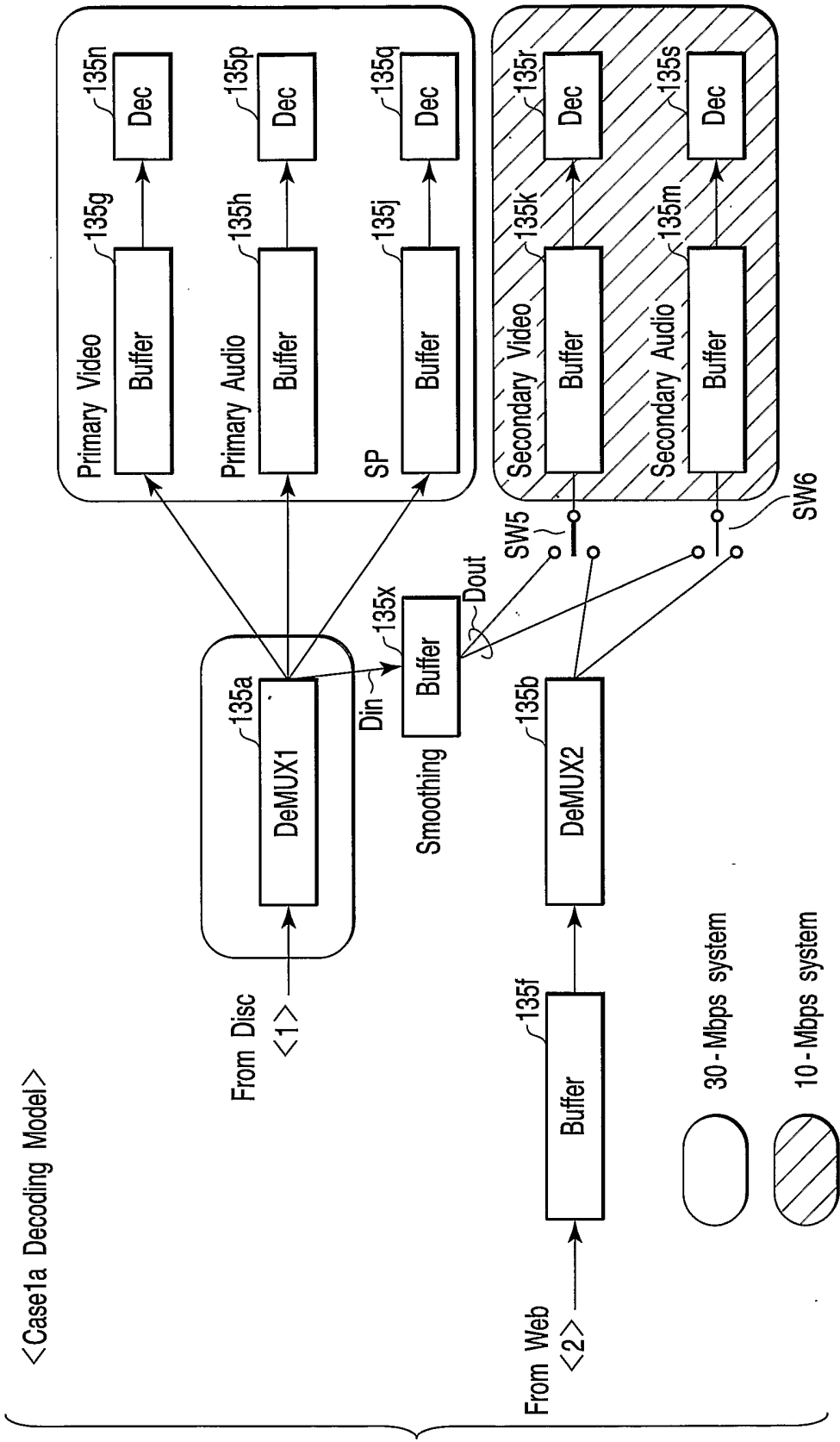


FIG. 135

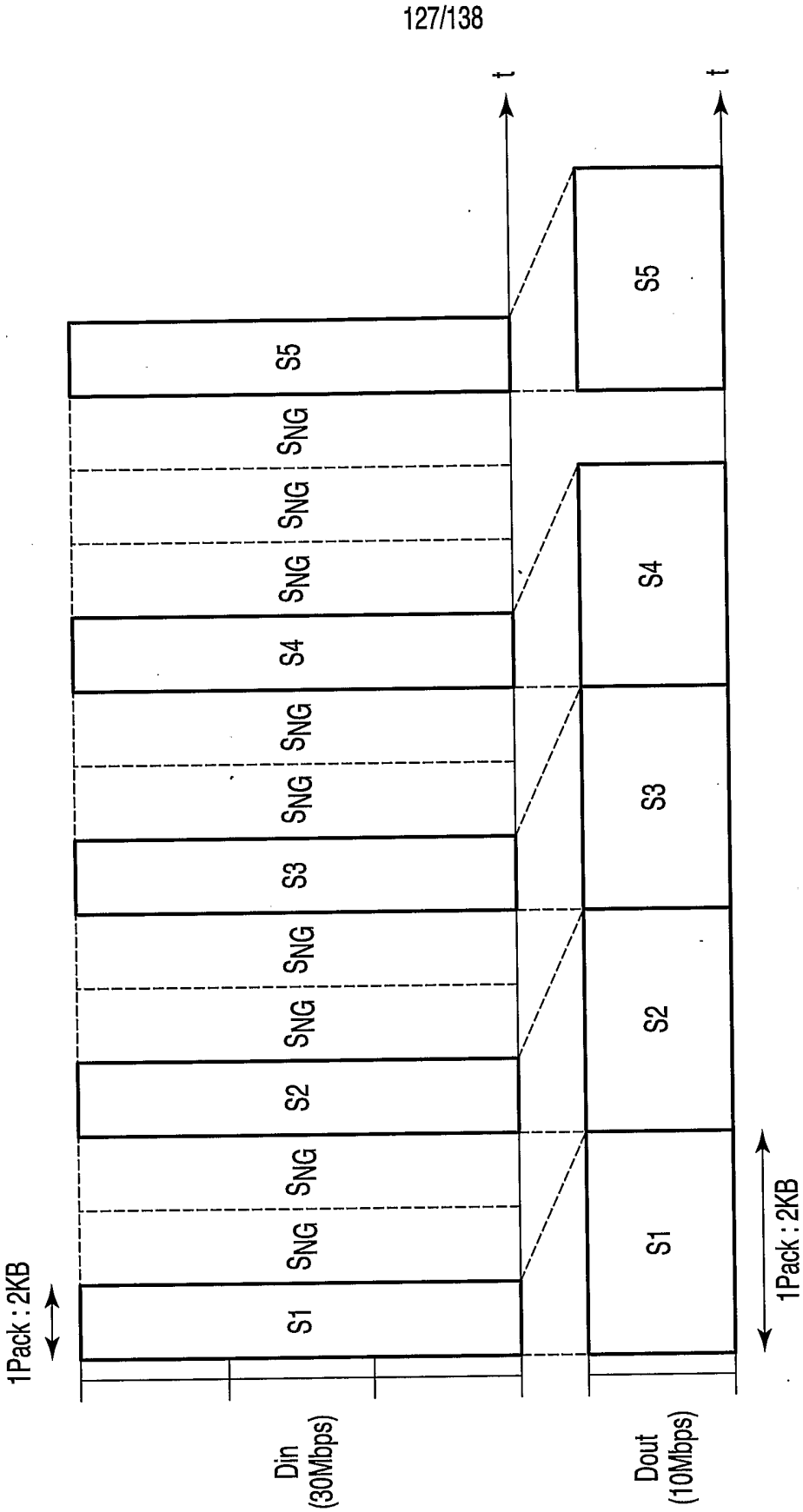


FIG. 136

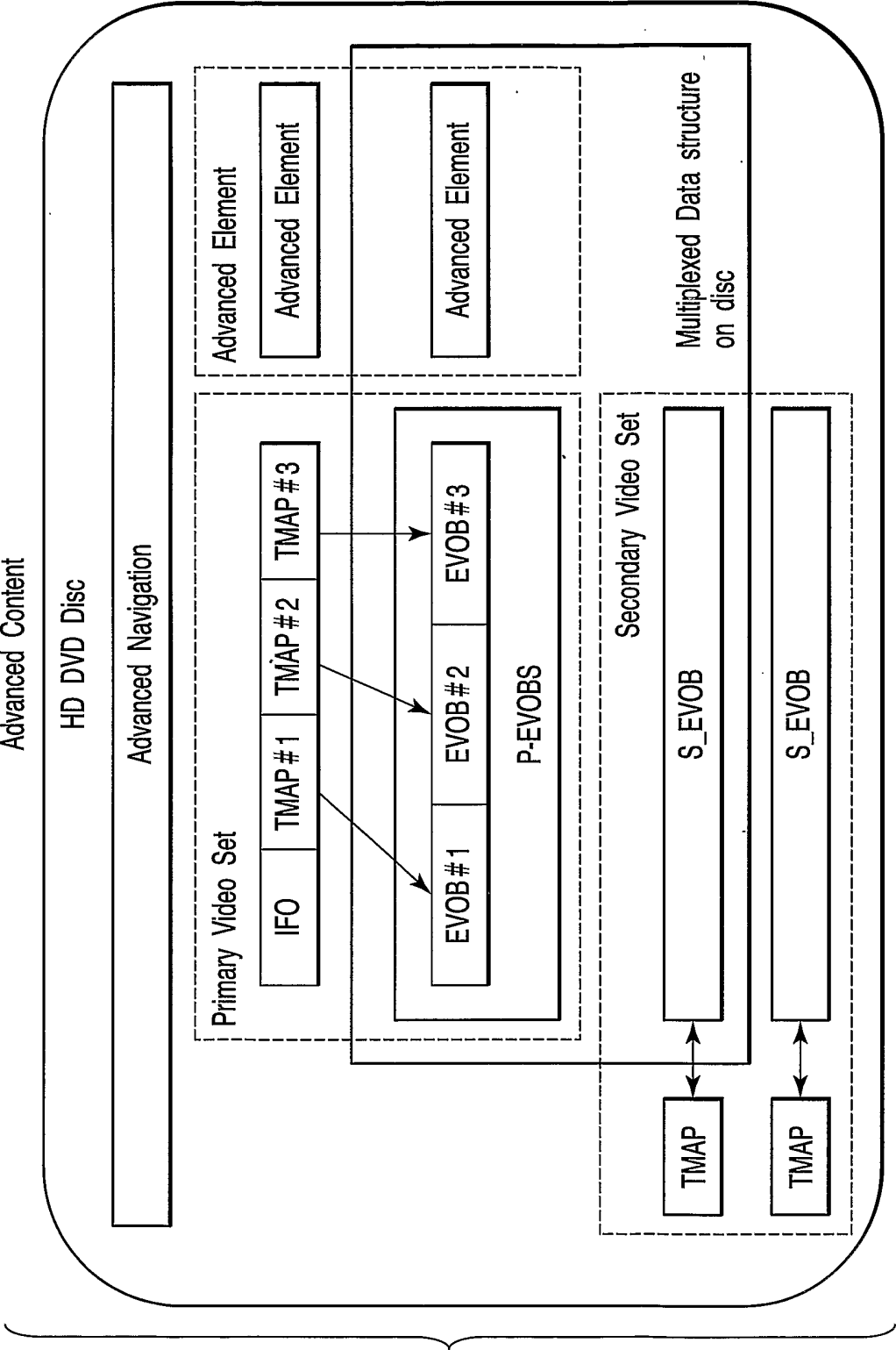


FIG. 137

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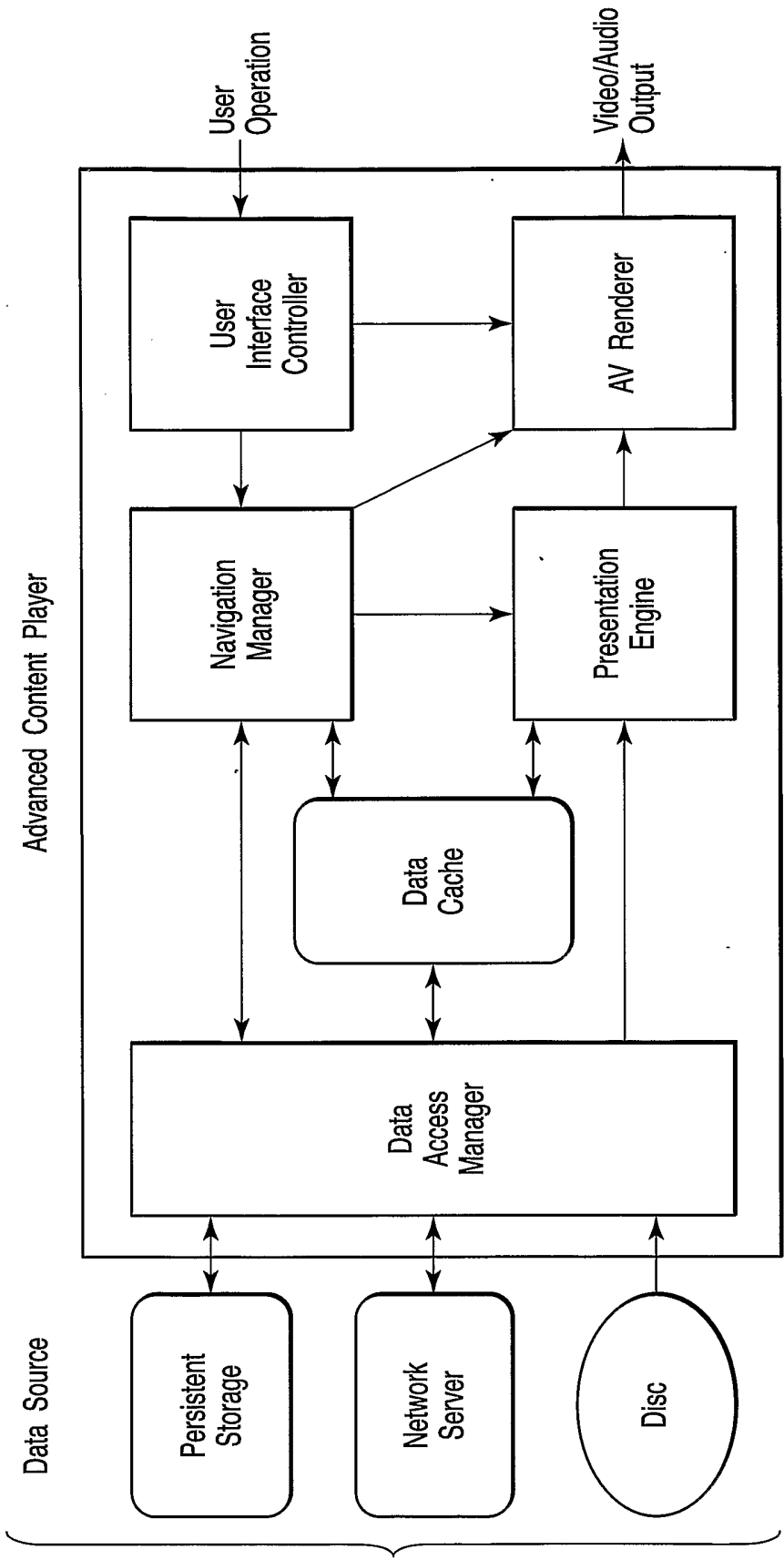
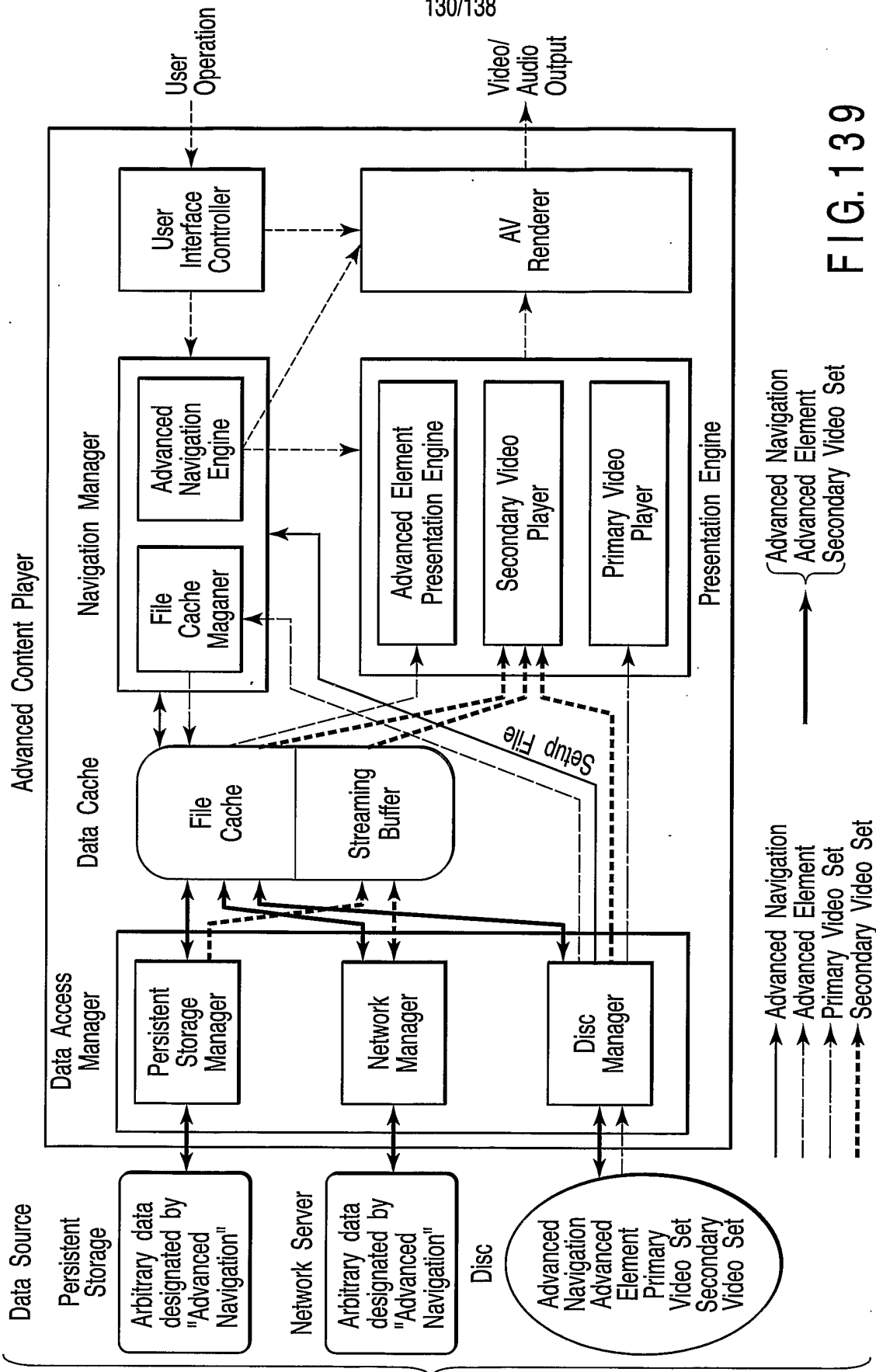


FIG. 138

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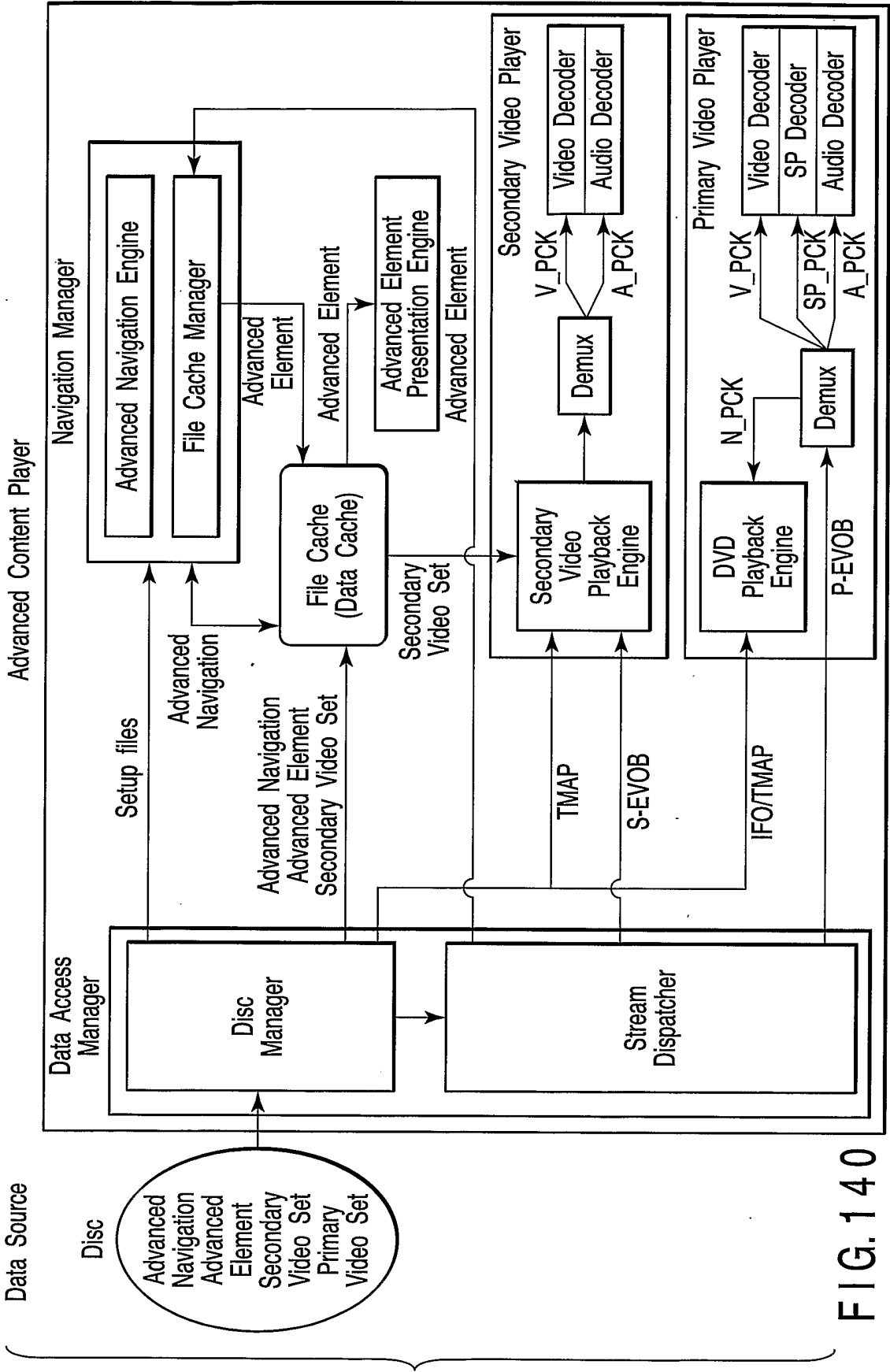


FIG. 140

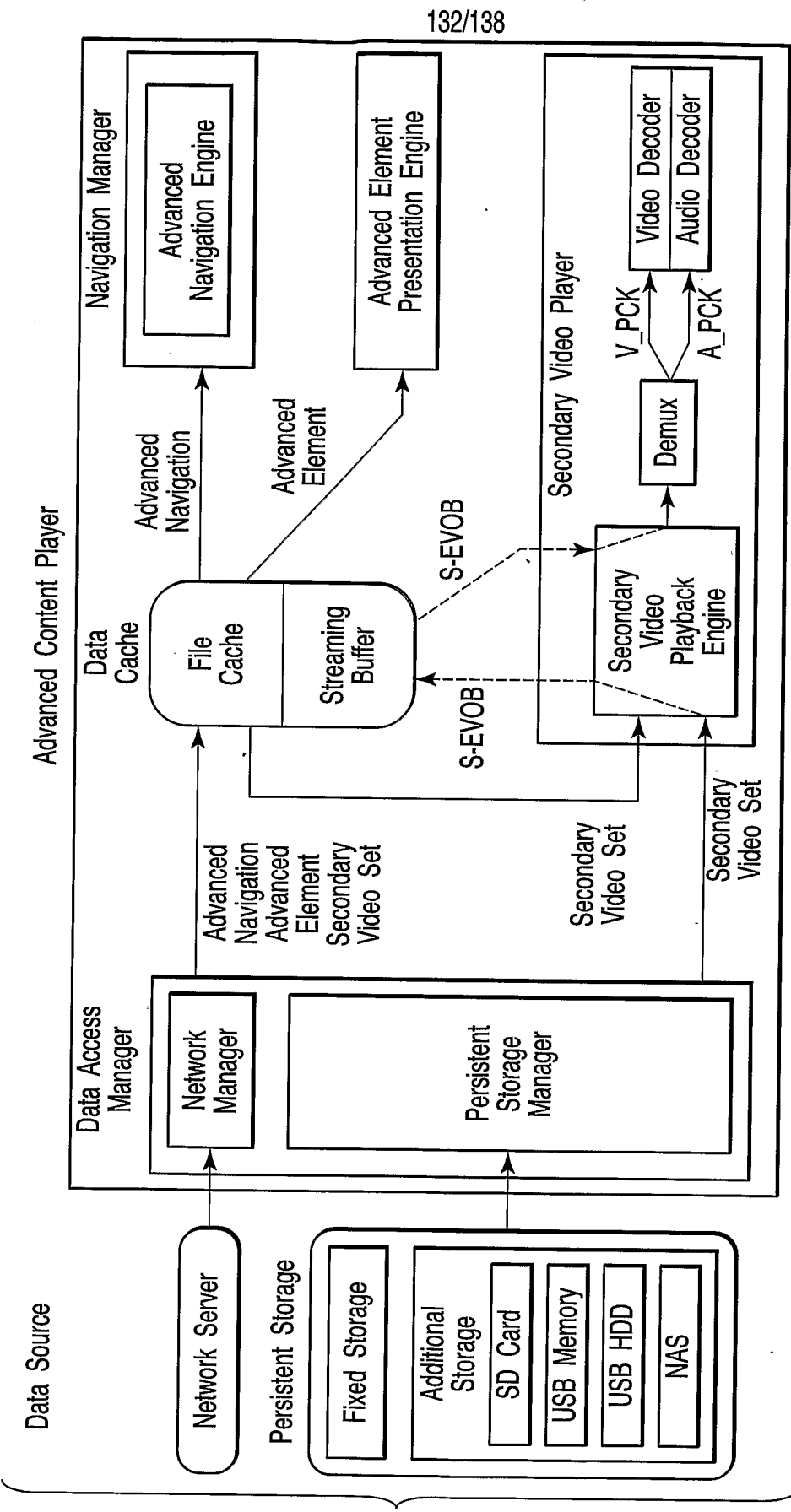


FIG. 141

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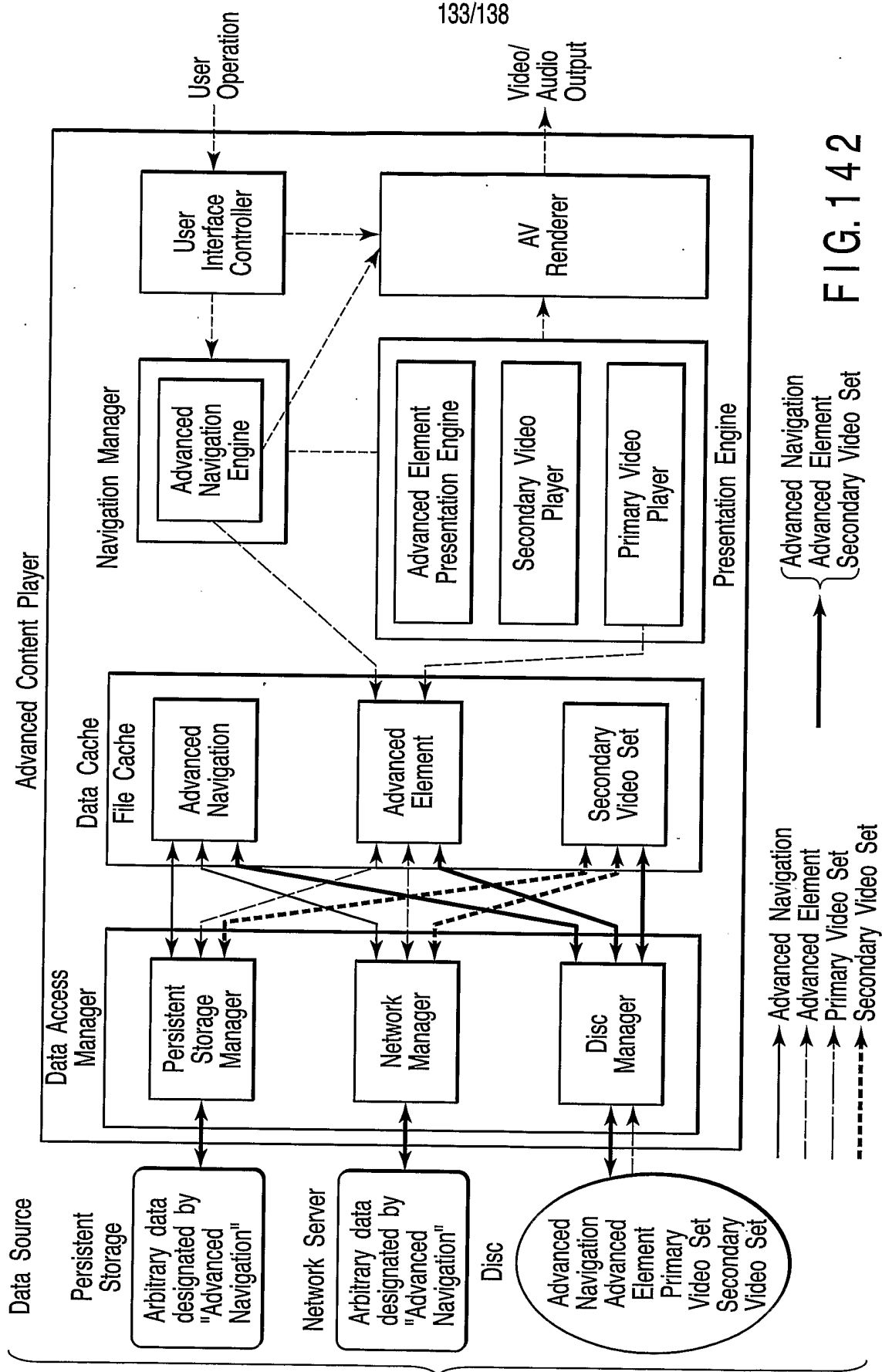


FIG. 142

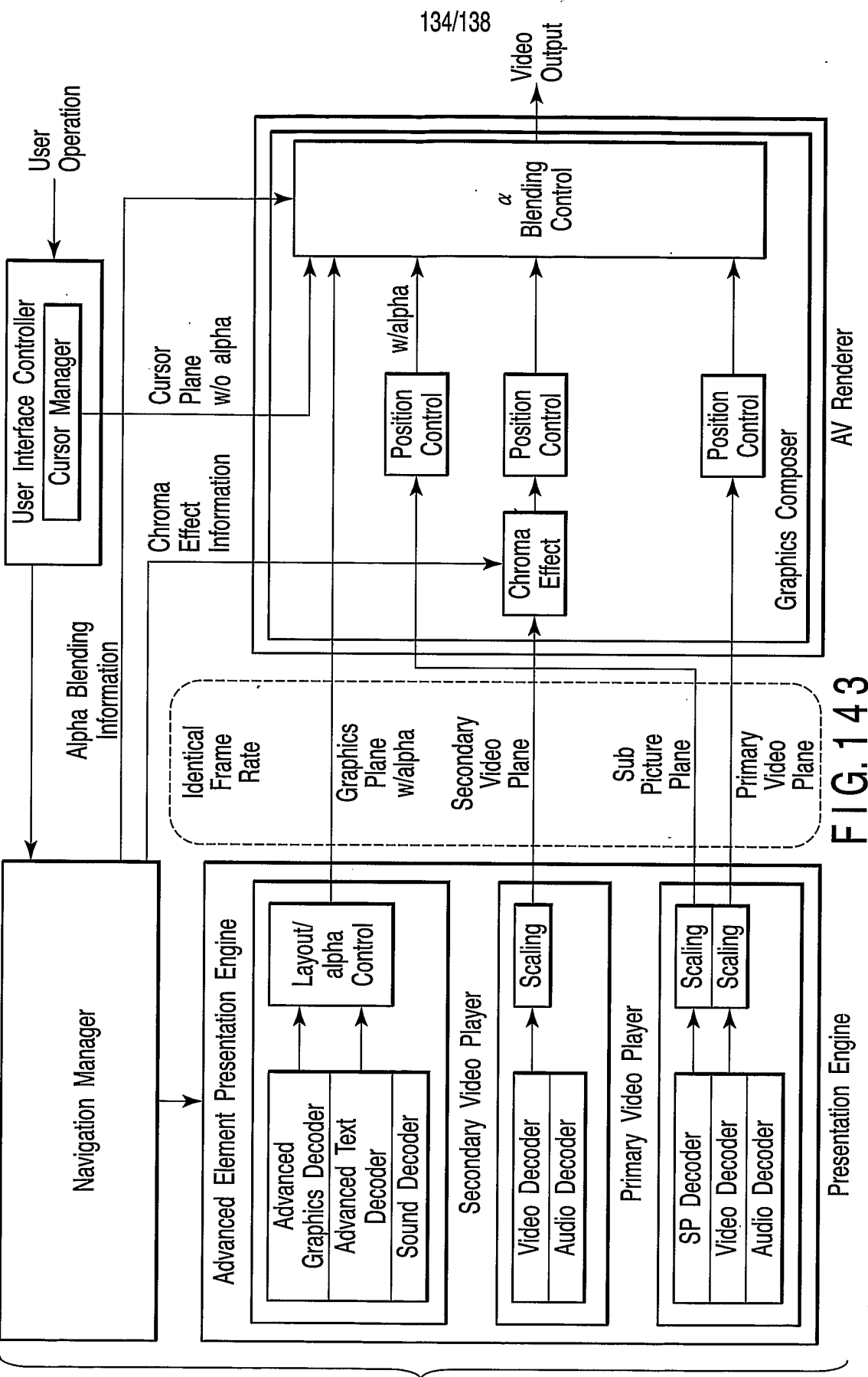


FIG. 143

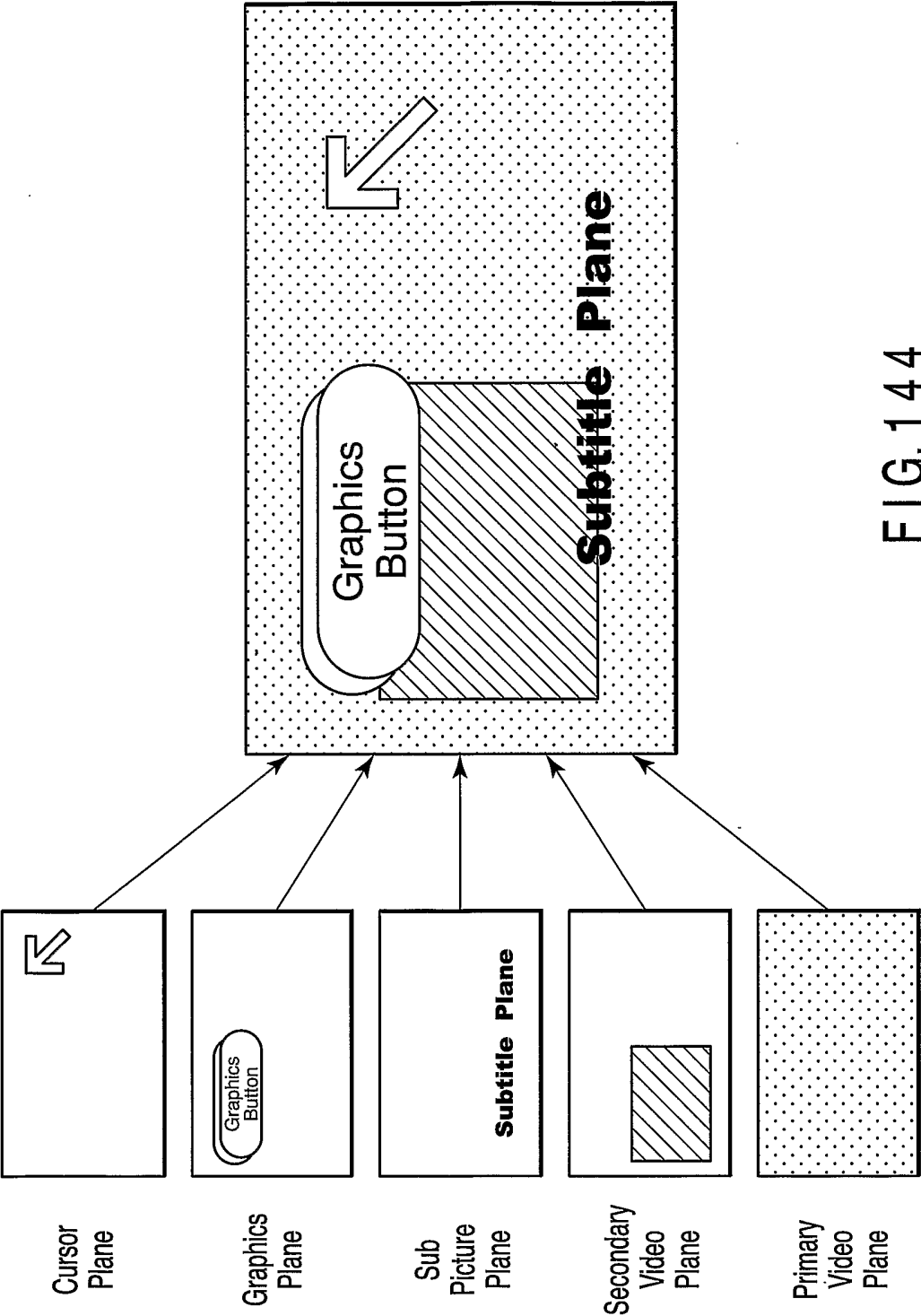


FIG. 144

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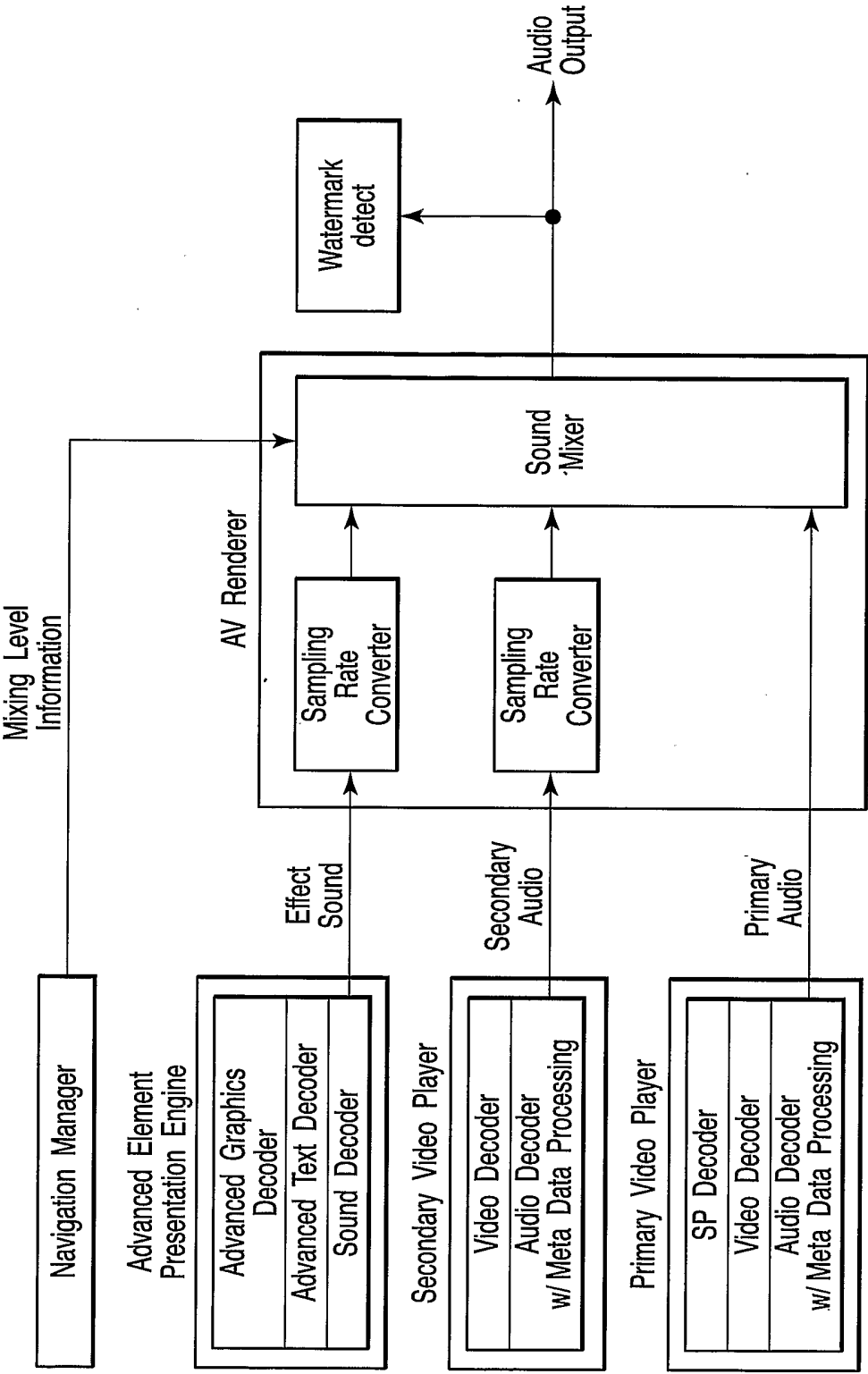


FIG. 145

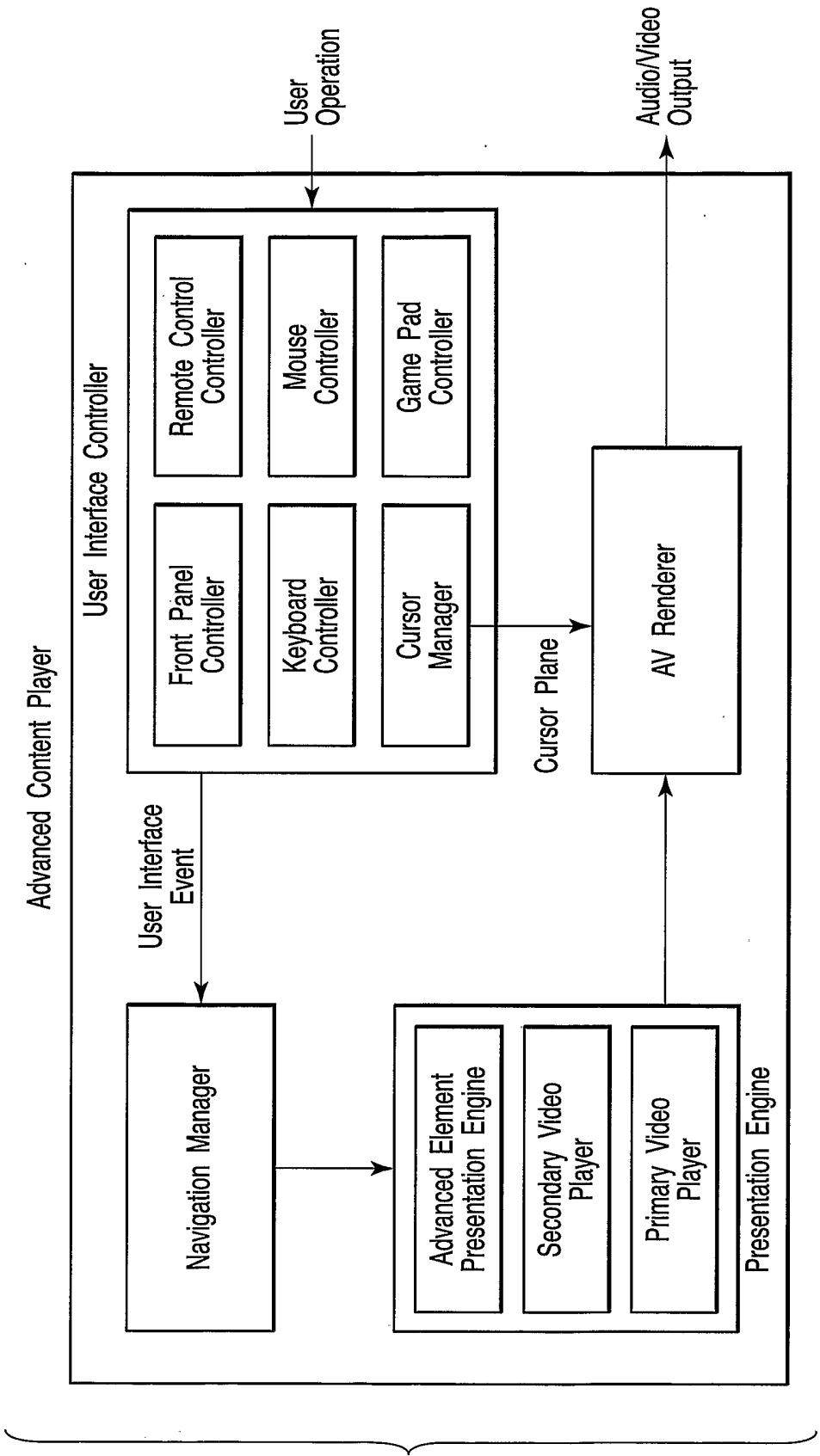


FIG. 146

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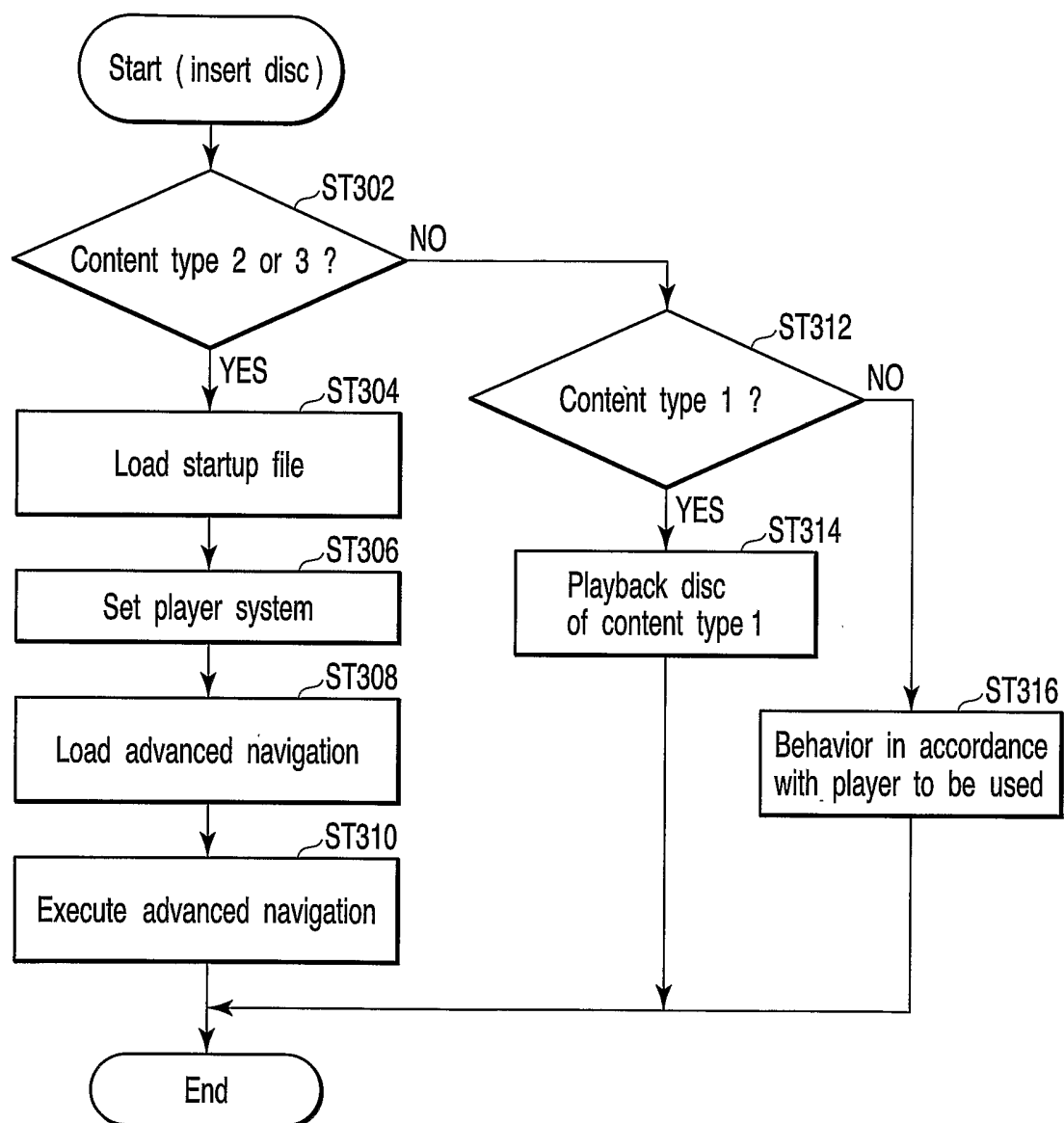


FIG. 147

INTERNATIONALSEARCHREPORT

International application No.

PCT/JP2005/024228

A. CLASSIFICATION OF SUBJECT MATTER

Int.Cl. *H04N5/91 (2006.01), G11B20/10 (2006.01), G11B20/12 (2006.01), G11B27/00 (2006.01), H04N5/85 (2006.01)*

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

Int.Cl. *H04N5/91 (2006.01), G11B20/10 (2006.01), G11B20/12 (2006.01), G11B27/00 (2006.01), H04N5/85 (2006.01)*

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Published examined utility model applications of Japan 1922-1996
 Published unexamined utility model applications of Japan 1971-2006
 Registered utility model specifications of Japan 1996-2006
 Published registered utility model applications of Japan 1994-2006

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	JP 2002-157862 A (Toshiba Corporation) 2002.05.31 Par.No. [0011] to [0026] ; Figs. 1 to 3 & US 2001/809 A1 & EP 1096489 A1 & WO 2000/062295 A1	1-7
A	JP 11-98467 A (Hitachi, Ltd.) 1999.04.09, Par.No. [0009] to [0012] , [0029] to [0035] (Family: none)	1-7

☒ Further documents are listed in the continuation of Box C.☐ See patent family annex.

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"A" document defining the general state of the art which is not considered to be of particular relevance

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"O" document referring to an oral disclosure, use, exhibition or other means

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"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

06.04.2006

Date of mailing of the international search report

25.04.2006

Name and mailing address of the ISA/JP

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INTERNATIONALSEARCHREPORT

International application No.

PCT/JP2005/024228

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
E, A	JP 2006-33067 A (Sony Corporation) 2006.02.02 Full Text; Figs. 1 to 34 (Family: none)	1-7